PRECISION CIRCUITS

CHAPTER 5

In the preceding chapters we dealt with many aspects of analog circuit design, including the circuit properties of passive devices, transistors, FETs, and op-amps, the subject of feedback, and numerous applications of these devices and circuit methods. In all our discussions, however, we have not yet addressed the question of the best that can be done, for example, in minimizing amplifier errors (nonlinearities, drifts, etc.) and in amplifying weak signals with minimum degradation by amplifier "noise." In many applications these are the most important issues, and they form an important part of the art of electronics. Therefore, in this chapter, we look at methods of precision circuit design (deferring the issue of noise in amplifiers to Chapter 8).

Chapter overview

This is a *big* chapter – and an important one. It deals with a range of topics, which need not be read in order. As guidance, we offer this outline: we start with a careful examination of errors in circuits made with operational amplifiers, and explore the use of an error budget. We explore issues of unspecified parameters and "typical" versus "worst-case" component errors, and discuss ways to deal with them. Along the way we deal with some neglected topics such as diode leakage at the sub-picoamp level, "memory effect" in capacitors, distortion and gain nonlinearity, and an elegant way to remedy *phase* error in amplifiers. We discuss op-amp distortion in detail, with comparative graphs and test circuits.

Next we discuss the dark side of rail-to-rail op-amps: their open-loop output impedance, and input common-mode crossover errors. We provide detailed selection tables for precision, chopper and high-speed op-amps, and comparative graphs charting their noise, bias current, and distortion. We show how to interpret the multitude of op-amp parameters, and we discuss the tradeoffs you'll have to make.

For those working in the low microvolt and nanovolt territory, we show the devastating effects of $1/f$ noise, and how auto-zero (AZ) op-amps solve this problem; but there's a tradeoff – the current noise of these devices that is often overlooked. As an interlude we look in some detail at the cleverness of the front-end stage of an exemplary precision digital multimeter.

Then we advance to difference and instrumentation amplifiers – these go to the head of the class both in terms of digging out a difference signal in the presence of common-mode input, and in terms of gain accuracy and stability. We show their internal designs and how they're used, with extensive tables and graphs comparing popular parts. Finally we take up fully differential amplifiers – these have differential inputs *and* outputs, and an output "common-mode control" input pin. Once again, we organize tables, internal circuit diagrams, and guidance for their use with high-performance ADCs.

For readers looking for the basics, this chapter can be skipped over in a first reading. Its material is not essential for an understanding of later chapters.

5.1 Precision op-amp design techniques

In the field of measurement and control there is often a need for circuits of high precision. Control circuits should be accurate, stable with time and temperature, and predictable. The usefulness of measuring instruments likewise depends on their accuracy and stability. In almost all electronic subspecialties we always have the desire to do things more accurately – you might call it the joy of perfection. Even if you don't always actually *need* the highest precision, you can still delight in a full understanding of what's going on.

5.1.1 Precision versus dynamic range

It is easy to get confused between the concepts of *precision* and *dynamic range*, especially because some of the same techniques are used to achieve both. Perhaps the difference can best be clarified by some examples: a 5-digit multimeter has high precision; voltage measurements are accurate to 0.01% or better. Such a device also has wide dynamic range; it can measure millivolts and volts on the same scale. A precision decade amplifier (one with selectable gains of 1, 10, and 100, say) and a precision voltage reference may

have plenty of precision, but not necessarily much dynamic range. An example of a device with wide dynamic range but only moderate accuracy might be a 6-decade logarithmic amplifier (log-amp) built with carefully trimmed op-amps but with components of only 5% accuracy; even with accurate components a log-amp might have limited accuracy because of lack of log conformity (at the extremes of current) of the transistor junction used for the conversion, or because of temperature-induced drifts.

Another example of a wide-dynamic-range instrument (greater than 10,000:1 range of input currents) with only moderate accuracy requirement (1%) is the coulomb meter described in §9.26 of the previous edition of this book. It was originally designed to keep track of the total charge put through an electrochemical cell, a quantity that needs to be known only to approximately 5% but that may be the cumulative result of a current that varies over a wide range. It is a general characteristic of wide-dynamic-range design that input offsets must be carefully trimmed in order to maintain good proportionality for signal levels near zero; this is also necessary in precision design, but, in addition, precise components, stable references, and careful attention to every possible source of error must be used to keep the sum total of all errors within the so-called error budget.

5.1.2 Error budget

A few words on *error budgets*. There is a tendency for the beginner to fall into the trap of thinking that a few strategically placed precision components will result in a device with precision performance. On rare occasions this will be true. But even a circuit peppered with 0.01% resistors and expensive op-amps won't perform to expectations if somewhere in the circuit there is an input offset current multiplied by a source resistance that gives a voltage error of 10 mV, for example. With almost any circuit there will be errors arising all over the place, and it is essential to tally them up, if for no other reason than to locate problem areas where better devices or a circuit change might be needed. Such an error budget results in rational design, in many cases revealing where an inexpensive component will suffice, and eventually permitting a careful estimate of performance.

To add some spice to the subject, we note that there is some controversy in the engineering community surrounding this business of error budgets. One camp (which we might characterize as strict constructionists) insists that you allow for the true worst case, or you are guilty of violating good engineering practice. For example, if there are 18 gain-determining resistors of $\pm 1\%$ tolerance in a circuit, then the guaranteed performance must be specified as $\pm 18\%$. The response of the other camp (which we might characterize as pragmatists) is "balderdash – it's overly constraining to allow an extremely unlikely possibility to limit the performance of a circuit design; and one can deal with such eventualities with component test procedures, finished circuit performance testing, and so on." We'll revisit this controversy (and take sides in the debate) after running through an introductory example.

5.2 An example: the millivoltmeter, revisited

To motivate the discussion of precision circuits, let's revisit a circuit from the previous chapter. There we flirted, briefly, with issues of precision in §4.4.3, mostly to illustrate the effects of input offset voltage V_{OS} and input bias current I_B in a low-level dc application (a 0–10 mV millivoltmeter with 10 MΩ input resistance[1]). Back then, with wide-eyed naivety, we were astonished to see that our trusty LF411 op-amp was wholly inadequate to the task; it had way too much offset, and too much input current as well. We found a solution in the form of either a precision low-bias op-amp (an OPA336) or a chopper (also known as "auto-zeroing") amplifier (an LTC1050).

As we'll soon see, our celebration of that "solution" was premature: we pronounced victory with an op-amp whose I_B alone caused the maximum allowable zero-input error of 1%. A careful design must take into account the cumulative effect of multiple sources of error.

5.2.1 The challenge: 10 mV, 1%, 10 MΩ, 1.8 V single supply

To make the problem more interesting, let's further constrain the specifications. This time we'll ask that the 0–10 mV meter operate from a single +3 V battery (either a lithium cell or a pair of alkaline AAA cells); that forces us to worry about "single-supply" operation, in which the op-amp must work down to zero volts at both input and output. Furthermore, it must work down to the end-of-life voltage of alkaline cells, which you see stated variously as 1.0 V/cell, or 0.9 V/cell; that means operation down to +1.8 V total supply voltage. And, as before, let's require an input resistance of 10 MΩ and insist that it indicates 0 mV ($\pm 1\%$ of full scale) when the input is either shorted

[1] Note that it can be used as a sensitive *current* meter: with its 10M input resistance and 1% accuracy, it can measure currents down to 10 pA ($1\% \times 10\,\text{mV}/10\,\text{M}\Omega = 10\,\text{pA}$).

or open. Note that this "zero-error" specification is different from a full-scale accuracy ("scale error") specification: we might be happy with ±5% full-scale accuracy, but we'd be most unhappy with a meter that reads 5% of full scale (here 0.5 mV) when there's nothing connected to it.

Following the suggestion from last chapter's design, let's use current-sensing feedback, so the design is independent of the internal resistance of the analog meter. Figure 5.1 shows the circuit.

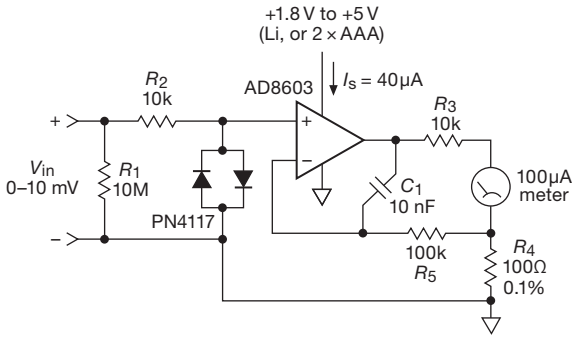


Figure 5.1. Accurate millivoltmeter powered with single lithium cell. The input protection clamp uses low-leakage diode-connected PN4117 JFETs.

5.2.2 The solution: precision RRIO current source

We use a precise current-sensing resistor R_4, in this case a 0.1% 100 Ω resistor. Sounds exotic, but in fact these things are commonplace: the ever-helpful DigiKey website shows over 100,000 in stock, from five different suppliers, at prices down to $0.20 (in quantities of 10). Note the routing of the input common ("−" terminal) connection directly to the low side of the sense resistor, a precaution that becomes increasingly important with small-value sense resistors, in which the wiring resistance of the ground return may add significant error.[2] Because the meter likely presents an inductive load (it's a moving coil in a magnetic field, which is both inductive in its own right and reactive through its motor-like electromechanical properties), we took the precaution of dividing the feedback path in the usual way (through R_5 at low frequencies, C_1 at high frequencies; see for example Figure 4.76). The 10k output resistor R_3 limits meter current for off-scale inputs.

The more challenging parts of this design are the input protection network (which we blithely ignored in Chapter 4's example), and, most critically, the choice of op-amp. First, the protection network: the requirements seem easy – clamp to a nondestructive op-amp input voltage (during input overvoltages), and draw less than ∼10 pA leakage current at the full-scale input voltage (10 mV), in both forward and reverse directions. (That amount of diode current would reduce the input resistance by 1%.) As it turns out, the datasheets don't ordinarily tell you how much current a diode draws at very low voltages. But if you go measure it, you'll be surprised at what you find (Figure 5.2). Everyone's favorite jellybean signal diode (1N914, 1N4148) is rather leaky, looking roughly like a 10 MΩ resistor at low voltages.[3] There are some specialized low-leakage diodes like the (somewhat hard to get) PAD-1 or PAD-5 that do much better; but you can do as well by using a diode-connected low-leakage JFET like the *n*-channel PN4117 (i.e., tie the source and drain together to form the cathode, and use the gate as the anode), or you can just use the diode terminal pairs from an ordinary *npn* transistor.[4] In this circuit the upstream 10k resistor R_2 limits the clamp current, while having no effect on the circuit accuracy.

And now for the op-amp. This was the stumbling block in Chapter 4, and it has gotten only more difficult here, with the single low-voltage supply. We can separate the errors into a "zero" error and an overall scale factor error. The latter is the easy part: the *circuit's* gain is accurately determined, so we merely require an accurate meter movement (if we don't want any trim adjustments; or we could reduce the sense resistor and add a resistor to produce a trimmable gain greater than unity). It's the "zero" requirement that is the tough part, because of the high sensitivity plus the high mandated input resistance. We require a worst-case combined effect of input offset and bias current of 100 µV and 10 pA *individually*. That is, each alone would cause a zero error of 0.1 mV ($V_{err}=V_{OS}+I_BR_1$), so each must be smaller so that the worst-case combination meets specifications.

We looked at some promising contemporary op-amp offerings, which we've listed (along with the usual suspects) in Table 5.1 on page 296. The inexpensive jellybeans in the

[2] Small-value sense resistors used for accurate measurement of high currents are available as *4-wire* resistors. This arrangement is known as a *Kelvin connection*, and the sense resistor is sometimes called a *shunt*.

[3] For all these diodes, the straight-line portion at low voltages represents a resistance in parallel with a non-conducting diode; so the low-leakage 1N3595 looks like 10,000 MΩ for $V \lesssim 10$ mV. These currents can be estimated from other diode parameters, namely reverse-bias leakage current at low reverse voltage or forward current at specified forward voltage. More on this in Chapter *1x*.

[4] Or our friend John Larkin's favorite, the collector-base junction of the BFT25, a low-cost 5 V *npn* microwave transistor. Its leakage is less than 10 fA when reverse biased, and <40 fA when forward biased up to 50 mV.

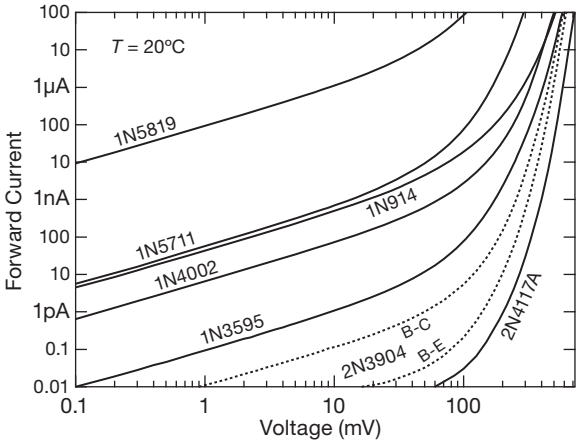


Figure 5.2. Diode datasheets are often skimpy with data like this, showing the low end of the current versus forward applied voltage. See Chapter *1x* for lots more detail.

first three rows are hopeless for this job, with both offset voltage and bias current far worse than required (listed in the bottom row). And they all fail to operate at +1.8 V; and the first two fail the "single-supply" requirement. So much for el-cheapo op-amps.

The LPV521 is characteristic of the new breed of "low-voltage, low-bias, low-power" op-amps. It does those things just fine, but, in common with many CMOS op-amps, its V_{OS} specs are only so-so. It's possible to do considerably better. For example, the CMOS precision OPA336 was our hero earlier; but look – its worse-case I_B consumes our entire error budget, and it would require V_{OS} trimming anyway, which is not so easily done in a single-supply setup (unlike our friend the LF411, this op-amp has no trim pins). We brushed off such niceties earlier; but this time let's accept full responsibility for producing products that meet specs. That means adopting some serious worst-case discipline.

What choices are left? Most designers would reach for an auto-zero (chopper) amplifier for a precision application that does not require bandwidth and that is tolerant of noise. The best candidate we found is the ADA4051, with excellent V_{OS} specs, but five times too much I_B, if you believe the "maximum" specs. (Competing auto-zero amplifiers had greater bias current, greater minimum supply voltage, or both.) The last two op-amps are contemporary CMOS op-amps that qualify as "precision," owing to careful design and production-line offset trimming. Both meet the error budget goals (but see below). We chose the AD8603 because it meets the 1.8 V supply spec, and as a bonus it operates at 35% the supply current of the LTC6078 (a dual op-amp, with no single version we could find).[5] As listed in Table 5.5, the AD8603 runs at 40 μA quiescent current, and has 1 pA (max) input current and 50 μV (max) offset voltage at 25°C.

Are we done? Not quite. The specs in the table are for operation at 25°C. It can get plenty warmer, with impressive increases in bias current for CMOS devices (for which the "bias" current is leakage current). Manufacturers will ordinarily provide worst-case (and sometimes typical) values at the high end of the operating temperature range (e.g., 85°C for "industrial" temperature range devices). For our chosen AD8603 they specify $I_B(\text{max}) = 50$ pA at 85°C. No worst-case data for intermediate temperatures are given; but that value is consistent with a doubling every 10°C, so we can be confident that the circuit will meet specs[6] up to 50°C. It's worth noting that manufacturers are sometimes rather lazy in setting worst-case leakage specs, as for example with the LPV521 in the table, whose ratio of "maximum" to "typical" I_B is 100:1. One industry guru attributes that to an unwillingness to test production parts to a tighter spec.

Exercise 5.1. We designed a $\pm$10 mV meter in Chapter 4 (Figure 4.56), but our design in Figure 5.1 is unipolar (i.e., 0 to +10 mV). Your boss has asked you to modify the design for $\pm$10 mV capability, using a $\pm$100 μA zero-center meter and retaining the feature of operating with either a single lithium cell (or pair of AAA alkaline cells). *Hint*: you may wish to look at the AD8607 dual member of the AD8603 op-amp family. *Extra credit*: after finishing, you decide to impress your boss by enhancing the design so it will work with a 0–100 μA meter.

5.3 The lessons: error budget, unspecified parameters

From this rather simple first example we've learned some important basic principles: (a) first, you've got to identify and quantify the sources of error within a circuit in order to create an error budget; and (b) strict worst-case design requires that all components (passive and active) be operated within their datasheet specifications, and that the effects of

[5] Five other parts we considered (see Tables 5.5, pages 320–321, and 5.6, page 335): the bipolar LT1077A, lower I_s and V_{OS} but too high V_s (min); the bipolar LT6003, I_s only 1 μA, but too much offset voltage (a 5% error); the CMOS LMP2232A, poor 1.5% offset voltage; the MAX9617 auto-zero op-amp, with 0.1% (versus the AD8603's 0.5%) offset voltage, but whoa, its input *current* through 10 MΩ amounts to a 1.4% offset; and the ISL28133 auto-zero, but its 3% offset (from input current) again reveals the input current weakness of auto-zero amplifiers.

[6] The datasheet does provide a plot of "typical" I_B versus temperature, confirming its exponential behavior of a doubling every 10°C.

Table 5.1 Millivoltmeter Candidate Op-amps

Part #	Input	V_{os} typ (μV)	V_{os} max (μV)	I_{bias} typ (pA)	I_{bias} max (pA)	V_{cm} neg lim (V)	V_{out}	V_s (total) min (V)	V_s (total) max (V)	I_s typ (μA)	Price 100 pc ($)	Notes
uA741	BJT	2000	6000	80k	500k	2	1.5V from rails	10	40	1500	0.27	old HV BJT
LF411	JFET	800	2000	50	200	3	1.5V from rails	10	40	1800	0.88	HV JFET
LM358A	BJT	2000	3000	45k	100k	0	0 to V_+–1.2V	3	32	500	0.21	old HV SS
LPV521	CMOS	100	1000	0.01	1	–0.1	R-R	1.8	5.5	0.5	1.05	low-bias LV RRIO
OPA336	CMOS	60	125	1	10	–0.2	R-R	2.3	5.5	20	1.70	chap 4 "solution"
ADA4051	CMOS	2	17	5	50	0	R-R	1.8	5	15	2.20	LV auto-zero RRIO
LTC6078	CMOS	7	25	0.2	1	0	R-R	2.7	5.5	110	1.75	LV low-I_B low-V_{os} RRIO dual
AD8603	CMOS	12	50	0.2	1	–0.3	R-R	1.8	5	40	1.40	LV low-I_B low-V_{os} RRIO
Ckt Limit		-	*100*	-	*10*	*0*	*0 to anything*	*1.8*	*>3.6*			*must budget contributions*

their guaranteed worst-case errors be added (as unsigned magnitudes) to determine the overall circuit's performance.

So far, so good; and in this example the op-amp choice allowed us to meet (and exceed) our target zero-error specifications (1% of full scale, with input open or shorted), even under guaranteed worst-case values of I_B and V_{OS}.

A. Unspecified parameters: a pragmatic approach

But, looking a little closer, we've also seen in this design example that an unspecified parameter (forward diode current at low voltages, 0–10 mV) figured into the overall error.[7] What should one do under such circumstances?

The authors belong to the camp of "pragmatists" on this matter: first, you may have to be creative in reading the datasheet (as we did in the case of op-amp input current), particularly when the manufacturer's worst-case numbers represent a statement that "I don't want to test this parameter, so I'll put a conservative guess in the datasheet"; this is particularly relevant with parameters like leakage current, where the limits of automated test equipment (ATE) and testing time constraints encourage conservative worst-case datasheet specifications. Second, you may have to do some testing[8] of poorly specified (or unspecified) parameters (as we did with forward diode current). It may be sufficient to establish that the circuit effects of the unspecified parameter are completely insignificant (as here, when the current through the clamp diodes was less than 0.01 pA, or three orders of magnitude below budget); or, if it's a closer call, you may have to set up a testing regime of incoming components to ensure that you meet specs. And third, you may have to deal with a situation in which there are many components contributing to the overall error budget by simply validating the performance of the overall circuit, subassembly, or complete instrument at final test.

This approach may appear cavalier. But the fact is that there are many situations in which you simply cannot meet challenging specifications while staying within the published worst-case specifications (or lack of specifications). Two examples help make this point: one of the authors designed and manufactured a line of battery-powered oceanographic instruments, intended for long-duration (from weeks to as long as a year) submerged observations and data logging. A typical instrument might have 200 or more 4000B-series CMOS ICs. The datasheet lists the 25°C quiescent current[9] as "0.04μA (typ), 10μA (max)." Great. So 200 of these puppies probably draw a total of 8μA, but they *could* draw (in a wildly improbable scenario) as much as 2 mA. A year's worth of operation would require 70 mAh (using typical values), but, under strict worst-case rules, we would have to allow for 17.5 Ah (amp-hours). Here's the rub: the substantial battery pack for these volume-constrained deep submergeable pressure housings provided only 5 Ah of capacity (with some safety margin of derating). And 80% of the battery capacity was budgeted for the sensors and recorders. So strict worst-case design would require quadrupling the battery pack (and

[7] As did the op-amp input current at modestly elevated temperatures.

[8] If you buy in large quantities the manufacturer may be willing to do these tests.

[9] Amusingly, the specs are the same for simple parts such as gates, or complex parts such as counters or arithmetic logic units.

expanding the pressure case), or, alternatively, removing a substantial volume of the instrumentation payload. The solution was (and still is) obvious: build the subcircuits, and test them for conforming quiescent current. They invariably worked just fine, and the testing served mostly to identify modules in which there was a defective component, usually caused by improper handling of the sensitive CMOS components.

A second example is a commercial instrument, namely a sensitive electrometer from Keithley. These things will measure currents down to *femto*amps (10^{-15}A), which requires a front-end stage of extraordinarily low bias current. They accomplish this with a JFET follower matched pair as input stage to a conventional precision op-amp, in a current-to-voltage configuration (the input is a summing junction, at zero volts). And to keep the gate current low, they operate the JFETs at a very low drain voltage of just +0.55 V, with the source terminal sitting just a fraction of a volt below the drain. Now, nowhere in the JFET datasheet will you find anything telling you what happens at such low voltages; and they won't tell you what the gate leakage is likely to be. You can throw up your hands and say that such an instrument cannot be made. Or you can do what Keithley did, which is to find a good source of JFETs and qualify them with in-house testing so that you can get on with the job.

Both examples illustrate that there are situations in which you simply cannot meet your design requirements while staying within the manufacturer's published worst-case specs. Having said that, we note that there are some engineers who will not deviate from strict worst-case specified component parameters in their circuit designs. They don't want to use special parts, and they won't touch such stuff with a 10-foot (3m) pole. We invite you to choose what you would do.

5.4 Another example: precision amplifier with null offset

Having warmed up with the millivoltmeter, let's tackle a more complex design, one in which there are multiple error challenges. We describe the design choices and errors of this particular circuit within the framework of precision design in general, thus rendering painless what could otherwise become a tedious exercise.

We designed a precision amplifier (Figure 5.3) that lets you "freeze" the value of the input signal, amplifying any subsequent changes from that level by gains of exactly 1, 10, or 100. This might come in particularly handy in an experiment in which you wish to measure a small change in some quantity (e.g., light transmission or RF absorption) as some condition of the experiment is varied. It is ordinarily difficult to get accurate measurements of small changes in a large dc signal, owing to drifts and instabilities in the amplifier. In such a situation a circuit of extreme precision and stability is required.

Here we show the example of a *strain gauge* sensor, which consists of a strain-sensitive resistive bridge whose elements change resistance (slightly!) in response to mechanical strain. A common resistance value is 350 Ω; and the sensitivity is such that, when biased with +5 V, the differential output voltage across the bridge changes by ±10 mV in response to the rated full-scale mechanical strain.[10] This small differential voltage sits on a dc level of +2.5 V, so you've got to begin with a good differential amplifier.

An important note at the outset: digital techniques offer an attractive alternative to the purely analog circuitry used here. A skilled designer would likely make use of precision analog/digital conversion techniques, perhaps in a hybrid implementation (in which a stable DAC is used to create the nulling signal within an analog circuit like ours), or perhaps in an all-digital scheme that relies on the intrinsic precision of a high resolution ADC alone.[11] Regardless, our all-analog example offers a smorgasbord of important lessons in precision design. But the reader can confidently look forward to exciting revelations in chapters to come.

5.4.1 Circuit description

The front-end begins with an *instrumentation amplifier* U_1, a configuration of three op-amps that we'll talk about later (§5.15); these are differential-input amplifiers that excel in achieving high common-mode rejection, and allow gain selection with a single resistor (one or more are often provided internally). Here we've selected one with a good combination of low input current, offset drift, and noise, for reasons we'll explain later. Its gain of ×100 is followed with a noninverting ×10 gain stage (U_2), for an overall gain of ×1000; that produces a full-scale output of ±10 V as input to the nulling circuitry (U_3–U_5). If the input signal were single ended (e.g., from a thermocouple, photosensor, microwave absorption detector, or whatever), you would omit U_1, bringing in the signal at point "X," and adjusting the gain of U_2 accordingly.

[10] The strain-gauge sensor sensitivity is "2mV-per-volt"; that's pretty low. There are semiconductor strain sensors with higher sensitivity, but they may not be as stable.

[11] There's an example of the latter in §13.9.11C.

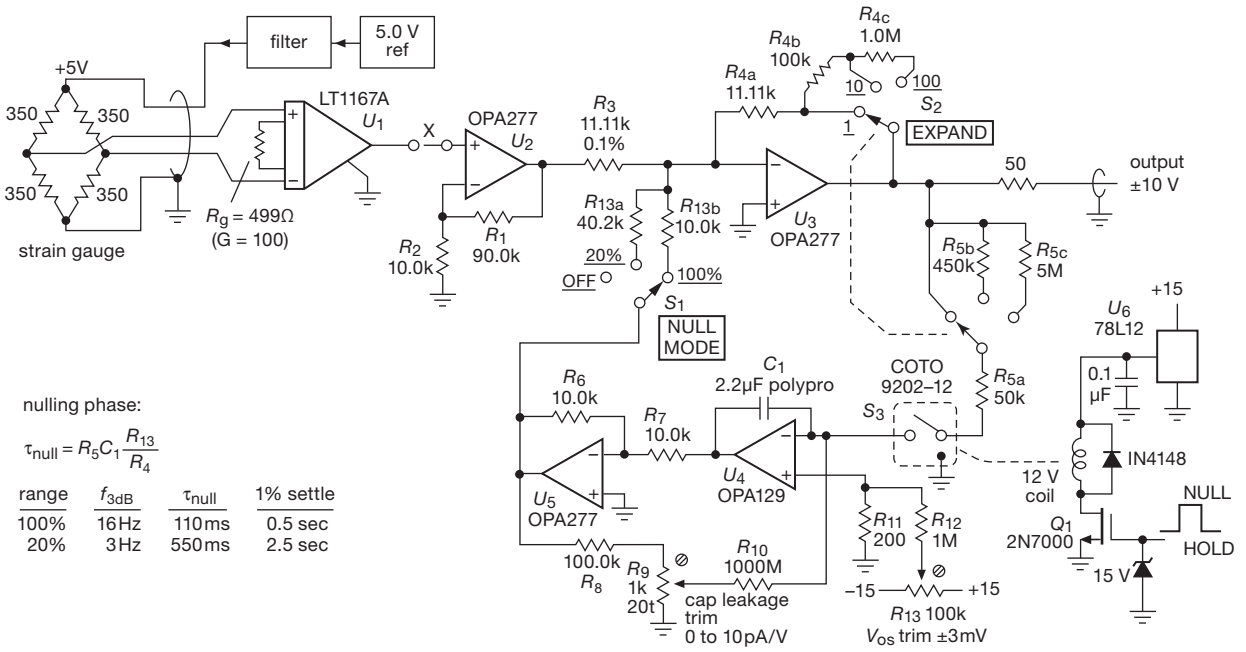


Figure 5.3. Autonulling dc laboratory amplifier. Gain-setting resistors are 0.1% tolerance.

The nulling circuitry works as follows: the amplifier stage U_3 is configured in the inverting configuration, permitting a dc offset according to a current added to the summing junction. Nulling takes place when relay (switch) S_3 is activated by a logic HIGH to coil driver Q_1's gate. Then U_4 charges the analog "memory" capacitor (C_1) as necessary to maintain zero output. No attempt is made to follow rapidly changing signals, because in the sort of application for which this was designed the signals are essentially dc, and some averaging is a desirable feature. When the switch is opened, the voltage on the capacitor remains stable, resulting in an output signal from U_3 proportional to the wanderings of the input thereafter. The gain about the input null level can be increased in decade steps (switch S_2) to expand changes; the gain of the nulling integrator is switched accordingly to keep the feedback bandwidth constant. Switch S_1 selects the full-scale nulling range (100%, 20%, or none).

There are a few additional features that we should describe before going on to explain in detail the principles of precision design as applied here: (a) U_5, in addition to providing the needed inversion of the nulling level, participates in a first-order leakage-current compensation scheme: the tendency of C_1 to discharge slowly through its own leakage ($\geq$100,000 MΩ, corresponding to a time constant of $\geq$3 days) is compensated by a small charging current through R_{10} proportional to the voltage across C_1; and (b) integrator U_4 was selected for very low input current I_B (to minimize droop during "hold"), for which the tradeoff is relatively poor offset voltage V_{OS}; so we added an external offset trim (R_{11}–R_{13}). This is not terribly critical, in any case, because an offset here merely causes a nonzero null of the same magnitude.

5.5 A precision-design error budget

For each category of circuit error and design strategy, we will devote a few paragraphs to a general discussion, followed by illustrations from the preceding circuit. Circuit errors can be divided into the categories of (a) errors in the external network components, (b) op-amp (or amplifier) errors associated with the input circuitry, and (c) op-amp errors associated with the output circuitry. Examples of the three are resistor tolerances, input offset voltage, and errors that are due to finite slew rate, respectively.

Let's start by setting out our error budget. It is based on a desire to keep input drift (from temperature and power-supply variations) down to the 10 μV level, and nulled drift (primarily from capacitor "droop," along with temperature and supply variations) below 1 μV/min (referred to the input, or RTI). As with any budget, the individual items are arrived at by a process of tradeoffs, based on what can be

done with available technology. In a sense the budget represents the end result of the design, rather than the starting point. However, it will aid our discussion to have it now.

It's important to understand that the items in such a budget come from several sources: (a) parameters that are specified in the datasheet; (b) estimates of poorly specified (or unspecified) parameters; and (c) parameters that you may not even realize are important.[12] We might paraphrase these, respectively, as the *knowns*, the *known unknowns*, and the *unknown unknowns*.

5.5.1 Error budget

These are all in the form of worst-case voltage errors (at 25°C) *referred to the instrument input*.

1. ×100 Difference amplifier (U_1: LT1167A)

Offset voltage	40 μV
Noise voltage (0.1–10 Hz)	0.28 μVpp (typ – no "max" spec)
Temperature	0.3 μV/°C
Power supply	28 nV/100 mV change
Input offset current ×R_s	0.11 μV/350 Ω of R_s

2. ×10 Gain amplifier (U_2: OPA277)

Offset voltage	0.5 μV
Temperature	10 nV/°C
Time	2 nV/month (typ – no "max" spec)
Power supply	1 nV/100 mV change
Bias current	0.3 μV
Load-current heating	5 nV at full scale (5 mW, 0.1°C/mW)

3. Output amplifier (U_3: OPA277)

Offset voltage	50 nV
Temperature	1 nV/°C
Time	0.2 nV/month (typ – no "max" spec)
Power supply	0.1 nV/100 mV change
Bias current	30 nV
Load-current heating	5 nV at full scale (1 kΩ load)

4. Hold amplifier (U_4: OPA129)

U_4 offset tempco	10 nV/°C
Power supply	10 nV/100 mV change
Capacitor droop (see current error budget)	0.4 μV/min
Charge transfer	1.1 nV

Current errors through C_1 (needed for the preceding voltage error budget) are as follows:

Capacitor leakage	
Maximum (uncompensated)	(100 pA)
Typical (compensated)	10 pA
U_4 input current	0.25 pA
U_4's nulled V_{OS}/R_{10}	0.1 pA
Relay S_3 OFF leakage	10 pA (1 pA typ)
Printed-circuit-board leakage	5.0pA

Not bad, though you *could* complain about the 40 μV input offset – but we'd reply that a few tens of microvolts of *static* offset is of no concern in a nulling instrument, it's only the *drift* (with time and temperature) that matters. The various items in the budget will make sense as we discuss the choices faced in this particular design. We organize these by the categories of circuit errors listed earlier: network components, amplifier input errors, and amplifier output errors.

We'll address these errors quantitatively, in the context of Figure 5.3, beginning in §5.7.6, after looking at error sources a bit more generally in the next section.

5.6 Component errors

The degrees of precision of reference voltages, current sources, amplifier gains, etc., all depend on the accuracy and stability of the resistors used in the external networks. Even where precision is not involved directly, component accuracy can have significant effects, e.g., in the common-mode rejection of a differential amplifier made from an op-amp (see §§4.2.4 and 5.14), where the ratios of two pairs of resistors must be accurately matched. The accuracy and linearity of integrators and ramp generators depend on the properties of the capacitors used, as do the performances of filters, tuned circuits, etc. As we will see presently, there are places where component accuracy is crucial, and there are other places where the particular component value hardly matters at all.

Components are generally specified with an initial accuracy, as well as the changes in value with time (stability) and temperature. In addition, there are specifications of voltage coefficient (nonlinearity) and bizarre effects such as "memory" and dielectric absorption (for capacitors). Complete specifications also include the effects

[12] We discovered an example of the latter while measuring femtoamp leakage currents in a nice shielded test enclosure: after the box had been opened to change anything inside, it took quite a while for the measurements to settle down. Turns out that the process of moving things caused some rearrangement of surface charge on the Teflon-insulated wires, with a long relaxation time. Pease talks about this in his article "What's All This Teflon Stuff, Anyhow?" – see the footnote references in §5.10.7. We've experienced a similar bizarre manifestation with analog panel meters, where a swipe of the hand across the glass face can cause the needle to move way upscale... and *stay* there!

of temperature cycling and soldering, shock and vibration, short-term overloads, and moisture, with well-defined conditions of measurement. In general, components of greater initial accuracy will have their other specifications correspondingly better, in order to provide an overall stability comparable to the initial accuracy. However, the overall error that is due to all other effects combined can exceed the initial accuracy specification. Beware!

As an example, RN55C 1% tolerance metal-film resistors have the following specifications: temperature coefficient (tempco), 50 ppm/°C over the range −55°C to +175°C; soldering, temperature, and load cycling, 0.25%; shock and vibration, 0.1%; moisture, 0.5%. By way of comparison, the legacy 5% carbon-composition resistors (Allen–Bradley type CB) have these specifications: tempco, 3.3% over the range 25–85°C; soldering and load cycling, +4%, −6%; shock and vibration, ±2%; moisture, +6%. From these specs it should be obvious why you can't just select (using an accurate digital ohmmeter) carbon resistors that happen to be within 1% of their marked value for use in a precise circuit, but are obliged to use 1% resistors (or better) designed for long-term stability as well as initial accuracy. For the utmost in precision it is necessary to use ultra-precise resistors or resistor arrays, such as Susumu's RG-series of SMT (surface-mount technology) resistors (tolerance to 0.02%, tempco to 5 ppm/°C), Vishay's MPM-series metal-film networks (absolute tolerance to 0.05%, matched to 0.01%; absolute tempco to 25 ppm/°C, matching tempco to 2 ppm/°C), or their even better "Bulk Metal Foil" types (absolute tolerance to 0.005%, matched to 0.001%; absolute tempco to 0.2 ppm/°C, matching tempco to 0.1 ppm/°C).

5.6.1 Gain-setting resistors

In the preceding circuit (Figure 5.3), 0.1% resistors are used in the gain-setting network, R_1–R_4, for accurately predictable gain. As we'll see shortly, the value of R_3 is a compromise, with small values reducing offset current error in U_3 but increasing heating and thermal offsets in U_2. Note that 1% resistors are used in the offset attenuator network, R_5–R_{13}; here absolute accuracy is irrelevant, and the stability of 1% metal-film resistors is altogether adequate.

5.6.2 The holding capacitor

A. Leakage

The largest error term in this circuit, as the error budget shows, is capacitor leakage in the holding capacitor, C_1. Capacitors intended for low-leakage applications give a leakage specification, sometimes as a leakage resistance, sometimes as a time constant (megohm-microfarads). In this circuit C_1 must have a value of at least a few microfarads in order to keep the charging rate from other current error terms small (see budget). In that range of capacitance, film capacitors (polystyrene, polypropylene, and polyester) have the lowest leakage. Polypropylene capacitors (from manufacturers such as Epcos, Kemet, Panasonic, Vishay, and Wima, generally with voltage ratings of 200–600 V) often have dc leakage specified in units of megohm–microfarads, with values in the range of 10,000–100,000 MΩ μF; thus for a capacitance of 2.2 μF this amounts to a parallel equivalent leakage resistance of at least 4.5–45 GΩ.

Even so, and adopting a plausible value of, say, 100 GΩ, that's equivalent to a leakage current of 100 pA at full output (10 V), corresponding to droop rates of roughly 3 mV/min at the output, the largest error term by far. For that reason we added the leakage-cancellation scheme described earlier. It is fair to assume that the effective leakage can be reduced to 10% of the capacitor's worst-case leakage specification (in practice, we can probably do much better). No great stability is required in the cancellation circuit, given the modest demands made of it. As we will see later when we discuss voltage offsets, R_{10} is kept intentionally large so that input voltage offsets in U_4 aren't converted to a significant current error.

B. Dielectric absorption

We're not done with the capacitor, yet. An important effect, quite apart from resistive leakage, is capacitor "memory," officially known as *dielectric absorption*.[13] This is the tendency of capacitors to return, to some extent, to a previous state of charge, as shown in the measured data of Figure 5.4 (each capacitor was held at +10 V for a day or more, then discharged to 0 V for 10 s, then open-circuited and observed while it did its thing); see also the discussion in §§*1x.3*, 4.5.1, 4.5.6, and 13.8.4.

5.6.3 Nulling switch

In the previous edition of this book, the analogous circuit (Figure 7.1, page 393) used MOSFETs (instead of relay S_3) to activate the nulling circuit. That choice provided plenty of education, because we had to worry about (a) MOSFET channel leakage, devastatingly large at about ∼1 nA, and (b) gate charge injection, of order 100 pC in that circuit. The solutions there were (a) the use of a series-connected

[13] It is not entirely clear that what is called "leakage" in high-quality capacitors is in fact distinct from dielectric absorption; see the footnote in §4.5.5.

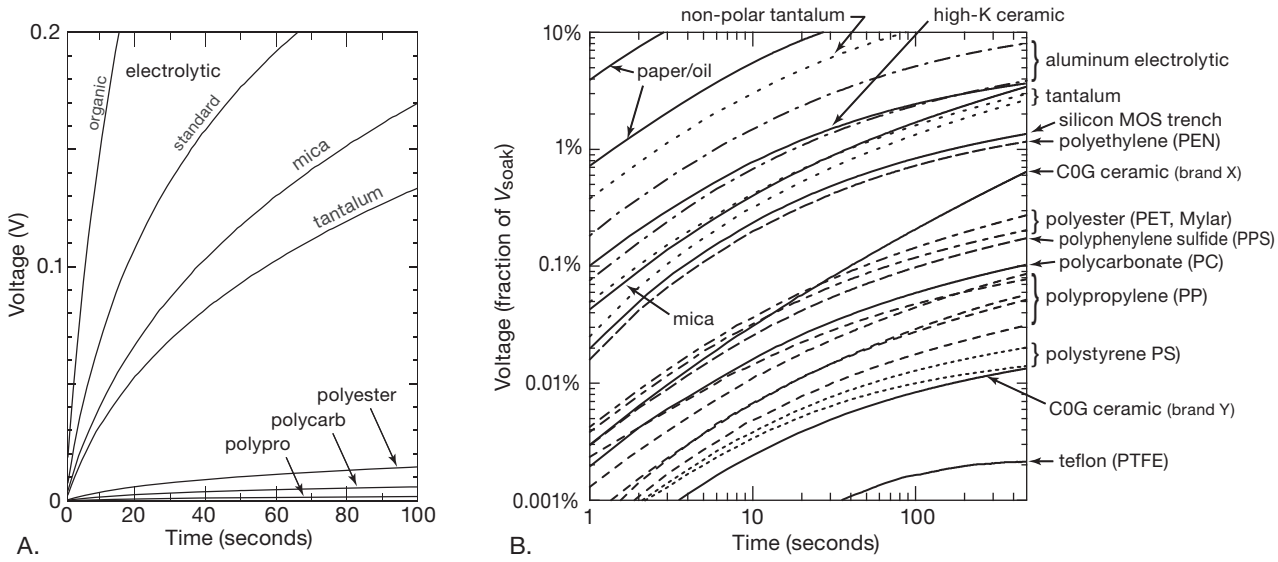


Figure 5.4. Capacitors exhibit *memory effect* (dielectric absorption), a tendency to return to a previous state of charge. This is highly unhelpful in applications (such as analog sample-and-hold) in which a capacitor is used to retain an analog voltage. A. linear plot, showing the basic effect; B. log–log plot, revealing four decades of dirty laundry. Teflon is the uncontested winner; but it's hard to find, so the plastic film types (PS and PP) are generally your best choice. Ceramic C0G can be excellent, but beware brand variations.

MOSFET pair, such that the downstream MOSFET had all four terminals (source, drain, gate, and substrate) ordinarily at zero volts, and (b) a sufficiently large holding capacitor such that the error was negligible, along with the observation that charge transfer was not of great concern because it resulted in a small offset of the auto-zero.

This time we have taken a more pragmatic (but less educational) approach, by using instead a small signal *relay*. The Coto 9202-12 is a small (4 mm×6 mm × 18 mm) shielded relay, energized by 12 Vdc at 18 mA, with a specified OFF resistance of $10^{12}\,\Omega$ minimum ($10^{13}\,\Omega$ typical). The worst-case $R_{\rm off}$ value corresponds to a droop rate of 0.3 mV/min, but ten times less for "typical" $R_{\rm off}$. Relays isolate better than transistor switches (higher $R_{\rm off}$ and lower $C_{\rm off}$, here < 1 pF), and they have better ON performance as well (lower $R_{\rm on}$ than a low-capacitance analog switch, here $<0.15\,\Omega$).

Of course, there *is* capacitance between the coil and the contacts, and thus an opportunity for the same sort of charge transfer as with a MOSFET switch (where the full-swing transitions at the gate couple capacitively to the drain and source). As we remarked in Chapter 3 (§3.4.2E), the total charge transferred is independent of the transition time and depends only on the total control-voltage swing and the coupling capacitance: $\Delta Q = C_{\rm coup}\Delta V_{\rm control}$. In this circuit, charge transfer results in a simple voltage error of the auto-zero, because the charge is converted to a voltage in the holding capacitor C_1. It's easy to estimate the error: the Coto relay specifies a coil-to-contact capacitance of 0.2 pF (for our grounded-shield configuration), and thus a corresponding charge transfer of $\Delta Q = 2.4$ pC when the 12 V coil is energized.[14] That produces a voltage step of $\Delta V_{\rm C} = \Delta Q/C_1 = 1.1\,\mu$V across the 2.2 μF capacitor C_1. This is completely within our error budget; in fact, we've likely overestimated the effect, because our calculation assumed that the entire coil undertook a 12 V step, whereas the average step is half that value.

For readers who harbor a deep-seated dislike of mechanical relays, we show in Figure 5.5 a switch implementation with series-connected JFETs. During HOLD the JFET gates are back-biased to -5 V, by the somewhat tortured level-shifting circuit Q_1–Q_4. The reader is invited to estimate the magnitude of droop (use the datasheet's $I_{\rm D(off)}$=0.1 pA) and of charge injection (use the datasheet's $C_{\rm rss}$=0.3 pF) for this circuit.

5.7 Amplifier input errors

The deviations of op-amp input characteristics from the ideal that we discussed in Chapter 4 (finite values of input impedance and input current, voltage offset, common-mode rejection ratio, and power-supply rejection ratio, and

[14] Assuming care is taken in the wiring layout to maintain the low 0.2 pF capacitance of the HOLD signal.

Table 5.2 Representative Precision Op-amps

Part #	# per pkg[a]	Supply[p] Range (V)	Supply[p] I_Q (mA)	Input Current @25°C typ (pA)	Input Current @25°C max (pA)	Offset Voltage V_{os} typ (μV)	Offset Voltage V_{os} max (μV)	Offset Voltage ΔV_{os} typ (μV/°C)	CMRR min dB	e_n @1kHz typ (nV/√Hz)	GBW typ (MHz)	Slew typ (V/μs)	Swing to Supply IN +	Swing to Supply IN −	Swing to Supply OUT +	Swing to Supply OUT −	null pins	DIP avail	Cost qty 25 ($US)	Comments
bipolar																				
LT1077A	**1**	2.2–44	0.05	7nA	9nA	9	40	0.4	97	27	0.23	0.08	-	●	-	●	●	●	3.84	single-supply
LT1013	**2**,4	3.4–44	0.35	12nA	20nA	40	150	0.4	100	22	0.7	0.4	-	●	-	●	-	●	3.13	single-supply
OPA277P	**1**,2,4	4–36	0.79	0.5nA	1nA	10	20	0.1	130	8	1	0.8	-	-	-	-	●	●	3.17	improved OP-27
LT1012AC	**1**	8–40	0.37	25pA	0.1nA	8	25	0.2	114	14	0.5	0.2	-	-	-	-	●	●	5.11	superbeta, comp
LT1677	**1**	3–44	2.8	2nA	20nA	20	60	0.4	109	3.2	7.2	2.5	●	●	●	●	●	●	3.07	or AD8675, 0.5nA
LT1468	**1**	7–36	3.9	3nA	10nA	30	75	0.7	96	5	90	23	-	-	-	-	●	●	4.26	0.7ppm distortion
JFET																				
LF412A[b]	***1,2***	*12–44*	*1.8*	*50*	*200*	*500*	*1000*	*-*	*80*	*25*	*4*	*15*	*-*	*-*	*-*	*-*	*-*	*●*	*4.47*	*'411A V_{os} <0.5mV*
OPA827	**1**	8–40	4.8	15	50	75	150	1.5	104	3.8	22	28	-	-	-	-	-	-	9.00	0.1–10Hz: 0.25μVpp
CMOS																				
LMC6482A[b]	***2,4***	*3–16*	*0.5*	*0.02*	*4*	*110*	*750*	*1*	*70*	*37*	*1.5*	*1.3*	●	●	●	●	-	●	*1.88*	*LMC7101 = SOT23*
MAX4236A	**1**	2.4–6	0.35	1	500	5	20	0.6	84	14	1.7	0.3	-	●	●	●	-	-	1.78	shdn, '37 decomp
OPA376	**1**,2,4	2.2–7	0.76	0.2	10	5	25	0.26	76	7.5	5.5	2	●	●	●	●	-	-	1.32	etrim
auto-zero																				
AD8628	**1**,2,4	2.7–6	0.85	30	100	1	5	0.002	120	22	2.5	1	●	●	●	●	-	-	1.92	SOIC-8, SOT23-5

Notes: (a) **boldface** indicated number in a package for the part # listed. (b) not precision, listed for comparison. (p) I_Q, typical, per amplifier.

their drifts with time and temperature) generally constitute serious obstacles to precision circuit design, and they force trade-offs in circuit configuration, component selection, and the choice of a particular op-amp. The point is best made with examples, as we will do shortly. Note that these errors, or their analogs, exist for amplifiers of discrete design as well.

While reading through the following discussion, you may find it helpful to refer to Tables 5.2 and 5.3 where we list ten favorite precision op-amps (plus two inexpensive and not-precision comparees). Those listings add quantitative flesh to a somewhat abstract set of bony teachings.

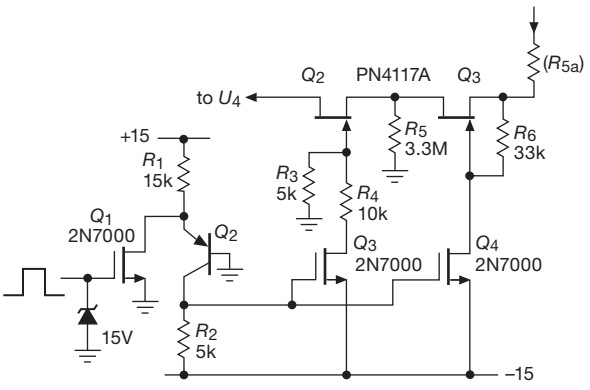


Figure 5.5. Electronic switch replacement of mechanical relay S_3 in Figure 5.3. A lot of work, and no improvement in performance.

5.7.1 Input impedance

Let's discuss briefly the error terms just listed. The effect of finite input impedance is to form voltage dividers in combination with the source impedance driving the amplifier, reducing the gain from the calculated value. Most often this isn't a problem, because the input impedance is bootstrapped by feedback, raising its value enormously. As an example, the OPA277P precision op-amp (with BJT-input, not FET-input stage) has a typical "differential-mode input impedance," of 100 MΩ. In a circuit with plenty of loop gain, feedback raises the input impedance to the datasheet's "common-mode input impedance" 250,000 MΩ. If that's not good enough, FET-input op-amps have astronomical values of R_{in}, for example $10^{13}\,\Omega$ (differential) and $10^{15}\,\Omega$ (common-mode) for the OPA129 that's used in this circuit.

5.7.2 Input bias current

More serious is the input bias current. Here we're talking about currents measured in nanoamps, and this already produces voltage errors of microvolts for source impedances as small as 1 kΩ. Again, FET op-amps come to the rescue, but with generally increased voltage offsets as part of the bargain. Bipolar superbeta op-amps such as the LT1012 can also have surprisingly low input currents. As an example, compare the OPA277 precision bipolar op-amp with the LT1012 (bipolar, optimized for low bias current), the OPA124 (JFET, precision and low bias), the OPA129 (ultra-low-bias JFET), and the LMC6001 (CMOS, lowest-bias op-amp); these are some of the best you can get at the

Table 5.3 Nine Low-input-current Op-amps

	Supply		Input current @25°C		V_{OS}	TCV_{OS}	
Part #	V_{total} (V)	I_Q (μA)	typ (pA)	max (pA)	max (μV)	typ (μV/°C)	max (μV/°C)
bipolar							
OPA277P	10–36	790	500	1000	20	0.1	0.15
superbeta							
LT1012AC	8–40	370	25	100	25	0.2	0.6
AD706	4–36	750	50	200	100	0.2	1.5
JFET							
OPA124PB	10–36	2500	0.35	1	250	1	2
OPA129B	10–36	1200	0.03	0.1	2000	3	10
MOSFET							
MAX9945	4.8–40	400	0.05	-	5000	2	-
CMOS, low-voltage							
LMP7721	1.8–6	1300	0.003	0.02	150	1.5	4
LMC6001A	5–16	450	0.01	0.025	350	2.5	10
ADA4530-1	4.5–16	900	<0.001	0.02	50	0.13	0.5

time of writing, and we've chosen the best grade of each one (Table 5.3; and see also Table 5.5 on pages 320–321 for greater detail, along with a wider selection of precision op-amps with low offset voltage; as well as relevant tables in Chapter *4x*).

Well-designed FET amplifiers have extremely low bias currents, but with much larger offset voltages, compared with the precision OPA277. Because the offset voltage can always be trimmed, what matters more is the drift with temperature. In this case the FET amplifiers are 4 to 20 times worse. The op-amp with the lowest input current uses MOSFETs for the input stage. MOSFET op-amps are popular because of the proliferation of inexpensive units like TI's TLC270 series, as well as the ultra-low-bias-current devices like National's LMC6000-series parts. However, in contrast to JFETs or bipolar transistors, MOSFETs can have very large drifts of offset voltage with time, an effect that is discussed below. So the improvement in current errors you buy with a FET op-amp can be wiped out by the larger voltage error terms. With any circuit in which bias current can contribute significant error, it is often wise to ensure that both op-amp input terminals see the same dc source resistance (see, e.g., Figure 4.55); then the op-amp's *offset current* becomes the relevant specification. But be aware that a number of precision op-amps use a "bias-compensation" scheme to cancel (approximately) the input current in order to make that error term smaller (look back at Exercise 2.24 on page 125 to see how it's done). With op-amps of this type you generally don't gain anything by matching the dc resistances seen by the two inputs, as the residual bias current and the offset current are comparable, in a bias-compensated op-amp.

A. Variation with temperature

One additional point to keep in mind when using FET-input op-amps is that the input "bias" current is actually gate leakage current, and it rises dramatically with increasing temperature: it roughly doubles for every 10°C increase in chip temperature, as seen in Figure 5.6. Because FET op-amps often run warm (our jellybean LF412, for example, dissipates 100 mW when run from ±15 V supplies), the actual input current may be considerably higher than the 25°C figures you see on the datasheet.[15] By contrast, the input current of a BJT-input op-amp is actual base current, relatively constant with temperature. So a FET-input op-amp with impressive input-current specs on paper may not give such an improvement over a good superbeta bipolar unit. As the graph shows, for example, the LT1057 JFET-input op-amp with its ∼3 pA input current (at 25°C) will have an input current of about 100 pA at 75°C chip temperature, which is higher than the input current of the superbeta LT1012 at the same temperature. And our jellybean LF412 JFET op-amp has an input current that is comparable to that of the LT1012 at 25°C and many times higher at elevated temperatures.

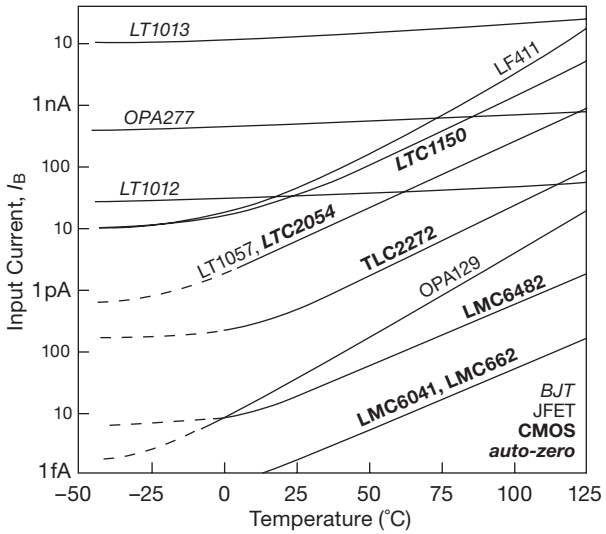


Figure 5.6. Op-amp input current versus temperature, plotted from datasheet values. See also Figures 5.38 and 3.48. JFET-input op-amps are indicated in plain ("roman") type, BJT op-amps are *italic*, CMOS op-amps are **bold**, and auto-zero op-amps are ***bold slant***.

[15] Making this quantitative, the LF412's maximum quiescent current is 6.5 mA, thus 195 mW dissipation when run from ±15 V supplies. In a DIP-8 package that produces a 22°C temperature rise (the thermal resistance $R_{\Theta JA}$=115°C/W), with a consequent quadrupling of the specified I_B=200 pA (max). If the op-amp were driving a load, you'd have yet more dissipation. To put this in perspective, though, note that the driving impedance seen at the op-amp's input would have to be greater than 1 MΩ in order for this current-induced error to exceed the 1 mV (typ) input offset-voltage error.

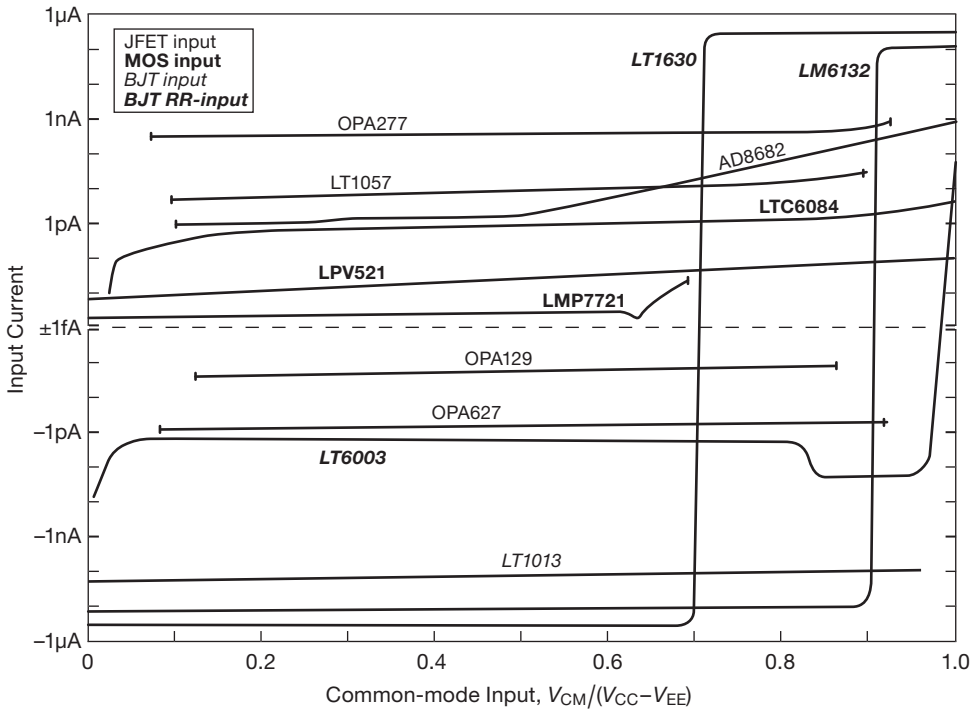


Figure 5.7. Op-amp input current (at 25°C) versus common-mode input voltage, plotted over their operating range from datasheet values; BJT op-amps with rail-to-rail input stages ("BJT RR-input") undergo an abrupt reversal of input-current polarity.

B. Variation with common-mode input voltage

Finally, a very important caution: when comparing op-amp input currents, watch out for some op-amp designs whose I_B depends on the input voltage. This behavior is common in op-amps that are designed to operate over a rail-to-rail input (RRI) range, both FET-input and (especially) BJT-input types. The spec sheet usually lists I_B only at zero volts (or mid-supply), but a good datasheet will show curves as well. See Figure 5.7 for some typical I_B versus V_{in} behavior. The good performance of the OPA129 and OPA627 in this regard is due in part to their use of cascode input stages. The LMP7721 stands out not only for its 20 fA maximum input current, but for its prize-winning trajectory on this graph.

5.7.3 Voltage offset

Voltage offsets at the amplifier input are obvious sources of error. Op-amps differ widely in this parameter, ranging from "precision" op-amps offering worst-case V_{OS} values generally in the 10s of microvolts to ordinary jellybean op-amps like the LF412 with V_{OS} values of 2–5 mV. At the time of writing,[16] the champion (by a slim margin) in the (nonchopper, see below) world of low offsets is the bipolar OPA277P ($\pm 20\,\mu$V, max), which, astonishingly, is equaled by the *CMOS* MAX4236A (though the latter's drift is 12× worse, as one might expect).

Although many good single op-amps (but not duals or quads) have offset-adjustment terminals, it is still wise to choose an amplifier with inherently low initial offset V_{OS} max, for several reasons. First, op-amps designed for low initial offset tend to have correspondingly low offset drift with temperature and with time. Second, a sufficiently precise op-amp eliminates the need for external trimming components (a trimmer takes up space, needs to be adjusted initially, and may change with time). Third, offset voltage drift and common-mode rejection are degraded by the unbalance caused by an offset-adjustment trimmer.

Figure 5.8 illustrates how a trimmed offset has larger drifts with temperature. It shows also how the offset adjustment is spread over the trimmer pot rotation, with best

[16] In the previous edition of this book we awarded that honor to the MAX400M, with its specified worst-case V_{OS} of 10 μV. With confidence we added that "we expect to see further incremental improvements in this area." That confidence was evidently misplaced: the Maxim website now says of the MAX400 "This product was manufactured for Maxim by an outside wafer foundry using a process that is no longer available. It is not recommended for new designs. The datasheet remains available for existing users." *Sic transit...*

resolution near the center, especially for large values of trimmer resistance. Finally, you'll generally find that the recommended external trimming network provides far too much range, making it nearly impossible to trim V_{OS} down to a few microvolts; even if you succeed, the adjustment is so critical it won't stay trimmed for long. Another way to think about it is to realize that the manufacturer of a precision op-amp has *already* trimmed the offset voltage, in a custom test jig using "laser-zapping" techniques; you may be unable to do any better yourself. Our advice is (a) to use precision op-amps for precision circuits, and (b) if you must trim them further, arrange a narrow-range trim circuit similar that shown in Figure 5.3, with values adjusted to produce a full-scale range of $\pm 50\,\mu$V, linear in trimmer rotation (e.g., R_{11}=33 Ω and R_{12}=10M). Figure 5.9 shows how to arrange narrow-range external offset trims for both inverting and noninverting amplifier configurations.

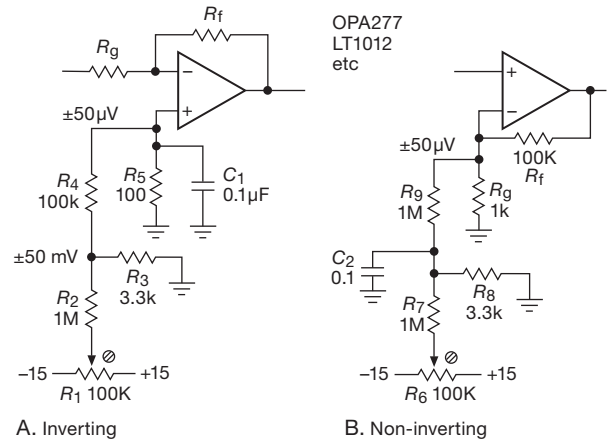


Figure 5.9. Narrow-range external trimming networks for precision op-amps.

Because voltage offsets can be trimmed to zero, what ultimately matters is the drift of offset voltage with time, temperature, and power-supply voltage. Designers of precision op-amps work hard to minimize these errors. You get the best performance from bipolar-input (as opposed to FET) op-amps in this regard, but input-current effects may then dominate the error budget. As shown in Table 5.2 on page 302 the best op-amps keep drifts below 1 μV/°C; the OPA277P typifies the best drift specification (for a nonchopper op-amp): ΔV_{OS}=0.2 μV/°C, max.

Another factor to keep in mind is the drift caused by self-heating of the op-amp when it drives a low-impedance load. It is often necessary to keep the load impedance above 10k to prevent large errors from this effect. As usual, that may compromise the next stage's error budget from the effects of bias current! We'll see just such a problem in this design example. For applications in which drifts of a few microvolts are important, the related effects of thermal gradients (from nearby heat-producing components) and thermal emf's (from voltages across junctions of dissimilar metals) become important. This will come up again when we discuss the ultraprecise *chopper-stabilized* amplifier in §5.11.

An important caution: when datasheets specify the particular measurement conditions for a parameter like V_{OS}, they mean it! A sobering example is shown in Figure 5.10, a plot of V_{OS} versus V_{CM} for the AD8615 op-amp, whose datasheet declares (on the front page) "Low offset voltage: 65 μV max," but whose tabular data reveal that the measurement conditions are "$V_{CM} = 0.5$ V and 3.0 V."

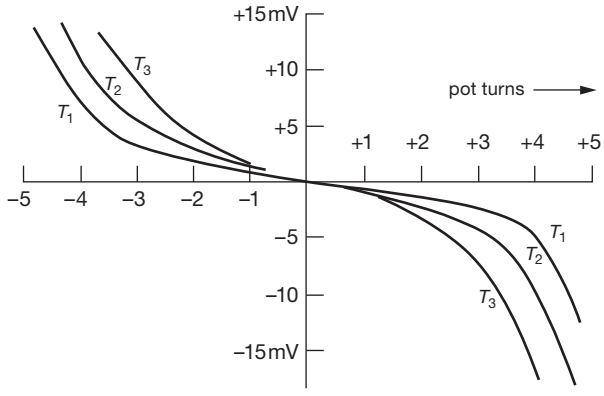


Figure 5.8. Typical op-amp offset versus offset-adjustment potentiometer rotation for several temperatures.

5.7.4 Common-mode rejection

Insufficient common-mode rejection ratio (CMRR) degrades circuit precision by effectively introducing a voltage offset as a function of dc level at the input. This effect is usually negligible, because it is equivalent to a small gain change, and in any case it can be overcome by choice of configuration: an inverting amplifier is insensitive to op-amp CMRR, in contrast with a noninverting amplifier. However, in "instrumentation amplifier" applications you are looking at a small differential signal riding on a large dc offset, and a high CMRR is essential. In such cases you have to be careful about circuit configurations and, in addition, you must choose an op-amp with a high-CMRR specification. Once again, a superior op-amp like the OPA277 can solve your problems, with a CMRR (min)

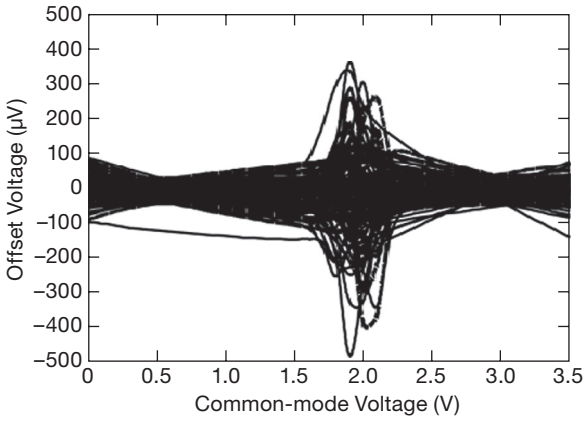


Figure 5.10. This op-amp specifies a maximum offset voltage of $\pm 60\,\mu$V. But it also specifies the following conditions: V_s=3.5 V, and V_{CM}=0.5 V or 3.0 V. Moral: don't ignore the footnotes!

at dc of 130 dB, compared with our jellybean LF411's meager specification of 70 dB. We discuss high-gain differential and instrumentation amplifiers later in the chapter, beginning at §5.13.

5.7.5 Power-supply rejection

Changes in power-supply voltage cause small op-amp errors. As with most op-amp specifications, the power-supply rejection ratio (PSRR) is referred to a signal at the *input*. For example, the OPA277 has a specified PSRR of 126 dB at dc, meaning that a 1 volt change in one of the power-supply voltages causes a change at the output equivalent to a change in the differential input signal of 0.5 μV.

The PSRR drops with increasing frequency, approximately tracking the behavior of the open-loop gain, and a graph documenting this scurrilous behavior is often given on the datasheet. For example, the PSRR (relative to the negative rail) of our favorite OPA277 begins dropping at 1 Hz and is down to 95 dB (typ) at 60 Hz and 50 dB at 10 kHz. This rarely presents a problem, because power-supply noise is also decreasing at higher frequencies if you have used good bypassing. However, 120 Hz ripple could present a problem if an unregulated supply is used.

It is worth noting that the PSRR will not, in general, be the same for the positive and negative supplies. Thus the use of dual-tracking regulators doesn't necessarily bring any benefits. Note also that PSRR is often specified for $G = 1$, and can be considerably worse at higher gains; in fact, op-amps have been found that exhibit *gain* (!) from one rail to the output at moderate gain settings.

5.7.6 Nulling amplifier: input errors

Now we're ready to embark on a detailed discussion of the most serious error issues in the amplifier of Figure 5.3. The circuit begins with an optional precision instrumentation amplifier front-end U_1 (more in §5.15), here chosen for its stable and accurate differential gain of 100×, low input current, and adequately low noise (9 nV/$\sqrt{\text{Hz}}$ typ at 10 Hz). Its worst-case offset voltage and tempco ($\pm 40\,\mu$V, 0.3 μV/°C) specs are a factor of two worse than a precision op-amp like the OPA277 (in the best grade), but its 120 dB (min) CMRR as a difference amplifier, combined with 0.08% worst-case gain accuracy, 50 ppm/°C (max) gain tempco, and low-voltage noise, make it a good front-end for a low-level bridge application like this. Although not important with the low source impedance in this example, its input current is satisfyingly low for a BJT-input amplifier, at just 0.35 nA max.[17]

For single-ended inputs U_1 is omitted, and the signal is brought in at point "×" (add a 470 Ω series resistor, with a pair of low-leakage clamp diodes – see Figure 5.2 – to the rails, for overdrive protection). The OPA277's accuracy and stability rule here, though it is tempting to consider substituting a FET-input part; but the $\gtrsim 10\times$ poorer V_{OS} tempco specification more than offsets the advantage of low input current, except with sources of very high impedance. The OPA277's 1 nA (max) bias current gives an error of 1 μV/1 kΩ source impedance, whereas the best-in-class JFET OPA627B (at $35 apiece!), though providing negligible current error with its 5 pA (max) input current, would exhibit voltage offset drifts as large as 3 μV/4°C (4°C is considered a typical laboratory ambient temperature range of variation). In this circuit one could happily add an offset trim to U_2, preferably in the manner of Figure 5.9. As mentioned earlier, feedback bootstraps the input impedance to 250 GΩ and eliminates any gain errors from finite source impedance, up to 25 MΩ (for a gain error less than 0.01%).

U_2 drives an inverting amplifier (U_3), with R_3 chosen as a compromise between heat-produced thermal offsets in U_2 and bias-current offset errors in U_3. The value chosen keeps heating down to 5 mW (at 7.5 V output, the worst case), which works out to a temperature rise of 0.8°C (the op-amp has a thermal resistance $R_{\Theta JA}$ of about 0.15°C/mW, see §9.4), with a consequent maximum voltage offset of $\Delta V_{OS} = TCV_{OS}\Delta T = 0.12\,\mu$V. The 11 kΩ

[17] In fact, if noise is of primary concern you could substitute the ×4 quieter INA103 instrumentation amplifier at the front-end, paying the price in input offset current: a whopping 1 μA (thus $\pm 350\,\mu$V) of static offset voltage created by the 350 Ω differential source resistance here.

source impedance seen by U_3 results in an error due to bias-current offset, but, with U_3 inside a feedback loop with U_4 and U_5 trimming the overall offset to zero, all that matters is the drift in the current error term. The OPA277 provides a graph of typical bias-current change with temperature (not often specified by manufacturers), from which the error result of 0.2 μV/4°C in the error budget is calculated. Reducing the value of R_3 would improve this term, at the expense of the heating term in U_2.

The dc input impedance of U_3 comes closer to presenting a problem. To estimate the error, we compare U_3's differential input impedance of 100 MΩ with the worst-case (i.e., with gain set for ×100) impedance seen driving its input. The latter is just the feedback resistance (1M) divided by the loop gain G_{OL}/G_{CL}, thus 10 Ω. So the worst-case loading effect is 1 part in 10^7, three orders of magnitude less than 0.01% error. This is one of the toughest examples we could think of, and even so the op-amp input impedance presents no problem, thus demonstrating that, in general, you can ignore the effects of op-amp input impedances.

Drifts in offset voltage in both U_2 and U_3 over time, temperature, and power-supply variations affect the final error equally and are tabulated in the budget. It is worth pointing out that they are all automatically cancelled at each "zeroing" cycle, and only short-term drifts matter anyway. These errors are all in the microvolt range, thanks to a good choice of op-amp. U_4 has larger drifts, but it must be a FET type to keep capacitor current small, as already explained. Note that errors in U_4's output are amplified by the gain setting of U_3; thus they are specified as *input* errors in the budget.

Note the general philosophy of design that emerges from this example: you work at the problem areas, choosing configurations and components as necessary to reduce errors to acceptable values. Tradeoffs and compromises are involved, with some choices depending on external factors (e.g., the use of a FET-input op-amp for U_2 would be preferable for source impedances greater than about 10 kΩ).

5.8 Amplifier output errors

As we discussed in Chapter 4, op-amps have some serious limitations associated with the output stage. Limited slew rate, output crossover distortion (§2.4.1A), and finite open-loop output impedance can all cause trouble, and they can cause precision circuits to display astoundingly large errors if not taken into account.

5.8.1 Slew rate: general considerations

As we mentioned in §4.4.1K, an op-amp can swing its output voltage only at some maximum rate. This effect originates in the frequency-compensation circuitry of the op-amp, as we will soon explain in a bit more detail. One consequence of a finite slew rate is to limit the output swing at high frequencies to a maximum of $V\text{pp} = S/\pi f$, as shown in §4.4.2 and plotted here in Figure 5.11.

A second consequence is best explained with the help of a graph of slew rate versus differential-input signal (Figure 5.12). The point to be made here is that a circuit that demands a substantial slew rate must operate with a substantial voltage error across the op-amp's input terminals. This can be disastrous for a circuit that pretends to be highly precise: the feedback loop is in error, more so as the output slews more rapidly, thus producing a distorted output waveform. (Look ahead at the measured distortion plots in Figure 5.19 on page 311 to see this effect, for example in the LT1013 for which S=0.8 V/μs.)

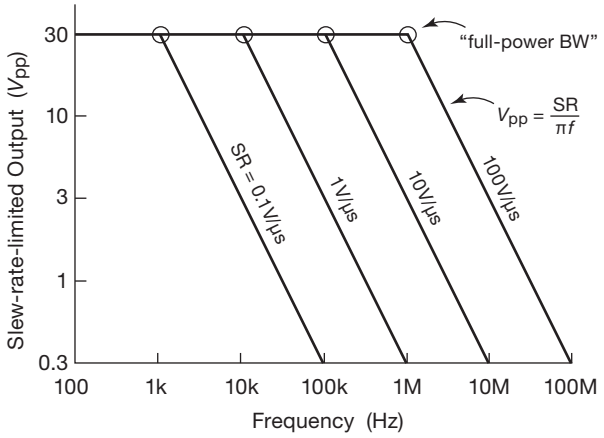


Figure 5.11. Maximum output swing versus frequency.

Let's look at the innards of an op-amp in order to get some understanding of the origin of slew rate (see §*4x.9* for a more extensive discussion). The vast majority of op-amps can be summarized with the notional "Widlar circuit" shown in Figure 5.13. A differential input stage,[18] loaded with a current mirror, drives a stage of large voltage gain with a compensation capacitor from output to input. The output stage is a unity-gain push–pull follower. The compensation capacitor C is chosen to bring the open-loop gain of the amplifier down to unity before the phase shifts

[18] We've simplified it slightly: the input stage of Widlar's original LM101 used a *pnp* differential pair, but it was configured as a common-base amplifier driven by an *npn* follower pair.

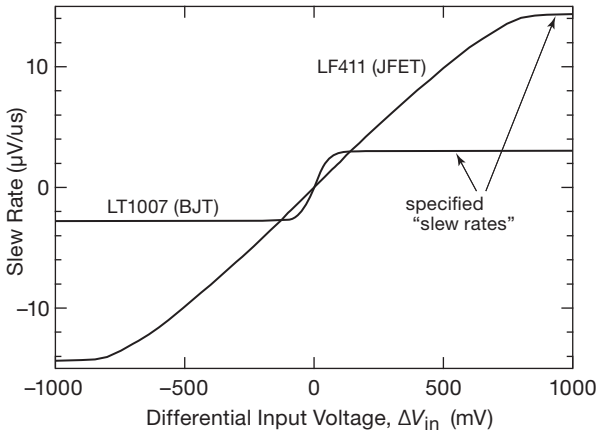


Figure 5.12. A substantial differential-input voltage is required to produce the full op-amp slew rate, as shown in these measured data. For BJT-input op-amps it takes ∼60 mV to reach full slew rate; for JFETs and MOSFETs it's more like a volt.

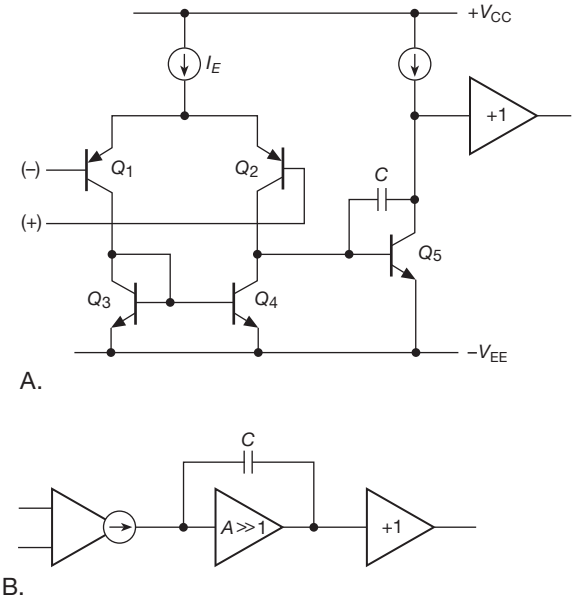


Figure 5.13. Typical op-amp internal compensation scheme.

caused by the other amplifier stages have become significant. That is, C is chosen to put f_T, the unity-gain bandwidth, near the frequency of the next amplifier rolloff pole, as described in §4.9. The input stage has very high output impedance, and it looks like a current source to the next stage.

The op-amp is slew-rate limited when the input signal drives one of the differential-stage transistors nearly to cutoff, driving the second stage with the total emitter current I_E of the differential pair. For a BJT input stage this occurs with a differential input voltage of about 60 mV, at which point the ratio of currents in the differential stage is 10:1. At this point Q_5 is slewing its collector as rapidly as possible, with all of I_E going into charging C. The transistor Q_5 and C thus form an integrator, with a slew-rate-limited ramp as output. It's not hard to derive an expression for the slew rate, knowing how bipolar transistors work – see the discussion in §*4x.9*. The bottom line is that the classic BJT-input op-amp circuit of Figure 5.13 has a slew rate S given by $S \approx 0.3 f_T$.

To get a higher slew rate, then, you can choose an op-amp with greater bandwidth f_T; if you are operating at closed-loop gains greater than unity, you can use a decompensated op-amp (with its higher f_T value). But there are ways (as explained in §*4x.9*) to beat the limit $S \approx 0.3 f_T$ (which assumed a unity-gain-compensated op-amp with a BJT differential input configured for maximum gain, i.e., with R_E=0). Namely: (a) use an op-amp with reduced input-stage transconductance (either a FET-input op-amp or a BJT-input op-amp with emitter degeneration); (b) use an op-amp with a different input-stage circuit, specifically designed for enhanced slew rate – examples are the "cross-coupled transconductance reduction" technique (used in the TLE2142 family; see the cross-coupled input stage circuit shown in §*4x.9*), and the Butler "wide dynamic range transconductance stage" (used for example in the OP275 and OP285; see the Butler input stage circuit in §*4x.9*); (c) use a current-feedback (CFB) op-amp, or a CFB variant (with a buffered inverting input) that mimics an ordinary voltage-feedback (VFB) op-amp.

These tricks work. If we define an enhancement factor m (i.e., S=$0.3 m f_T$), the LF411 (with JFET input) plotted in Figure 5.12 has $m = 12$, compared with the bipolar LT1007 (m=1.0); the TLE2141 (with cross-coupled BJT input stage) has $m = 25$, and the OP275/285 (with Butler input stages) have $m = 8$; the LT1210 (a CFB op-amp) has $m = 55$ with the recommended feedback resistor; and the LT1351 (a CFB in VFB's clothing) has $m = 220$.

For a deeper look at slew rate, turn to the extended discussion in Chapter *4x* (§*4x.9*).

5.8.2 Bandwidth and settling time

Slew rate measures how rapidly the output voltage can change. The op-amp slew-rate specification usually assumes a large differential input voltage (60 mV or more), which (in spite of its potential for creating output distortion) is not unreasonable, given that an op-amp whose output isn't where it's supposed to be will have its input driven

hard by feedback, assuming a reasonable amount of loop gain. Of perhaps equal importance in high-speed precision applications is the time required for the output to get where it's going following an input change. This *settling-time* specification (the time required to get within the specified accuracy of the final value and stay there; see Figure 5.14) is always given for devices such as digital-to-analog converters, where precision is the name of the game, but it is not normally specified for op-amps.

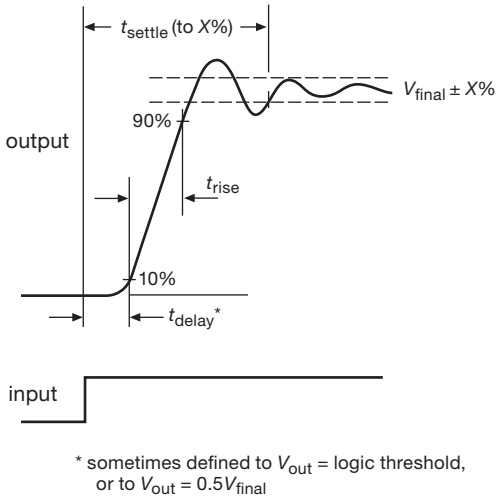


Figure 5.14. Settling time defined.

We can estimate op-amp settling time by considering first a different problem, namely, what would happen to a perfect voltage step somewhere in a circuit if it were followed by a simple *RC* lowpass filter (Figure 5.15). It is a simple exercise to show that the filtered waveform has the settling times shown. This is a useful result, because you often limit bandwidth with a filter to reduce noise (more on that later in the chapter). To extend this simple result to an op-amp, just remember that a compensated op-amp has a 6 dB/octave rolloff over most of its frequency range, just like a lowpass filter. When connected for closed-loop gain G_{CL}, its "bandwidth" (the frequency at which the loop gain drops to unity) is approximately given by

$$f_{3dB} = f_T/G_{CL} \ .$$

As a general result, a system of bandwidth B has response time $\tau \approx 1/(2\pi B)$; thus the equivalent "time constant" of the op-amp is

$$\tau \approx G_{CL}/2\pi f_T.$$

The settling time is then roughly 5–10τ.

Let's try our prediction on a real case. The TLE2141

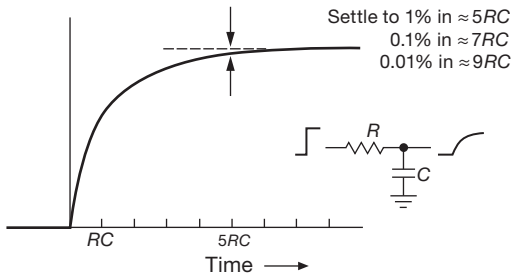


Figure 5.15. Settling time of an *RC* lowpass filter.

from TI is a precision fast-settling op-amp, with an f_T of 5.9 MHz. Our simple formula then estimates the inverting-configuration (i.e., G=2) response time to be 54 ns, thus a settling time of 378 ns (7τ) to 0.1%. This is in good agreement with the datasheet's value of 340 ns.

There are several points worth making: (a) our simple model gives us only a lower bound for the actual settling time in a real circuit; you should always check the slew-rate-limited rise time, which may dominate. (b) Even if slew rate is not a problem, the settling time may be much longer than our idealized "single-pole" model, depending on the op-amp's compensation and phase margin. (c) The op-amp will settle more quickly if the frequency compensation scheme used gives a plot of open-loop phase shift versus frequency that is a nice straight line on a log–log graph (as in Figure 5.17); op-amps with wiggles in the phase-shift graph are more likely to exhibit overshoot and ringing, as in the upper waveform shown in Figure 5.14. (d) A fast settling time to 1%, say, doesn't necessarily guarantee a fast settling time to 0.01%, since there may be a long tail (Figure 5.16). (e) There's no substitute for an actual settling-time specification from the manufacturer.

Table 5.4 lists a selection of high-speed op-amps suitable for applications that demand high f_T, high slew rate, fast settling time, and reasonably low offset voltage.

5.8.3 Crossover distortion and output impedance

Some op-amps (for example the classic single-supply LM324/358) use a simple push–pull follower output stage, without biasing the bases two diode drops apart, as we discussed in §2.4.1. This leads to "class-B" distortion near zero output, because the driver stage has to slew the bases through $2V_{BE}$ as the output current passes through zero (Figure 5.18). This crossover distortion can be substantial, particularly at higher frequencies where the loop gain is reduced; see the measured data in Figure 5.19. It is greatly reduced in op-amp designs that bias the output

Table 5.4 Representative High-speed Op-amps[x]

Part #	# per pkg[a]	Supply[p] range (V)	Supply[p] I_Q (mA)	I_{in} @25°C typ (pA)	Offset Voltage V_{os} typ (mV)	Offset Voltage V_{os} max (mV)	Offset Voltage ΔV_{os} typ (μV/°C)	e_n typ[r] ($\frac{nV}{\sqrt{Hz}}$)	GBW typ (MHz)	Slew typ (V/μs)	I_{out} typ (mA)	C_{in} (pF)	dist. graph	Swing to Supply IN +	Swing to Supply IN −	Swing to Supply OUT +	Swing to Supply OUT −	null pins	DIP avail	cost qty 25 ($US)	Comments
bipolar																					
LT1468	**1**	7–36	3.9	3nA	0.03	0.08	0.7	5	90	23	22	4	-	-	-	-	-	●	●	4.26	0.7 ppm dist
LT1360	**1**,2,4	5–36	4	0.3μA	0.3	1	9	9	50	800	34	3	-	-	-	-	-	●	●	2.75	C-Load™
LM6171	**1**,2	5–36	2.5	1μA	1.5	3	6	12	100	3600	90	-	●	-	-	-	-	-	●	2.57	VFB+CFB
AD844	**1**	9–36	6.5	0.2μA	0.05	0.3	1	9	330[g]	2000	60	2	-	-	-	-	-	●	●	5.23	CFB, comp pin
AD8021[b]	**1**	4.5–26	7	7.5μA	0.4	1	0.5	2.1	925	420	60	1	●	-	-	-	-	-	-	2.42	comp pin, 16-bits
JFET																					
OPA604A	**1**,2	9–50	5.3	50	1	5	8	10	20	25	36	10	-	-	-	-	-	●	●	2.93	3ppm, dual '2604
OPA827A	**1**	8–40	4.8	15	0.08	0.15	1.5	3.8	22	28	30	9	-	-	-	-	-	-	-	9.00	quiet, accurate
ADA4637	**1**	9–36	7.0	1	0.12	0.3	1	6.1	80	170	45	8	-	-	-	-	-	-	d	10.12	decomp, G>7
low-voltage bipolar																					
LT6220	**1**,2,4	2.2–13	0.9	15nA	0.07	0.35	1.5	10	60	20	35	2	-	●	●	●	●	-	-	1.75	SOT-23
LMH6723	**1**,2,4	4.5–13	1	2μA	1	3	-	4.3	370	600	110	1.5	-	-	-	-	-	-	-	2.03	CFB, SOT23-5
ADA4851	**1**,2,4	3–12.6	2.5	2.2μA	0.6	3.4	4	10	125	200	85	1.2	-	-	●	●	●	-	-	1.40	SOT23-5, shdn pin
LT1818	**1**,2	4–12.6	9	2μA	0.2	1.5	10	6	400	2500	70	2	-	-	-	-	-	-	-	1.35	VFB+CFB, fast
LT6200	**1**,2	3–12.6	16.5	10μA	0.2	1.2	8	0.95	165	50	70	4	-	●	●	●	●	-	-	2.99	1% dist at 50MHz
LT6200-10	**1**	3–12.6	16.5	10μA	0.2	1.2	8	0.95	1600	450	70	4	-	●	●	●	●	-	-	2.99	fastest RRIO
OPA698[e]	**1**	5–13	16	3μA	2	5	15[m]	5.6	450	1100	55	1	n	-	-	●	●	-	-	4.14	clipping
low-voltage JFET																					
OPA656	**1**	9–13	14	2	0.25	1.8	2	7	230	290	50	2.8	●	-	-	-	-	-	-	5.59	low $e_n \cdot C_{in}$ noise
OPA657	**1**	9–13	14	2	0.25	1.8	2	7	1600	700	50	4.5	-	-	-	-	-	-	-	10.01	decomp, G>7
ADA4817	**1**,2	5–10.6	19	2	0.4	2	7	4	1050	870	70	1.5	-	●	-	-	-	-	-	4.93	lowest $e_n \cdot C_{in}$
CMOS																					
AD8616	**2**	2.7–6	1.7	0.2	0.02	0.06	1.5	7	24	12	150	7	-	●	●	●	●	-	-	1.52	'8615 SOT23-5
LMP7717	**1**,2	1.8–6	1.15	0.05	0.01	0.15	1	6.2	88	28	15	15[c]	-	-	●	●	●	-	-	2.18	decomp, G>10
OPA350	**1**,2,4	2.5–7	5.2	0.5	0.15	0.5	4	7	38	22	40	6.5	-	●	●	●	●	-	●	1.67	6ppm

Notes: (a) **boldface** indicates number in a package for the part number listed. (b) for G<20 use ext C_C chosen to set f_{3dB}=200MHz. (c) 15mA sinking, 47mA sourcing. (d) for DIP-8 see OPA637. (e) OPA699 decomp. (g) at G=10. (m) max. (n) distortion plot for OPA699. (p) I_Q, typical, per amplifier. (r) at 1kHz. (x) see also the fast op-amp table in *Chapter 4x*.

push–pull pair into slight conduction ("class-AB"), for example the LT1013, which is an improved version of the LM324. The right choice of op-amp can have enormous impact on the performance of low-distortion audio amplifiers. Perhaps this problem has contributed to what the audiophiles refer to as "transistor sound." Some modern op-amps, particularly those intended for audio applications, are designed to produce extremely low crossover distortion. Examples are the LT1028, the AD797, and the excellent "LME49000" series from NSC, e.g., the LME49710. The latter, for example, has less than 0.0001% distortion over the full audio band of 20–20 kHz. (That's the claim, anyway; we may be overly gullible!) These amplifiers all have very low noise voltage, as well; the LT1028, for example, vies for the title of world noise-voltage champion, with $e_n = 1.7\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ (max) at 10 Hz. See the expanded plots of op-amp distortion in Figures 5.43 and 5.44, where various op-amps compete for the title of king of low distortion. High-voltage op-amps have an advantage below 10 kHz, whereas the low-voltage types have an advantage above 200 kHz.

The open-loop output impedance of a typical split-supply op-amp is highest when the output is near ground, because the output transistors are operating at their lowest current into the (ground-returned) load. The output impedance also rises at high frequency as the transistor gain drops off, and it may rise slightly at very low frequencies because of thermal feedback on the chip.

It is easy to neglect the effects of finite open-loop output impedance, thinking that feedback will cure everything. But when you consider that some op-amps have open-loop output impedances of a few hundred ohms, it becomes clear that the effects may not be negligible, especially at low to moderate loop gains. Figures 5.20 and 5.21 shows some typical graphs of op-amp output impedance, both with and without feedback.

The finite output impedance contributes also to instability when driving capacitive loads, as we discussed in

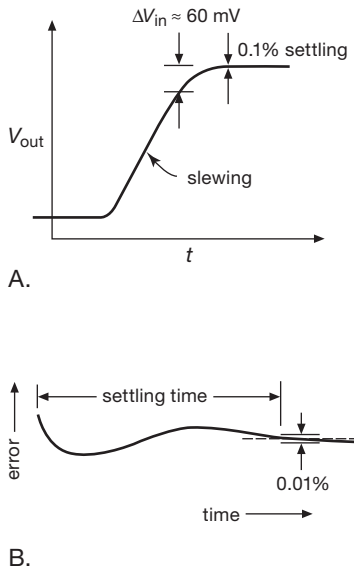


Figure 5.16. A. The slewing decreases when input error approaches 60 mV. B. Settling to high precision can be surprisingly lengthy.

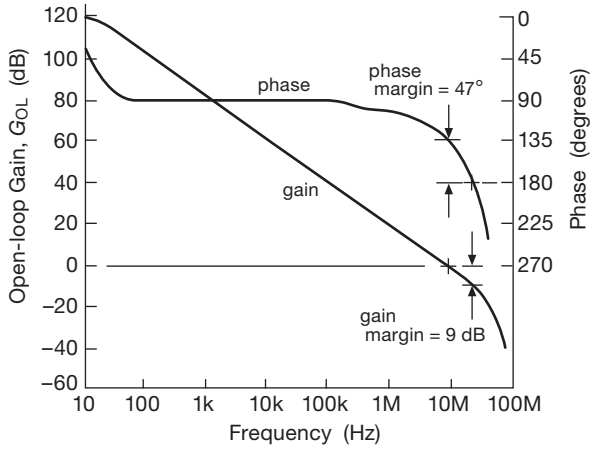


Figure 5.17. OP-42 gain and phase versus frequency.

§4.6.2, owing to the additional lagging phase shift within the feedback loop that is created by R_{out} in combination with C_{load}. Several common solutions were shown in Figure 4.78, including a split feedback path or the inclusion of a unity-gain buffer within the loop. The latter is worth a mention here.

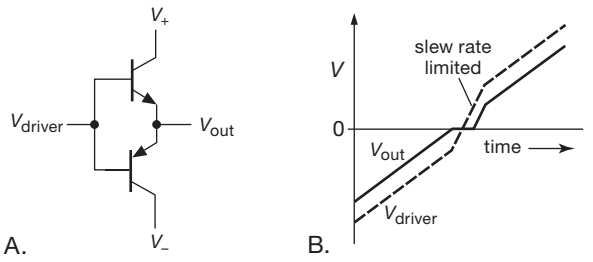


Figure 5.18. Crossover distortion in class-B push–pull output stage.

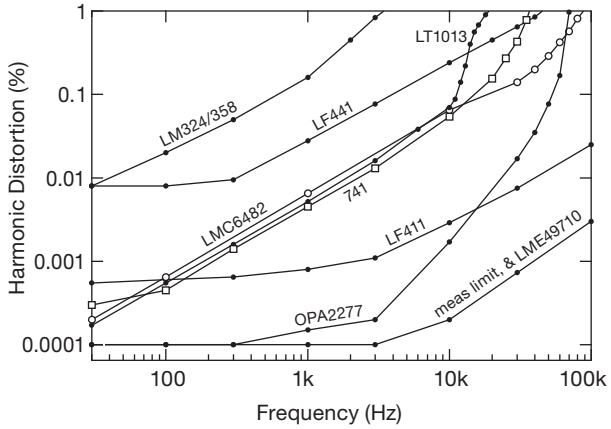


Figure 5.19. Measured harmonic distortion versus frequency for several popular op-amps (1 Vrms output, unloaded). See also Figures 5.43 and 5.44.

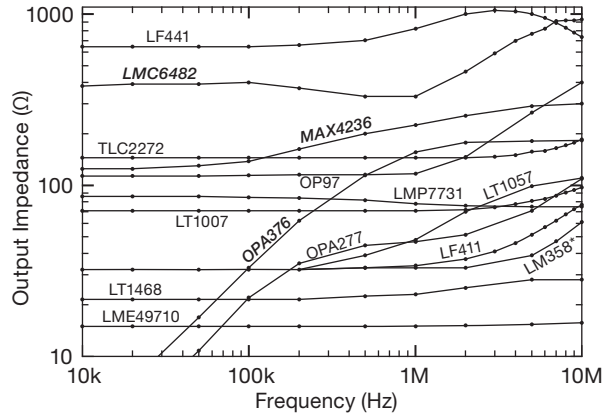


Figure 5.20. Measured open-loop output impedance versus frequency for a selection of op-amps. Parts shown in **bold** have CMOS output circuits. * With output pulldown resistor.

5.8.4 Unity-gain power buffers

If the technique of split feedback paths is unacceptable, one solution is to add a unity-gain high-current buffer inside

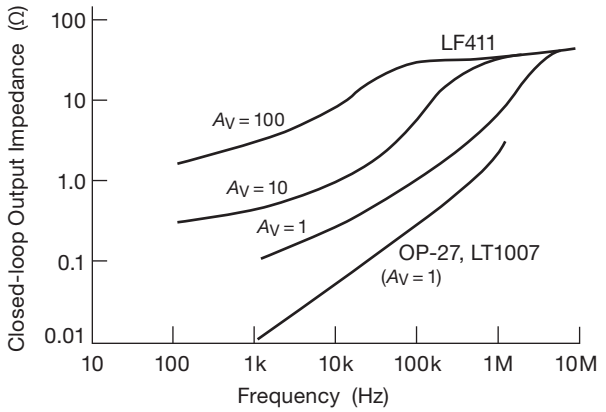


Figure 5.21. Closed-loop output impedance versus frequency for the LF411 and LT1007 op-amps, from manufacturers' datasheets.

the loop, as, for example, in the general-purpose laboratory amplifier of Figure 4.87. The LT1010 in that circuit has adequate bandwidth (>10 MHz) in which it adds little phase shift; thus it can be within the feedback loop with a small amount of external compensation.

These "power boosters" can, of course, be used for loads that require high current (for example, driving a terminated coax cable), regardless of whether or not there are problems with capacitance. And unity-gain buffers are useful even with loads of only moderate current, in the context of precision circuit design, because they prevent thermal drifts by keeping the heat out of the low-offset amplifier. You can see a couple of examples of power boosters in Figures 5.47 and 13.119, as well as in the discussion in Chapter *4x*.

5.8.5 Gain error

There's one more error that arises from finite open-loop gain, namely, an error in closed-loop gain owing to finite loop gain.

We calculated in Chapter 2 (§2.5.2) the expression for closed-loop gain in a feedback amplifier, $G = A/(1+AB)$, where A is the open-loop gain and B is the "gain" of the feedback network. You might think that the $A \geq 100$ dB or so of op-amp open-loop gain is plenty, but when you try to construct extremely precise circuits you are in for a surprise. From the preceding gain equation it is easy to show that the "gain error," defined as

$$\delta_G = \text{gain error} \equiv \frac{G_{\text{ideal}} - G_{\text{actual}}}{G_{\text{ideal}}},$$

is just equal to $1/(1+AB)$ and ranges from 0 for $A = \infty$ to 1 (100%) for $A = 0$.

Exercise 5.2. Derive the foregoing expression for gain error.

The resulting frequency-dependent gain error is far from negligible. For instance, an LF411 with its 106 dB of low-frequency open-loop gain will have a gain error of 0.5% at low frequencies when configured for a closed-loop gain of 1000. Worse yet, the open-loop gain drops 6 dB/octave above 20 Hz, so our amplifier would have a gain error of a whopping 10% at 500 Hz! Figure 5.22 plots calculated gain error versus frequency for the OPA277, with its extraordinary 140 dB of low-frequency open-loop gain, when configured for closed-loop gains of 100 and 1000. It should be obvious that you need plenty of gain and a high f_T to maintain accuracy at even moderate frequencies.

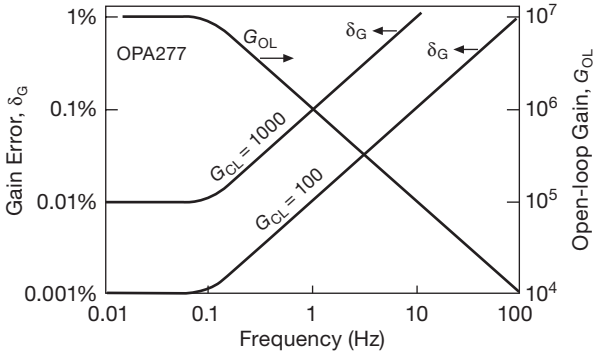


Figure 5.22. OPA277 gain error.

We plotted these curves from the graph of open-loop gain versus frequency given in the datasheet. Even if your op-amp datasheet provides a curve, it's best to work backward from the specified f_T (i.e., the datasheet's GBW; see Figure 5.42 and associated discussion) and dc open-loop gain, figuring the open-loop gain at the frequency of interest, and thus the gain error (as above) as a function of frequency. This procedure yields

$$\delta_G = \frac{1}{1 - jBf_T/f} \approx \frac{f}{Bf_T},$$

where B is, as usual, the gain of the feedback network, and the approximation is valid for the useful case $Bf_T/f \gg 1$. Of course, in some applications, such as filters, B may also depend on frequency.

Exercise 5.3. Derive the foregoing result for $\delta_G(f)$.

5.8.6 Gain nonlinearity

Op-amps have lots of open-loop gain at low frequencies, and the excess (G_{OL}/G_{CL}) is the loop-gain feedback mechanism that contributes to accuracy and the reduction of the op-amp's intrinsic nonlinearities, as discussed first

in §2.5.3. Ideally, then, we want lots of open-loop gain in a precision circuit. And that's why auto-zero amplifiers (§5.11) and precision op-amps are built with high open-loop gains, for example ∼160 dB for the auto-zero LMP2021, and ∼150 dB for the precision LT1007.

For *accuracy*, then, we want lots of loop gain. For purposes of *linearity*, however, it's OK to have less loop gain – what matters more is the intrinsic linearity of the op-amp, combined with an open-loop gain characteristic that changes linearly (if at all) with output swing. The intrinsic linearity is strongly influenced by the output-stage design, particularly when the amplifier is driving a load: crossover distortion is always bad, as is an output stage that is asymmetric in its source/sink capabilities (like the LM358, with a Darlington *npn* pullup and single *pnp* pull-down). And a poor layout within the chip can create nonlinearities from the thermal offsets produced by local heating when driving a load.

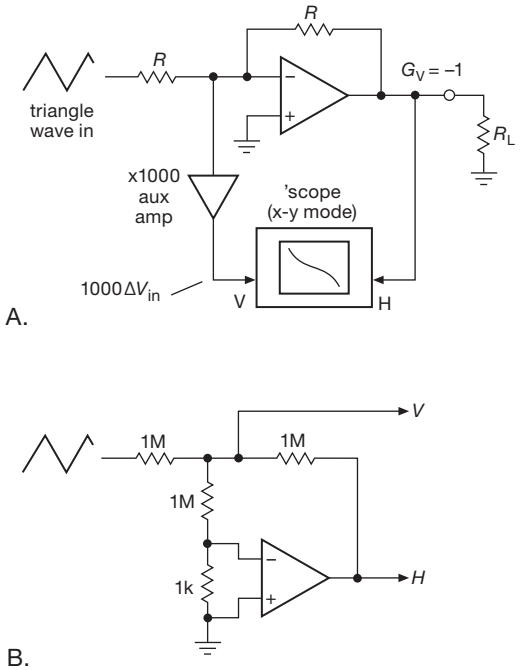


Figure 5.23. Low-frequency gain nonlinearity test circuit. A. Notional: an auxiliary amplifier makes visible the μV-scale differential input voltage versus output swing. B. Circuit used by Pease for measurements in AN-1485 (see footnote on the current page).

In a nice set of measurements, Bob Pease[19] explored the low-frequency gain nonlinearity of a selection of op-amp types (sadly, none from other manufacturers), operating as unity-gain inverters with a full-swing output; he made measurements both when the op-amps were unloaded, and when they were driving a 1 kΩ load. The basic scheme is shown in Figure 5.23A, where a 'scope looks at the amplifier input error versus output swing. For Pease's actual measurements he used the subtle variant in Figure 5.23B, in which the op-amp amplifies its error by ×1000, delivering the bad news directly.

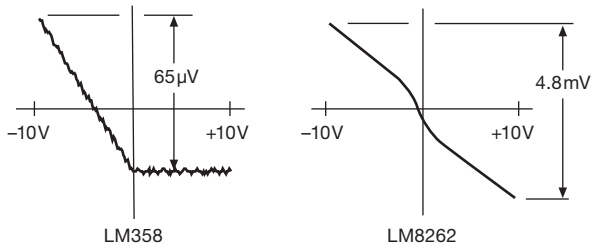


Figure 5.24. Gain nonlinearity traces for two op-amps with output-stage deficiencies. In these *x*–*y* displays the vertical axis shows the (small) differential input signal required to produce the (full-swing) output signal indicated on the horizontal axis. To estimate gain error, divide the vertical deviation from a best-fit straight line by the full-swing output.

The sorts of things you see (with a loaded op-amp) are shown in Figure 5.24, where we've sketched Pease's traces for the aforementioned LM358 (afflicted with asymmetric source/sink) and the LM8262 (fast, but afflicted with some crossover distortion). An exemplary op-amp like the very low distortion LM4562 presents an ideal nearly-horizontal straight line. The LF411 (single) and LF412 (dual), our JFET jellybeans, show an interesting contrast: according to Pease, the LF411 chip layout is sub-optimal (in terms of gain and thermal effects), with great effort rewarded by better results in the dual LF412.

Here are some of his summary results for op-amps driving a somewhat lighter load (4 kΩ). In general, the measured gain nonlinearity, when *unloaded*, was far smaller than these listed values. Keep in mind that these measurements were made at very low frequencies (generally just a few hertz), where the loop gain is maximum.

[19] National Semiconductor App Note AN-1485: *The Effect of Heavy Loads on the Accuracy and Linearity of Operational Amplifier Circuits (or, "What's All this Output Impedance Stuff, Anyhow?")*. Gain nonlinearity data can be found in some datasheets, for example the AD620 instrumentation amplifier.

HV BJT (V_{sig}=±10 V)		
LM8262	12 ppm	xover dist.
LM358	1 ppm	asym. output stage
LF411	1.4 ppm	poor layout – thermal
LF412	0.3 ppm	better layout
LM4562	0.025 ppm	pro-audio, $G_{OL} = 10^7$

CMOS RRO ($V_{sig} = \pm 4$ V)		
LMC6482	1.1 ppm	jellybean
LMC6062	0.2 ppm	precision

CMOS auto-zero ($V_{sig} = \pm 2$ V)		
LMP2012	0.2 ppm	precision

5.8.7 Phase error and "active compensation"

We've talked mostly about the *gain* error caused by limited op-amp bandwidth (and therefore falling loop gain with increasing frequency). But limited loop gain also produces *phase* error, which can be important in applications such as video, interferometry, and so on. And the effect is not at all negligible – recall (§1.7.9) that a single *RC*-like rolloff creates a phase shift of ∼6° at a frequency of $f_C/10$, and ∼0.6° at $f_C/100$; the latter is two full decades below the −3 dB breakpoint. If we model an op-amp's rolloff of open-loop gain similarly (a "single-pole" rolloff), we can expect comparable phase shifts.

In this approximation, the resultant phase shift for an op-amp voltage amplifier is thus given by

$$\phi = \tan^{-1}\left(\frac{f}{f_C}\right) \approx \frac{f}{f_C} \text{ (radians)},$$

where the −3 dB breakpoint f_C is the frequency at which the loop gain has fallen to unity: $f_C = f_T/G_{CL}$. Here G_{CL} is the closed-loop gain (as set by the feedback network), and f_T is the gain–bandwidth product (GBW) of the op-amp (for a single-pole rolloff that's the same as the frequency at which the open-loop gain is unity; but for typical op-amps, with more complicated rolloffs, you want to use the GBW figure). Multiply by 57.3 (180/π) to get the answer in degrees. The approximate result (the last expression) is reasonably accurate for small-to-moderate phase shifts, up to 0.5 radian, say.

There are several ways to address this problem. The simplest is to use an amplifier of greater bandwidth. If you don't want to (or cannot) do that, another possibility is to introduce an *RC* network in the feedback path to cancel the phase error (in *s*-plane language, you are introducing a zero to cancel a pole). This can be effective, but requires "tuning" of the compensation network to match the frequency response of the particular op-amp specimen itself; and because the op-amp's characteristics change with temperature, the network must do likewise. A third possibility is to cascade two stages, each configured for lower gain (and therefore smaller phase error).

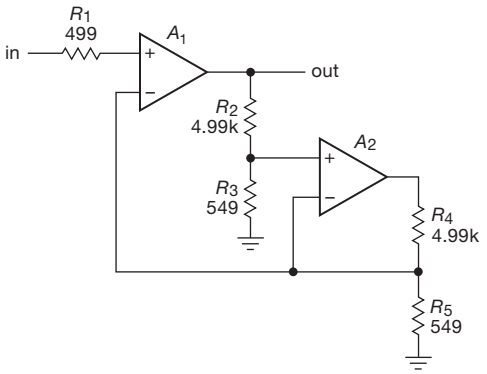


Figure 5.25. Phase-error reduction by means of "active compensation," exploiting the closely matched frequency responses of dual op-amp pair A_1 and A_2.

But an elegant solution is *active compensation*, a clever technique that uses a second matching op-amp to create a replica of the error, which can then be subtracted from the main amplifier. Figure 5.25 shows how this can be done.[20] The bandwidth of the main amplifier is unchanged, but its phase error is reduced dramatically, as shown in the SPICE simulation and measured data of Figure 5.26. There is some peaking in the amplitude response – about +3 dB at the frequency at which the phase error is 45° – but generally insignificant within the frequency range over which the phase error is small (e.g., +0.1 dB at f=0.1f_T/G_{CL}). A circuit configured for low closed-loop gain will generally exhibit greater peaking.[21] Under the assumption that the

[20] See "Active Feedback Improves Amplifier Phase Accuracy," by J. Wong, *EDN Magazine*, 17 Sept 1987; reprinted as Analog Devices AN-107. Wong credits the idea to Soliman in a 1979 paper, and Soliman credits the idea to Brackett and Sedra in a 1976 paper. But Wong's paper is the most useful reference for understanding the configuration of Figure 5.25.

[21] In SPICE simulations we found that the peaking increased to ∼7 dB for the LF412 model configured for G=2; this can be tamed by adding a compensation capacitor C_c across feedback resistor R_2. Choosing C_c to match the op-amp's f_T (i.e., $C_c = 1/2\pi f_T R_2$) reduced the peaking to 4 dB, at the expense of tripling the (pretty small) phase error. In his article, James Wong warns that the technique may result in an unstable amplifier for low gains, below G=5 for example. He shows also how the technique can be further improved if A_2 is made from two amplifiers.

amplifiers are matched, it can be shown that this technique produces a phase shift given approximately by

$$\phi \approx \left(\frac{f}{f_C}\right)^3 \text{ (radians)},$$

again accurate for small-to-moderate phase shifts ($\lesssim 30°$, say, i.e., the small-angle approximation).

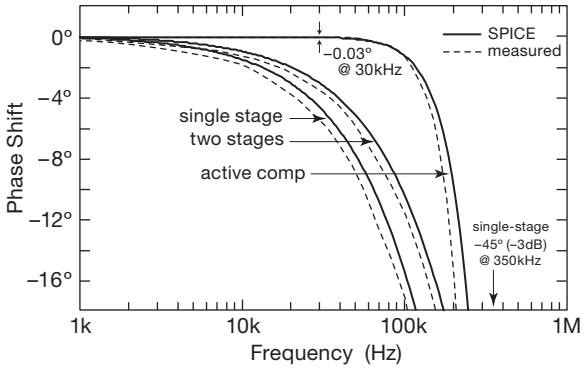


Figure 5.26. SPICE simulation and measured data of the phase shift versus frequency for the circuit of Figure 5.25, implemented with an LF412 dual JFET op-amp. For comparison the analogous data are plotted for both a single G=10 stage and for a cascade of two stages, each with G=$\sqrt{10}$. The measured part's f_C was 295 kHz, somewhat lower than the SPICE model's 350 kHz.

Real op-amps are not perfectly matched. To see how a mismatch in f_T affects the phase compensation, we ran a SPICE simulation with an f_T mismatch of $\pm 10\%$ (Figure 5.27). Evidently our test-bench part, chosen quite at random (dip fingers into parts bin, grasp first part touched, extract, measure), has a considerably better f_T matching, as suggested by Wong: "Monolithically matched dual or quad op-amps can provide the frequency-matching characteristics (to within 1% to 2%) necessary for the success of the active-feedback approach."[22]

It's interesting to compare predicted phase shifts for several scenarios mentioned at the outset: (a) a single amplifier of given bandwidth (call it f_{T0}, 3 MHz for the LF412), configured for a closed-loop gain $G = 10$; (b) two cascaded stages, each with $G = \sqrt{10}$; (c) the active compensation method of Figure 5.25; and (d) a single amplifier of greater bandwidth ($10 f_{T0}$, say). Here are the calculated results:

[22] We went back to the bench and measured a handful of LF412 dual op-amps. Among different specimens the f_T values ranged over $\pm 20\%$, but within any single part the f_T's of its two op-amps matched typically to 0.1%, with one outlier showing a 1.5% mismatch.

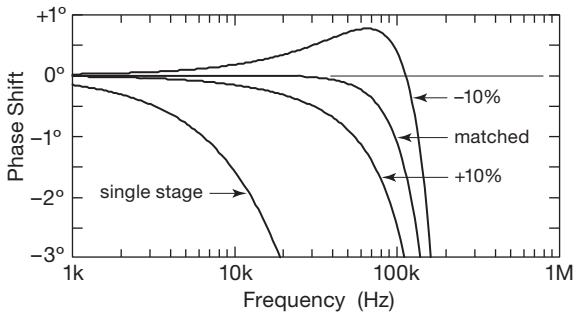


Figure 5.27. Active compensation of phase error requires matched op-amp bandwidths, as seen in this SPICE simulation for which the f_T of the compensation op-amp A_2 has been varied $\pm 10\%$ relative to that of the signal path op-amp A_1.

	Single, G=10	2-stage, each G=$\sqrt{10}$	Active, comp	Single, f_T=10f_{T0}
$0.001 f_{T0}$	−0.57°	−0.36°	−0.00006°	−0.006°
$0.003 f_{T0}$	−1.7°	−1.1°	−0.0015°	−0.17°
$0.01 f_{T0}$	−5.7°	−3.6°	−0.06°	−0.57°
$0.03 f_{T0}$	−17.2°	−10.9°	−1.5°	−1.7°
$0.1 f_{T0}$	−45°	−36°	−45°	−5.7°

It's clear that the remarkable (and underutilized) technique of active compensation represents an efficient use of resources. The noninverting $G = 2$ case looks especially useful, e.g., to drive backterminated 75 Ω video cables.[23]

5.9 RRIO op-amps: the good, the bad, and the ugly

In Chapter 4 (§§4.4.1, 4.4.2, and 4.6.3) we introduced rail-to-rail op-amps, including (a) op-amps that operate properly with common-mode inputs over the full supply voltage range (RRI), (b) op-amps that can swing their outputs over the full supply range (RRO), and (c) op-amps that can do both (RRIO). With lower supply voltages increasingly in vogue, you see many new op-amps with these desirable capabilities.

Desirable, but to be used with caution. These benefits come at a cost, which we'll discuss here in the context of precision design (with further discussion in Chapter *4x*). In circuits that strive for accuracy there are some hidden compromises in the designs of these op-amps about which the datasheet may be, uh, understated (or completely silent). Here are the important ones.

[23] Or sometimes called "double-terminated", as in Figure 12.110. We suggest trying your amplifier of choice, taking care to terminate the second op-amp with 150 Ω.

5.9.1 Input issues

A. Input-current crossover

Most RRI op-amps use a complementary pair of differential input stages, with their inputs driven in parallel, to handle the full supply voltage range (Figure 5.28). This causes a shift in input current because the signal path changes from one pair to the other, as seen clearly in Figure 5.7 (particularly the BJT-input RRI op-amps: LT1630, LM6132). An abrupt change in input current causes input errors from finite driving impedance. Some RRI op-amps avoid this problem by using an on-chip charge pump to generate a supply voltage beyond the rail, so a single input amplifier allows rail-to-rail inputs. Examples are the OPA360-series,[24] the AD8505 and ADA4505, the MAX4162-series, and the MAX4126-series. Except for the BJT-input MAX4126, these all use MOS inputs.

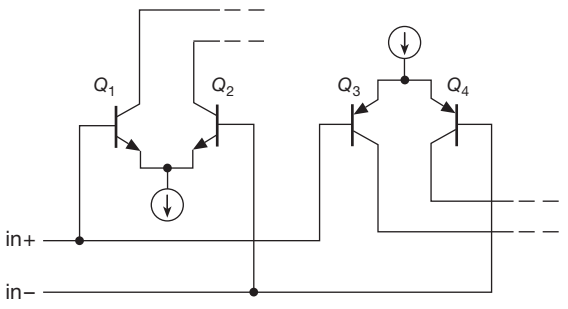


Figure 5.28. A typical rail-to-rail input circuit consists of a pair of complementary differential amplifiers, with downstream circuitry to select the active pair's output.

In situations where you need RRO but don't need full rail-to-rail *input* (a voltage amplifier with $G>2$, say), be sure to consider an RRO op-amp with input extending to the negative rail only (sometimes called "ground sensing"). Note also that, by using an op-amp in an inverting circuit configuration, you avoid this problem completely (but you probably would not choose an RRI op-amp for such a configuration anyway).

B. Input offset-voltage crossover

The dual input stages of RRI op-amps cause similar mischief in terms of their input offset voltage V_{OS}, as seen in Figure 5.29. The abrupt change can occur close to either end of the supply range, as seen in the LMP7701 and LMP7731 op-amps from the same manufacturer. These curves were adapted from their respective datasheets, which typically display a figure showing a tangle of overlapping curves measured on multiple op-amp samples (if they're willing to show any data at all about this seamy topic). Here you can see by comparison the uncomplicated (and downright *boring*) behavior of an RRI op-amp with an on-chip charge pump powering a single input amplifier. This variation of V_{OS} with V_{CM} is not only undesirable, it is also unpredictable, as you can see in Figure 5.30.

This problem is nicely circumvented by the use of an inverting configuration, which holds constant the common-mode input voltage. More generally, always consider using an inverting configuration to prevent *any* circuit misbehavior caused by op-amp dependence on V_{CM}.[25]

The OPA350 datasheet shows a nice example (Figure 5.31) of input crossover effects in RRI op-amps, namely a 17 dB increase in audio distortion in a G=1 follower when the 3 Vpp sinewave input is shifted upward to enter the crossover region.[26] The same graph illustrates nicely how increased closed-loop gain causes increased distortion owing to decreased loop gain.

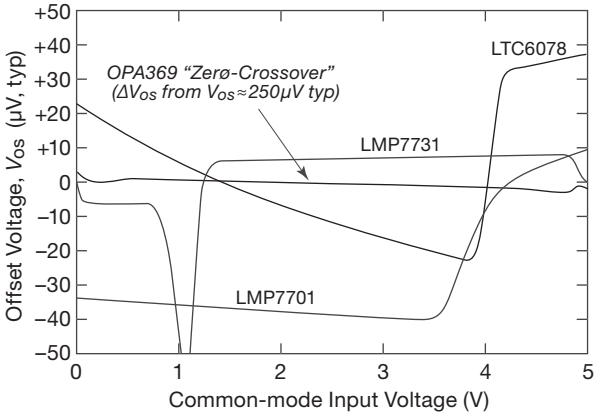


Figure 5.29. Op-amps with rail-to-rail inputs usually exhibit a shift of V_{OS} as the input voltage passes control from one input pair to the other. The OPA369 circumvents this by using a single input pair, powered beyond the rail by an on-chip charge pump.

5.9.2 Output issues

A. Output impedance

The output stage of a conventional (not RRO) op-amp is ordinarily a complementary push–pull follower (or some

[24] Playfully named "Zer∅-Crossover" amplifiers, or ZCOs.

[25] As Jim Williams liked to say, "Use an inverting configuration, unless you can't."

[26] See also Bonnie Baker's article (in the *Baker's Best* series) "Where did all that racket come from?" in *EDN Magazine*, 23 April 2009, available at edn.com.

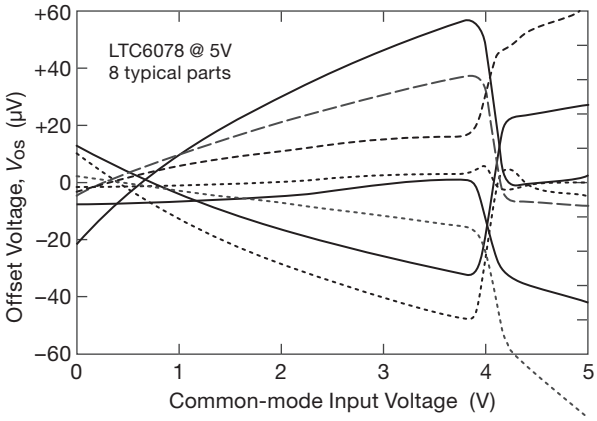


Figure 5.30. The shift of offset voltage in an RRI op-amp can be unpredictable (even as to the *sign* of the effect!), as seen in these data, adapted from the unusually forthcoming manufacturer's datasheet.

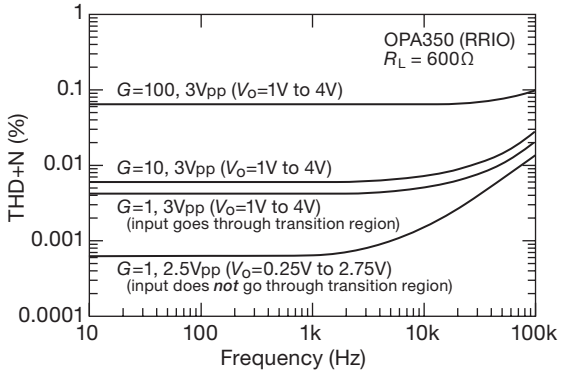


Figure 5.31. Distortion versus frequency for the OPA350 RRIO op-amp. The two lowest curves show the dramatic increase in distortion when the input signal enters the input crossover region. Increasing closed-loop gain causes further distortion because of reduced loop gain.

variation thereupon), biased with some conduction overlap to prevent crossover distortion at mid-supply (see §5.8.3). By contrast, the output complementary pair in an RRO op-amp is configured as a push–pull common-source *amplifier*; see Figure 5.32. That's necessary for the output to reach the rails (absent a second set of beyond-the-rails supply voltages). But it creates problems, owing to its inherently high output impedance.

The high Z_{out} means that the output-stage gain (and therefore the loop gain) depends on the value of load resistance; and a capacitive load creates large phase shifts, compromising the loop stability (see, for example, Figure 4.79). These problems are addressed in part by use of

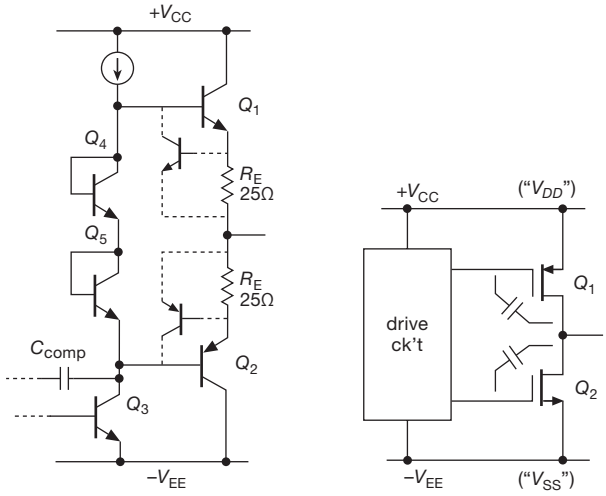


Figure 5.32. The classic (not rail-to-rail) op-amp output stage is a push–pull unity-gain follower with inherently low output impedance, biased (via Q_4Q_5) to suppress crossover distortion; it has straightforward biasing and current limiting. By contrast, a rail-to-rail output stage (usually implemented in CMOS) is a push–pull common-source amplifier ($G > 1$) with inherently high output impedance; it requires considerable trickery in its biasing and current limiting.

internal feedback around the output stage (the capacitors in Figure 5.32B), so that the gain and output impedances are reasonably well controlled except at low frequencies – see for example Figures 5.33 and 5.34.[27]

B. Saturation at the rails

Some "rail-to-rail output" op-amps (in particular, those with a BJT output stage) don't quite make it the last few millivolts; that's because the output transistor's saturation voltage is not zero. (This is not usually a problem with MOSFET outputs, which look like an R_{on} to one rail or the other when driven full range.) Usually this doesn't matter, because what you care about most is getting full use of a limited supply voltage (when operating with low-voltage supplies). But it does matter, for example, if you've got a single-supply setup in which the op-amp is driving an ADC whose conversion range goes clear down to ground.

In such a case be sure to check the specifications. Some

[27] It's unusual to see plots (or even tabulated values) of *open-loop* output impedance on datasheets; and in cases where a graph is shown, it rarely extends to very low frequencies. It is likely that other op-amps, including some with conventional (follower) output stages, also exhibit a rise in open-loop output impedance at very low frequencies. This is rarely of concern, though, owing to the very high loop gain down there.

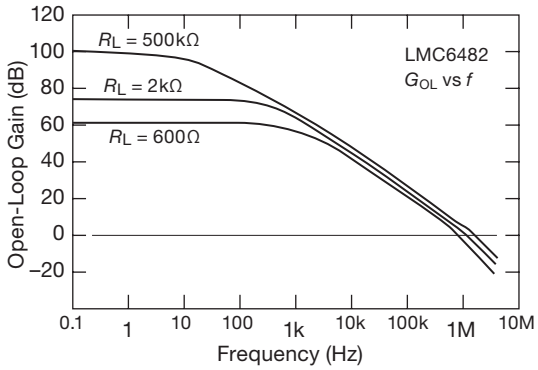


Figure 5.33. The low-frequency gain of rail-to-rail output op-amps may depend strongly on load resistance, as seen here for the LMC6482.

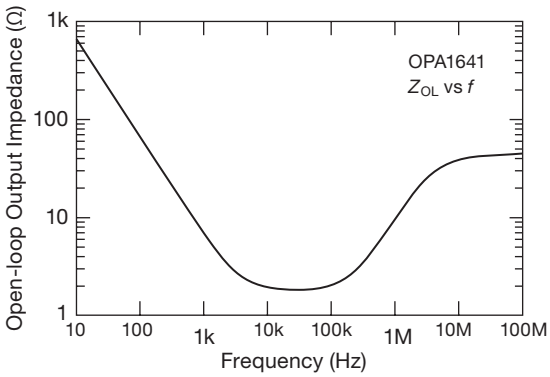


Figure 5.34. For some RRO op-amps the open-loop output impedance rises markedly at low frequencies, owing to internal capacitive negative feedback around the output stage that becomes ineffective at low frequencies. But, not to worry, there's lots of loop gain at low frequencies in typical op-amp applications.

RRO op-amps will warn you that the output will not reach the negative rail (e.g., 10 mV for the bipolar LT6003); others will instruct you to add an external pull-down resistor or current sink (e.g., the bipolar LT1077, which saturates to 3 mV with no pull-down, and 0.1 mV with a 5kΩ pull-down). Op-amps with clean MOSFET saturation will tell you not to worry – the unloaded output will go all the way to ground (e.g., $\lesssim$0.1 mV for the CMOS AD8616 or AD8691).

C. Distortion

The rail-to-rail output stage (Figure 5.32B) presents real challenges to the chip designer when it comes to quiescent biasing and reduction of crossover distortion. Despite heroic efforts, these amplifiers generally perform some 20–40 dB worse than their conventional (non-RRO) counterparts in terms of distortion, as seen in the pair of plots in Figures 5.43 (non-RRO) and 5.44 (mostly RRO);[28] see also the SPICE plots in §*4x.11*.

D. Monticelli's output circuit

An elegant RRO circuit solution was devised by Monticelli,[29] and it is shown here in simplified form in Figure 5.35 (there's a full discussion in §*4x.11*). It has the effect of biasing the push–pull pair Q_1Q_2 in such a way that there is current overlap at crossover, and, better still, there is continuing current through both transistors *throughout the output swing*. We might call this "push–pull class-A" mode (though it seems to have been named already: "class-AA"). It is used, for example, in the CMOS OPA365 and in the BJT OPA1641. And it works – these parts have −114 dB and −126 dB harmonic distortion, respectively.

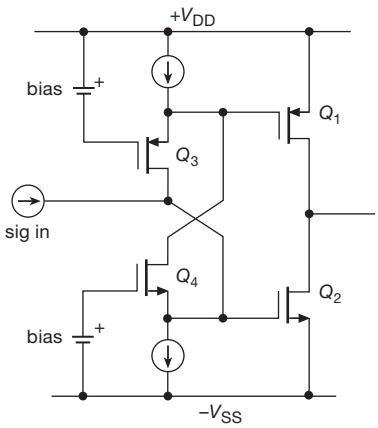


Figure 5.35. The Monticelli rail-to-rail output circuit.

Here's a capsule description of the Monticelli circuit's operation: first, think of Q_3 and Q_4 each as unity-gain current amplifiers whose source terminal is the "summing

[28] In fairness, we note that some of the poorer "distortion" results (which are actually THD+N – distortion plus noise) may be due to the lower supply voltages of the RRO op-amps, necessitating lower signal levels, thus causing noise to loom larger.

[29] See his patent US4570128, and his IEEE *JSSC* paper (SC-21, #6, 1986), in which he says "The output stage (Figure 8) must solve a level shifting problem that has plagued rail-to-rail designs for some time. Elaborate solutions have been proposed that combine multiple embedded feedback loops that are in effect op amps within op amps. To succeed as a general-purpose quad, a simpler solution had to be found." Although originally developed at NSC, this circuit (or close variations) is popular with op-amp designers at Analog Devices and at TI (even before it swallowed NSC).

junction" (because the gate is held at fixed voltage). Now imagine an increasing input signal current, which reduces the net current sunk at Q_4's source. This reduces its V_{GS}, which increases Q_2's V_{GS}, thereby increasing the output pull-down current. Meanwhile, the reduced drain current in Q_4 causes less of Q_3's source current to be diverted, thus increasing the V_{GS} of Q_3; that causes a reduction in Q_1's V_{GS}, and therefore a lower output pullup current. The overall quiescent current is set by the dc bias applied to Q_3 and Q_4. So it's a nicely balanced circuit, with a single-ended current input and a push–pull current output.

This is one cool circuit! In §*4x.11* there's a more complete description, including SPICE simulations of a BJT implementation, and comparison with a conventional (not rail-to-rail) class-AB push–pull emitter follower. This inherently symmetrical circuit also works well with differential current drives to the drains of both Q_3 and Q_4, a configuration you'll often see.

5.10 Choosing a precision op-amp

If there's no such thing as a perfect operational amplifier, then that's especially true for precision op-amps. Although sufficient perfection may be achieved in a few parameters, the design tradeoffs required for achieving this invariably degrade other parameters. For example, if we need a very quiet medium-frequency op-amp, a world-class quiet IC, we won't be able to enjoy world-class low-input-bias currents.[30] That's because the amplifier will use bipolar input transistors, which will have to be operated at fairly high collector currents, and you know what that means for the base currents (e.g., look at the LT1028). Another example: if we want micropower operating current, we won't be able to enjoy world-class fast settling time, because we won't be able to have a high f_T and fast slew rates; that takes power, and lots of it.

In this section we take an in-depth look at the process of choosing a precision op-amp that is right for the job at hand, linked closely to a broad selection of exemplary parts in Tables 5.5 (pages 320–321) and 5.6 (on page 335). If you've got a circuit design that you've been struggling with, this section should hit the spot. The nitty-gritty level of detail that follows is essential to the careful design that distinguishes an excellent circuit from a compromise-ridden also-ran. For the casual reader, on the other hand, the level of detail in the following treatment may be, well, "not superficial enough."[31]

As we begin our tour of precision op-amp parameters and their significance, we invite you to bury yourself in the data. With your circuit design goals in mind, start with an important op-amp parameter and look for the best choices. After zeroing in on a winning value, you can examine other parameters for that op-amp: do some of the other parameters for our winning op-amp now look like poor choices? Maybe your op-amp isn't a winner after all. Or perhaps you've got to return to your design goals and adjust them in accordance with reality, and repeat the process. Remember always that "engineering is the art of compromise."

5.10.1 "Seven precision op-amps"

Seven is a nice number, and, in preparation for the extended discussion of the very practical issue of choosing a precision op-amp, we provide in Table 5.5 (pages 320–321) a comparison of the important specifications for an updated listing of seven of our favorite precision op-amps. The problem is, we just couldn't restrict ourselves to a mere seven – it's closer to seven *dozen*! Spend some time with it (and check off your own seven faves!) – it will give you a good feeling for the trade-offs you face in high-performance design with op-amps. Note particularly the trade-offs of offset voltage (and drift) versus input current for the best bipolar and JFET op-amps. You also get the lowest noise voltage from bipolar op-amps, trending downward with increasing bias current; we'll see why that happens later in Chapter 8 when we discuss noise. The awards for low-input *current*, however, always go to the FET op-amps, again for reasons that will become clear later. In general, choose FET op-amps for low-input current and current noise; choose bipolar op-amps for low-input voltage offset, drift, and voltage noise.

Among FET-input op-amps, those using JFETs dominate the scene, particularly where precision combined with low noise is needed (but not *all* JFET op-amps: note that our jellybean favorites, the LF411/412, are not precise enough to qualify for membership in the table). That dominance is being challenged, though, by some low-voltage CMOS parts like the factory-trimmed MAX4236A and OPA376, and by parts like the TLC4501A that use tricks such as auto-zeroing at power-up.[32]

[30] In §8.6.3 we show a discrete op-amp circuit where both of these goals are achieved.

[31] A phrase lifted from a student's reply in an end-of-course questionnaire: "This course was not superficial enough for me."

[32] There was traditionally a problem peculiar to MOSFETs, which has been largely solved through process improvements. MOS transistors

Table 5.5 "Seven" precision op-amps (page 1: high voltage)

	Part #	# per pkg[a]	Supply[p] Range (V)	Supply[p] I_Q[t] (mA)	Input Current @25°C typ (pA)	Input Current @25°C max (pA)	Offset Voltage V_{os} typ (µV)	Offset Voltage V_{os} max (µV)	Offset Voltage ΔV_{os} typ (µV/°C)	Offset Voltage ΔV_{os} max (µV/°C)	CMRR min dB	Noise[t] $e_n p\text{-}p$ dc[b] (µV)	Noise[t] e_n 1kHz (nV/√Hz)	Noise[t] i_n[k,o] 1kHz (fA/√Hz)	GBW typ (MHz)	Slew typ (V/µs)	Settle[d] (µs)	dist. graph	C_{in} pF	Swing to Supply IN +	Swing to Supply IN −	Swing to Supply OUT +	Swing to Supply OUT −	null pins	comp pin	shdn pin	DIP avail	Cost qty 25 ($US)	Comments
	HV bipolar																											
❑	LT1077A	1	2.2-44	0.05	7 nA	9 nA	9	40	0.4	1.6	97	0.5	27	65	0.23	0.08	-	-	low	-	●	-	●	●	-	-	●	3.84	single-supply
❑	LT1490A	2,4	2.5-44	0.04	1 nA	8 nA	110	500	2	4	84	1	50	15	0.2	0.07	-	-	4.6	●	●	●	●	-	-	-	●	3.25	over-the-top
❑	AD8622A	2,4	4-36	0.22	45 nA	200 nA	10	125	0.5	1.2	125	0.2	11	150	0.56	0.48	-	●	5.5	-	-	●	●	-	-	-	-	5.33	
❑	LT1013[f]	2,4	3.4-44	0.35	12 nA	20 nA	40	150	0.4	2	100	0.55	22	70	0.7	0.4	-	-	low	-	●	-	●	-	-	-	●	3.13	single-supply
❑	OPA277	1,2,4	4-36	0.79	0.5 nA	1 nA	10	20	0.1	0.15	130	0.22	8	200	1	0.8	16	●	low	-	-	-	-	●	-	-	●	3.17	improved OP-27
❑	TLE2141A	1,2,4	4-44	3.5	0.7 µA	1.5 µA	175	500	1.7	-	85	0.5	10.5	1900	5.9	45	0.4	-	low	-	●	-	●	●	-	-	●	1.15	cross-coupled slew
❑	LT1677	1	3-44	2.8	2 nA[e]	20 nA[e]	20	60	0.4	2	109	0.09	3.2	1200	7.2	2.5	5	-	4.2	●	●	●	●	●	-	-	●	3.07	an "RRIO LT1007"
❑	AD8675	1,2	9-36	2.5	0.5 nA	2 nA	10	75	0.2	0.6	114	0.1	2.8	300	10	2.5	-	●	-	-	-	●	●	●	-	-	-	2.22	0.6 ppm dist, dual='76
❑	OPA2209	1,2,4	4.5-40	2.2	1 nA	4.5 nA	35	150	1	3	120	0.13	2.2	500	18	6.4	2.1	-	4	-	-	-	-	-	-	-	-	4.04	0.25 ppm dist, SOT23
❑	LT1007[g]	1	4-44	2.7	10 nA	35 nA	10	25	0.2	0.6	117	0.06	2.5	400	8	2.5	-	●	-	-	-	-	-	●	-	-	●	2.48	'37 decomp 60MHz
❑	ADA4004	1,2,4	9-36	2.2	40 nA	90 nA	40	125	0.7	1	110	0.15	1.8	3500	12	2.7	-	-	-	-	-	-	-	-	-	-	-	4.20	family, SOT23
❑	LT1468	1	7-36	3.9	3 nA	10 nA	30	75	0.7	2	96	0.3	5	600	90	23	0.8	●	4	-	-	-	-	●	-	-	●	4.26	0.7ppm dist
❑	AD8597	1,2	9-36	4.8	40 nA	210 nA	10	120	0.8	2.2	120	0.08	1.1	4300	10	16	2.0	●	12	-	-	-	-	-	-	-	●	3.71	1ppm dist
❑	LT1028A[h]	1	8-44	7.4	25 nA	90 nA	10	40	0.2	0.8	108	0.04	0.85	4700	75	15	-	-	5	-	-	-	-	●	●	-	●	6.48	lower I_B than AD697
	HV superbeta																											
❑	LT6010A[n]	1,2,4	2.7-40	0.14	20	110	10	35	0.2	0.8	107	0.4	14	100	0.35	0.11	45	●	4	-	-	●	●	●	-	●	-	2.22	replace LT1012, w/RRO
❑	LT1012AC[m]	1	2.4-40	0.37	25	100	8	25	0.2	0.6	114	0.5	14	*(20)*	0.5	0.2	-	-	low	-	-	-	-	●	●	-	●	5.11	has overcomp pin
❑	OP97E	1,2,4	4.5-40	0.40	30	100	10	25	0.2	0.6	114	0.5	14	*(20)*	0.9	0.2	-	●	low	-	-	-	-	●	●	-	●	5.52	dual='297, quad='497
❑	AD706	2,4	4-36	0.8	50	200	30	100	0.2	1.5	106	0.5	15	50	0.8	1.5	-	-	2	-	-	-	-	-	-	-	●	4.05	AD704 quad
❑	LT1884A	2,4	3.5-40	0.85	150	400	25	50	0.3	0.8	114	0.4	9.5	50	2	0.9	10	-	-	-	-	●	●	-	-	-	●	5.23	LT1882 slower, lower I_B
	HV JFET																											
❑	AD795	1	8-36	1.3	1	2	100	500	3	10	90	1	9	0.6	1.6	1	11	-	2.2	-	-	-	-	-	-	-	-	7.97	replace OPA111
❑	OPA124PB	1	10-36	2.5	0.35	1	100	250	1	2	100	1.6	8	0.5	1.5	1.6	10	-	3	-	-	-	-	●	-	-	●	6.40	substrate pin
❑	OPA140	1,2,4	4.5-40	1.8	0.5	10	30	120	0.35	1	126	0.25	5.1	0.8	11	20	0.9	-	10	-	●	●	●	-	-	-	-	3.75	0.5ppm dist
❑	AD711C	1,2	9-36	2.5	15	25	100	250	2	5	86	2	16	10	4	20	1.0	●	5.5	-	-	-	-	●	-	-	●	2.19	AD712 low-cost dual
❑	LT1055C[x]	1	20-40	2.8	10	50	120	700	3	12	85	2	15	1.8	4.5	12	1.8	-	4	-	-	-	-	●	-	-	●	2.52	1057 dual, 1058 quad
❑	ADA4000	1,2,4	5-36	2	5	40	200	1700	2	-	80	1	16	10	12	2.7	-	-	5.5	●	-	-	-	-	-	-	-	1.46	jellybean, AD711 subst
❑	OPA192	1,2,4	4.5-40	1	5	20	5	25	0.2	0.5	120	1.3	5.5	1.5	10	20	0.9	-	0.6	●	●	●	●	-	-	-	-	3.87	CMOS, e-Trim™
❑	OPA134	1,2,4	5-36	4	5	100	500	2000	2	-	86	-	8	3	8	20	1	●	-	-	-	-	-	●	-	-	●	1.60	0.8 ppm dist, jellybean
❑	OPA1641	1,2,4	4.5-40	1.8	2	20	1000	3500	-	-	120	-	5.1	0.8	20	11	-	●	8	-	●	●	●	-	-	-	-	2.76	0.5 ppm dist, family
❑	AD8620A	1,2	10-27	2.5	2	10	85	250	0.5	1	90	1.8	6	5	25	60	0.6	●	15	-	-	-	-	●		-	-	11.86	caution, ±12V max
❑	OPA827	1	8-40	4.8	15	50	75	150	1.5	-	104	0.25	3.8	2.2	22	28	0.55	-	9	-	-	-	-	-	-	-	-	9.00	cheaper than 627
❑	OPA627B	1	10-36	7	1	5	40	100	0.4	0.8	106	0.6	4.5	1.6	16	55	0.55	●	7	-	-	-	-	●	-	-	●	30.00	ADA4627B 2nd-source
❑	OPA637B	1	10-36	7	1	5	40	100	0.4	0.8	106	0.6	4.5	1.6	80	135	0.45	●	7	-	-	-	-	●	-	-	●	30.00	ADA4637B 2nd-source
❑	AD549KH	1	10-36	0.6	75 fA	0.1	150	250	2	5	90	4	35	0.5	1	3	5	-	1	-	-	-	-	●	-	-	-	28.00	-L=60fA, TO-99 pkg
❑	OPA129B	1	10-36	1.2	30 fA[z]	0.1	500	2000	3	10	80	4	17	0.1	1	2.5	-	-	2	-	-	-	-	-	-	-	●	12.00	lowest I_B for HV part
	HV chopper																											
❑	LTC1150	1	4.8-32	0.8[s]	10	100	0.5	10	0.01	0.05	110	1.8	high	*(1.8)*	2.5	3	-	-	n	-	●	●	●	●	-	-	●	6.00	only HV AZ w/int caps
❑	ADA4638	1	4.5-33	0.85	45	90	0.5	4.5	-	0.08	130	1.2	66	100	1.5	1.5	4	-	9	-	●	●	●	●	-	-	-		5kHz noise spike
❑	OPA2188	1,2,4	4-40	0.42	160	850	6	25	0.03	0.09	120	0.25	8.8	750	2	0.8	27	-	9.5	-	●	●	●	-	-	-	●	new	

Table 5.5 "Seven" precision op-amps (page 2: low voltage)

	Part #	# per pkg[a]	Supply[p] Range (V)	Supply[p] I_Q[t] (mA)	Input Current @25°C typ (pA)	Input Current @25°C max (pA)	Offset Voltage V_{os} typ (µV)	Offset Voltage V_{os} max (µV)	Offset Voltage ΔV_{os} typ ($\frac{\mu V}{°C}$)	Offset Voltage ΔV_{os} max ($\frac{\mu V}{°C}$)	CMRR min dB	Noise[t] V_{npp} dc[b] (µV)	Noise[t] e_n 1kHz ($\frac{nV}{\sqrt{Hz}}$)	Noise[t] i_n[k] 1kHz ($\frac{fA}{\sqrt{Hz}}$)	GBW typ (MHz)	Slew typ (V/µs)	Settle[d] (µs)	dist. graph	C_{in} pF	Swing to Supply IN +	Swing to Supply IN −	Swing to Supply OUT +	Swing to Supply OUT −	null pins	comp pin	shdn pin	DIP avail	Cost qty 25 ($US)	Comments
	LV bipolar																											
❑	**LT6003**	**1**,2,4	1.5-18	0.001	40	140	175	500	2	5	73	3	325	12	0.003	0.001	slow	-	6	●	●	●	●	-	-	-	-	1.60	SOT-23, '6004 dual
❑	**EL8176**	**1**	2.4-6	0.055	0.5 nA	2 nA	25	100	0.7	-	90	1.5	28	160	0.4	0.13	-	-	-	●	●	●	●	-	-	-	●	-	2.30	SOT-23, charge pump
❑	**LMP7731**	**1**,2	1.8-6	2.2	1.5 nA	30 nA	6	500	0.5	5.5	101	0.08	2.9	1100	22	2.4	-	●	-	●	●	●	●	-	-	-	-	-	1.98	SOT-23, 3nV/√Hz
❑	**LT6220**	**1**,2,4	2.2-13	0.9	15 nA	150 nA	70	350	1.5	5	85	0.5	10	800	60	20	0.3	-	2	●	●	●	●	-	-	-	-	1.75	SOT-23
❑	**LT6230**	**1**,2,4	3-12.6	3.3	5 µA	10 µA	100	500	0.5	3	96	0.18	1.1	2400	215	70	0.05	-	7	-	-	●	●	-	-	●	-	2.72	-10=decomp, 1.3GHz
	LV JFET																											
❑	**OPA656**	**1**	9-13	14	2	20	250	1800	2	12	80	-	7	1.3	230	290	0.02	●	2.8	-	-	-	-	-	-	-	-	5.59	'657 for 1.6GHz *G*>5
	LV CMOS																											
❑	**LMP2232A**	**1**,**2**,4	1.8-6	0.01	0.02	1	10	150	0.3	0.75	81	2.3	60	0.1[c]	0.13	0.06	slow	●	-	●	●	●	●	-	-	-	-	3.29	'31 single, '34 quad
❑	**TSV611A**	**1**,2	1.8-6	0.01	1	10	-	800	2	-	61	-	105	-	0.12	0.04	-	-	-	●	●	●	●	-	-	-	-	0.82	'612 dual
❑	**AD8603**	**1**,2,4	1.8-6	0.04	0.2	1	12	50	1	4.5	85	2.3	25	50	0.4	0.1	23	-	23	●	●	●	●	-	-	-	-	1.36	SOT-23, 1pA
❑	**LTC6078**	**2**,4	2.7-6	0.055	0.2	1	7	25	0.2	0.7	95	1	18	0.25[c]	0.75	0.05	24	-	18	●	●	●	●	-	-	-	-	3.54	*V*os degrades near V_+
❑	**LTC6081**	**2**,4	2.7-6	0.33	0.2	1	-	70	0.2	0.8	93	1.3	13	0.5	3.6	1	6	●	7	●	●	●	●	-	-	-	-	4.75	degrade V_{cm}>V_{cc}–1.5V
❑	**MAX4236A**	**1**	2.4-6	0.35	1	500	5	20	0.6	2	84	0.2	14	0.6	1.7	0.3	1	●	7.5	-	●	●	●	-	-	●	-	1.78	not AZ, '4237 decomp
❑	**LMC6482A**	**2**,4	3-16	0.5	0.02	4	110[v]	750	1	-	70	-	37	30	1.5	1.3	-	-	3	●	●	●	●	-	-	-	●	1.88	'7101=SOT-23
❑	**OPA376**	**1**,2,4	2.2-7	0.76	0.2	10	5	25	0.26	1	76	0.8	7.5	0.25[c]	5.5	2	2	●	13	●	●	●	●	-	-	-	-	1.32	CMOS etrim™
❑	**OPA364**	**1**,2	1.8-5.5	0.85	1	10	-	500	3	-	74	10	17	0.6	7	5	1.5	-	3	●	●	●	●	-	-	●	-	2.18	chg-pump, 20ppm dist
❑	**TLC4501A**	**1**,2	4-7	1	1	60	10	40	1	-	90	1.5	12	0.6	4.7	2.5	2.2	●	8	-	●	●	●	-	-	-	-	3.06	0.3s self-cal at pwr-on
❑	**OPA743**	**1**,2,4	3.5-13	1.1	1	10	1500	7000	8	-	66	11	30	2.5	7	10	15	●	4	●	●	●	●	-	-	-	-	1.58	jellybean
❑	**LMP7715**	**1**,2	1.8-6	1.15	0.05	1	10	150	1	4	85	-	5.8	10	17	9.5	-	-	15	-	●	●	●	-	-	-	-	2.05	
❑	**LMP7717**	**1**,2	1.8-6	1.15	0.05	1	10	150	1	4	85	-	6.2	10	88	28	-	-	15	-	●	●	●	-	-	-	-	2.18	decomp, *G*>10
❑	**AD8692**	**1**,2,4	2.7-6	0.85	0.2	1	400	2000	0.3	6	68	1.6	8	50	10	5	1	-	5	-	●	●	●	-	-	-	-	1.36	0.6ppm dist
❑	**AD8616**	**1**,2,4	2.7-6	1.7	0.2	1	23	60[f]	1.5	7	80	2.4	7	50	24	12	0.5	-	7	●	●	●	●	-	-	-	-	1.52	jellybean, '15 SOT-23
❑	**OPA350**	**1**,2,4	2.5-7	5.2	0.5	10	150	500	4	-	74	-	7	4	38	22	0.5	-	6.5	●	●	●	●	-	-	-	●	1.67	
❑	**OPA380**	**1**,2	2.7-7	7.5	3	50	4	25	0.03	0.1	100	3	5.8	10	90	80	2	-	3	-	●	●	●	-	-	-	-	5.39	transimp, w/auto-zero
❑	**LMC6001A**	**1**	5-16	0.45	10 fA	25 fA	-	350	2.5	10	83	-	22	0.13	1.3	1.5	-	-	-	-	●	●	●	-	-	-	●	12.19	
❑	**LMP7721**	**1**	1.8-6	1.3	3 fA[y]	20 fA	26	150	1.5	4	83	-	7	10	17	10	-	●	-	-	●	●	●	-	-	-	-	11.89	very low e_n for 20fA!
	LV chopper																											
❑	**MAX9617**	**1**	1.6-6	0.06	10	140	0.8	10	0.01	0.12	116	0.42	42	100	1.5	0.7	-	-	-	●	●	●	●	-	-	●	-	1.60	charge pump
❑	**AD8638**	**1**,2	5-16	1	1.5	40	3	9	0.01	0.06	118	1.2	60	noisy	1.35	2.5	3	●	4	-	●	●	●	-	-	-	-	3.31	auto-zero; SOT23
❑	**LTC2050H**	**1**	2.7-11	0.8	7	50	0.5	3	-	0.03	120	1.5	-	-	3	2	-	-	1.7	-	●	●	●	-	-	●	-	2.19	replaces LTC1050
❑	**AD8551**	**1**,2,4	2.7-6	0.7	10	50	1	5	0.01	0.04	120	1	42	*(2)*	1.5	0.4	50	-		●	●	●	●	-	-	-	-	2.34	
❑	**OPA735**	**1**,**2**	2.7-13	0.6	100	200	1	5	0.01	0.05	115	2.5	135	40	1.8	1.5	-	-	10	-	●	●	●	-	-	●	-	3.38	auto-zero, "chopper"

Notes: (a) **boldface** indicates number in a package for the part number listed. (b) 0.01Hz–10Hz or 0.1Hz–10Hz. (c) calculated. (d) usually to 0.01%.
(e) for V_{EE}+1.4V < V_{CM} < V_{CC}–0.7V. (f) LTC suggests LT1490/1. (g) LTC suggests LT1677. (h) LTC suggests LT6200/30. (k) at 1kHz or 10kHz (i.e., above the 1/*f* corner), except 10Hz for chopper op-amps. (m) LT1097 cheaper. (n) dual & quad have degraded I_B and V_{OS}; '6013=decomp. (o) values in parenthesized *(thin italics)* should not be relied upon; measured values often are as much as 5x–100x larger; see discussion in Chapter 8. (p) per amplifier. (s) can be reduced to 200µA. (t) typical. (v) V_{OS} is insensitive to V_{CM}.
(x) the (hard-to-get) -A version has V_{OS}=50µV typ, 150µV max. (y) special pinout for guard. (z) special pinout for guard+substrate pin.

Finally, the so-called *chopper-stabilized* (here and in Table 5.6 on page 335) amplifiers form the most important exception to the generalization that FET op-amps, particularly MOSFET types, suffer from larger initial offsets and much larger drifts of V_{OS} with temperature and time than do bipolar-transistor op-amps. In fact, these devices (known also as *auto-zero* or *zero-drift* amplifiers) are the amplifiers with the *smallest* offset voltage and drift, typically in the $\pm 1\,\mu$V and $\pm 0.05\,\mu$V/°C range. They use MOSFET analog switches and amplifiers to sense, and correct, the residual offset error of an ordinary op-amp (which itself is often built with MOSFETs, on the same chip). This is not without compromise, however: chopper-stabilized amplifiers have some unpleasant characteristics that make them unsuited for many applications, as we'll see in §5.11.

5.10.2 Number per package

The first column in Table 5.5 gives the choices available for the number of devices per package (the number in **boldface** shows which choice matches the part number). We generally list single op-amp parts, even though in practice the dual op-amps are more useful and popular (in some cases distributors don't even stock the single types). Special features such as pins for external offset-nulling, compensation, and shutdown are available only for the single op-amp package types and are indicated in the right-hand columns. Generally the specs are identical for the different dies and appear on the same datasheet, but not always.

5.10.3 Supply voltage, signal range

It's likely your first consideration will be supply-voltage range and signal levels. High-voltage parts (able to operate from ± 15 V, i.e., 30 V total) are listed first in the table, with parts ordered more or less by I_Q, the quiescent supply current, in each category. Battery-powered applications benefit from low supply currents, but some low-drift applications do as well, because op-amp self-heating temperature effects will be less. Some parts offer a version with a power-shutdown (SHDN) pin. For example, the LT6010 current drops from 135 μA to 12 μA in shutdown (but a *gotcha*: the shutdown pin itself takes another 15 μA) and it takes 25 μs to turn ON or OFF. Other parts do better, e.g., the OPA364 draws 0.9 μA when off.

Circuits operating with high-voltage supplies benefit from using high signal levels, such as ± 10 V full scale. An offset voltage, say V_{OS}=40 μV, is a smaller fraction of 20 Vpp than it is of 0 to 4 V. With the exception of chopper op-amps, you don't get any offset-voltage improvement for low-voltage parts.

Low-voltage parts finish the table (most are 5.5 V maximum total supply, but some permit 11 V or higher, suitable for ± 5 V operation), but it's important to realize that many "high-voltage" parts are designed and specified to work well at low voltages, even under 3 V. Some work well with ± 3 V to ± 5 V supplies and should not be rejected simply because they can also work at higher voltages. But be warned, you need to examine the common-mode input range and output-swing range. For example, although a 44 V op-amp like the LT1490 works with 3 V supplies and allows rail-rail inputs and outputs, another fine 44 V LTC op-amp, the low-noise LT1007 (which works down to 4 V) is limited to inputs and outputs no closer than 2 V from the rails – nearly useless when running from ± 2 V. Clearly it's not meant to be a low-voltage part. Your quick guide to these issues is the "swing to supply" columns; the LT1490 has all four checks whereas the LT1007 has no checks.

5.10.4 Single-supply operation

If you're running with low supply voltages, you may want to use a single-supply power arrangement. Op-amps capable of single-supply use have at a minimum the ability to operate their inputs and outputs to the negative rail (i.e., ground). Many permit operation with outputs also to the positive rail, and claim rail-to-rail outputs on the front page of the datasheet. But be warned, there's usually a performance degradation when outputs are near the supply rails. Some op-amps offer zero-volt or below-the-rail operation if you add a pulldown resistor.[33]

Seven high-voltage op-amps on our list offer single-

are susceptible to a unique debilitating effect that neither FETs nor bipolar transistors have. It turns out that sodium-ion impurity migration and/or phosphorus polarization effects in the gate insulating layer can cause offset voltage drifts under closed-loop conditions, in extreme cases as much as 0.5 mV over a period of years. The effect is increased for elevated temperatures and for a large applied differential-input signal, with some datasheets showing a typical 5 mV change of V_{OS} over 3000 hours of operation at 125°C with 2 V across the input. This sodium-ion disease can be alleviated by introducing phosphorus into the gate region. Texas Instruments, for example, uses a phosphorus-doped polysilicon gate in its "LinCMOS" series of op-amps (TLC270-series) and comparators (TLC339 and TLC370-series). These popular inexpensive parts come in a variety of packages and speed/power selections and maintain respectable offset voltages with time (50 μV eventual offset drift per volt of differential input).

[33] Many op-amps can do this, but without saying so. That's because their pullup transistor and drivers work all the way down to the negative

supply operation; some, like the LT1013, excel at it. Two offer full rail-to-rail input-output, or RRIO, operation. The fast-slewing TLE2141 is especially interesting (fast settling, but high bias current), as is the low-noise LT1677 (lower bias current, but slow slewing and settling). All but two of the low-voltage parts offer single-supply operation.

There are some precision op-amps with low supply currents, down to 10–60 μA, although this will severely limit your choices for other parameters. There's even a respectable 0.85 μA (and 1.8 V) op-amp, the LT6003. Some op-amp types, such as the JFETs, don't offer any low-power parts. Nonetheless you might choose one for its low noise and low bias current.

5.10.5 Offset voltage

Perhaps the single parameter most often associated with precision amplifiers is input-voltage error. To measure small offset voltages, use the op-amp's gain to magnify the effect, as shown in Figure 5.36. Offset voltage was our required parameter to gain entry into the table; few op-amps with maximum offset voltages above 250 μV made the grade. There are plenty of parts with $<10\,\mu$V *typical* offset voltages, but "typical" isn't a reliable spec when you're in the business of manufacturing precision instruments. Bipolar input stages have an advantage over JFET and CMOS in the table, but they suffer from higher input bias currents. The "superbeta" parts are a pleasant exception, especially at high temperatures (see Figure 5.38), but none of these parts has inputs that operate to either supply rail.

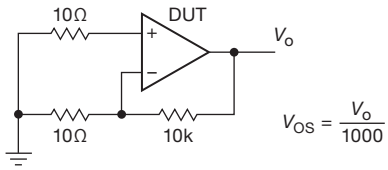


Figure 5.36. Offset-voltage test circuit. The $\times$1000 voltage gain makes submillivolt offsets in the device under test (DUT) easily measurable. Input *current* effects are negligible, owing to the small (10 Ω) resistances seen at the op-amp's inputs. Add a 200 Ω resistor at the output if you want to drive a cable.

Many low-offset parts have disappeared since the second edition of our book, especially in the JFET category. They've become too expensive for the market and have lost the competition with low-voltage chopper and auto-zero op-amps. The latter have a few token representatives in this table, but they have their own selection table (Table 5.6 on page 335) that you should examine if they look good for your design. They have their own problem issues, such as current noise, which we discuss in §5.11.

Offset voltage variation with common-mode input voltage is a serious issue for some parts, especially for RRIO op-amps (see Figure 5.29), and is not addressed on the table. It's always important to follow up an initial choice from the table with a careful examination of the datasheet. For example, the OPA364 and MAX9617 use internal charge pumps to power their input stages, eliminating this problem completely.

Offset voltage drift with temperature is an important parameter when stability of measurements matters. This parameter is not production tested. The maximum drift spec may not be very reliable, and some manufacturers have stopped providing such a spec.

Offset voltage drift with time is a parameter that used to appear in precision op-amp datasheets, with values of order 300–400 nV/month; a few high-performance parts like the LT1007 claimed 200 nV/month, and chopper-stabilized op-amps generally claim drifts of 50 nV/month.[34] This is somewhat uncharted territory, and some people claim that drift slows with time, or perhaps it is more akin to a random walk, in either case suggesting that a drift specification should perhaps have units of $\mathrm{nV}/\sqrt{\mathrm{month}}$.

5.10.6 Voltage noise

Voltage noise is the in-band variation of op-amp input–offset voltage that's indistinguishable from signal. It's useful to view it as a "noise spectral density" function $e_n(f)$, which tells you the rms noise voltage in a 1 Hz bandwidth (see §8.2.1) centered at frequency f. Figure 5.37 shows an idealized plot of input voltage-noise density as a function of frequency for some of the op-amps in the table. For most op-amps $e_n(f)$ is essentially flat for frequencies above its "$1/f$ corner frequency," with e_n increasing below the corner frequency, approximately as $1/\sqrt{f}$. (Auto-zero, or "chopper-stabilized," op-amps are not shown. They behave differently, because low-frequency "noise" is removed by the auto-zero process, so their e_n is flat at low frequencies. We'll discuss them shortly; see §5.11.)

The voltage–noise column in Table 5.5 (pages 320–321) shows e_n at its commonly specified frequency of 1 kHz,

rail, without sourcing current to the output. Some require a minimum pull-down current, e.g., 0.5 mA for the OPA364.

[34] Although it's possible to do considerably better, for example the measured 6 nV/month reported by Bob Pease for the LMP2011. See Table 8.3 and Figures 8.60, 8.61, 8.110, and 8.110. The IF3602 is a large-geometry dual JFET available from InterFet, shown for comparison.

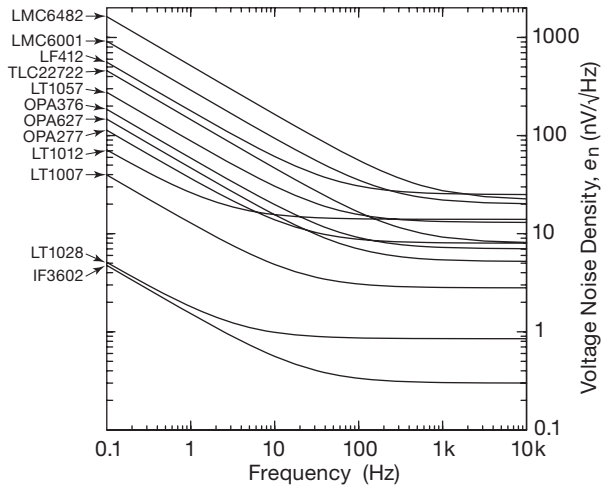


Figure 5.37. Voltage noise density e_n for a selection of representative op-amps, showing the increase of noise power below the $1/f$ corner frequency. Some op-amps that have good specs at 1 kHz don't look so good at 0.1 Hz. Figure 5.54 shows the resulting voltage noise v_n when such a noise density is integrated over frequency.

comfortably in the flat region above the $1/f$ corner of most op-amps; e_n varies from $0.85\,\text{nV}/\sqrt{\text{Hz}}$ for the high-current LT1028 to $325\,\text{nV}/\sqrt{\text{Hz}}$ for the current-starved LT6003.

We discuss noise in greater detail below, and in Chapter 8, but let's start with the simple relationship relating voltage-noise *density* e_n (given in units of $\text{nV}/\sqrt{\text{Hz}}$) to the value of *integrated* total voltage noise V_n (in units of nV, or μV; and either rms or peak-to-peak) over some frequency passband. In the flat (i.e., white noise) region of e_n at frequencies above the $1/f$ corner, the integrated noise voltage is simply $V_n = e_n\sqrt{\text{BW}}$.

As we'll see, for circuits with bandwidths of 1–10 kHz or more, the noise voltage V_n is dominated by the noise density at high frequencies. Datasheets for most op-amps give an e_n value at 1 kHz, but some also specify it at 10 kHz, 100 kHz, or even 1 MHz; and they usually provide graphical plots of e_n versus frequency. Because the integrated noise voltage is dominated by e_n at the high-frequency end of the operating range, be sure to use that value (or a frequency-weighted eyeball average) in the simple formula above. Using a high-frequency e_n value is particularly important for transimpedance amplifiers, which suffer from "$e_n\omega C$" current noise ($i_n = e_n 2\pi f C_{in}$) at high frequencies.

A. "1/*f*" noise

We discuss $1/f$ noise in greater detail in Chapter 8 (where we show how to determine the $1/f$ noise corner, etc., in §8.13.4, but here we address the practical question "What effect do the scary-looking rising noise-density curves in Figure 5.37 have on my circuit noise?" The noise density e_n is indeed higher at low frequencies, but that density gets multiplied by a smaller frequency span. Put another way, the total noise *voltage* (as contrasted with the noise *density* e_n) contributed by an op-amp depends both on its e_n and on the circuit's bandwidth. More precisely, the mean square noise voltage is the integral of e_n^2 over the bandpass:

$$v_n^2 = \int_{f_a}^{f_b} e_n^2(f)df,$$

where $e_n(f)$ is the noise spectral density (often plotted in datasheets), and the bandpass (or observation band) extends from f_a to f_b. We then get the rms noise voltage by taking the square root of v_n^2.

We perform the integrations and show the devastating effects of $1/f$ noise in the integrated-noise plots later, in Figure 5.54 (in the context of auto-zero op-amps). More bandwidth means more noise, and all of the curves rise as $\sqrt{f}$ at the high end. The op-amps are ranked in order by their high-frequency e_n values; it's interesting to compare their positions in Figure 5.37 with the rankings in Figure 5.54. At the low frequency end, the noise voltage of conventional op-amps levels off, because their rising $1/f$ noise density makes up for the reduced bandwidth,[35] whereas the noise voltage of the auto-zero amplifiers continues its downward trend.

Let's use Figure 5.54 to explore a revealing example. The LT1012 has an e_n (at 1 kHz) of $14\,\text{nV}/\sqrt{\text{Hz}}$, and a 2.5 Hz corner frequency.[36] If it were used in a precision amplifier with a high-frequency cutoff beyond 1 Hz, for example, it would be less noisy than an OPA277, even though the latter has a lower noise density at 1 kHz of $8\,\text{nV}/\sqrt{\text{Hz}}$, because the latter has a 20 Hz noise corner (indicated by black dots). But the OPA277 wins back some respect when we see that it's dramatically quieter than a competing $9\,\text{nV}/\sqrt{\text{Hz}}$ part, the TLC2272, which suffers from a much higher 330 Hz noise corner.[37]

[35] Well, not exactly: if the noise power density were truly to continue rising as $1/f$, the integral would diverge at zero frequency (dc). For Figure 5.54 we set the low frequency limit to be 0.01 Hz.

[36] How to determine the corner frequency? See the discussion in §8.13.4.

[37] The LT1012 and OPA277 are BJT parts, and the TLC2272 is a CMOS op-amp. The '2272 boasts a miniscule 60 pA max bias current, far better than the '277's 1 nA, but not much better than the '1012's amazing 100 pA.

The integrated-noise plots are revealing, but it's helpful to have a single number in the table to evaluate our op-amps. The "$V_n$pp" parameter in the table is the peak-to-peak voltage noise over a 0.1-10 Hz band. This shows an op-amp's "dc noise" as seen in the flat part of the curves in Figure 5.54. The values range up to 11 μVpp (but parts like the LMC6482 avoid the competition by not listing any spec at all). The LT1028 is the winner with 35 nVpp, but the LMP7731 is a notable part at 80 nVpp, with its RRIO capabilities and its SOT-23 package. The ADA4075, with its 1 mV offset, didn't make the precision table, but its 60 nVpp noise level is attractive.

Precision op-amps that suffer from $1/f$ noise (i.e., all except auto-zero types) have a $V_n$pp spec whose lower frequency limit is usually 0.1 Hz. If you need a lower starting frequency (such as 0.01 Hz, used in the plots), multiply the low-frequency noise-voltage value listed by the square root of the number of additional low-end decades you want (that's an interesting $1/f$ noise factoid). As long as the $1/f$ corner is multiple decades higher, you can ignore the white-noise contribution to the spectrum.

The $V_n$pp parameter is your primary clue about an op-amp's long-term drift performance.

5.10.7 Bias current

The available input bias currents range from femtoamps to microamps (nine orders of magnitude!). In some applications this is the parameter that rules out entire classes of op-amp choices. Parts featuring very low typical input currents often have unimpressive *maximum* specs; this is due to the difficulty and expense of automated testing at currents below 10 pA or so. For example, the $1.88 low-cost LMC6482A CMOS op-amp has a 20 fA "typical" spec, but the datasheet shows a 4 pA maximum value – that's 200× worse.[38] However if you're willing to pay over $10, you can get an LMP7721 with a 20 fA maximum spec.

As we've said earlier (and will say again), the "bias" current of JFET and CMOS op-amps is a *leakage* current, and it increases exponentially with temperature; see Figure 5.38. That's the bad news. The *good* news (as we've also said before) is that there are some op-amps (like the LT1012 and AD706) that have low JFET-like input currents, but which have BJT inputs and therefore enjoy better high-temperature performance and improved offset voltages and drift, see Figure 5.6.

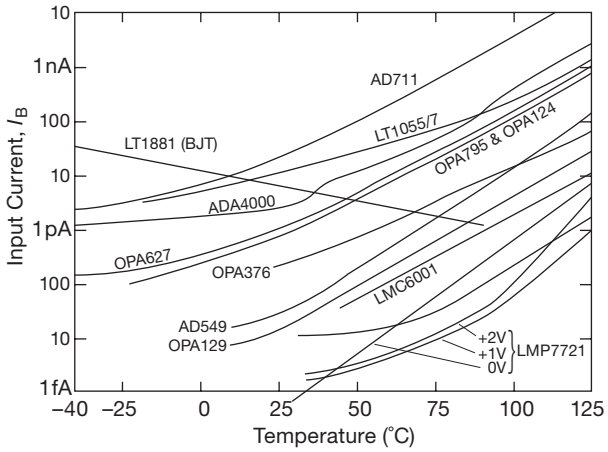


Figure 5.38. Input current versus temperature for a representative set of op-amps from Table 5.5 on pages 320–321, taken from manufacturers' datasheets. See also Figures 5.6 and 3.48.

Low-input-current op-amps generally have higher offset voltage and offset voltage drift, and they're usually more noisy. The OPA627 and ADA4627 JFET op-amps are exceptions, and they've served us well, but they're expensive. Happily the new JFET OPA827 has lower noise and offers some price relief. The AD743, not in the table because of its 1 mV offset voltage, sports a 2.9 nV/$\sqrt{\text{Hz}}$ spec. Looking at CMOS parts (which are low voltage), we find that the LMP7715 is the best low-noise contender, at 5.8 nV/$\sqrt{\text{Hz}}$, but a low-cost part like the AD8616, with 1 pA bias and 7 nV/$\sqrt{\text{Hz}}$ noise, may offer a good compromise.

High-speed op-amps often have high input bias currents, typically 200 nA to as much as 20 μA. They also tend to have high offset voltages, above 0.5 mV, so most parts in this category didn't even make it into the precision table, but appear in the high-speed op-amp category in Table 5.4 on page 310. TI's fast OPA656 and '657 low-voltage JFET op-amps, with 2 pA typical bias current, 290 V/μs slew rate, and 20 ns settling time, demanded and were granted residence in both tables. The OPA380 is a 90 MHz op-amp meant for fast transimpedance applications, featuring 50 pA and 25 μV maximum offset. This part uses an auto-zero circuit to achieve 25 μV offset, but avoids excess current noise with an isolating filter (Figure 5.41).

A. Measuring bias current

To measure input currents (or offset currents) down to the nanoamp level or so, you can use the simple circuit in Figure 5.39. For *really* small currents, though, you've got

[38] Note 13 on the LMC6482 datasheet says "Guaranteed limits are dictated by tester limitations and not device performance. Actual performance is reflected in the typical value." That's illuminating, but not entirely helpful for the designer of a mass-production instrument.

to play some clever tricks: a current down in the femtoamps (10^{-15} A) develops only *micro*volts across a gigaohm! (And you'll never see that, because offset voltages are much larger.) Instead, accumulate the tiny input current in an integrator (itself constructed with an ultra-low-input-current op-amp), as shown in Figure 5.40A.[39] Here the short length of shielded cable (with Teflon dielectric) serves as the integrator's feedback capacitance C_1 (standard 50 Ω coax has a capacitance of almost exactly 1 pF/cm, see Appendix H). You can watch the ramp directly, or, if you want to get fancy, add a differentiator as shown.

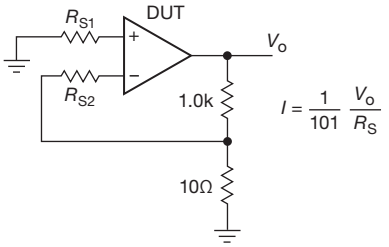
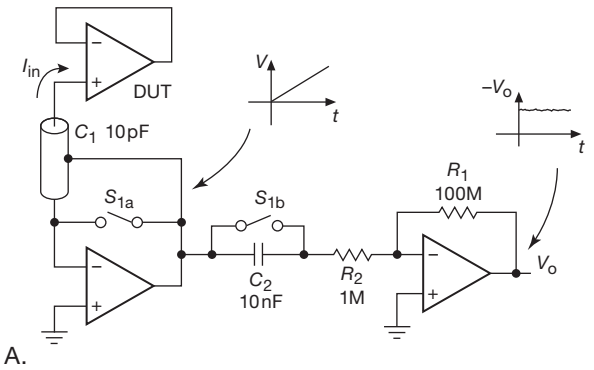


Figure 5.39. Input current (set one of the R_s's to zero) and input offset current ($R_{s1} = R_{s2}$) test circuit. Use R_s values large enough so that the voltage developed across them is at least some 10s of millivolts, so that errors due to offset voltage can be ignored. Add a 200 Ω resistor at the output if you want to drive a cable.

A somewhat simpler method, which has worked well for us, is to wire the op-amp as a follower, with a small capacitor from the (+) input to ground (Figure 5.40B); the op-amp's input current then generates an input ramp, faithfully reproduced at the output. At first we struggled with memory effects in mica and film capacitors, but we finally settled on an air (variable) capacitor, of the sort that were used to tune AM radios in the good ol' days. With the capacitor set to 365 pF we got an output ramp of 0.20 mV/s from an LMC6482, and thus an input current of 73 fA. You can "reset" this circuit by collapsing the supply rails. Be sure to put the whole business in a metal box: these open inputs are really sensitive beasts!

[39] Based on nice techniques worked out by Paul Grohe and Bob Pease at National Semiconductor. See Paul Rako's article "Measuring Nanoamperes" (*EDN*, 26 April 2007), and two riffs by Pease from his series in Electronic Design: "What's All This Teflon Stuff, Anyhow?" (14 Feb 1991) and "What's All This Femtoampere Stuff, Anyhow" (2 Sept 1993). Nicely readable versions currently available at `http://electronicdesign.com/test-amp-measurement/whats-all-teflon-stuff-anyhow` and `.../whats-all-femtoampere-stuff-anyhow`.



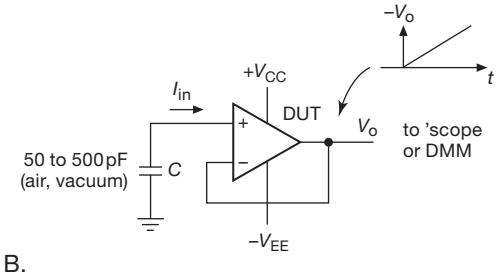


Figure 5.40. Integrate the input current to measure in the picoamp range (and below). A. With a separate MOS-input op-amp integrator whose input current is in the low femtoamps (e.g., an LMP7721, I_B=3 fA typ, 20 fA max). Use electromechanical relays (not MOS switches) for S_1, for example the COTO 9202 series shown in Figure 5.3. B. More simply, let the device's input current charge a small capacitor, and observe the ramp at the G=1 output.

5.10.8 Current noise

The op-amp's input current noise density i_n flows through the source impedance seen at the amplifier's input terminals, contributing an equivalent noise voltage density $i_n Z_s$; this is often negligible compared with the amplifier's e_n. We can define a "noise impedance" for the op-amp, $Z_n \equiv e_n/i_n$, so that we can safely ignore current noise when the source impedance $Z_s \ll Z_n$.

Typical i_n values range from 0.1 fA/$\sqrt{\text{Hz}}$ to 50 fA/$\sqrt{\text{Hz}}$ for CMOS and JFET op-amps, and up to 5 pA/$\sqrt{\text{Hz}}$ for low-noise BJT-input op-amps that operate at relatively high input currents. The superbeta BJT LT1012 (with its low input current) does considerably better, at 20 fA/$\sqrt{\text{Hz}}$. But note that the high-i_n LT1028 is the *voltage*-noise winner, for which we pay the penalty of high current noise. Its noise impedance $Z_n = 850\,\Omega$, which means that unusually low circuit resistances must be used, say 300 Ω or less, to obtain the full benefit from its low voltage noise. By contrast, the BJT LT1013 single-supply op-amp has $Z_n = 315$ kΩ, a comfortably high value.

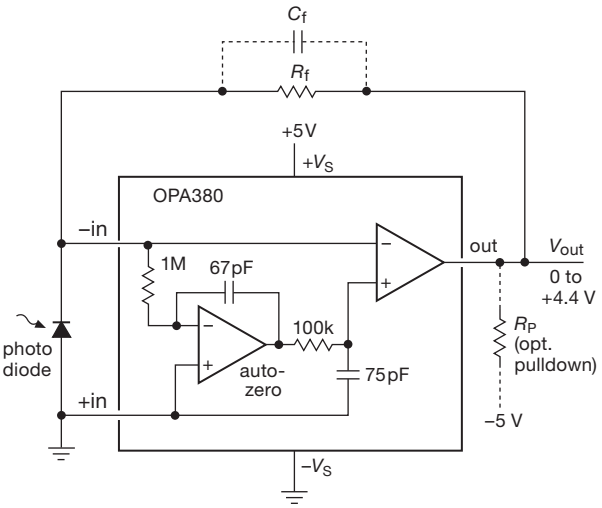


Figure 5.41. The OPA380 achieves 4 μV typical voltage offset using an auto-zero feature, yet it has only 10 fA/$\sqrt{\text{Hz}}$ current noise at 10 kHz. It is ideal for transimpedance applications such as the photodiode preamp shown here.

The manufacturer's spec gives the current-noise density at a high frequency like 1 kHz or 10 kHz, chosen to be well above the current-noise $1/f$ corner frequency. The current-noise $1/f$ corner typically comes at much higher frequencies than the voltage-noise $1/f$ corner. For example, the OPA277 has a 20 Hz voltage-noise corner, but a 200 Hz current-noise corner; for an LT1007 it's 2 Hz and 120 Hz. The differing $1/f$ corner frequencies mean that we'll get different Z_n values at low frequencies. Returning to the LT1028, for example, its relatively higher current noise at low frequencies lowers Z_n to 212 Ω at 10 Hz. This requires that we push down further our circuit resistances for optimum performance, to 100 Ω or less.

At high frequencies the current noise may consist largely of the fundamental (and unavoidable) *shot noise*, the statistical fluctuations of electron flow (eq'n 8.6). For an input bias (or leakage) current I_B that lower bound is $i_n = \sqrt{2qI_B}$; for a bias current of 10 pA that evaluates to i_n=1.8 fA/$\sqrt{\text{Hz}}$ (from which you can conveniently scale up or down by the square root of I_B). The typical and maximum bias-current specs vary widely, as we saw; evidently many manufacturers simply list the calculated shot-noise value corresponding to the typical bias-current spec. For example, the LT1013 has I_B=12 nA typ, from which we can calculate a current-noise density of 62 fA/$\sqrt{\text{Hz}}$; the manufacturer's spec is 70 fA/$\sqrt{\text{Hz}}$.

An important exception comes in the case of BJT-input op-amps with bias-current cancellation circuits: that greatly reduces the dc input current, but not the current noise. For example, the LT1007, with its quiet e_n=2.5 nV/$\sqrt{\text{Hz}}$, has a 10 nA bias-current spec, from which we calculate a shot-noise current of 56 fA/$\sqrt{\text{Hz}}$, but the manufacturer's spec is i_n=400 fA/$\sqrt{\text{Hz}}$. That's seven times too high! What's going on? To achieve low-noise voltage they run the input transistors at high collector currents. That creates a high base current, which would be the op-amp's input current if they hadn't used the old base-current-cancellation trick (*§4x.10*). So the dc bias current is small, but the current noise is large. The ultralow-e_n LT1028 also uses bias-current cancellation, keeping its bias current down to 25 nA, but with a current noise 10 times higher than the calculated shot-noise value. And for the LT6010, whose input current is brought down to just 20 *pico*amps (the kind of low currents you see in FET-input op-amps), the current noise is 40 times larger than calculated shot noise.

In other words, it's important to realize that the input *noise* current of a bias-cancelled op-amp will be considerably larger than you would expect if you were to calculate the shot noise arising from the net (i.e., cancelled) input bias current. Rather, you need to calculate the shot noise from the uncancelled base currents (and then apply a factor of $\sqrt{2}$ to account for the additional noise in the cancellation current). For example, using the LT6010's value of $I_B = \pm 20$ pA (typ), you'd incorrectly estimate a shot noise current of $i_n \approx 2.5$ fA/$\sqrt{\text{Hz}}$, whereas the datasheet lists a typical value (at 1 kHz, well above the $1/f$ corner) of 100 fA/$\sqrt{\text{Hz}}$; similarly for the LT1028 (which specifies 1000 fA/$\sqrt{\text{Hz}}$, versus the 90 fA/$\sqrt{\text{Hz}}$ you would incorrectly estimate from the net I_B). Table 5.5 does not tell you if a BJT amplifier employs bias-current cancellation, but there's a convenient bias-cancel column in the low-noise BJT op-amp Table 8.3a on page 522. Op-amps with bias-current cancellation generally do not have rail-to-rail input stages.[40]

A caution: some datasheets list greatly optimistic values for i_n, evidently making exactly this error. For example, the bias-cancelled LT1012's datasheet shows a typical i_n leveling off to 6 fA/$\sqrt{\text{Hz}}$ (beyond the $1/f$ corner), which is what you'd calculate from the specified net (i.e., cancelled) input current of ±100 pA max, whereas you would expect a value about 10 times larger (assuming the uncancelled

[40] The LT1677 listed in Table 8.3a is an exception. Its datasheet has a graph labelled "Input Bias Current Over the Common Mode Range," showing that the bottom 1.4 V and top 0.7 V of the common-mode range suffer from high bias currents. The op-amp's offset voltage is also degraded in these regions. But, hey, we warned you about that in §5.9.1!

base current is about 100× larger). We were skeptical of the datasheet's claimed i_n, so we measured it (along with others that appeared to be similarly in error), and found[41] $i_n \approx 55\,\text{fA}/\sqrt{\text{Hz}}$. This error shares some of the characteristics of an *epidemic*, having infected also the datasheets of auto-zero op-amps. For example, the exemplary AD8628A (listed in Table 5.6 on page 335) specifies an input noise-current density of $5\,\text{fA}/\sqrt{\text{Hz}}$; imagine our surprise when we measured a value 30 times larger. Not to be outdone, the MCP6V06 auto-zero op-amp's specification of $0.6\,\text{fA}/\sqrt{\text{Hz}}$ is rather at odds with its measured $170\,\text{fA}/\sqrt{\text{Hz}}$. See the discussion in §§5.11.

It's important to note that, in the case of chopper and auto-zero op-amps, the input current noise spec is usually given at a low 10 Hz frequency, because that's below the region of very high switch charge-injection current noise (see §3.4.2E). If you take the trouble to measure an auto-zero's input noise, you'll see something like the plots shown in Figure 5.52. This is an unfortunate situation, with insufficient guidance (and perhaps deliberate obfuscation) from manufacturers. We discuss this further in §5.11.

5.10.9 CMRR and PSRR

The common-mode rejection ratio, CMRR, tells how much the input offset voltage V_{OS} varies with common-mode input voltage. The problem, of course, is that such a change in V_{OS} masquerades as a change in the input signal voltage.

The CMRR values vary from 70 dB (min) for our favorite LMC6482 (an inexpensive CMOS dual op-amp), up to 130 dB for the precision OPA277. Degradation of CMRR at high frequencies often matters, and there's usually a plot in the op-amp's datasheet (check out a few for yourself, for example these two; we show CMRR plots for other types of op-amp in Figures 5.73 and 5.82). For example, the LMC6482 typical CMRR starts falling upward of 1 kHz, and it's down to 80 dB at 10 kHz. It's interesting that both the OPA277 and the AD8622 (another expensive high performer at dc) degrade to about 80 dB by 10 kHz, joining our jellybean CMOS friend. Other parts do better, such as the LT1007 (114 dB typ at 10 kHz). And to repeat a warning we made elsewhere: the CMRR spec often applies to only a limited common-mode range; read the datasheet carefully.

Note this universal cure: a time-honored way to avoid CMRR troubles is to use an inverting configuration.

[41] Datasheets for the closely similar OP-97 and LT1097 make the same error, evidently corrected in that of the later LT6010 (the recommended successor to the LT1012).

The power-supply rejection ratio, PSRR (not shown in Table 5.5 (pages 320–321) tells how much V_{OS} varies with power-supply voltage. Typical dc values are 60–80 dB for the LMC6482, up to 130 dB for the OPA277 (but only 100 dB for the AD8622). Study your datasheets!

Frequently one rail is much worse than the other, especially for ac PSRR, because of the op-amp's compensation capacitor (see Figure 4.43, where Q_5 and Q_6 are referenced to the negative rail). For example, the OPA277 suffers by an extra 25 dB on its negative rail. The ac PSRR is significant in two regions, 100–120 Hz (and harmonics) for power-supply ripple, and at high frequencies for cross-talk from other circuitry.

A common defense against PSRR problems, in sensitive applications such as low-level input stages, is to add an *RC* filter in the supply rails.

5.10.10 GBW, f_T, slew rate and "*m*," and settling time

It's tempting to think one can never have too much GBW (gain–bandwidth product, or f_T, its original name that we favor, see Figure 5.42). After all, a higher GBW means greater loop gain, and higher loop gain means lower error (gain, phase, distortion). What's more, with higher f_T we're well on our way to having a faster slew rate, via the formula $S{=}0.32mf_T$ as discussed in detail in §*4x.9*.

Furthermore, a faster slew rate means a greater full-power bandwidth (FPBW): a sinewave $V(t) = A\sin\omega t$ has a peak slew rate $S = \omega A$, so $\text{FPBW} = S/\pi V\text{pp}$. Finally, because the first step on the way to waveform settling is the slewing delay $t = \Delta V/S$, a higher f_T is an important step (and often the primary determinant), toward a faster settling-time spec. The data in Tables 5.4 (page 310), and 5.5 (pages 320–321) let you explore the essential question "what price higher bandwidth?"

A. An aside: GBW and f_T

First, a short riff on "GBW" and "f_T." Figure 5.42 shows a plot of open-loop gain versus frequency for a THS4021 wideband op-amp. This is a "decompensated" op-amp, stable for closed-loop gains ≥ 10, with a textbook Bode plot. The term GBW properly describes the product of open-loop gain times frequency in the region in which the gain is dropping at 6 dB/octave (i.e., $G_{OL} \propto 1/f$). Its extrapolation crosses the G_{OL} =0 dB axis at a frequency equal to GBW. At that frequency, however, the gain is less than unity, owing to the effect of additional higher-frequency poles in the amplifier. Strictly speaking, the symbol f_T is used for the (lower) frequency at which $G_{OL} = 1$.

But we *like* the simpler variable f_T, and so do plenty of other folks; so it's loosely used in place of GBW. Perhaps this is excusable, given that f_T is pretty close in value to GBW for op-amps that are compensated for stability at unity gain (this describes most op-amps). In any case, unless stated otherwise, we will use f_T to mean GBW.

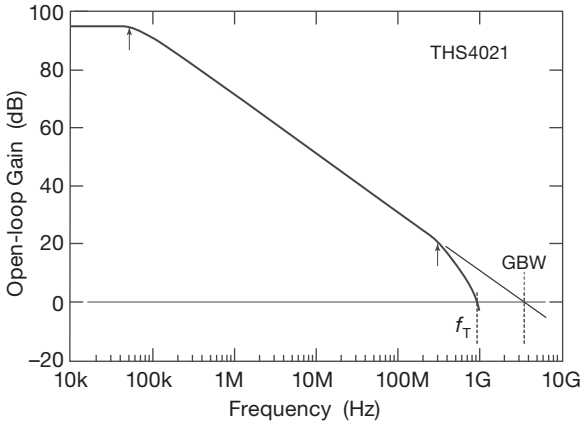


Figure 5.42. An op-amp's gain–bandwidth product (GBW) is the frequency at which the extrapolation of the open-loop gain curve crosses the unity-gain axis. It's often loosely called "f_T," though the latter is properly the frequency at which the closed-loop gain is unity. The arrows indicate the dominant and second poles. The data here are taken from the THS4021 datasheet, which also shows the phase shift reaching 180° by 400 MHz.

B. Milking stools and barstools

If offset voltage is one leg of the precision op-amp stool,[42] bandwidth and speed certainly comprise another. Many of the fast op-amps we wanted to include in the precision Table 5.5 (pages 320–321) were passed over because they had too much offset voltage – so some of them have gotten their own Table 5.4 (page 310). For example, in that table we have the LT6200, with 165 MHz GBW, 0.95 nV/$\sqrt{\text{Hz}}$, and 1.0 mV of offset. It's a conventional voltage-feedback (VFB) op-amp offering 50 V/μs slewing and 140 ns settling time, so whatever is wrong with it? Only 50 V/μs, with $m = 1$? A limited 10 V ($\pm$5 V) supply with a high 16.5 mA current? And a *very* high 40 μA (!) max bias current? The point here is that there's a price to pay for high f_T, and maybe this part is not so attractive after all. But at least the LT6200 has less than 1 nV/$\sqrt{\text{Hz}}$ of noise, 1% distortion at 50 MHz, and it's even RRIO to boot.[43] And there's the LT6200-10 variant, with 1.6 GHz of GBW. Nice!

We did include some admirable high-speed op-amps in Table 5.5, (pages 320–321), overlooking in one case an, uh, *underwhelming* V_{OS}. One such favorite of ours is the OPA656, a member of a small family of highly useful op-amps offered by the Burr–Brown division at TI: it combines 230 MHz GBW, 290 V/μs slew rate, and 20 ns settling time. With its 2 pA JFET inputs having less than 3 pF of input capacitance, we readily forgive its 1.8 mV offset voltage, and we're happy enough with its 7 nV/$\sqrt{\text{Hz}}$ of input voltage noise. It is excellent for a transimpedance amplifier of the sort used with photodiodes – see §§*4x.3* and *4x.9*. The OPA656 even has a distortion plot in Figure 5.44 (see next subsection), where we see it has less than 0.1% distortion to 10 MHz and beyond. And it has a 1.6 GHz cousin, the OPA657. When we need higher operating voltage, and crave lower e_n, we turn to the 4.5 nV/$\sqrt{\text{Hz}}$ OPA637; it has somewhat greater capacitance (7 pF), and less bandwidth (80 MHz). And there's the 90 MHz OPA380 shown above. Very fine products from BB/TI.

On the subject of low distortion, Linear Technology offers the LT1468 (90 MHz, 75 μV, $\pm$15 V supply), which claims 0.7 ppm distortion with a 10 V signal; and with its 0.8 μs settling time, it's a good candidate to feed hungry ADCs. Not to be outdone, National Semiconductor offers the LMP7717, a CMOS op-amp with an 88 MHz f_T, yet drawing just 1 mA, working down to a 1.8 V total supply (!), and offering 1 pA I_B, 6 nV/$\sqrt{\text{Hz}}$ e_n, and 150 μV V_{OS} input specs, along with rail-to-rail outputs. Parts like these suggest that maybe we can have our speed + precision cake and eat it too.

5.10.11 Distortion

Although much of the precision analog-design field concerns dc and low frequencies, there are applications that require accuracy at higher speeds: audio and video, communications, scientific measurement, and so on. With falling op-amp loop gain, input errors are rising, output impedance is rising, and slew-rate limitations may come into play. We need a way to evaluate an op-amp's performance at mid to high frequencies. Some manufacturers help by providing harmonic distortion curves in their datasheets. If

[42] Three-legged, or four? We've found that city folks don't know why it is that milking stools have three legs, whereas barstools have four. Those with rural upbringing can tell you, in a flash.

[43] Yeah, OK, but read carefully: on page 10 of the datasheet you'll see that there are ~1 mV shifts of offset voltage when the input is within 1.5 V of the rail! A powerful incentive to use inverting mode!

it's painful going through hundreds of datasheets looking for tabulated specs to compare, it's doubly painful leafing through their back pages looking for distortion curves.

Here we provide some "value added" (hey, this book isn't cheap!) by compiling distortion plots from the datasheets of fifty selected high-performance op-amps: Figure 5.43 (for high-voltage op-amps) and Figure 5.44 for low-voltage and RR-output op-amps (including some HV types). The op-amps listed in Tables 5.4 (page 310) and 5.5 (pages 320–321) have a check in the "dist graph" column if they appear in one of these graphs. We've also measured the distortion and made plots for some popular older op-amps that don't have datasheet plots; see Figure 5.19.

Burr–Brown/TI's OPA134 and OPA627, along with National Semi's LME49990 and other LME49700-series op-amps, are the winners in the high-voltage category. LTC's LT1468 stands out as well. Analog Devices' AD8021 stands out at high frequencies, and they often recommend it for driving ADCs. TI's THS3061, which has an impressive 7000 V/μs slew rate, looks pretty good above 100 kHz, and as a bonus it can deliver 145 mA into 50 Ω. The OPA1632 and LME49724 are fully differential amplifiers, see §5.17.

The low-voltage and RR op-amp table comprises mostly parts running from $\pm$5V supplies, or lower. Most of these have rail-to-rail outputs, demonstrating that RRO op-amps can compete in the precision arena. Some low-voltage op-amps are at a disadvantage relative to their HV cousins, because they have to use abnormally low signal levels. Consider the graph's OPA1641 JFET winner, at 0.5 ppm. It's a high-voltage part with RR outputs, and it gets tested with 8.5 Vpp signals, a luxury not available to the low-voltage RRO parts. The OPA376 is the low-voltage winner in this category, at 3 ppm, while being tested with 2.8 Vpp. It's interesting that both of these op-amps use the Monticelli output stage (see Figure 5.35 and §*4x.11*).

A. Distortion: some caveats

Some caution is advised here: it's tempting to look at the distortion plots and think that you know how the op-amps compare. But some of the distortion measurements need to be taken with more than a grain of salt, and a few caveats are in order. First, there are no de facto standards for op-amp distortion, and manufacturers have chosen different operating conditions.[44] Some use THD, others THD+N (total harmonic distortion plus noise), and still others may concentrate on specific distortion products, e.g., the 2nd or 3rd harmonic. These hidden choices affect an op-amp's standing on the charts.

Second, the distortion plots sometimes reveal artifacts of the measurement process – for example, the curves in Figures 5.43 and 5.44 begin with a flat distortion profile starting at dc, which, however, often continues well past the frequency at which we know the op-amp's open-loop gain is falling (i.e., past the primary pole). This is contrary to expectations, and likely reveals a noise-floor instrumental limitation, rather than reality; i.e., the op-amp is better than advertised.

Third, the curves ultimately show the expected rising distortion at higher frequencies; this is due to internal op-amp nonlinearities and to loss of loop gain, both within and outside the op-amp. But this region is heavily dependent on signal size and on load, with varying choices by different manufacturers. At some higher frequency the upward curve may steepen sharply. This is usually due to third-harmonic distortion. We cannot generalize about where second- or third-harmonic distortion will dominate for any given op-amp, but it seems that second-order distortion is the most common culprit. This is surprising, given an effort in many op-amps to fully balance the design.

Fourth, when you're in the $<$10 ppm territory all kinds of strange things can bite you. As wise guru Jim Williams once remarked, "If you think you've measured something to 1 ppm, you're probably wrong." Figure 5.45 illustrates an often-overlooked issue. Here we have a precision OPA1641 op-amp that's capable of distortion performance below 1 ppm at 1 kHz, and in noninverting mode the op-amp has just over 20 ppm of distortion at 100 kHz. It's a JFET op-amp with 8 pF of input capacitance (reasonably low, especially considering the op-amp's low 5 nV/$\sqrt{\text{Hz}}$ noise spec). In an admirable display of candor, the datasheet warns us that "The *n*-channel JFETs in the FET input stage exhibit a varying input capacitance with applied common-mode input voltage," and it provides plots of increasing distortion arising from an input source resistance driving the op-amp's dynamically-changing capacitance. For example, with $R_s = 600\,\Omega$ the 100 kHz distortion increases dramatically, to 100 ppm. They suggest carefully matched input impedances to reduce this type of distortion (an effect that is not confined to this particular op-amp: be warned!). Better yet, use the inverting configuration.

Finally, a look at the test circuit used by many manufacturers to make sub-100 ppm distortion measurements (Figure 5.46). The trick is to reduce op-amp's loop gain by a factor of 100, thus increasing the distortion by the same factor; the reported distortion is then gotten by dividing the

[44] Manufacturers use different voltage levels (2 Vpp, 3 Vrms, 10 V peak, and 20 Vpp), different loads (100 Ω, 600 Ω, 2k, 10k and open circuit), different common-mode voltages, different analyzer filters, and even different gains for their measurements.

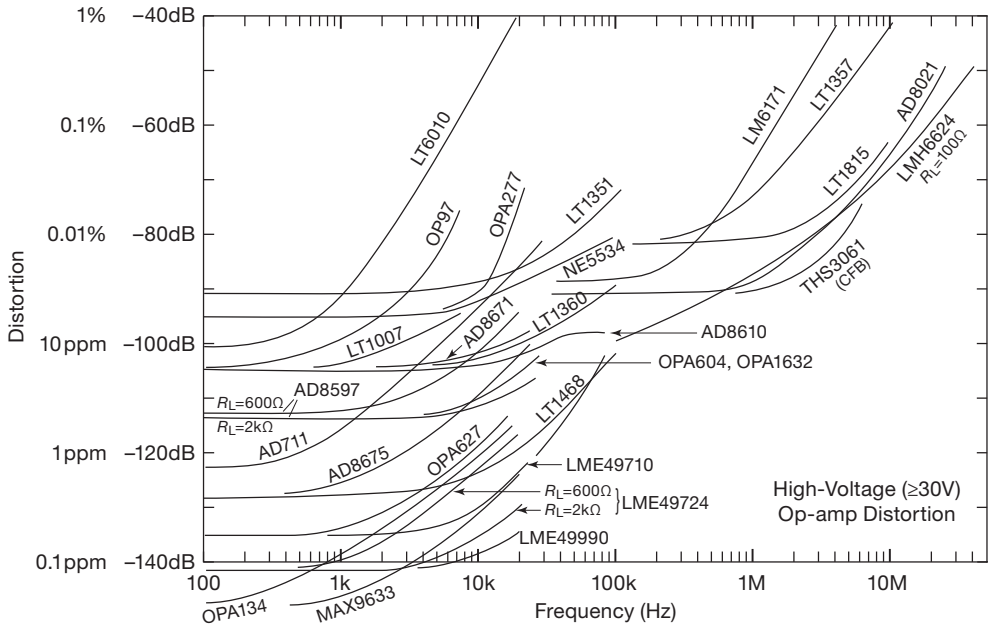


Figure 5.43. Harmonic distortion versus frequency for a selection of "high-voltage" ($\geq$30 V total supply) op-amps, from manufacturers' datasheets.

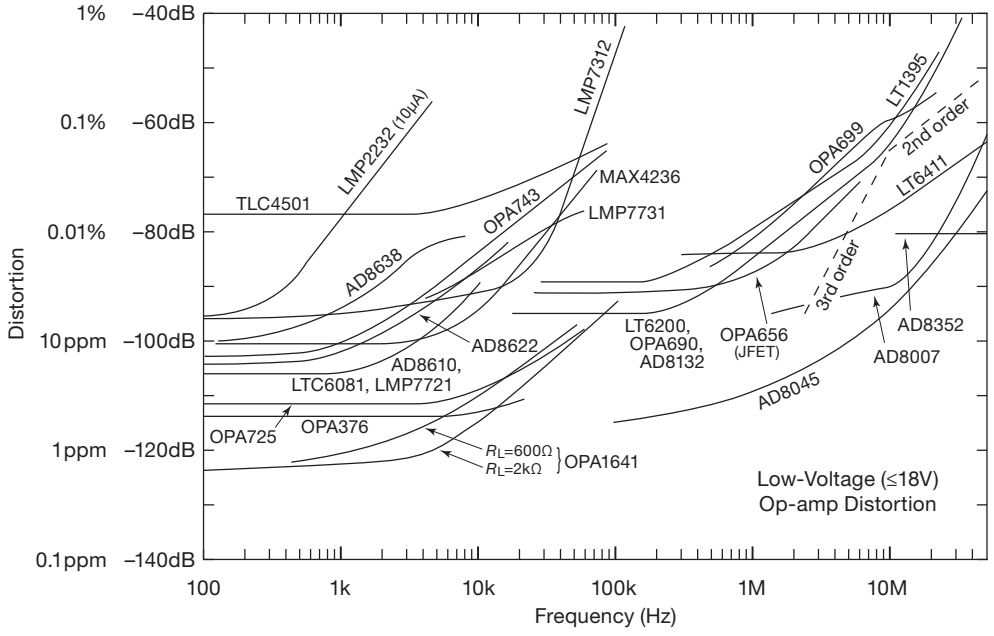


Figure 5.44. Harmonic distortion versus frequency for selection of "low-voltage" ($\leq$18 V total supply) op-amps, from manufacturers' datasheets. Most of these have rail-to-rail output stages. See also Figure 5.19.

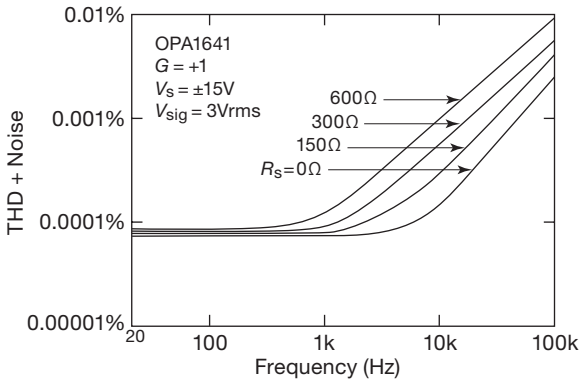


Figure 5.45. The variation of input capacitance with signal voltage causes additional distortion at higher frequencies, dependent on source resistance.

measured distortion by 100. But an immediate concern is the artificial nature of the test, with the op-amp seeing an artificially low source impedance. It's safe to conclude that we all can likely benefit from further work in this area.[45]

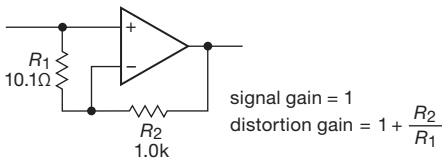


Figure 5.46. Distortion test circuit. With the values shown, the op-amp's effective loop gain is reduced by ×100. Add a series resistor at the output if driving a cable.

5.10.12 "Two out of three isn't bad": creating a perfect op-amp

We recognized at the outset that there's no such thing as a perfect operational amplifier – but not to fear, there is usually a workaround. If you find the input performance specs you need in one op-amp, and the output specs in another, it may be possible to combine them into a "composite amplifier" circuit that acts as a single op-amp (combining the best features of each) in your feedback loop. Or you can create a composite op-amp by adding a discrete input or output stage to the op-amp IC of your choice. If your overall feedback circuit has very high gain, such as a G=10,000 amplifier, you may not need to worry about compensation (e.g., see Figure 5.61). However, it's not terribly difficult to deal with the high loop gain of a G=1 circuit, as we show in §*4x.5*.

In Chapter *4x*'s discussion of composite amplifiers we show a robust amplifier configuration in which the second op-amp's gain is reduced to unity at a frequency well above the first op-amp's f_T, allowing loop-gain flexibility. In addition, no restrictions are placed on common-mode voltages or amplifier input connections in this approach. It's a good configuration to consider, although as often as not you'll find plenty of variations in real-world composite amplifier implementations.

A. Design example: precise high-current piezo positioner

A nice application for a composite amplifier is a precision microscope stage positioner, implemented with a pair of multilayer piezo elements. These devices are both swift and stiff – our chosen part, for example, is good to tens of kilohertz and tens of kilograms – and they provide stable and accurate positioning (at the nanometer scale) over their limited motion (here 6 μm). On the down side, they present a difficult load, being highly capacitive (here 0.75 μF) and requiring relatively high drive voltage (here 100 V full-scale).

A suitable circuit is shown in Figure 5.47. We'd like a moderately fast response, say 1 V/μs slewing speed, which requires $I = CdV/dt = 1.5$ A of drive capability into the piezo pair's 1.5 μF of capacitance. Our signal source is a 16-bit DAC8831, with a +5 V full-scale output (R-2R ladder, R_{out}=6.25 kΩ) with a fast SPI interface. Running the DAC with a 5 V reference produces a least-significant bit (LSB) step size of 76 μV. Its R_{out} of 6.25k requires an op-amp whose input current is less than 12 nA in order not to add error greater than an LSB. And the op-amp has to swing to +100 V while driving 1.5 μF.

So, we're evidently looking for a muscular 150 V, 1.5A op-amp with an input offset less than 75 μV, and bias current less than 10 nA. Looking, looking... no joy! No such part is available, so we'll solve the problem by rigging up a composite amplifier with a gain of 20. We'll specify a frequency response to 25 kHz (which is about 20% of the piezo's mechanical self resonant frequency).

For the input op-amp we chose the AD8675 from Table 5.5 (pages 320–321). It has low input errors (75 μV and 2 nA max), and enough output swing to drive a high-voltage $G = 20$ stage. For our output amplifier we chose the Apex PB51, a power driver capable of 300 V and 1.5 A (but subject to a safe-operating-area constraint, for example limited to 130 V drop when driving 2 A for 100 ms). Its maximum input errors are 1.75 V (!) and 70 μA (!) – that's

[45] OK, we plead guilty-as-charged to that conclusion favored by all academics – "needs more study (and a grant proposal is in the mail)."

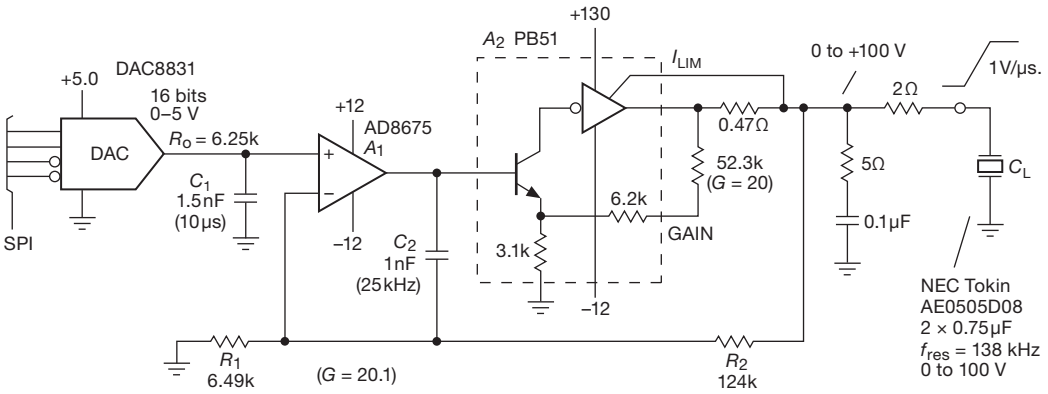


Figure 5.47. Precision composite amplifier for driving a 1.5 μF piezo positioner: output to 100 V and 1.5 A, with 75 μV maximum offset, 2 nA maximum input current, and response to 25 kHz. The specified piezo actuator moves 6 μm per 100 V.

why they call it a *driver* rather than an op-amp! Its gain is set with an external resistor, here 52.3 kΩ (for $G = 20$) to match the desired overall gain.

The single feedback path R_1R_2 sets $G = 20$ for the composite amplifier. A general scheme for compensating composite amplifiers is suggested in §*4x.5*. Here we use a different scheme, with C_2 isolating the input op-amp A_1 from the output amplifier A_2 at frequencies above 25 kHz. That way we don't have to worry about A_2's response at high frequencies, where it's struggling with its capacitive load. This is really a "custom" configuration, forced on us by the, uh, *challenging* load; what is has in common with other composite amplifier configurations is the single overall feedback path that determines its gain over most of its operating regime.

To get an understanding of circuit operation, imagine a 2 V input step from the DAC. This causes a 2 V step at the output of A_1 (which acts like a follower at high speeds, owing to C_2), which is approximately the right signal to tell A_2 to drive current into C_L and move its output by 40 volts (that's why we chose G=20 for the output stage). As amplifier A_2 approaches this goal, op-amp A_1 takes charge and presents correction signals to cause the output to hone in on the precise value.

A few additional circuit details: the DAC's response is slowed by C_1 to a 10 μs time constant – no point in jolting the amplifier, given its limited bandwidth. The series *RC* across the output promotes stability, both by reducing A_2's open-loop gain at high frequencies and by providing lossy damping; it's widely used in audio power amplifiers. Note that substantial bypass capacitors (as much as 100 μF, not shown) are needed with such large signal currents.

Further examples of the composite amplifier technique can be found in §*4x.5* and in Figures 5.58, 5.59, 5.61, 8.49, 8.50, 8.78, 8.80, and 13.48.

And other examples of techniques that exploit the "two-out-of-three" concept of adding external amplifier blocks are (a) discrete JFET front-end for a BJT op-amp (e.g., Figure 5.58), (b) output unity-gain buffer (§5.8.4), and (c) bootstrapped power supplies to extend voltage range, or to improve CMRR (e.g., Figure 5.79).[46]

5.11 Auto-zeroing (chopper-stabilized) amplifiers

Even the best of precision low-offset op-amps cannot match the stunning V_{OS} performance of the so-called "chopper-stabilized" or "auto-zero" (also called "zero-drift") op-amps. Ironically, these interesting amplifiers are built with CMOS, otherwise famous for its mediocrity when it comes to offset voltage or drift. The trick here is to put a second *nulling* op-amp on the chip, along with some MOS analog switches and offset-error storage capacitors. One of several possible configurations is shown in Figure 5.48. The main op-amp functions as a conventional (imperfect) amplifier. The nulling op-amp's job is to monitor the input offset of the main amplifier, adjusting a slow correction signal as needed in an attempt to bring the input offset exactly to zero. Because the nulling amplifier has an offset error of its own, there is an alternating cycle of operation in which the nulling amplifier corrects its own offset

[46] If you're considering piezo positioners for a precision application, be aware that they exhibit some nonlinearity and hysteresis when driven from a voltage source. These issues are said to be ameliorated when the drive signal is quantified by *charge* instead of voltage. See Chapter *3x* for a precision current-drive circuit that circumvents this problem and makes fast linear piezo steps.

voltage. Both amplifiers have a third "nulling" input terminal, analogous to the offset trim seen in some op-amps.

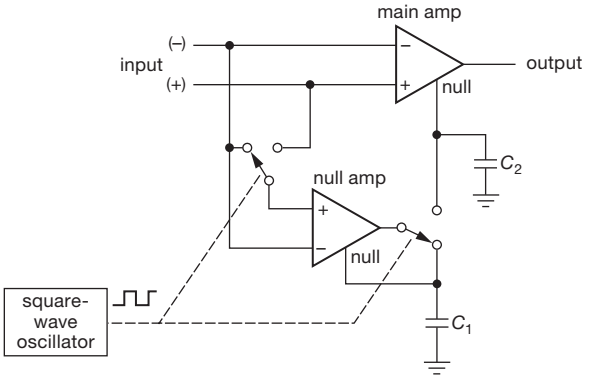


Figure 5.48. The original ICL7650 and ICL7652 auto-zero ("chopper-stabilized") op-amps. Capacitors C_1 and C_2 are external.

The auto-zeroing cycle goes like this. (a) Disconnect the nulling amplifier from the input, short its inputs together, and connect its output to C_1, the holding capacitor for its correction signal; the nulling amplifier now has zero offset. (b) Now connect the nulling amplifier across the input and connect its output to C_2, the holding capacitor for the main amplifier's correction signal; the main amplifier now has zero offset (assuming that the nulling amplifier has not drifted). The MOS analog switches are controlled by an on-board oscillator, typically running in the range of 1–50 kHz.

5.11.1 Auto-zero op-amp properties

Auto-zero op-amps do best what they are optimized for, namely delivering V_{OS} values (and tempcos) 5–50 times better than the best precision bipolar op-amp (see Table 5.6 on the next page). What's more, they do this while delivering full op-amp speed and bandwidth.[47] They also have extraordinarily high open-loop gain at low frequencies (typically 130–150 dB, a consequence of their "composite amplifier architecture"); and, happily, they are inexpensive, particularly when compared with conventional precision op-amps.

That's the good news. The bad news is that auto-zero amplifiers have a number of diseases that you must watch out for. Being CMOS devices, most of them have a severely limited supply voltage – often 6 V total supply, with a smaller selection that can run up to 15 V, and just one[48] older generation part (the LTC1150) that can operate at ±15 V.

Of greater importance is the problem of clock-induced noise. This is caused by charge coupling from the MOS switches (see §3.4.2E) and can cause wicked spikes at the output. The specifications are often misleading here, because it is conventional to quote input-referred noise with $R_S = 100\,\Omega$ and also to give the specification for only very low frequencies. For example, a typical input-referred noise voltage might be 0.3 μVpp (dc to 1 Hz, with $R_S = 100\,\Omega$). However, with zero input signal the output waveform might consist of a train of 5 μs-wide 10 mV spikes of alternating polarity!

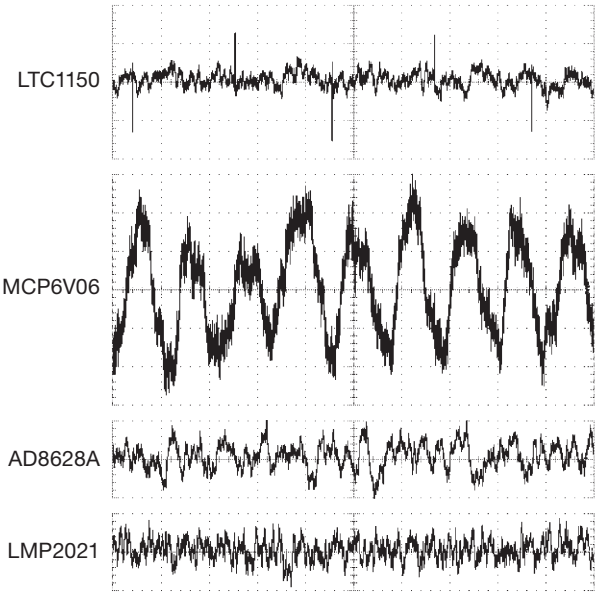


Figure 5.49. Output waveforms from four auto-zero op-amps, configured for $G = 100$, with the input connected to ground through a 100 Ω resistor. Vertical: 2 mV/div; horizontal: 100 μs/div.

The internal switching also causes spikes of *input* current, which means that input signals of high source impedance R_s will exhibit larger input-referred spikes. Figures 5.49 and 5.50 show this behavior, measured with R_s of 100 Ω and 1 MΩ, in several auto-zero op-amps configured

[47] Unlike an earlier generation of synchronous amplifiers that were also called "chopper amplifiers," but that had bandwidth limited to a fraction of the chopping clock frequency.

[48] Some old high-voltage types that require external (correction-signal) capacitors may still be available.

Table 5.6 Chopper and Auto-zero Op-amps

Part #	# per pkg	Supply (V)	I_Q typ (µA)	I_{bias}[a] @25°C typ (pA)	I_{bias}[a] @25°C max (pA)	Offset Voltage @25°C typ (µV)	Offset Voltage @25°C max (µV)	Offset Voltage Tempco typ (nV/°C)	Offset Voltage Tempco max (nV/°C)	CMRR min (dB)	Noise[t,u] V_{npp} dc[d] (µV)	Noise[t,u] e_n 1kHz ($\frac{nV}{\sqrt{Hz}}$)	Noise[t,u] i_n 10Hz ($\frac{fA}{\sqrt{Hz}}$)	HF Noise (>kHz)	GBW typ (MHz)	Slew (V/µs)	t_{settle} typ (µs)	$t_{recovery}$ typ (µs)	Swing to Supplies? IN +	IN −	OUT +	OUT −	Cost qty 25 ($US)	Comments
ADA4051-1	1,2	1.8-6	15	20	70	2	15	20	100	110	2	95	100	40	0.13	0.04	110	120	●	●	●	●	1.86	auto-zero, -2=dual
OPA333	1,2	1.8-7	17	70	200	2	10	20	50	106	1.1	55	100	>20	0.35	0.16	45	80	●	●	●	●	2.57	auto-cal, dual=2333
ISL28133	1,2	1.65-6.5	18	30	300	2	8	20	75	118	1.1	65	79	8	0.4	0.1	35	?	●	●	●	●	1.69	dual=28233
OPA330	1,2	1.8-7	21	200	500	8	50	20	250	100	1.1	55	100	>20	0.35	0.16	45	80	●	●	●	●	2.56	auto-cal, dual=2330
MAX9617	1,2	1.6-6	59	10	140	0.8	10	5	120	122	0.42	42	100	50	1.5	0.7	-	?	●	●	●	●	1.60	RRIO, charge pump
OPA378	1,2	2.2-6	125	150	550	20	50	100	250	100	0.4	20	200	15	0.9	0.4	9	4	c	●	●	●	2.16	auto-cal, '2378=dual
LTC2054	1,2	2.7-6	140	1	150	0.5	3	20	30	120	1.6	85	-	-	0.5	0.5	-	4000	-	●	●	●	2.25	dual=2055
AD8538	1,2	2.7-6	150	15	25	5	13	30	100	115	1.2	52[h]	-	1,15	0.43	0.4	5	50	●	●	●	●	1.80	self-calibrating, duals
MCP6V06	1,2	1.8-6.5	200	1	-	-	3	-	50	140	1.7	82	*(0.6)*	1	1.3	0.5	300	100	●	●	●	●	1.42	low-cost, dual = 6V07
LTC1049	1	5-16	200	15	50	2	10	20	-	110	3	100	*(2)*	10	0.8	0.8	-	6000	-	●	●	●	2.85	miniDIP pkg avail
OPA335	1,2	1.8-7	285	70	200	1	5	20	50	110	1.4	55	20	10	2	1.6	6	50	-	●	●	●	2.56	auto-cal, dual=2335
MAX4238	1	2.7-6	600	1	-	0.1	2	10	-	120	1.2	30	-	?	1	0.35	1000	3300	-	●	●	●	1.62	'4239=decomp, *G*=5
OPA734	1,2	2.7-13	600	100	200	1	5	10	50	115	2.5	135	40	17	2	1	-	8	-	●	●	●	2.56	auto-cal, dual=2734
AD8551	1,2,4	2.7-6	700	10	50	1	5	5	40	120	1	42	*(2)*	3.5	1.5	0.4	50	50	●	●	●	●	2.34	auto-zero
AD8572	1,2,4	2.7-6	700	10	50	1	5	5	40	120	1.3	51	*(2)*	2.3	1.5	0.4	-	50	●	●	●	●	3.34	auto-zero
LTC2050HV	1	2.7-11	750	25	75	0.5	3	-	30	115	1.5		-	?	3	2	-	2000	-	●	●	●	3.20	OK for ±5V supplies
AD8628	1,2,4	2.7-6	850	30	100	1	5	2	20	120	0.5	22	*(5)*	15	2.5	1	-	50	●	●	●	●	1.92	self-cal, dual, quads
LMP2011	1,2,4	2.5-5.8	930	3	-	0.12	25	15	-	95	0.85	35[h]	-	25	3	4	-	50	-	●	●	●	2.55	copper leadframe
TLC4501A	1,2	4-6	1000	1	60	10	40	1000	-	90	1.5	12	0.6	none	4.7	2.5	2	0	-	●	●	●	2.88	self-zero at powerup
AD8638	1,2	5-16	1000	1.5	40	3	9	10,40	60	118	1.2	60	-	8	1.35	2.5	3	50	-	●	●	●	2.29	self-calibrating, duals
LTC1050	1	4.75-16	1000	10	30	0.5	5	10	50	114	1.6		*(1.8)*	2.5	2.5	4	-	3000	-	●	●	●	2.85	favorite
LMP2021	1,2	2.5-5.8	1100	25	100	0.4	5	4[g]	20	105	0.2	11	350	30	5	2.5	-	50	-	●	●	●	3.38	EMI-rej, '2022=dual
MAX4236A[b]	1	2.4-6	350	1	500	5	20	600	2000	84	0.2	14	0.6	none	1.7	0.3	1	0	-	●	●	●	1.78	comp low V_{os} CMOS
AD8616[b]	1,2,4	2.7-6	1.7	0.2	1	23	60[f]	1500	4000	80	2.4	7	2	none	24	12	0.5	0	●	●	●	●	1.52	comp CMOS
LT1028[b]	1	8-44	7400	25nA	95nA	10	40	200	800	108	0.04	0.9	4700	none	75	15	-	0	-	-	-	-	6.31	comp low-noise BJT
high-voltage																								
LTC1150	1	4.8-32	800	10	60	0.5	10	10	50	110	1.8		-	0.5	1.8	1.5	-	20ms	-	●	n	n	5.84	OK for ±15V supplies
LT1012A[b]	1	2.4-40	380	25	100	8	25	200	600	114	0.5	14	20	none	0.8	0.2	-	0	-	-	-	-	5.11	comp low I_B bipolar
LT1007A[b]	1	7-44	3000	10nA	35nA	10	25	200	1000	117	0.06	2.5	1500	none	8	1.7	-	0	-	-	-	-	5.83	comp low e_n bipolar

Notes: (a) check datasheets for plots of I_{bias} vs common-mode input voltage. (b) conventional (not auto-zero) precision op-amps, for comparison. (c) crossover region. (d) 0.01Hz to 10Hz. (f) at V_{CM}=0.5V and 3.0V (with V_S=5.5V), but as much as 400µV near V_{CM}=2V. (g) 1nV/°C at V_S=2.5V. (h) 150nV/√Hz hump above 2kHz. (n) near to rails w/o load. (t) typical. (u) current noise values indicated with *light italics* should not be relied upon; measured values appear to be greater by factors of 100 or more, see discussion in Chapter 8.

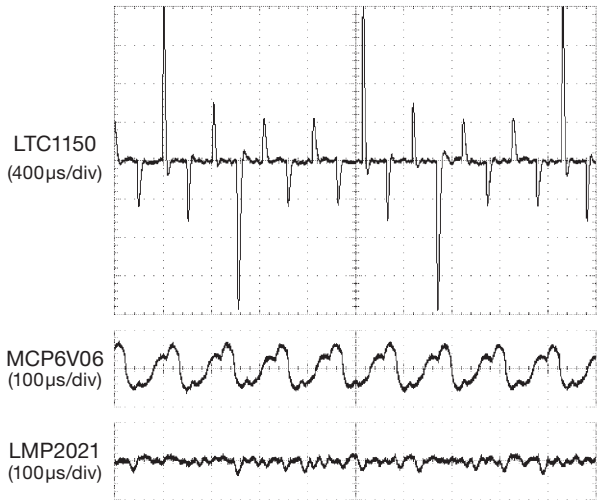


Figure 5.50. Output waveforms from three auto-zero op-amps, configured for $G = 100$, with the input connected to ground through a 1 MΩ resistor. Vertical: 100 mV/div.

for a voltage gain of 100.[49] There is considerable variation among these parts, with the conventional auto-zero configuration (Figure 5.48, used in the LTC1150 and MCP6V06) exhibiting greater clock feedthrough compared with that of alternative designs (as in the AD8628A and LMP2021) that are intended to reduce these undesirable effects.[50]

The datasheets *do* reveal this unseemly behavior, indirectly, in plots of voltage noise versus frequency.[51] Figure 5.51 shows a pair of such plots for two auto-zero products from Analog Devices: the AD8551 has a ∼4 kHz fixed-frequency oscillator, whereas their AD8571 has a deliberately variable (spread-spectrum) oscillator to eliminate sharp spectral lines (which can create undesirable intermodulation with nearby signal frequencies). Note, by the way, that these plots specify an input signal of *zero* source impedance.

It's always instructive to make some actual measure-

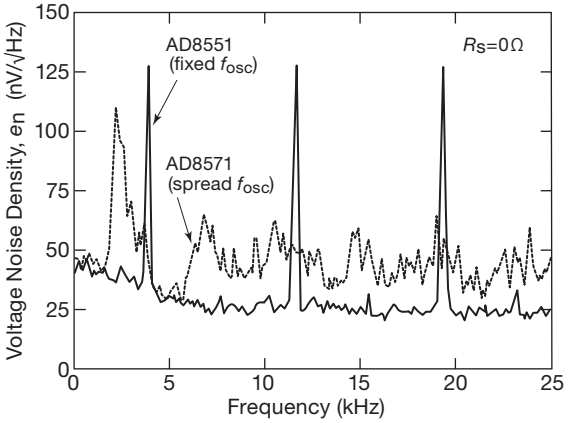


Figure 5.51. Spectra of noise voltage, adapted from their datasheets, for a pair of auto-zero op-amps. The AD8571 varies its oscillator frequency in order to suppress sharp spectral features.

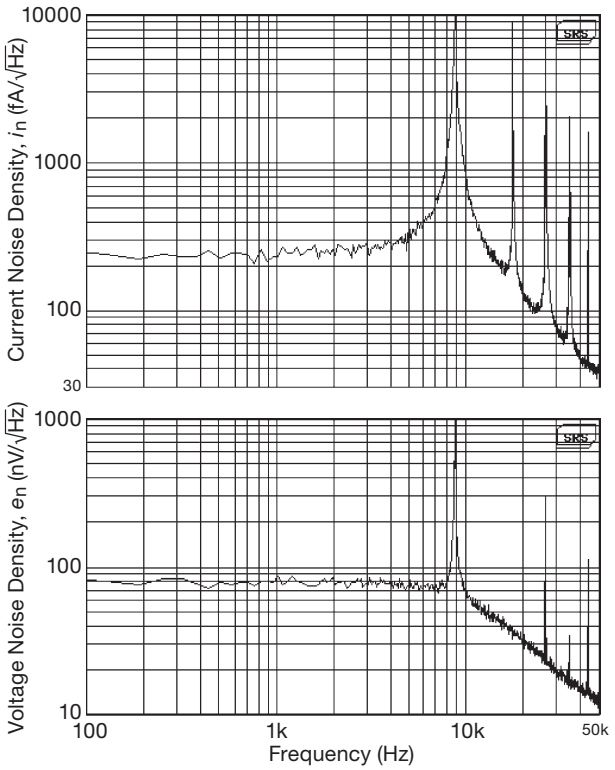


Figure 5.52. Measured voltage-noise density (bottom) and current-noise density (top) for an MCP6V06 auto-zero amplifier. Switch-induced clock noise at 9 kHz (and harmonics) is prominent.

[49] The waveforms in the latter show 8 nApp current spikes for the LTC1150, 1 nApp noise for the MCP6V06 (despite its impressive spec: 0.6 fA/$\sqrt{\text{Hz}}$ at 10 Hz), and 0.2 nApp "rumble" for the LMP2021 (which sports a 0.35 pA/$\sqrt{\text{Hz}}$ current-noise spec).

[50] From the AD8628A datasheet: "The AD8628/AD8629/AD8630 family uses both auto-zeroing and chopping in a patented ping-pong arrangement to obtain lower low frequency noise together with lower energy at the chopping and auto-zeroing frequencies, maximizing the signal-to-noise ratio for the majority of applications without the need for additional filtering. The relatively high clock frequency of 15 kHz simplifies filter requirements for a wide, useful noise-free bandwidth."

[51] Watch out, though, for claimed values of noise *current* – the low values listed in many datasheets are completely incorrect, sometimes by factors of ×10 to ×100, evidently having been calculated a priori as the shot noise corresponding to the dc input current; see the discussion in §5.10.8 and in §8.9.1F.

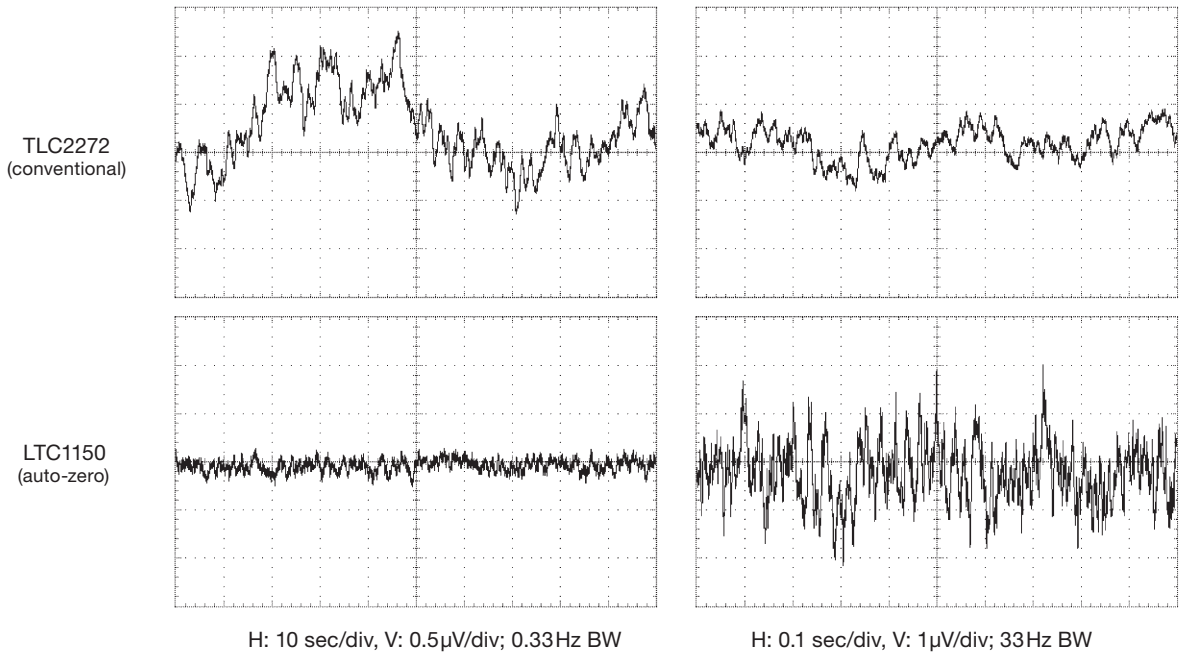


Figure 5.53. At very low frequencies a chopper-stabilized op-amp has lower noise than a conventional op-amp, but with 100× greater bandwidth it has more noise, as seen in these measured traces. See also Figure 5.54

ments yourself, if for no other reason than for "fact checking" the manufacturer. We ran some spectral noise plots for a half-dozen auto-zero amplifiers, with a particular interest in chopper-induced narrowband noise at the clock frequency and its harmonics. For these measurements we took data with $R_s = 0$ (to reveal the input voltage noise e_n), and then with R_s=1M Ω (to reveal input current noise i_n). Figure 5.52 shows the results, for one specimen from our collection of auto-zero amplifiers. The low-frequency measured e_n agrees well with the datasheet's value of $82\,\text{nV}/\sqrt{\text{Hz}}$, but, as noted above, the measured current-noise density i_n is far greater than the specified value of $0.6\,\text{fA}/\sqrt{\text{Hz}}$ – a factor of ×400 in this case.

For low-frequency applications you can (and should) *RC*-filter the output to a bandwidth of a few hundred hertz, which will suppress output spikes. This spiky input-current noise is also of no importance in applications with low input impedances, in integrating applications (e.g., integrating ADCs; see §13.8.3), or in applications in which the output is intrinsically slow (e.g., a thermocouple circuit with a meter at the output). In fact, if you want only very slow output response, and therefore lowpass-filter the output to extremely low frequencies (below 1 Hz), a chopper amplifier will actually have *less* noise than a conventional low-noise op-amp; see Figures 5.53 and 5.54.

Another way to put it is that auto-zero amplifiers have lots of *wideband* voltage noise ($\sim 50\,\text{nV}/\sqrt{\text{Hz}}$ at 1 kHz, compared with just a few $\text{nV}/\sqrt{\text{Hz}}$ for a good low-noise op-amp), but their noise density holds constant at very low frequencies, as contrasted with the $\sim 1/f$ ("flicker-noise") divergence of conventional op-amps (and everything else; see Chapter 8). For example, a conventional low-noise BJT op-amp like the LT1007 has $e_n = 2.5\,\text{nV}/\sqrt{\text{Hz}}$ (typ) at 1 kHz, but its noise power density rises as $1/f$ below its "corner frequency" of 2 Hz, thus $e_n \sim 100\,\text{nV}/\sqrt{\text{Hz}}$ at 0.001 Hz. Compare that with an auto-zero like the AD8551, with roughly flat $e_n = 42\,\text{nV}/\sqrt{\text{Hz}}$: the latter will have far smaller fluctuations on time scales of minutes. In fact, the AD8551's datasheet even specifies a peak-to-peak noise voltage from "0 Hz to 1 Hz" of 0.32 μV (typ); no conventional op-amp would dare to project its drift out to infinite time!

A final problem with auto-zero amplifiers is their unfortunate overload recovery. What happens is this: the auto-zeroing circuit, in attempting to bring the input difference voltage to zero, implicitly assumes there is overall feedback operating. If the amplifier's output saturates (or if there is no external circuit to provide feedback), there will be a large differential input voltage, which the nulling amplifier sees as an input offset error; it therefore blindly

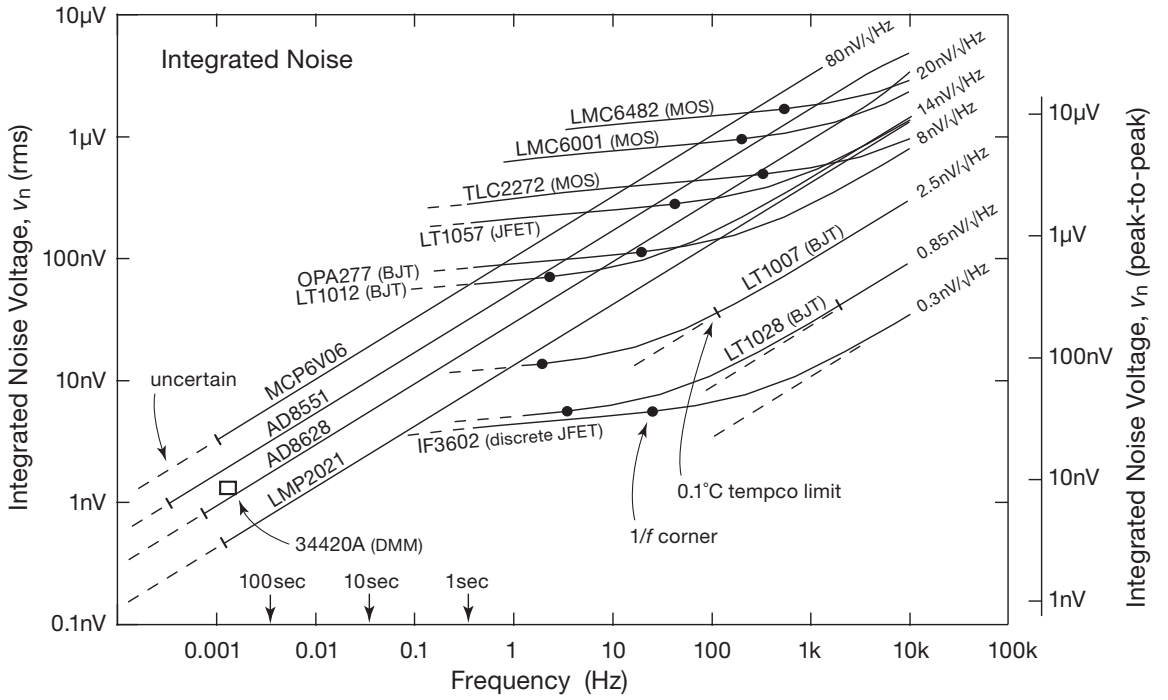


Figure 5.54. Integrated RMS voltage noise versus amplifier bandwidth. The "zero-drift" aspect of auto-zero amplifiers causes their low-frequency integrated noise voltage v_n to fall at low frequencies, proportional to the square root of the lowpass bandwidth. In contrast, the rising noise density $e_n \propto 1/\sqrt{f}$ of conventional op-amps causes a plateau in the integrated noise voltage v_n below the $1/f$ corner frequency, as seen in these computed curves. (We chose a lower limit of 0.01 Hz, when integrating the latter because the unbounded integral is divergent. This plot lets you see the region where the auto-zero op-amp wins or loses, compared with the op-amp of your choice. You can draw in estimated plots for other parts if you know e_n and the $1/f$ corner frequency. See §8.13.4.) See Figure 5.37 for the spectral noise-density plots behind these curves, and see Figure 8.63 for graphs of three dozen more op-amp types.

generates a large correction voltage that charges up the correction capacitors to a large voltage before the nulling amplifier itself finally saturates. Recovery is slow – t_r can extend up to several milliseconds. One "cure" is to sense when the output is approaching saturation, and clamp the input to prevent it. You can prevent saturation in chopper amplifiers (and in ordinary op-amps, as well) by bridging the feedback network with a bidirectional zener (two zeners in series), which clamps the output at the zener voltage, rather than letting it limit at the supply rail; this works best in the inverting configuration.

Alternatively, you can do an end run around this problem by choosing a part with fast recovery time, for example the OPA378 or OPA734 (with $t_r = 4\,\mu$s and $8\,\mu$s, respectively).

5.11.2 When to use auto-zero op-amps

- Slow but accurate measurements from transducers: weigh scales, thermocouples, current shunts
- Accurate in-circuit dc conditioning, for example creating precise sets of voltages from a voltage reference
- "Normal-bandwidth" applications that want low-voltage and low-I_B CMOS, can tolerate broadband noise, require low offset voltages ($\ll$1 mV), and don't want to pay the cost premium of precision CMOS op-amps.

5.11.3 Selecting an auto-zero op-amp

Table 5.6 on page 335 lists a nice selection of currently available auto-zero op-amps, plus a few conventional op-amps for comparison. This is a good place to go first when you need an auto-zero amplifier. It's also a good place to learn about some of the common properties, and some of the quirks, of these amplifiers. Here are some comments, to get you started.

Supply voltage All but one of the parts are low voltage, 5.5–6 V max, and many will operate down to 2 V or less. Five can operate from ±5 V supplies. The supply

currents range from 15 μA to 1.1 mA. The parts are listed approximately by supply current.

Input current Auto-zero amplifiers are built with CMOS, so the input currents are typically in the picoamps. We might expect the input currents to be in the single picoamp territory like other CMOS op-amps. Although there are a few parts for which this is true (e.g., MAX4238, MCP6V06 and LTC2054), most have considerably higher currents, up to 0.5 nA max, no doubt because of input-switch charge coupling. Even most conventional JFET op-amps do better except at high temperatures, where auto-zeros are usually much better. For example, I_B of the auto-zero LMP2021 typically stays below 75 pA at 125°C (for any common-mode voltage), compared with the (conventional) JFET OPA124 (0.5 nA) and LF412 (10 nA). Auto-zero input currents are not as low as the best conventional CMOS parts (with their femtoamp currents), but considerably better than conventional precision BJT-input parts like the LT1028 or LT1007 on the table.[52]

Offset voltage This is where auto-zero amplifiers really shine, with maximum offset voltages ranging from 0.1–5 μV typical (and 2 μV–25 μV maximum – in the precision design business you've often got to pay serious attention to "maximum" specifications). A few conventional (not auto-zero) parts can approach this figure (20 μV for the CMOS MAX4236A, 25 μV for the BJT LT1012A and LT1007), but they cannot begin to match the excellent tempco of the auto-zero parts (generally in the range of 5–20 nV/°C), gained, of course, by continual offset correction. Conventional op-amps also suffer from the devastating effects of $1/f$ noise, which sets performance floors in the 10–100 nV region; see Figure 5.54 and associated discussion.

The auto-zero amplifiers have typical offset-voltage tempco drifts from 4–100 nV/°C. The maximum specs range up to 250 nV/°C (and beyond? many parts don't list a maximum spec). The AD8628 and LMP2021 are the winners in this category. But these parts consume about 1 mA and thus can be expected to have more self-heating of the die than say the 60 μA MAX9617 with its 5 nV/°C spec. No manufacturer can afford to perform temperature tests on production parts, so these specs should be taken with a grain of salt.

Is this performance believable? At the level of nV/°C you have to worry seriously about thermocouple effects in external connections, and even within the chip's lead frame itself: typical thermal EMFs are of order 5–40 μV/°C – that's 1000× (or more) the specified tempco of these auto-zero op-amps!

Voltage noise Auto-zero amplifiers exhibit higher broadband noise than conventional op-amps, owing to their CMOS inputs and associated switching elements. The voltage noise density e_n at 1 kHz (the usual benchmark) is of order 50–100 nV/$\sqrt{\text{Hz}}$, bested by many conventional CMOS parts, and all BJT parts. But, unlike conventional op-amps, the noise density does not rise at low frequencies, so the low-frequency integrated noise voltage (which you can think of as fluctuations, or drift) is better than even the best low-noise op-amps (as seen in Figures 5.53 and 5.54). Speaking approximately, the integrated-noise voltage falls as the $1/\sqrt{t}$ (or proportional to the square root of lowpass frequency).

In addition to e_n, a useful parameter in the table is the 0.1–10 Hz peak-peak voltage noise (v_n) specification. The AD8628, MAX9617, and OPA378[53] do very well with 0.5 μV to 0.4 μVpp noise specs, but the LMP2021 is the clear winner with an e_n of 11 nV/$\sqrt{\text{Hz}}$ and a v_n of 0.26 μVpp. This part is available in a convenient SOT23 package.

A note of caution, here: the very low-frequency extrapolation (i.e., long-term drift) must eventually become dominated by other drift sources (e.g., diffusion of impurities); see Bob Pease's column, "What's All This Long-Term Stability Stuff, Anyhow?" (published in *Electronic Design*, 20 July 2010).

Noise current The noise current density i_n must be at least the shot noise value (given by $i_n = \sqrt{2qI_B}$, with $q = 1.6 \times 10^{-19}$ C; thus 1.8 fA/$\sqrt{\text{Hz}}$ for a bias current of I_B=10 pA) corresponding to the input current I_B, generally in the range of a few fA/$\sqrt{\text{Hz}}$. In fact, for most of these parts it is much larger – by as much as factors of 10–100.

The current noise is 125× higher than shot noise for

[52] Some parts warn that for high source impedances the bias current may change dramatically as a function of input *capacitance*! For example, the input current of an LMP2021 with $R_s = 1$ GΩ varies from −25 to +25 pA for an input shunt capacitance C_s ranging from 2 to 500 pF. Note that such input currents create large offsets with such high source resistances: 25 pA into 1 GΩ is 25 mV. A graph in the datasheet shows that the input current I_B goes through zero for C_s =22 pF. Other manufacturer's parts show similar effects. In a transimpedance amplifier with high R_F, using a large feedback capacitor C_F can dramatically reduce the bias-current error.

[53] A favorite of Phil Hobbs, who gushes "The OPA378 is a really beautiful zero-drift chopper-CAZ mutation that doesn't exhibit switching noise, and has constant 35 nV [per root hertz] noise down to DC. I've used it in an etalon-locked diode laser for downhole applications, and it's a thing of great charm."

the LMP2021, which was the winner of the voltage-noise competition. Parts that claim to do well in this regard (i.e., with input noise current approximately equal to the calculated shot noise[54]) include the AD8572, AD8551, and LTC1050. The MCP6V06 is the winner of the current-noise competition, with 0.6 fA/$\sqrt{\text{Hz}}$. This spec predicts 2 μV from current noise through a 1 GΩ resistor in a 10 Hz bandwidth, about equal to the part's 1.7 μV voltage noise v_n. The TLC4501A does equally well because it does its auto-zero only once, at powerup. But we know that the TLC4501A, unlike the MCP6V06, will do well at higher frequencies and with wider bandwidths because it has no busy auto-zero oscillator and switches. But it will do poorly at long time scales because of $1/f$ noise and multiple sources of drift.

A caution, loyal reader, as you make your choices – seven otherwise-attractive parts in the table, like the AD8538 and LMP2011, don't have any current-noise specs or plots. You may need to go to the bench to get your answer.

Slew rate and settling time For the listed parts the slew rates range from 0.04 to 2.5 V/μs, and the gain–bandwidths range from 0.13 to 4.7 MHz. The faster parts are meant to compete for use in ordinary op-amp sockets. For these parts, the settling time t_s is dominated by slew rate. But there are anomalies, for example the MCP6V06[55] and MAX4238, whose settling times are one or two orders of magnitude longer than the competition. This may be related to recovery time – the parts with recovery times in the milliseconds have very long settling times (the MAX4238), or they aren't willing to say (five other parts).

Input voltage range Most auto-zero op-amps do not support input voltages to the positive rail (though they all are rail-to-rail *output*). The MCP6V06, OPA333, ISL28133, MAX9617, and most of the Analog Devices parts are notable for full rail-to-rail input operation, without V_{OS} or CMRR degradation. The MAX9617 achieves this using an internal above-the-rail charge-pump power supply. Note also that the V_{OS} specifications may be conditioned on a restricted range of input voltage – most of the way to V_+ for some, others only partway. *Be sure to read the fine print in the specification!* For example, the OPA335 datasheet says "$(V_-) - 0.1\text{V} < V_{cm} < (V_+) - 1.5\text{V}$" next to its 130 dB CMRR spec, and "$V_{cm} = V_s/2$" next to its 1 μV offset-voltage spec.

Packages A few of the (older) Linear Technology types are available in DIP-8 packages for easy breadboarding. Otherwise you can use a SOIC-to-DIP or SOT23-to-DIP adapter (check out offerings from Aries or Bellin Dynamic Systems).

[54] At 10 Hz, who knows about higher frequencies?!

[55] The curious part number reminds us old timers of a favorite vacuum tube of yesteryear.

5.11.4 Auto-zero miscellany

A. ac-coupled "chopper amp"

When considering auto-zeroing chopper amplifiers, be sure you don't confuse this technique with another "chopper" technique, namely the traditional low-bandwidth chopper amplifier in which a small dc signal is converted to ac ("chopped") at a known frequency, amplified in ac-coupled amplifiers, then finally demodulated by multiplying with the same waveform used to chop the signal initially (Figure 5.55). This scheme is quite different from the full-bandwidth auto-zeroing technique we've been considering, in that it rolls off at signal frequencies approaching the clock frequency, typically just a few hundred hertz. You sometimes see it used in chart recorders and other low-frequency instrumentation.

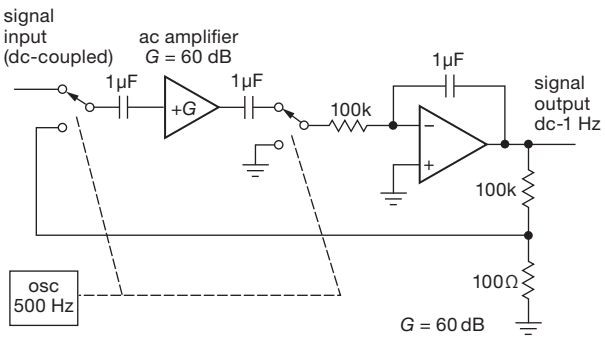


Figure 5.55. An ac-coupled chopper amplifier.

B. Thermal offsets

When you build dc amplifiers with submicrovolt offset voltages, you should be aware of *thermal offsets*, which are little thermally driven batteries produced by the junction of dissimilar metals. You get a Seebeck-effect "thermal EMF" when you have a pair of such junctions at different temperatures. In practice you usually have joints between wires with different plating; a thermal gradient, or even a little draft, can easily produce thermal voltages of a few microvolts. Even similar wires from different manufacturers can produce thermal EMFs of 0.2 μV/°C, 10 to a 100 times the

typical drift spec of the auto-zero amplifiers in Table 5.6 on page 335! The best approach is to strive for symmetrical wiring and component layouts, and then avoid drafts and gradients.

Here, for approximate guidance when worrying about parasitic thermocouples, Peltier shifts, and the like, are some thermoelectric-pair voltages (from Agilent AN-1389-1):

Copper-to-	Approx μV/°C
Copper	<0.3
Cd–Sn solder	0.2
Sn–Pb solder	5
Gold	0.5
Silver	0.5
Brass	3
Be–Cu	5
Aluminum	5
Kovar	42
Silicon	500
Copper oxide	1000

C. Power-up self-calibrating op-amps

Texas Instruments has an interesting approach to circumventing clocking noise in auto-zero amplifiers, namely to do it only once! Their TLC4501 family of "Self-calibrating (Self-Cal™) precision CMOS rail-to-rail output operational amplifiers" comes to life at power-up, performing an auto-zero and holding the correcting offset in an on-chip DAC. The good news is that you get no chopping noise and pretty good offset specs (10 μV typ, 40 μV max), at least compared with those of typical CMOS op-amps. The less good news, as you might expect, is that the drift of offset with temperature is unspectacular (you might say "worst in class"), at $\pm$1000 nV/°C typ, compared with $\sim$20 nV/°C for true auto-zero op-amps (see Table 5.6 on page 335).

D. The non-chopper competition

You can't beat the $\sim$1 μV (typ) offset specs of these auto-zero and chopper amplifiers, but you can do pretty well with the very best factory-trimmed precision op-amps. Bipolar op-amps like the LT1007 or LT1012 do quite well, at 10 μV (typ), and the remarkable CMOS MAX4236A has an offset spec of 5 μV typ (and 20 μV max). Note, however, that these factory-trimmed op-amps cannot come close to the drift specs of a true auto-zero: $\pm$200 nV/°C (typ) for the bipolar types, and $\pm$600 nV/°C for the CMOS, compared with $\sim$20 nV/°C for true auto-zero.

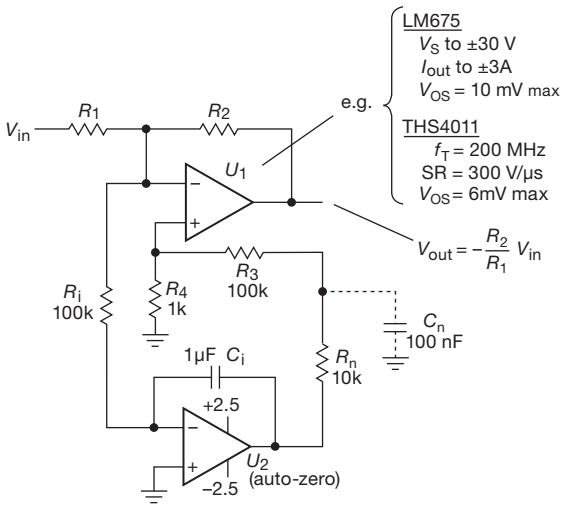


Figure 5.56. External auto-zeroing. But watch out for U_1's high bias current.

E. External auto-zero

You can use an auto-zero op-amp as an external offset trim for a conventional op-amp. This can be handy when you need a high-voltage, high-power, or high-speed op-amp whose input offset is too large. Figure 5.56 shows the scheme, which works most naturally with the inverting configuration, as shown.

The (low-voltage) auto-zero op-amp U_2 is configured as an integrator, looking at the error voltage at the inverting input of the conventional op-amp (U_1). The integrator's output is attenuated by $\times$100 to trim the voltage at the noninverting accordingly. With the values shown, the integrator's output responds according to $dV/dt = -\Delta V/R_iC_i$; thus a 10 μV error produces a $-100\,\mu$V/s integrator output and a $-1\,\mu$V/s correction at the noninverting terminal of U_1. A long time constant is both desirable (op-amp offsets drift only very slowly), and necessary (to prevent loop oscillation). Here the correction range is set by divider R_3R_4, so that a $\pm$1 V integrator output produces a $\pm$10 mV correction. The LM675 is a nice high-power op-amp (3 A output current, supply voltage to $\pm$30 V, with sophisticated on-chip safe-operating-area and thermal protection), but with a maximum offset voltage of $\pm$10 mV. The auto-zero reduces that by a factor of 1000. Similarly, the THS4011 is a fast op-amp ($f_T = 200$ MHz, SR = 300 V/μs) with a maximum offset voltage of $\pm$6 mV. An additional noise filter R_nC_n at the output of the auto-zero, as shown, may be needed to suppress switching noise in the (slow) correction loop when this technique is used with small signals and low-noise parts like the THS4011 (7.5 nV/$\sqrt{\text{Hz}}$). You can think

of this technique as a discrete implementation of the integrated scheme of Figure 5.41.

F. Auto-zero instrumentation amplifiers

In §5.13 we discuss *instrumentation amplifiers*, which are differential-input amplifiers with very high input impedance (10 MΩ–10 GΩ), wide gain range (G_V=1–1000, set by internal or external gain-setting resistors), and very high CMRR at higher gains (110–140 dB at G_V = 100). These are largely built with conventional (not auto-zero) circuitry; but some are of the CMOS auto-zeroing flavor, with very low offset voltage and drift (to 10 μV and 20 nV/°C max). Table 5.8 (page 363) lists a good selection of instrumentation amplifiers, which include auto-zero types such as the AD8553, AD8230, AD8293, INA333, LTC2053, and MAX4209. Some *conventional* instrumentation amplifiers that compete in this low-drift arena are the LTC1167/8 (40 μV, 50 nV/°C max) and the AD8221 (25 μV, 300 nV/°C max).

G. Do it yourself

If you like to get down into the guts of circuits, take a look at the LTC1043 "Precision Instrumentation Switched Capacitor Building Block." It lets you make your own high-CMRR differential amplifier. That's just one of its many tricks, which include switched-capacitor filters, oscillators, modulators, lock-in amplifiers, sample-and-hold, frequency-to-voltage conversion, and "flying capacitor" voltage inversion, multiplication, and division. The datasheet makes great bedtime reading.

5.12 Designs by the masters: Agilent's accurate DMMs

This is another in the series of featured "Designs by the Masters," in which we take a close look at some exemplary circuit designs. Think of these as *master classes* in circuit design. You can learn a lot by cracking open a well-designed instrument. A fine example is provided by the excellent digital multimeters from Agilent, specifically their 34401A (6.5-digit) and 34420A (7.5-digit) benchtop meters. In Chapter 13 (§13.8.6) we discuss the precision "multislope ADC" technique they use. Here, in the context of precision *analog* design, we look in some detail at the clever front ends they've designed, from schematics helpfully provided in their service manuals.[56] Let's see how the real pros do it!

5.12.1 It's *impossible!*

At first glance, the task is impossible. Here's why.

Accuracy We need accuracy and linearity at the parts-per-million level, in a meter whose full-scale ranges go down to a fraction of a volt (100 mV for the 34401A, 1 mV for the 34420A). That's far down in the nanovolt range.

Low noise Accuracy is useless if the instrumental noise makes successive measurements bounce around by many LSBs. So we need input voltage noise levels down at the nanovolt level for the most sensitive ranges.

High input impedance A voltmeter should have high input impedance to minimize circuit loading. So, for measurements at the ppm level, you'd like R_{in} to be something like a million times as large as typical circuit impedances. That puts you in the gigaohm range, with input currents down in the picoamps.

Hence the quandary: gigaohms and picoamps means FETs. Conventional FET op-amps won't deliver this performance, though, owing to relatively large offset voltage, drift, and voltage noise. AZ op-amps (see Table 5.6 on page 335) are considerably more accurate, but they suffer from plentiful current noise. And discrete JFETs (those with large area can have very low voltage noise, less than 1 nV/$\sqrt{\text{Hz}}$) with their uncertain I_D versus V_{GS} characteristics (§3.1.5) would seem to be hopeless at the fractional microvolt level. End of discussion.

5.12.2 Wrong – it *is* possible!

But it *can* be done. The trick is to realize that a digital instrument (with its on-board microprocessor brain) can calibrate away offsets (with a "zero" measurement) and scale errors (with a "full-scale" measurement), so that what matters is not the presence of offsets *per se*, but their stability (drift) during the measurement time.[57] That allows the use

[56] In the good ol' days, manufacturers proudly published their circuits. Nowadays it's far less common – for example, circuit diagrams are absent from the service manual for the Agilent 34410A (successor to the 34401A). Happily, some manufacturers (e.g., Stanford Research Systems) continue to display their circuit ingenuity, with full schematics and parts lists.

[57] A further trick is to average many such calibrate-measure cycles (up to 2 minutes' worth, in the 34420A) to beat down the scatter.

of discrete dual JFETs, with their unbeatable combination of low e_n and low I_B, in a JFET-enhanced op-amp configuration. That way you can get the high loop gain you need for linearity, particularly in the sensitive ranges where the front-end gain is 1000 or 10,000.

That's not the end of the story. You need precise resistor networks with low voltage coefficient, a circuit configuration that maintains its accuracy over a large common-mode input range (to ± 10 V), and of course a voltage reference whose stability determines the instrument's overall accuracy.

5.12.3 Block diagram: a simple plan

These instruments leverage the power of "embedded control" (an on-board microcontroller) to deliver great performance from an architecture of great simplicity. The basic scheme (Figure 5.57) is simplicity itself: it consists of a single amplifier, configured in the familiar noninverting op-amp connection, with floating ground referenced to the (−) input jack. The microcontroller is the boss, here: its code implements the high-accuracy ADC (§13.8.6), and takes care of the multiple on-the-fly calibrations that are needed to wring part-per-million (or better) performance out of a collection of inexpensive parts. Let's dive into the guts of these two DMMs to see how it all works.

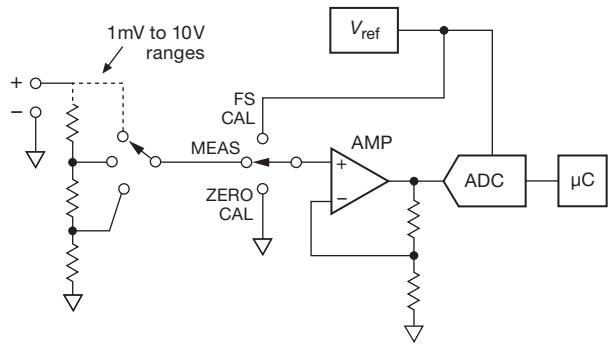


Figure 5.57. The Agilent DMMs: extreme simplicity... at the block diagram level.

5.12.4 The 34401A 6.5-digit front end

The 34401A made its debut in 1991, surprising the T&M (test and measurement) world with astonishingly good performance (resolution to 6.5 digits, measurements to 1000/s, accuracy to 20 ppm) at an affordable price (∼$1k). The input amplifier (preceded by protection circuitry, and attenuators for the 100 V and 1000 V ranges[58]) provides gains of $\times 100$ (100 mV range), $\times 10$ (1 V range), and $\times 1$ (10 V range) with $R_{in} > 10\,G\Omega$; the input attenuator kicks in for the 100 V and 1000 V ranges, for which $R_{in} = 10\,M\Omega$.

The basic structure is a low-noise precision op-amp (an OP-27), driven by a JFET source-follower pair, as shown in Figure 5.58A. (The configuration in Figure 5.58B, where a JFET common-source differential amplifier replaces the follower, is used in the 34420A to provide the additional loop gain and lower voltage noise needed for its more sensitive 1 mV and 10 mV full-scale ranges). The BJT-input op-amp provides lots (120 dB) of stable ($0.2\,\mu V/°C$) low-noise ($3\,nV/\sqrt{Hz}$) gain, but at a price: an unacceptable input current of ± 15 nA, with correspondingly high input current noise ($1.7\,pA/\sqrt{Hz}$). The JFET follower cures problems of input current and noise, at the expense of offset stability ($40\,\mu V/°C$!) and significant added voltage noise ($10\,nV/\sqrt{Hz}$). The tradeoff sounds bad – but it's good enough for this instrument. (It's *not* good enough for the more accurate and sensitive 34420A, as we'll see presently.)

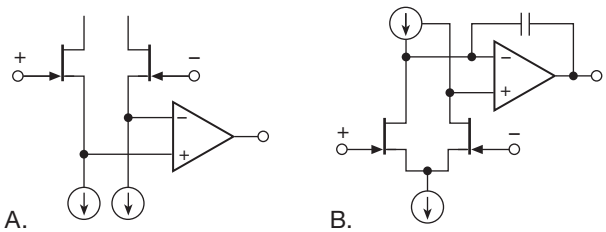


Figure 5.58. Basic JFET-enhanced op-amp configurations for the Agilent DMMs: A. source follower, used in the 34401A; B. common-source differential amplifier, used in the 34420A.

The full circuit is shown in Figure 5.59. Note first the bootstrapped drain supply for the JFET pair: Q_2 maintains a constant drain-source voltage across Q_1 (equal to Q_2's V_{GS} at the operating current, the latter held constant by the complicated-looking current sinks on Q_1's source terminals). This is essential, because the JFET pair Q_1 is anything but precise (would you believe, $V_{OS}(max)$=40 mV?!), and so the variation of this mediocre offset voltage with varying input signal (thus varying V_{DS}) would surely torpedo any expectation of accuracy. But by bootstrapping the drains to follow the sources, the transistors don't even know that there's any input signal variation; they don't know, so they can't mess things up. Furthermore, the low operating voltage (1–2 V) keeps the gate leakage small

[58] All have 20% "overrange," e.g., ±12 V on the "10 V" range.

and unchanging with input voltage variations over the full ±15 V input signal range. Clever![59]

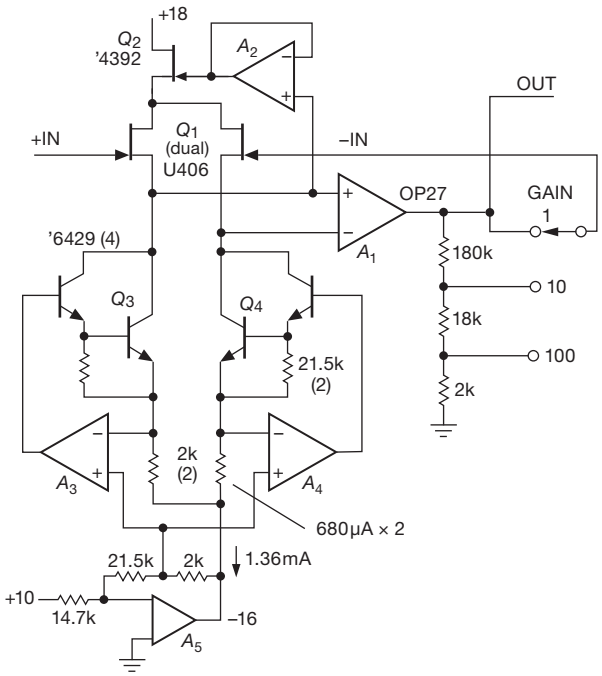


Figure 5.59. Agilent 34401A frontend amplifier, capable of measurements with 0.1 μV resolution. The input is single-ended, amplified and measured with respect to the instrument's input common terminal.

The circuit's voltage gain is set accurately by the analog switch and matched resistor network, implemented in a custom gain-switching IC.

The source pull-down circuitry is a current sink pair, based on the stable +10 V reference that is used also for the downstream ADC (see §13.8.6). It's easier to understand in the redrawn form of Figure 5.60, in which only one of the current sink pair is shown and the Darlington is replaced with a single *npn* transistor. The left-hand opamp generates a voltage across R_2 of $V_{\rm REF}R_2/R_1$; hence the sink current shown in the figure. In its DMM, Agilent uses a matched network for R_2 and the pair of R_3's (one for each source pull-down). The extra resistor R_4 offsets the emitter voltage downward, to $V_{\rm E} = -V_{\rm REF}R_4/R_1$, to provide the needed compliance for input signals that range over ±15 V (±12 V operating range, plus an additional 3 V to accommodate ripple and noise). If you plug in the resistor values from Figure 5.59, you'll find that the compliance extends down to −14 V (emitter is at −14.6 V), and that the individual source pull-down currents are 680 μA. The designers used Darlington transistors to keep the base current error small (roughly $I_c/4500$, assuming a transistor beta of 200).

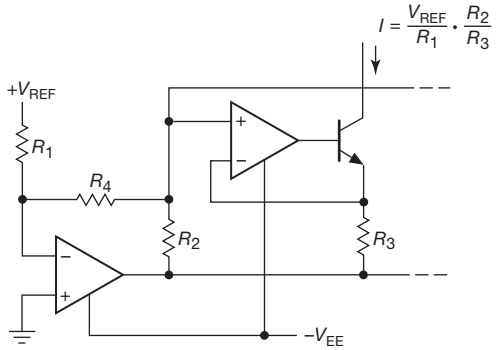


Figure 5.60. Agilent 34401A reference-based current sink.

5.12.5 The 34420A 7.5-digit frontend

With the 6.5-digit 34401A as a warmup, let's look at its wiser sibling, the 34420A 7.5-digit DMM. It boasts both improved resolution and greater sensitivity (1 mV fullscale), putting real demands on the accuracy, stability, and noise of the front end. On its most sensitive range the frontend amplifier has a gain of 10,000 (to bring the ±1 mV input to the ±10 V ADC span), requiring lots of open-loop gain to maintain accuracy and linearity. With sensitivity and resolution comes a demand for low noise; for example, the specifications list a "DC Voltage Noise" (with 2-minute averaging) of 1.5 nV(rms) on the 10 mV range – that's 0.15 ppm.

To meet these demands, the designers used the configuration of Figure 5.58B, in which the JFET pair is configured as a common-source differential amplifier for greater loop gain and reduced noise. The full amplifier circuit is shown in Figure 5.61.

Once again the JFETs are operated at constant current (2 mA each), with bootstrapped drains (held $V_Z - 2V_{BE} - 1$ V above the source, i.e., $V_{\rm DS} \approx 2.5$ V). They chose JFETs with much larger geometry for greatly reduced noise voltage (an impressively low $e_{\rm n}$=0.4 nV/$\sqrt{\rm Hz}$ at 10 Hz). These are *monster* JFETs: $I_{\rm DSS}$=50 mA min, 1000 mA max (how's that for a parameter spread?!), with an input capacitance of order 500 pF, and an unenviable offset voltage spec of ±100 mV (with no tempco

[59] It's necessary that Q_1's $V_{\rm GS}$ at 0.7 mA be less than Q_2's $V_{\rm GS}$ at 1.4 mA, because the difference is Q_1's $V_{\rm DS}$ operating voltage. It's likely that Agilent has an incoming batch inspection to ensure that this condition is met.

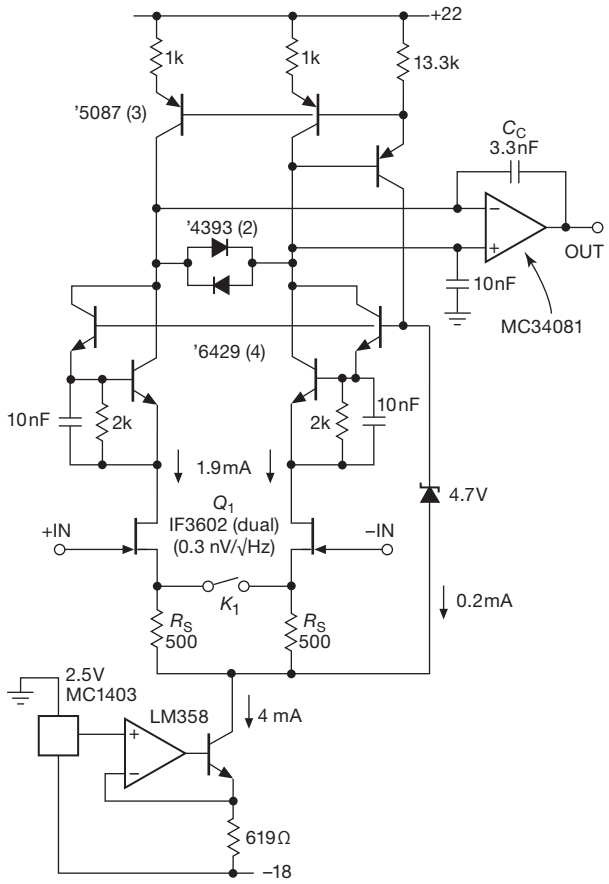


Figure 5.61. Agilent 34420A frontend gain block, used for measurements of 0.1 nV resolution with G=10,000 (gain-switching feedback shown in Figure 5.62).

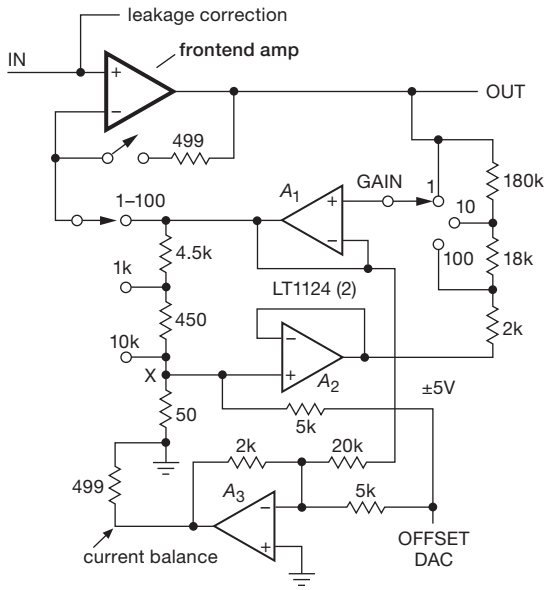


Figure 5.62. Agilent 34420A range-switching feedback circuit, wrapped around the "amplifier" of Figure 5.61. See §13.3.3 and Figure 13.15 for the leakage-correction circuit.

specified). This latter parameter would not appear to bode well for nanovolt measurements! (As we'll see presently, there's provision for continually trimming the measured offset.) These things are brutes, but they sure are quiet. We'll return to this circuit shortly, to deal with issues of gain, bandwidth, and noise. First, though, a look at the overall gain-setting loop.

A. Two-stage gain-setting loop

What's shown in Figure 5.61 is the bare amplifier, which sits in the feedback and trimming circuit of Figure 5.62. The gain selection is made with two stages of precision attenuators of high stability (these are custom modules), isolated by precision op-amp A_1 whose offset can be read and cancelled as part of the measurement cycle (by comparing the amplifier's output with the two possible G=1 configurations). The offset of A_2 can also be measured, from the amplifier's output when the offset DAC is set to zero and the gain to $\times 100$.

The "offset DAC" is used to dispatch the big elephant in the room (the JFET pair's offset, which can range to $\sim$50 mV). This it does by generating a voltage output ($\pm$5 V range), which offsets the node marked "**x**" over a range of $\pm$50 mV. Follower A_2 replicates that offset at the bottom of the right-hand gain-setting divider, establishing an effective ground-reference point for the first-stage divider. Here's a subtle point: with a circuit of this precision (in the 1 mV range, the LSB is just 1 nV) you have to worry about stuff you normally ignore – for example the effect of voltage drops through ground path resistance. Here the offset DAC may sink or source up to 1 mA (5 V across 5 kΩ), which would push "ground" around by an unacceptable 100 nV if its path had, say, 0.1 mΩ of resistance. That's why the designers added A_3, whose output pushes a balancing current of opposite sign into the same ground node; if that current is matched to 1%, the error is reduced to 1 nV.

This raises a related point: when talking nanovolt stability, don't we have to worry about A_1's drift? Quite so – but the effect of splitting the gain-setting attenuator into two sections is to reduce that drift by the attenuation of the left-hand divider, i.e., $\times 100$ in the most sensitive range.

B. Care and feeding of the JFETs

An important rule to follow when doing low-distortion precision design using discrete transistors: use a circuit configuration that keeps the transistor's operating conditions (V_{ds} and I_d) unchanged as the input signal changes. Both of these amplifier designs carefully follow this rule, but in different ways, to achieve their good performance. Both designs also operate the JFETs at low drain-source voltages to reduce gate leakage and to minimize self-heating (see §3.2.8).

This same rule is followed (in the second design) with respect to the MC34081 JFET op-amp's input voltages, which are both pinned to $2V_{BE}$+1.9 V below the +22 V rail. Likewise the mirror transistors see no change in voltage for a signal-input swing from −15 to +15 V. Only the feedback capacitor C_c sees a change in voltage.

Finally, despite the low V_{DG} operating voltage for the Q_1 JFETs, there is still an issue of small gate leakage currents to worry about. Agilent has added an input-bias-current correction circuit, with an 8-bit DAC, to solve this problem. We discuss how this interesting circuit works in §13.3.3.

C. Amplifier gain: ×1 to ×10,000, stable to 0.1 ppm

Turning back to the amplifier (Figure 5.61), we can understand some nice subtleties. A closed-loop gain of ×10,000 is needed in the 1 mV range, for which lots of open-loop gain is needed. The JFET differential amplifier provides gain ahead of the op-amp; though the gain is not easily calculated at dc (it depends on the impedance of the current-mirror drain load), we can estimate its gain–bandwidth product f_T by noting that compensation capacitor C_c makes the differential gain roll off according to $G = g_m X_C/2$, where g_m is the transconductance of each JFET at the operating current (relay K_1 is closed for the 1 mV and 10 mV ranges, removing the 500 Ω source degeneration resistors). f_T is the frequency at which the gain of the composite amplifier has fallen to unity, i.e., $f_T = g_m/4\pi C_c$. To estimate g_m, we note that these JFETs are operating well down in the subthreshold region (their I_{DSS} is typically 300 mA), where FETs behave more like BJTs (I_D exponential in V_{GS}; see Figure 3.15), with their transconductance proportional to drain current and with g_m only somewhat less than a BJT operating at the same current. For the IF3602 JFET running at I_D=2 mA, then, we can estimate $g_m \approx 60$ mS (a BJT would have $g_m = 40I_c$ mS), thus f_T=1.5 MHz.

Running this backward, we find that the open-loop gain is about 10^6 at 1 Hz, as seen in Figure 5.63. Source degeneration is enabled for the low-gain ranges, to maintain stability. The rolloff is easy to calculate, because the differential transconductance is reduced to $1/2R_s$ (1 mS), thus f_T=50 kHz.

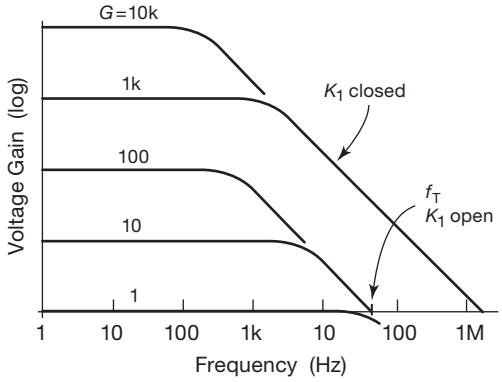


Figure 5.63. Differential gain for the amplifier of Figure 5.61.

D. Sub-nanovolt amplifier noise

Finally, the important issue of noise. This is a big deal when you're talking nanovolts; it's the reason the designers chose huge-geometry JFETs, in spite of their, uh, less-than-ideal characteristics (offset voltage, input capacitance). Noise matters most in the most sensitive range, where full scale is 1 mV (1.2 mV, to be precise, owing to the 20% overrange), and the 6.5-digit LSB is 1 nV.

There are several noise sources here. The JFETs contribute about 1 nVrms in a 3 Hz band around 1 Hz (e_n=0.4 nV/$\sqrt{\text{Hz}}$ each, multiplied by 1.4 for uncorrelated noise). To explore further the measurement fluctuations, take a look at Figure 5.54, where we show the voltage noise of various op-amps, of both conventional and chopper-stabilized varieties. The traces show the integrated rms noise voltage up to a cutoff frequency (x-axis), including the effect of the component's $1/f$ noise. The IF3602 is the lowest-noise part on the graph. If we assume an integration time of 100 PLC (powerline cycles) or 1.67 s, to achieve 7.5-digit performance, that interval corresponds to a 0.6 Hz cutoff frequency, and about 3 nV rms of noise. If we average 64 such measurements over a two-minute period, we could hope for the rms fluctuations to be reduced by 8×, to about 0.4 nV. Agilent claims 1.3 nV in their datasheet, evidently allowing for some nonrandom variations and other errors.

Noise in the current sink is of lesser importance because the differential stage cancels it to a high degree; that's a good thing, because this design uses a noisy voltage reference! (the MC1403 is an early bandgap design,

with unspecified noise voltage).[60] The 34420A's digitizing capability drops from 7.5 to 6.5 digits when operating faster than 20 PLC, or 1.5 readings/sec, and further drops to 5.5 digits above 25 rd/s, and 4.5 digits above 250 rd/s, so rising high-frequency amplifier noise wouldn't be noticed.

E. Going beyond the specifications

When pushing the limits of the possible, you often find that the job cannot be done while respecting worst-case component specifications. Here, for example, the critical JFET transistor pair has a worst-case specified gate leakage current of 500 pA (at 25°C), whereas the instrument specifies a maximum input current of 50 pA. What to do?

If you are a major manufacturer, you can often persuade the supplier to screen parts to a tighter specification. In any case, you can do the job yourself. Be aware, though, that there's generally no guarantee of process continuity, and the availability of better-than-specified parts; worse still, the special parts you need may be discontinued altogether! One possibility, if you're willing to hazard a guess as to an instrument's long-term popularity, is to buy a lifetime supply of a critical part.

5.13 Difference, differential, and instrumentation amplifiers: introduction

These terms describe a class of dc-coupled amplifiers that accept a differential signal-input pair (call them V_{in+} and V_{in-}), and output either a single-ended signal or a differential-output pair that is accurately proportional to the difference: $V_{out} = G_V \Delta V_{in} = G_V(V_{in+} - V_{in-})$. Their shared claim to fame is high common-mode rejection, combined with excellent accuracy and stability of voltage gain. Here are their distinctive features, as commonly understood among circuit designers.

Difference amplifier differential in, single ended out; op-amp plus two matched resistor pairs (Figure 4.9, §4.2.4, and Figure 5.65); CMRR 90–100 dB; accurate but low gain (G_V=0.1–10); input impedance 25–100k, intended to be driven by a low impedance; inputs typically can go beyond rails.

Instrumentation amplifier differential in, single ended out; very high input impedance (10 MΩ–10 GΩ), wide gain range (G_V=1–1000), and very high CMRR at higher gains (110–140 dB at G_V=100); §5.15, e.g., Figure 5.77.

Differential amplifier differential or single ended in, differential out; most are low voltage, fast settling, and wideband; ideal for twisted-pair cable drivers and fast differential-input ADCs; §5.17, e.g., Figure 5.95.

An obvious application is the recovery of a signal that is inherently differential, but that rides on some common-mode level or that is afflicted by common-mode interference. Figure 5.64 shows an example of each.

The first example is the strain gauge we saw earlier (§5.4), a bridge arrangement of resistors that converts strain (elongation) of the material to which it is attached into resistance changes; the net result is a small change in differential-output voltage when powered by a fixed dc-bias voltage. The resistors all have approximately the same resistance, typically 350 Ω, but they are subjected to differing strains. The full-scale sensitivity is typically ±2 mV per volt, so that the full-scale output is ±10 mV for 5 V dc excitation. This small differential-output voltage (proportional to strain) rides on a +2.5 V dc level. The differential-input amplifier must have extremely good CMRR in order to amplify the millivolt differential signals while rejecting the ∼2.5 V common-mode signal and its variations. For example, suppose you want a maximum error of 0.1% of full scale. That's ±0.01 mV, riding on 2500 mV, which amounts to a CMRR of 250,000:1, or 108 dB. This overestimates the needed CMRR: in practice you would perform a "zero calibration," so that the CMRR need be adequate only to reject *variations* in the +5 V bridge bias; something like 60 dB would suffice here.[61]

The second example (Figure 5.64B) comes from the world of professional audio, where you encounter some pretty impressive challenges. In a concert recording situation, for example, you may have microphones hanging from a high ceiling, with connecting cables 100 m or more in length. The peak signal levels may be around a volt, dropping to a millivolt during quiet portions of the music. But you've got to keep powerline pickup and

[60] For single-ended amplifiers we would want the current-source noise i_n to be less than $e_n(\text{amp})g_m$. We can use the expression $e_n(\text{ref})/e_n(\text{amp}) = g_m R_S$ to determine the allowable voltage noise in the current-source reference. For this circuit that ratio is 37, thus only 11 nV/$\sqrt{\text{Hz}}$ for a noise contribution comparable to that of the 0.3 nV/$\sqrt{\text{Hz}}$ JFET. The MC1403 reference is about 20× worse than this. Evidently the Agilent engineers are relying on the matched noise currents in the two JFETs to cancel to better than 5%, enforced by the 1% current-mirror resistors. At frequencies above about 10 Hz, however, the 10 nF capacitor defeats this cancellation.

[61] We'll see the strain gauge again, in connection with analog-to-digital converters, in §13.9.11C (Figure 13.67). A similar biased bridge arrangement is used in the platinum resistance temperature detector (RTD), which is the sensor used in the microcontroller-based thermal controller in §15.6.

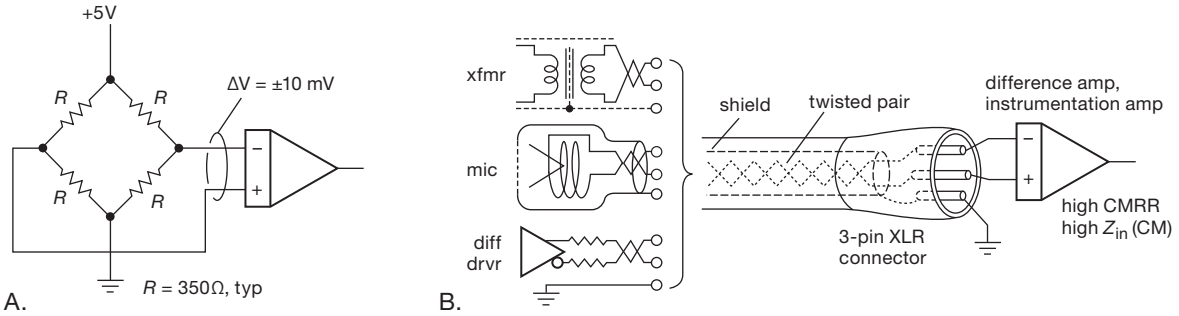


Figure 5.64. Inherently differential signals, for which good common-mode rejection is required. (a) Strain gauge. (b) Audio balanced line pair.

other annoyances (e.g., switching noise from lighting dimmers) another 40 dB below that – the human ear is distressingly sensitive to extraneous sounds. Add up the dBs – we need 100,000:1 suppression of pickup ($<10\,\mu$V)! Seems impossible; but recording engineers have been doing this successfully for decades by the simple expedient of transporting audio signals on a balanced differential pair (with a standard signal impedance of 150 Ω or 600 Ω). For this they use a well-shielded twisted-pair cable, terminated in the legendary 3-pin XLR latching connector (which can take plenty of abuse – it's got a tough metal shroud, good strain relief, and so on). And to keep the signal fully balanced they use either a high-quality audio transformer or a well-designed differential-output driver (at the transmit end), and another transformer or a well-designed differential-input amplifier (at the receive end).

Lest we leave a misleading impression, we hasten to note that differential-input amplifiers are helpful also in situations in which the signals themselves are not inherently differential. Two common examples are accurate low-side current sensing (Figure 5.68a), and the use of differential-input amplifiers when sending signals between instruments (Figure 5.67). In the latter, the flexibility provided by a difference amplifier's output REF pin enables us to avoid ground loops while transporting a signal between a pair of instruments whose local grounds are not identical.

The tricks involved in making good instrumentation amplifiers and, more generally, high-gain differential amplifiers are similar to the precision techniques discussed earlier. Bias current, offsets, and CMRR errors are all important. Let's begin by discussing the design of difference amplifiers for less critical applications, working up to the most demanding instrumentation requirements and their circuit solutions.

5.14 Difference amplifier

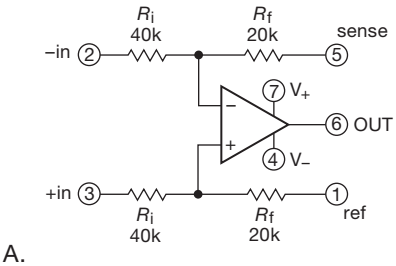
Let's look first at the difference amplifier: its basic operation, some applications, a closer look at its performance parameters, and finally some clever circuit variations.

5.14.1 Basic circuit operation

The classic difference amplifier (Figure 5.65) consists of an op-amp with matched resistor pairs R_f and R_i, for which the differential gain is

$$G_{diff} \equiv \frac{V_{out}}{V_{in+} - V_{in-}} = R_f/R_i.$$

Assuming an ideal op-amp, the common-mode rejection



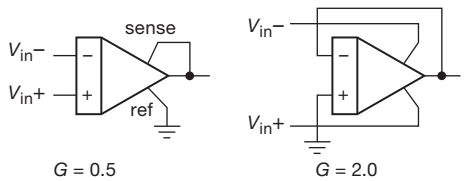


Figure 5.65. Classic difference amplifier, with the resistor values used in the AD8278/9 (G=0.5 or 2).

is limited by the matching of the R_f/R_i ratio in the two paths; with discrete resistors of 1% tolerance, for example,

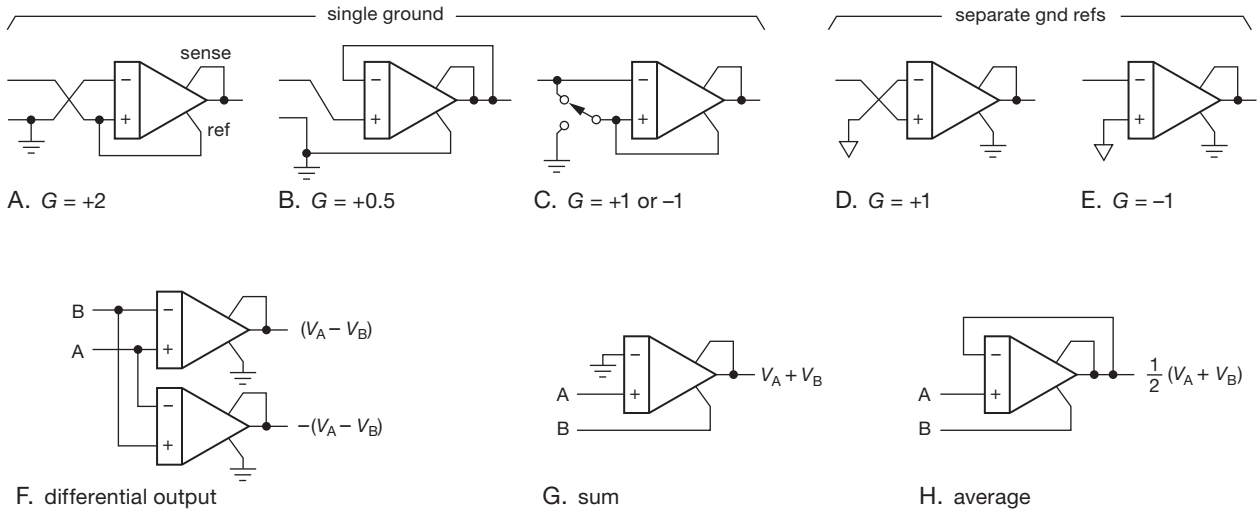


Figure 5.66. The tricks that a unity-gain difference amplifier can play. Note the separate ground symbols for (D) and (E).

you could expect a CMRR of ∼40 dB at low frequencies (where the op-amp itself has a higher CMRR and where effects of capacitive imbalance are negligible). That's adequate for situations in which only a modest CMRR is needed, for example low-side current sensing. For better performance you could trim one of the resistors or use resistors of tighter tolerance (for example the commonly available 0.1% types, typically $0.10–$0.20 in 100-pc quantity, or the inexpensive Susumu RG-series, with 0.05% tolerance and 10 ppm/°C tempco, about $1); or, better, use matched resistor pairs (e.g., the not-inexpensive Vishay MPM-series, with ratio tolerances down to 0.01% and ratio tempcos to 2 ppm/°C or the LT5400 matched quad, with similar ratio tolerance and with ratio tempco of 1 ppm/°C max).

But don't get carried away. . . because there are plentiful offerings of complete integrated difference amplifiers of excellent performance, costing a lot less than you'll spend rolling your own. We've listed many of these in Table 5.7 on page 353. In the "normal" configuration the SENSE line is connected to the output and the REF line is connected to circuit ground. But you can run it in reverse, as shown in Figure 5.65B. To give a sense of the performance you get with these integrated difference amplifiers, the worst-case specifications for the AD8278B in the figure are gain accuracy of ±0.02% (for G=0.5 or G=2); gain tempco of 1 ppm/°C; offset voltage and tempco of 100 μV and 1 μV/°C; and a CMRR of 80 dB. It costs about $3.

5.14.2 Some applications

A. Single-ended input

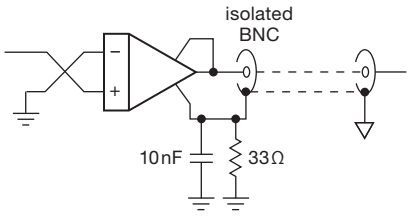
It's perfectly OK to use a difference amplifier with a single-ended input, for example, to get a precise and stable gain. Figure 5.66 shows some simple configurations. Note that differential nature of its inputs is not "wasted" when a difference amplifier is used as in (D) and (E), because its independent REF pin accommodates small differences in ground potential at input and output. Put another way, the output voltage, relative to *its* ground, is precisely ±1.0 (in this case) times the input voltage, relative to *its* ground.

B. Ground-loop isolation

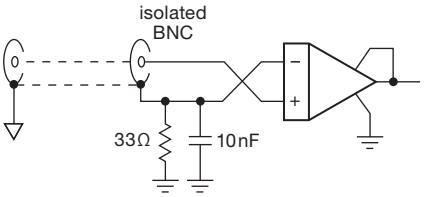
This property is just what you need for the ground-loop isolating application of Figure 5.67. In the first circuit, the driver side permits its output reference to float to the potential of the (non-floating) receive side. The small-value resistor and bypass capacitor allow a small voltage difference when forced by the receive end. Both sides are happy. In the second circuit we've allowed the receive side's input common to float, as forced by the (non-floating) driver. These circuits resolve minor ground-loop problems in single-ended cable connections; but they are no substitute for the fully isolated and/or balanced approach that's needed in demanding applications like professional audio or video.

C. Current sensing

Figure 5.68 shows difference amplifiers used for low-side and high-side current sensing, perhaps as part of a constant-

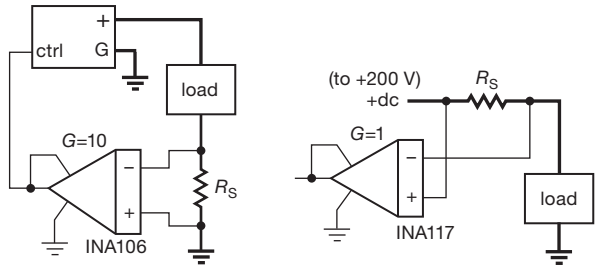


A. floating driver



B. floating receiver

Figure 5.67. Exploiting the REF pin to prevent powerline-frequency ground loops in connected instruments.



A. low-side B. high-side

Figure 5.68. Current sensing for measurement or control. Low-side sensing (A) is forgiving of CMRR, unlike high-side sensing (B), for which the difference amplifier shown would introduce significant error in a high-voltage and high-current supply (where only a small drop across R_s can be tolerated).

current control, or simply for precise load-current monitoring. At first glance it may seem unnecessary to use a difference amplifier for the low-side sensing configuration, because the sense resistor returns to circuit ground. But imagine that we're dealing with lots of power, say up to 10 A load current. We'd use a low-value precision sense resistor, perhaps 0.01 Ω, to keep its power dissipation below 1 W. Even though one side of it is connected to ground, it would be unwise to use a single-ended amplifier, because a connection resistance of just a milliohm would contribute 10% error! A differential-input amplifier is the solution, connected as shown to a 4-wire "Kelvin-connected" sense resistor. Note that the difference amplifier need not have particularly good CMRR, because the low side isn't moving far from ground.

The same is not true of the high-side configuration (Figure 5.68B), where the common-mode voltage is far greater than the differential voltage. Here we've specified a unity-gain difference amplifier intended for high-voltage applications, which permits common-mode input voltages to ±200 V (its internal circuit uses a 20:1 resistive divider at the front end, see §5.14.3). Let's look at the numbers: the dc voltage can range from 0 to 200 V; so the specified minimum CMRR of 86 dB (1:20,000) would interpret that as equivalent to a differential input variation of 10 mV. Is that bad? You bet! To maintain 1% accuracy of sensed current we'd need to size R_s to drop 1 V at full load current. That's a lot of "voltage burden," and it's a whopping 10 W of dissipated power, if this were a powerful 10 A dc supply as in the other example. So this scheme is adequate for a low-current 200 V supply; but there are better ways to do high-side sensing, for example by floating the amplifier and relaying the output down to earth as a current or (via an opto-isolator) as a digitized quantity.

D. Current sources

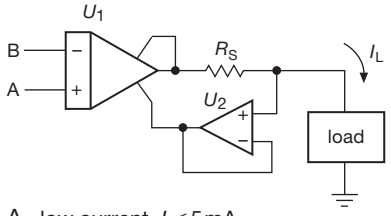
Figure 5.69 shows how to rig up a difference amplifier so that the (differential) input signal controls the voltage drop across a series sense resistor R_s; in other words, it's a current source. You're welcome to use a single-ended control input, if you want. The output current can be of either polarity, and these circuits don't know, or care, whether the load returns to ground or to some other potential.

The op-amp follower U_2 should be chosen for an input bias current that is small compared with the minimum load current, and an offset voltage that is small compared with the drop across R_s at minimum load current. You might begin by choosing R_s for ∼1 V drop at I_L(max), or smaller for high I_L(max) or for low supply voltage, then see if R_sI_L(min) is at least 100 μV or so. For a large dynamic range, say $I_L(\text{max})/I_L(\text{min}) \geq 10^4$, you'll need low-offset amplifiers for both U_1 and U_2; candidates are listed in Tables 5.2 (page 302), 5.3 (page 303), 5.5 (pages 320–321) and 5.6 (page 335).

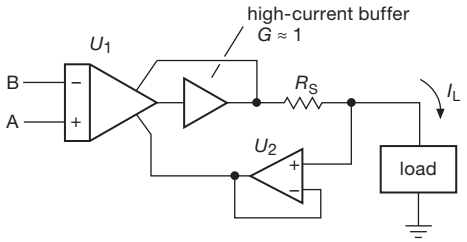
The circuit in Figure 5.69A is good for low currents. For load currents greater than ∼5 mA use a power buffer (Figure 5.69B); that can be a wideband (for stability) unity-gain IC buffer like the LT1010, or (if you need only one polarity of output current) a MOSFET or BJT follower. A favorite trick is to use a 3-terminal adjustable regulator (like an LM317) as a "power buffer," thus taking advantage of

its internal thermal and overcurrent protection. To use it this way, you drive the ADJ pin, and the OUT pin "follows" 1.25 V higher (should this be called a voltage *leader*?!). As always, feedback takes care of the offset.

We like using $G = 10$ difference amplifiers, such as INA106 or INA143, connected "backwards" for $G = 0.1$, because the sense voltage is then one-tenth of the programming voltage and thus doesn't eat up so much of the output compliance range. In common with other current sources whose output comes from an op-amp, these configurations become more like *voltage* sources at high frequencies, where op-amp compensation and slew-rate effects dominate (see §§4.2.5 and 4.4.4).[62] Better performance at high frequencies can be gotten with an active current source based on an instrumentation amplifier, configured with an inherently high impedance output terminal, as shown in Figure 5.87 in §5.16.9.



A. low current, $I_L \lesssim 5$ mA



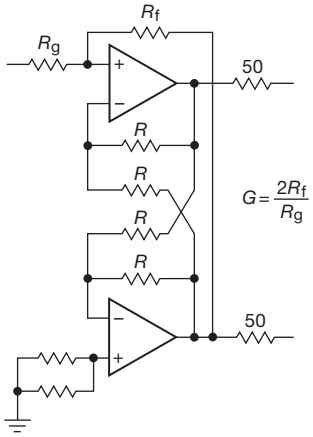
B. "any" current, R_S: mΩ to GΩ

Figure 5.69. Precision current sources: $I_L = G_{diff}(V_A - V_B)/R_S$. The (bipolarity) output current in (A) is limited to U_1's I_{out} (max). Adding a unity-gain power buffer (an IC wideband buffer, or a transistor follower) in (B) allows large output currents (follower U_2 can be omitted if R_S is less than 0.2 Ω or so). These circuits don't know, or care, where the load returns.

[62] You can define an effective current-source output *capacitance* $C_{eff} = I_{out}/S$ (where S is the output slew rate) as a way to characterize this shortcoming.

E. High-level line driver

Professional audio lives and breathes *differential* analog signaling, in the form of balanced lines terminated (usually) in a nominal 600 Ω bridging resistance. And the levels are substantial: pro-audio equipment adopts a "0 dB" standard of 1.23 Vrms,[63] and you'll generally see additional specified headroom of 16 dB to 20 dB without clipping. So a +20 dB level is 12.3 Vrms, or a differential amplitude of 17.4 V (34.8 Vpp). That requires some serious attention to line drivers, which should not compromise the low noise and distortion qualities of the program material.



	LM 7372	LME 49713	LME 49990	
I_o	±150	±100	±25	mA
e_n	14	1.9	0.9	nV/√Hz
THD	100	4	0.1	ppm
@freq	1000	1	1	kHz
GBW	120	132	110	MHz
slew	3000	1900	22	V/μS

Figure 5.70. Differential high-level line driver for professional audio.

Figure 5.70 shows a nice $G = 2$ circuit for the job, based around a pair of unity-gain difference amplifiers. These are implemented here with wideband op-amps of substantial output-current capability, allowing for an overall gain (differential-output voltage divided by single-ended input voltage) other than ×2. The particular listed op-amps all run from supply voltages to ±18 V, producing output swings (on each line of the pair) to ±15 V or so (significantly greater than the ±9 V corresponding to a 20 dB pro-audio headroom). Three op-amp choices are shown, of comparable bandwidth (∼100 MHz) but aimed at different applications. The LM7372 is well specified out to 10 MHz and intended for video and broadcast applications, whereas the LME49xxx parts are optimized for audio bandwidths, with pretty impressive specifications out to 20 kHz. It's not often (more like *never!*) that you see amplifiers with total harmonic distortion (THD) specifications of 0.1 ppm

[63] The base unit for audio level is "0 dBu," an rms voltage corresponding to 1 mW into a 600 Ω load; that works out to 0.775 Vrms. Pro-audio's 0 dB level is +4 dBu, hence 1.23 Vrms; home audio is considerably less muscular, at −10 dBu, or 0.25 Vrms.

(−140 dB), here combined with very low voltage noise ($1.4\,\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz).[64]

Some interesting alternatives are the DRV134 and the LME49724, which integrate a fully differential output circuit, capable of comparable performance, into one IC. The latter are examples of *fully-differential amplifiers*, with differential inputs as well; one input can be grounded to work with single-ended signal sources. We'll see these later, in §5.17.

F. Wideband analog over twisted pair

Twisted-pair network cable ("Cat-5e," etc.) is ordinarily used for digital data transmission on local area networks (LANs), but it can be used successfully for analog signals as well. The ubiquitous Cat-5e and Cat-6 cables contain four unshielded pairs (hence "UTP" – unshielded twisted pair), which interestingly are twisted with differing (incommensurate) pitches to minimize normal-mode coupling.

However, there's plenty of common-mode coupling, both between pairs and to the outside world. So you need to use a differential driver (§5.17) combined with a difference amplifier at the far end. And, for wideband transmission over anything longer than a few inches you need to terminate the pair in its characteristic impedance of 100 Ω (see Appendix H).

Figure 5.71 shows how to use a difference amplifier as an analog "line receiver" for such signals (we'll show the driving end later, in §5.17.2). This circuit comes from the AD8130's datasheet, illustrating the use of some "peaking" to compensate for the signal attenuation at high frequencies in the rather long (300m!) cable. The amplifier has $G = 3$ at low frequencies, with R_4C_1 kicking in around 1.6 MHz (where C_1's reactance equals R_3) to boost the sagging response. The result of this simple "equalizer" is to produce an overall response flat to ±1 dB from dc to 9 MHz. The ×3 low-frequency gain is needed (a) to compensate for the signal that is lost because of the resistive divider consisting of R_1 and the cable pair's 50.5 Ω round-trip resistance (a factor of 1.5), and (b) to double the output signal so it can drive a "back-terminated" video cable (see Appendix H).

5.14.3 Performance parameters

In the preceding sections we glossed over some important issues: input impedance (both differential and common mode) and its effect on gain when driven with finite source impedance or its effect on CMRR when driven from an unbalanced source impedance; common-mode input range; and amplifier bandwidth and its effect on CMRR. It's time to peel the onion.

A. Input impedance

From Figure 5.65 it might seem that the input impedance of a difference amplifier is R_i (or maybe some multiple of that), and that all is well if the signal driving it has a source (Thevénin) impedance R_S that is much less, perhaps our usual seat-of-the-pants criterion of $R_S \lesssim 10R_i$.

Not so! These amplifiers revel in their precise gain (the selections in Table 5.7 on the next page have worst-case gain errors of 0.1% or better), and that gain accuracy is compromised unless R_S is smaller than the amplifier's R_I by at least that factor. That's because the signal's source impedance is effectively in series with R_I, reducing the gain by the factor $R_I/(R_I + R_S)$.

That gain reduction is actually the lesser of two problems. If you read the fine print on the datasheet, you'll discover that the excellent CMRRs of these amplifiers (surely their greatest claim to fame) is invariably specified for (drumroll) $R_S = 0\,\Omega$! You could reasonably expect that the CMRR would be maintained if you drive both inputs with equal source impedances, because that should maintain the resistor ratio match. You could expect that, but you would be disappointed: the CMRR degrades rapidly with rising source impedances, even if they are precisely matched.

Why is that? During manufacture the internal resistors are laser trimmed, so that the ratio R_F/R_I of the upper resistor pair precisely matches the corresponding ratio of the lower pair. The *ratios* are what are matched, at the expense of the absolute values; the two input resistors R_I may differ somewhat.[65] So, if you drive it with a signal pair of matched source impedances R_S, you get a mismatch in the feedback ratios, hence degraded CMRR. Bottom line: drive these puppies from an op-amp output, or from a signal source of very low R_S (e.g., a low-value current-sense resistor).

That's not always possible, of course. What can be done to raise the input impedance? The first thought might be simply to raise all the resistor values by some large factor. That has several drawbacks, the most serious of which are (a) the Johnson noise contribution

[64] You've got to keep source impedances quite low in order not to degrade such a low e_n; even a 100 Ω resistor has an open-circuit voltage noise of $1.3\,\text{nV}/\sqrt{\text{Hz}}$. See Chapter 8.

[65] And the overall resistor scaling is typically good to only ±20% of the nominal value on the datasheet: absolute resistance value is sacrificed on the altar of resistor ratio matching.

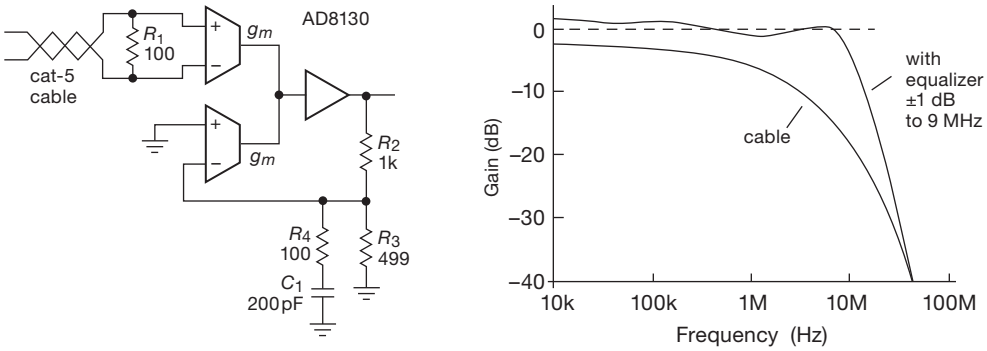


Figure 5.71. Differential line receiver for wideband analog over twisted pair. The "treble boost" provided by C_1R_4 compensates for the high-frequency rolloff of a 300m length of Cat-5 cable, as shown. See also Figure 5.101.

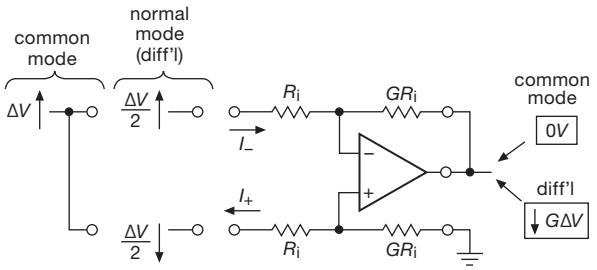
Table 5.7 Selected Difference Amplifiers

Part #	Config	Gain	ΔG max (%)	V_{CM}[a] ±max (Vpp)	Offset Voltage typ (μV)	Offset Voltage max (μV)	CMRR typ (dB)	CMRR min (dB)	Curve	Diff'l Z_{in} (kΩ)	Noise[k,t] $V_{n(pp)}$ dc[b] (μV)	Noise[k,t] e_n[r] ($\frac{nV}{\sqrt{Hz}}$)	BW −3dB (MHz)	Settle 0.01% (μs)	Filter?	Supply Range (V)	Supply I_s (mA)	Cost ($US)	DIP?	Comments
INA105K	A	1.0	0.025	20	50	500	100	72	1	50	2.4	60	1	5	-	10–36	1.5	8.88	•	legacy
AMP03G	A	1.0	0.008	20	25	750	95	75	1	50	2	20[c]	3	1	-	10–36	2.5	5.86	•	
INA134	A	1.0	0.075	26	75	250	90	74	4	50	7[d]	52	3.1	3	-	8–36	2.4	2.36	•	audio, < 5ppm dist
INA154	A	1.0	0.1	25	75	1500	90	74	4	50	2.6	52	3.1	3	-	8–36	2.4	2.36	-	low cost
INA132P	A	1.0	0.075	28	75	250	90	76	5	80	1.6	65[c]	0.3	88	-	2.7–36	0.16	4.62	•	
AD8271B	A[m]	0.5,2	0.02	>18	300	600	92	80	10	20	1.5	38	15	0.55	-	5–36	2.6	3.50	-	G_{diff} = 0.5,1,2
AD8273	A	0.5,2	0.05	>18	200	1400	86	77	7	24	7[d]	52	20	0.75	-	5–36	2.5	3.13	-	dual audio, 6ppm dist
THAT1206	F	0.50	0.5	26	-	10mV	90	70	-	48	-	28[n]	34	-	-	24–40	4.7	4.75	•	Z_{CM}=10M, <6ppm dist
AD8278B	A	0.5,2	0.02	>18	50	100	100	80	1	80	1.4	47	1	9	-	2–36	0.2	2.72	-	G_{diff} = 0.5,2; dual '79
INA106	A	10	0.025	11	50	200	100	86	1	20	1	30	5	10	-	10–36	1.5	11.00	•	0.10, legacy
INA143U	A	10	0.1	15	100	500	96	86	4[h]	20	1	30	0.15	9	-	4.5–36	1.0	3.36	-	0.10, dual=INA2143
LT1991A	A[m]	1,4,10	0.06	27	15	50	100	77	8[h]	90	0.25	46	0.11[f]	48	f	2.7–40	0.10	2.50	-	3,9,12, includes filter
LT1996A	A[m]	9-117	0.07	27[q]	15	50	100	80	-	33[q]	0.25	18	0.04	85	f	2.7–40	0.10	5.72	-	9,27,81, includes filter
LT1995	A[m]	1-7	0.2	31[q]	600	4000	87	75	-	4[q]	-	14	32	0.1	-	2.7–40	7	3.78	-	1,2,4,6
INA146	B	0.1[o]	0.1	100	1000	5000	80	70	11	200	10	550	0.55	80	•	4.5–36	0.57	4.25	•	high-voltage inputs
INA117P	C	1.0	0.02	200	120	1000	94	86	8	800	25	550	0.2	10	-	10–36	1.5	5.54	•	high-voltage inputs
AD629B	C	1.0	0.03	270	100	500	96	86	6	800	15	550	0.5	15	-	5–36	1.2	7.21	•	high-voltage inputs
INA149	C	1.0	0.02	275	350	1100	100	90	1	800	20	550	0.5	7	-	4–40	0.8	6.00	-	high-voltage inputs
AD8479	C	1.0	0.01	600	500	1000	96	90	u	2000	30	1600	0.13	11	-	5–36	0.55	9.66	•	highest CM voltage
AD628A	C	ext	0.1	120	-	1500	-	75		220	15	300	0.6	40	•	4.5–36	1.2	3.36	-	HV inputs, filter+gain
low-voltage																				
AD8275B	D	0.2	0.024	27	150	500	96	80	2	108	1[c,e]	40[e]	15	0.45	-	3.3–15	1.9	4.23	-	5, with V_{ref}/2 offset
AD830	E[g]	1-10	0.6	24	1500	3000	100	90	12	370	-	27	85	0.025	p	8–36	14	4.86	-	G=1+R_2/R_1, V_{in} < 2V
AD8129	E[g]	10-100	0.6	20	200	800	105	92	12	1000	-	4.6	200	0.02	p	4.5–25	10	2.90	-	G=1+R_2/R_1, V_{in} < 2V
AD8130	E[g]	1–20	0.6	20	400	1800	105	90	12	6000	-	12.3	250	0.02	p	4.5–25	11	2.90	-	G=1+R_2/R_1, V_{in} < 2V
EL5172	E[g]	1–20	1.5	20	7mV	25mV	95	75	-	300	-	26	250	0.01	p	4.7–12	5.6	1.27	-	G=1+R_2/R_1, V_{in} < 4V
INA152EA	A	1.0	0.1	18	250	1500	94	80	3	40	2.4	87	0.8	25	-	2.7–20	0.5	3.60	-	0.5,1,2, RRIO
MAX4198	A	1.0	0.1	0,5	30	500	90	74	9[h]	50	7.8	58	0.175	34	-	2.7–7	0.05	2.38	-	

Notes: (a) maximum common-mode voltage. (b) 0.01–10Hz or 0.1–10Hz. (c) at 100Hz. (d) 20 to 20kHz. (e) RTO (referred to output). (f) includes 4pF filter caps. (g) circuit E from Fig. 5.89. (h) CMRR curve flattens at BW frequency. (k) RTI (referred to input) unless noted. (m) multiple resistors. (n) noise = –107dBu, 20kHz BW. (o) includes extra opamp filter stage. (p) for G>1. (q) depends on gain. (r) at 1kHz. (t) typical. (u) between 5 & 6.

(e_n=0.13 $R^{\frac{1}{2}}$ nV/$\sqrt{\text{Hz}}$, thus a devastating 130 nV/$\sqrt{\text{Hz}}$ for resistor values of ∼1 MΩ; see Chapter 8), and (b) the bandwidth penalty caused by distributed stray capacitances. The second thought would be to put a pair of precision opamp followers at the inputs. That's OK, but there's a better way still, in circuit configurations such as the "three-opamp instrumentation amplifier," in which a front-end differential stage of high CMRR and high impedance drives a

difference amplifier output stage. We'll see these wonderfully useful amplifiers a bit later.



common mode: $Z_{in}\,(\text{each}) = (G+1)R_i$

$Z_{in}\,(\text{combined}) = \frac{1}{2}(G+1)R_i$

differential mode: $Z_{in}\,(-) \equiv \frac{\Delta V}{\Delta I} = 2\,\frac{G+1}{2G+1}\,R_i$

$Z_{in}\,(+) \equiv \frac{\Delta V_-}{\Delta I_+} = \frac{G+1}{2}\,R_i$

$\text{“}Z_{in}\text{”} \equiv \frac{\Delta V}{\Delta(I_- - I_+)} = \frac{G+1}{G}\,R_i$

Figure 5.72. The several input impedances of the difference amplifier.

Circling back around to the initial question, what exactly is the "input impedance"? There are several answers – look at Figure 5.72. By *common-mode* input impedance we mean the incremental impedance seen at either input[66] when both are driven together. With common-mode drive the two impedances are equal (apart from minor mismatching, as described above), because the output is fixed.

The *differential*-input impedance is a longer story. Taken individually (by grounding the other input), the two inputs exhibit different R_{in}: the inverting input connects to a virtual ground by way of R_i, so its $R_{in} = R_i$, whereas the noninverting input sees $R_{in} = R_i + R_f$. For a difference amplifier with G=10, for example, these differ by a factor of eleven. This is a useful result, particularly if you plan to use a difference amplifier in a single-ended configuration like Figures 5.66D and E. A purist could argue, though, that we've done it wrong: a single-ended input change is really a combination of a purely differential input plus a common-mode offset of half that value. To satisfy such a person we've calculated the expressions in Figure 5.72, based upon a "pure" (symmetrical) differential input. Even when defined this way the input impedances seen at the two inputs are different. This leads us to a final definition of differential input impedance based on the *net* input-current change, as shown. Datasheets most often list this value, without explanation: you've been warned!

A final observation: you won't generally see difference amplifiers with gains greater than 10, because R_i, and therefore the differential input impedance, becomes unmanageably small. For example, to get $G = 1000$ you might choose R_i=100 Ω and R_f=100k. OK, maybe you can live with ∼100 Ω input impedance; but you'd have to match the signal source impedances to 0.001 Ω in order not to degrade the CMRR. The take-away: use an *instrumentation amplifier* (§5.15), not a difference amplifier, for high-gain differential-input applications.

B. Common-mode input range

The voltage divider formed by R_i and R_f permits the standard difference amplifier to accept input signals *beyond* the supply rails: protection diodes are at the internal op-amp inputs, so the input signals could in principle swing as far as $\pm V_S(G+1)/G$ with supply voltages of $\pm V_S$ (figure out why). For example, the AD8278 has $R_i = 40$k and $R_f = 20$k, so it can be connected for $G = 0.5$ ("normal") or $G = 2$ (reversed). It specifies the common-mode input range for both gains: $-3(V_S{+}0.1)$ to $+3(V_S{-}1.5)$ for $G = 0.5$ (that is, about ±40 V with ±15 V supplies, which can be very useful indeed!), and $-1.5(V_S{+}0.1)$ to $+1.5(V_S{-}1.5)$ for $G = 2$ (i.e., ±20 V with ±15 V supplies). But check the specs – not all difference amplifiers let you go that far.

A few of the difference amplifiers in Table 5.7 on the preceding page) do better still; for example, the INA117 has a common-mode range of ±200 V while maintaining a differential gain of unity. This it does by using a pair of 200:1 voltage dividers at the input to bring the ±200 V signal within the op-amp's common-mode range of ±10 V (the circuit is shown in Figure 5.75C). The price it pays is degraded offset and noise: typical values of 120 μV and 550 nV/√Hz, compared with 25 μV and 20 nV/√Hz for the conventional AMP03.[67]

An important point: when using difference amplifiers with large input voltage ranges, beware of the large equivalent input errors created by imperfect common-mode rejection. For example, the AD629B specifies a typical dc CMRR of 96 dB, but you've got to consider the worst-case (minimum) value of 86 dB. With that CMRR, a 200 V

[66] Some manufacturers specify half that value, i.e., the impedance with both terminals tied together; the datasheet usually tells you which they mean.

[67] There's another way to boost V_{CM} without such compromise, by using a second op-amp to cancel the common-mode signal; see Figure 7.27 in the previous edition of this book. We are unaware of any commercial difference amplifiers that use this trick.

common-mode input has an input-referred differential error of 10 mV, completely swamping the 0.5 mV specified maximum offset voltage. Put another way, the error that is due to finite CMRR is larger than the specified V_{OS} for input $|V_{CM}| > 10$ V. And the situation is worse still at signal frequencies: you might imagine using such a difference amplifier for powerline current monitoring. At 60 Hz the CMRR of the INA117 (similar to that of the AD629B at dc) degrades to 66 dB (min). So the 160 V peak powerline signal produces a huge 80 mV input error. And ideally you'd want to monitor currents on powerlines at higher voltages still, perhaps to 400 V. There are better ways to do this; take a look at §13.11.1 if you're curious.

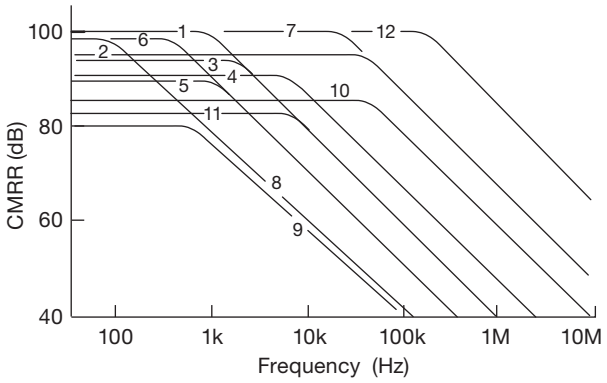


Figure 5.73. Common-mode rejection ratio versus frequency for the difference amplifiers in Table 5.7 on page 353 (identified in the "Curve" column).

C. Bandwidth

Difference amplifiers are built with op-amps, frequency compensated for stability with the by-now familiar $1/f$ open-loop gain rolloff. As with any op-amp circuit, loop gain is responsible for good behavior and the loss of loop gain at higher frequencies not only limits the bandwidth of a difference amplifier (and its linearity, constancy of gain, low output impedance, etc.), it also degrades the all-important CMRR. Figure 5.73 shows this behavior, for the difference amplifiers listed in Table 5.7 (page 353). Not surprisingly, amplifiers with greater closed-loop bandwidth (therefore with op-amps of greater f_T) maintain high CMRR to higher frequencies.

Note that some amplifiers perform well near dc but poorly at high frequencies, for example classic parts like the INA105, AD829 and LT1991, whereas others not so well rated at dc may look better at high frequencies, such as the AD8271 and the MAX4198. A few are outstanding performers from dc to high frequencies, according to the plots, such as the AD8273[68] and our favorite AD8275. The transconductance-balancing AD8129 does very well, but is limited to small inputs and high gains. This information is not shown in the datasheet specifications; it's necessary to look for performance plots to make the comparisons.

Common-mode rejection at higher frequencies is degraded also by the effects of stray inductance and by asymmetrical capacitive loading. It is necessary to balance the circuit capacitances to achieve good CMRR at high frequencies. This may require careful mirror-image placement of components. Even when so symmetrized, the decreasing input capacitive shunt reactance at higher frequencies creates an increasing sensitivity to any unbalance in signal source impedance.[69]

5.14.4 Circuit variations

A. Filter node

The difference amplifiers in Table 5.7 (page 353) include one (the INA146) with a node brought out for noise (low-pass) filtering of the difference stage ($G = 0.1$) by a capacitor to ground (Figure 5.75B). It includes a second single-ended stage, with gain set by a pair of external resistors; so you can have overall gains from 0.1 to 100. The low first-stage gain gives you a large ± 100 V common-mode range, though at the expense of noise and offset.

B. Offset trim

Integrated difference amplifiers are factory trimmed to pretty good precision, with typical values in the 25–100 μV range (but with worst-case offsets an order of magnitude larger). As with any op-amp circuit, you can rig up an external trim, as in Figure 5.74A. Here R_2R_3 divides the trimmer's ± 15 V range to ± 1 mV at the REF pin; R_1 balances the 10 Ω added resistance to ground, to preserve the CMRR. The amplifier's gain is slightly reduced by the ratio $R_f/(R_f + R_2)$; for a typical R_f of 25k, that's 0.04%, in the same range as the amplifier's specified gain accuracy. If this bothers you, use a smaller R_2.

[68] Although the datasheet *graphs* for the AD8273 show a best-in-class CMRR of 100 dB below 40 kHz, the *tabulated* typical CMRR is shown as only 86 dB. We like the plot better, but the designer may be stuck with the worst-case spec of 77 dB. Some users may want to perform incoming inspections to settle the issue.

[69] The professional audio people are keenly aware of these effects, and they don't mince words. As stated pithily by Whitlock and Floru in an Audio Engineering Society paper, "**Noise rejection in a balanced system has absolutely NOTHING to do with signal symmetry** (equal and opposite signal voltage swings). It is the balance of common-mode **impedances** that defines a balanced system!" [emphasis in the original].

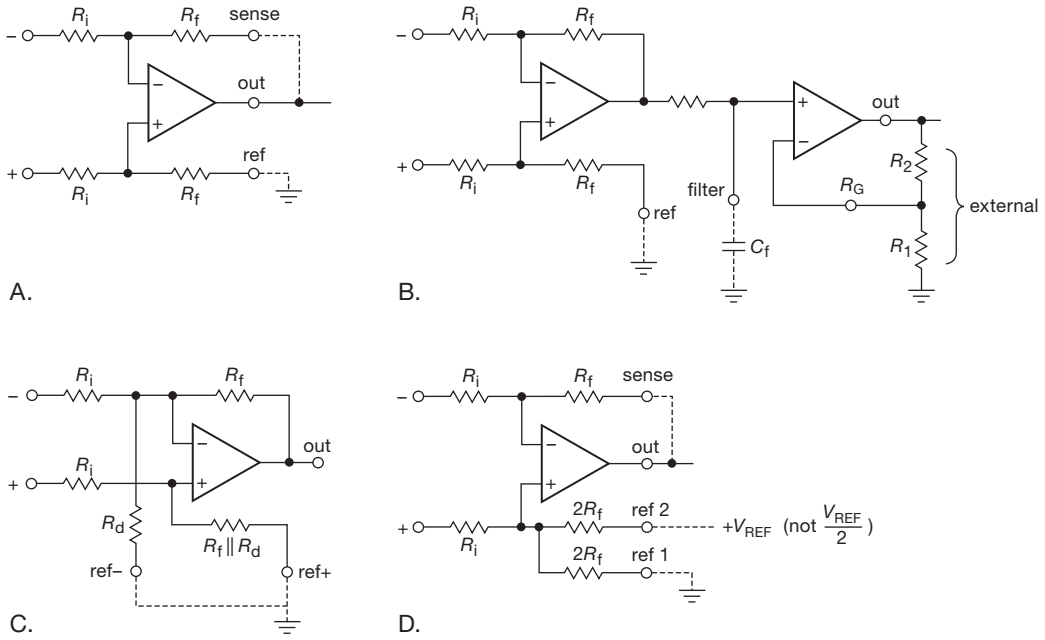


Figure 5.75. Circuit configurations for the difference amplifiers in Table 5.7 on page 353 (identified in the "Config" column). The "E" form is shown in Figure 5.89. The "C" form is used for high voltages (e.g., ±270 V for the AD629B).

C. CMRR trim

Likewise, you can trim out residual CMRR (caused by slight mismatch of resistor ratios R_f/R_i in the two paths) with the circuit of Figure 5.74B. It's important to limit the trim range, to permit an accurate and stable trim to something considerably better than the off-the-shelf 80 dB (worst-case) CMRR specification. You can't get any old value of trimmer, and it's best not to use values less than 100 Ω (even if you can find them) if you care about stability. Here we've chosen standard resistor values and a 100 Ω trimmer to produce a resistance range of 20–30 Ω from the REF terminal to ground, providing a ±5 Ω symmetrical variation around R_1. For the 25k resistor values of this unity-gain difference amplifier (rather typical; see Table 5.7 on page 353), this corresponds to a trim range adequate to null an initial CMRR of 75 dB. You can, of course, add an offset null to this circuit, as indicated.

D. Single-supply offset

One of the difference amplifiers in Table 5.7 (page 353) helpfully splits the reference feedback resistor into a parallel pair (Figure 5.75D), so it's easy to offset the output voltage range. For example, you could run the amplifier on a single +5 V supply, with REF2 driven from a clean reference of that same voltage. With no difference signal the output will be +2.5 V. The amplifier can accommodate input signals over a ±10 V range, and its gain of 0.2 takes a ±10 V differential input to a 0–4 V output. You can, of course, use a lower reference voltage. It's often convenient to use $V_{ref} = 4.096$ V when driving an ADC; this makes the step size come out in round numbers, e.g., 1 mV/step for a 12-bit conversion.

5.15 Instrumentation amplifier

The difference amplifiers of the previous section are inexpensive, and fine for many applications; and they have the nice feature of accepting inputs beyond the rails. But they have limited gain ($\leq$10) and CMRR ($\lesssim$85 dB min), their resistors make them somewhat noisy (20 nV/$\sqrt{\text{Hz}}$ to 50 nV/$\sqrt{\text{Hz}}$), and their relatively low input resistance ($\sim$10k to $\sim$100k) limits their utility to situations where the driving signals are of low impedance (op-amp outputs, low-Z balanced lines, low-R sense resistors).

If you need lots of gain, or a high input impedance, or superior CMRR, you need something different. It's called an *instrumentation amplifier*. These impressive devices have input impedances upward of 10^9 Ω, gains from unity to 1000 or so, low voltage noise (down to $\sim$1 nV/$\sqrt{\text{Hz}}$), and worst-case CMRRs of 100–120 dB (look ahead to Table 5.8 on page 363).

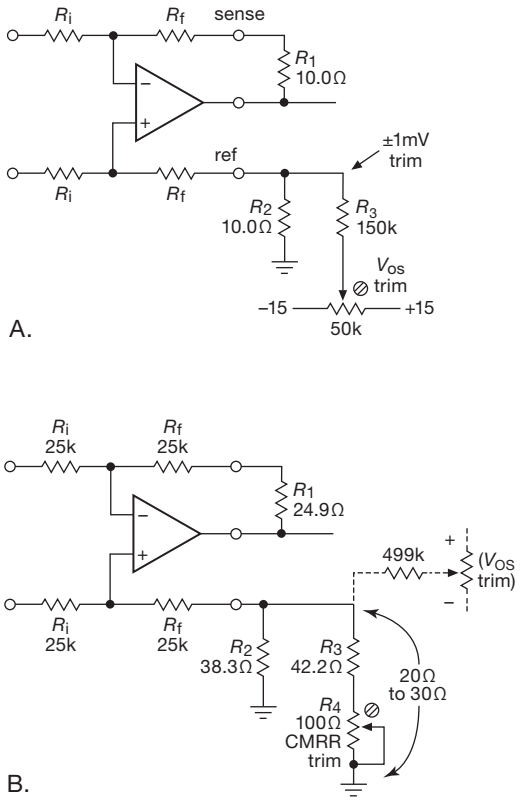


Figure 5.74. Trimming the offset and CMRR of a difference amplifier.

5.15.1 A first (but naive) guess

High input impedance – that's easy, just add op-amp followers to the difference amplifier (Figure 5.76); and then the resistors R_i and R_f can be smaller, reducing their Johnson noise contribution.

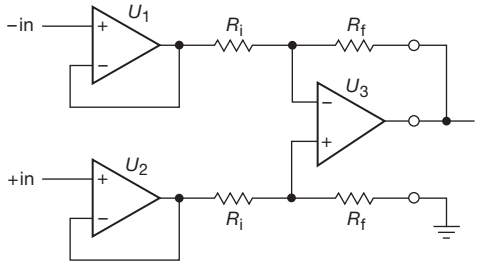


Figure 5.76. A first stab at improving the difference amplifier.

Indeed, this circuit has the enormous input impedance we expect from an op-amp follower, so there is no longer a problem from any reasonable source impedance.[70] But it does not improve the CMRR, which is still limited by the resistor ratio matching of R_f/R_i: it's really hard to do better than 100,000:1 with on-chip laser trimming (both initial trim and stability with time and temperature). In fact, this circuit degrades the CMRR somewhat, with two more amplifiers in the signal path.

5.15.2 Classic three-op-amp instrumentation amplifier

The circuit in Figure 5.77 is much better. It is the standard "three-op-amp instrumentation amplifier," one of several configurations that provide the desirable combination of high CMRR, high R_{in}, low e_n, and plenty of gain when you need it. The input stage is a clever configuration of two op-amps that provides high differential gain and unity common-mode gain without requiring close resistor matching. Its differential output represents a signal with substantial reduction in the comparative common-mode signal (when configured for $G_{diff} \gg 1$), and it is used to drive a conventional differential amplifier circuit. The latter is usually arranged for unity gain and is used to generate a single-ended output while removing the common-mode signal.

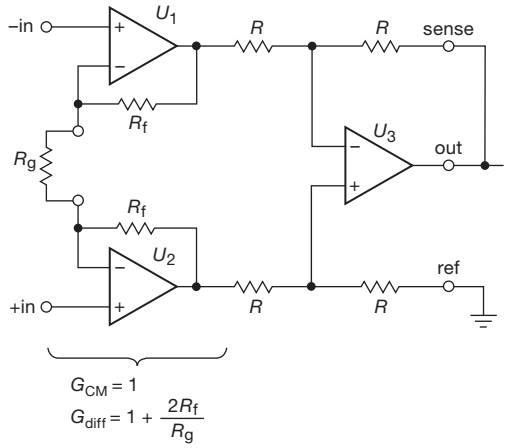


Figure 5.77. The classic three-op-amp instrumentation amplifier.

It's worth looking more closely at this circuit. We've hinted that it can deliver very high CMRR and very low e_n. But that is true *only when configured for large differential*

[70] At least at dc. At higher frequencies it again becomes important to have matched source impedances relative to the common-mode signal, because the input capacitance of the circuit forms a voltage divider in combination with the source resistance. "High frequencies" may even mean 60 Hz and its harmonics, because common-mode ac powerline pickup is a common nuisance.

gain. To see why, imagine we configure it for $G_{\rm diff} = 1$, by omitting the gain-setting resistor $R_{\rm g}$. Then we've just got the previous circuit (Figure 5.76), i.e., a buffered unity-gain difference amplifier. It has the same limitations of CMRR (set by resistor matching) and noise (from U_3's resistors).

Now imagine we set $G_{\rm diff} = 100$. We would do this by choosing $R_{\rm g}$ so that $1 + 2R_{\rm f}/R_{\rm g} = 100$, i.e., $R_{\rm g} = 2R_{\rm f}/(G-1)$. For the INA103, for example, $R_{\rm f} = 3\,{\rm k}\Omega$, so we'd use $R_{\rm g} = 60.6\,\Omega$. The INA103 conveniently includes a resistor of that value,[71] so for $G_{\rm diff} = 100$ you need only tie a pair of pins together. Let's look again at the CMRR and noise scene.

First, the CMRR: the front end has a differential gain of 100 and a common-mode gain of unity. In other words, it passes on to the difference amplifier stage a signal that has received 40 dB of CMRR blessings. Another 80 dB of CMRR in the output stage, and we've got the promised 120 dB CMRR. These numbers are typical of available instrumentation amplifiers, as listed in Table 5.8 (page 363) and graphed in Figure 5.82. For the INA103, for example, the datasheet lists CMRR=86 dB/72 dB (typ/min) for $G = 1$, and 125 dB/100 dB (typ/min) for $G = 100$.

Second, the noise voltage: the output stage still contributes Johnson noise from its resistor array, along with noise inherent in its amplifier. That cannot be helped. But that noise is combined with the *already-amplified* input signal, so the effect, *relative to the input signal* (RTI), is $100\times$ less. For the INA103, for example, the datasheet lists the noise density (typical, at 1 kHz) as $e_{\rm n}$=65 nV/$\sqrt{\rm Hz}$ for $G = 1$ and 1.6 nV/$\sqrt{\rm Hz}$ for $G = 100$.[72]

5.15.3 Input-stage considerations

Several preliminary comments here on the all-important input stage (about which a well-known circuit guru remarked "instrumentation amplifiers are all about their inputs"), with more to come later in §5.16.

A. Resistor matching

The circuit looks handsome with its symmetrical matched $R_{\rm f}$'s, but that requirement did not intrude in the discussion above. What is the effect of mismatched feedback resistors in the first stage? The common-mode gain remains unity (if you tie both inputs together, both outputs follow); and the differential gain expression is the same as before, but with $2R_{\rm f}$ not surprisingly replaced with the sum $R_{\rm f1} + R_{\rm f2}$. What changes, though, is that a purely differential input causes a differential output (amplified by $G_{\rm diff}$, as before) combined with some common-mode output.

You can see how this goes by imagining that U_2's feedback resistor is replaced with a short and U_1's resistor with $2R_{\rm f}$, and a symmetrical dc input signal $\pm\Delta V$ is applied to the inputs: U_2's output goes down ΔV, while U_1's goes up $(1 + 4R_{\rm f}/R_{\rm g})\Delta V$. That's the correct differential output, but with a common-mode offset of $(2R_{\rm f}/R_{\rm g})\Delta V$. This is not of great concern, providing that the $R_{\rm f}$'s are reasonably matched; they do not require the precise matching needed for the output stage.

B. The input amplifiers

It is essential that U_1 and U_2 have excellent CMRRs in order that a purely common-mode input signal is not converted to a differential signal (which would then be passed on to the output). Stated more precisely, they must have *matched* CMRRs, so that the voltage across $R_{\rm g}$ accurately tracks the differential input voltage. Viewing the circuit operation more generally in this light, the input amplifiers need not have extremely low individual offset voltages – what matters is that their offset voltages are accurately *matched* and remain so with changes in common-mode voltage. This gives rise to several circuit variants in which the "op-amps" U_1 and U_2 are configurations with well-matched base–emitter drops between each input and the corresponding $R_{\rm g}$ pin; see for example Figure 5.88C below.

C. Input-stage overload

The input stage amplifiers U_1 and U_2 will clip if their outputs are forced close to their supply rails, even though the output of the full circuit (U_3's output) would be expected to stay within safe bounds. Put another way, $V_{\rm CM} \pm 0.5V_{\rm diff}(1 + 2R_{\rm f}/R_{\rm g})$ must not reach either supply rail.

D. Signal guards

Instrumentation amplifiers are used with low-level signals, often conveyed by shielded cables to minimize noise. This adds input capacitance, thereby limiting the bandwidth (particularly with signals of moderate to high source impedance). Of perhaps greater importance, it degrades the CMRR at signal frequencies: the cable's shunt capacitance forms a voltage divider with the signal's source impedance, separately for each input; so if you have a

[71] More precisely, it includes an on-chip resistor, ratio matched to $R_{\rm f}$, to produce an overall guaranteed gain of 100.0±0.25%.

[72] The datasheet separates out the front-end and second-stage contributions, 1 nV/$\sqrt{\rm Hz}$ and 65 nV/$\sqrt{\rm Hz}$, respectively. From these you can calculate the input-referred noise $e_{\rm n}({\rm RTI}) = \left\{e_{\rm n}({\rm in})^2 + [e_{\rm n}({\rm out})/G]^2 + 4kTR_{\rm g}\right\}^{1/2}$. The last term is the square of the Johnson noise voltage $e_{\rm n} = 0.13\sqrt{R_{\rm g}}\,{\rm nV}/\sqrt{\rm Hz}$.

signal pair with unbalanced source impedances (a common situation), common-mode signal variations will create some differential signal input.[73] Finally, leakage currents become significant with signals of very high (MΩ to GΩ) source impedance. A nice technique for greatly reducing both the effective cable capacitance and any leakage current is to drive the shield actively with a "guard" voltage (Figure 5.78).

If there is a common shield surrounding the signal pair, the idea is to drive it with a buffered replica of the common-mode signal, as in circuit (A); a small series resistor is generally a good idea, for stability. A grounded outer shield can be used to eliminate any noise coupling to the guard, if that becomes necessary. This circuit requires access to the first-stage outputs, which you rarely get in an integrated instrumentation amplifier. Some amplifiers oblige by including this circuitry internally and providing a "data-guard" output pin, as in (B). If not, you can derive a common-mode signal yourself, as in (C).

Common-mode guarding reduces greatly the capacitive loading of the signal pair and therefore improves the CMRR (by minimizing the conversion of common-mode to normal-mode signals). But it does not reduce the effects of cable capacitance (and leakage) on *normal-mode* (differential) signals themselves. To accomplish that you need to shield the signals individually, driving each shield with a replica of the signal it shields, as in circuit (D). This is the familiar "bootstrap," acting here to reduce both the capacitance and leakage current seen by each signal. It thus minimizes high-frequency rolloff, and dc error in signals of high source impedance.[74] And, as with common-mode guarding, it also minimizes the degradation of CMRR by effectively eliminating the cables' capacitances.

Some instrumentation amplifiers provide these individual guard outputs, for example the INA116 shown, which is evidently intended for very low current measurements (it boasts a 3 *femto*amp typical input current). If not, you can roll your own, as in circuit (E), exploiting the fact that the R_g nodes follow the inputs. A caution, though: there is no guarantee that the signals at the R_g pins are not offset from the input signals, as for example in the configurations we'll see later in Figures 5.88C and 5.89F. With such an offset the bootstrap would be effective in minimizing capacitance, but much less so for leakage currents.

[73] Put another way, the cable's capacitance degrades the CMRR by creating differential phase shifts between the two signals, owing to their unbalanced source impedances.

[74] Instruments for measuring very low currents – "electrometers" and "source measure units" – include guard outputs and (usually) special BNC-like "triax" connectors for use with triaxial shielded cables.

E. Bootstrapped power supply

Figure 5.79 shows a trick that is analogous to signal guarding (you could call it "power-supply guarding"), occasionally helpful if you need to enhance further the CMRR of an instrumentation amplifier. U_3 buffers the common-mode signal level, driving the common terminal of a small floating split supply for U_1 and U_2. This bootstrapping scheme effectively eliminates the input common-mode signal from U_1 and U_2, because they see no swing (due to common-mode signals) at their inputs relative to their power supplies. U_3 and U_4 need not be bootstrapped if this is a discrete implementation. This scheme can do wonders for the CMRR, at least at dc. At increasing frequencies you have the usual problems of presenting matched impedances to the input capacitances.

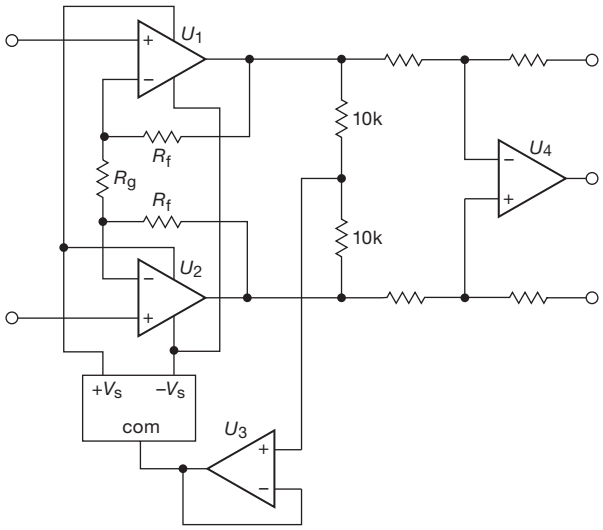


Figure 5.79. Instrumentation amplifier with bootstrapped input power supply for high CMRR.

5.15.4 A "roll-your-own" instrumentation amplifier

Integrated instrumentation amplifiers are excellent performers, and you usually can save a lot of work (and expense, and PCB real estate) by taking advantage of the broad selection of available parts, for which Table 5.8 (page 363) is a good starting point. But sometimes you need additional capability, for example a wider range of gains, or precise trimming of offsets and CMRR, or protection against outrageous abuse visited upon the input terminals.

Figure 5.80 is an example (another in the series of Designs by the Masters), based upon a commercialized

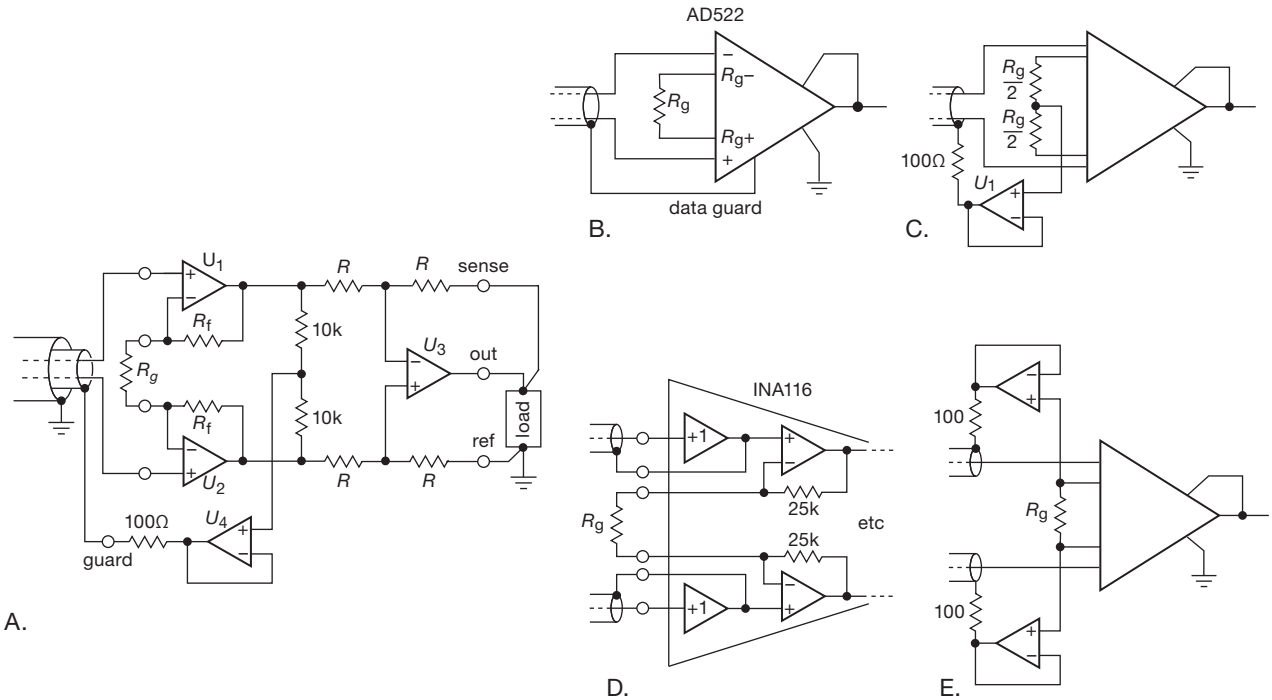


Figure 5.78. Signal guarding for greatly reduced cable capacitance. (A)–(C) are common-mode guards; (D) and (E) are individual signal guards.

design[75] from the talented John Larkin, to which we've added a few decorations. He needed a flexible front end that combined (a) overvoltage protection to ± 250 V, (b) logic-switchable gains from 1/16 to 256 by factors of four, (c) precision low offsets, (d) common-mode input range to ± 10 V for $G \geq 1$ and ± 140 V for $G < 1$, and (e) common-mode rejection of 120 dB at high gain.

The overall structure is the familiar three-op-amp configuration, with U_1 and its symmetrical twin (not shown) as the differential front end, which drives the unity-gain difference amplifier output stage U_3. The gain is set by analog switches U_2 and its twin, which select a tap on the resistor string R_6–R_{10}. For example, when the "$\times 64$" position is selected, $R_g = 201.1\,\Omega$ and $R_f = 6.411\,\text{k}\Omega$, thus a gain $G_{\text{diff}} = 1 + 2R_f/R_g = 64.8$.[76]

There's plenty more going on, here. Let's take it from left to right. The lossy chip inductor L_1 (these are often rated by their impedance, mostly resistive, at 100 MHz) combined with capacitor C_1 suppresses high-frequency interference that is outside the amplifier's bandwidth but able to cause nonlinear mischief in op-amp input stages. C_1's value is not critical, but its tight tolerance keeps the input impedance balanced in order not to compromise high-frequency CMRR. Relays K_1 and K_2 are used for frequent in-system offset and gain calibrations, essential for establishing and maintaining gain accuracies better than 0.1% (with a circuit using 1% resistor values) and zero offsets of $10\,\mu$V or better.[77] Relay K_3 switches in a $\times 16$ gain attenuation for input signals beyond ± 10 V. The input impedance is then set by R_1+R_2, with R_3 balance trim for good CMRR. R_1's value is a compromise between high R_{in} and bandwidth: here the 33.2 k, loaded by a downstream capacitance of $\sim$10 pF, puts the 3 dB rolloff at 500 kHz, about right for the product's 200 kHz specified bandwidth. But a higher input impedance would be nice, perhaps a nice round-number 1 MΩ (preferably with R_1 as a series pair

[75] Highland Technology's V490 VME Multi-Range Digitizer.

[76] The gain-setting resistors are chosen from standard "E96" 1% resistor values, so the actual gains differ from round-number values (with $\pm 1\%$ tolerance they would never be perfect, anyway). This front end would form part of a data-acquisition system, with overall gain and offset data held in software, from a calibration procedure carried out by relays K_1 and K_2.

[77] Larkin sings the praises of Fujitsu FTR-B3GA4.5Z DPDT relays, with their sub-picofarad capacitances.

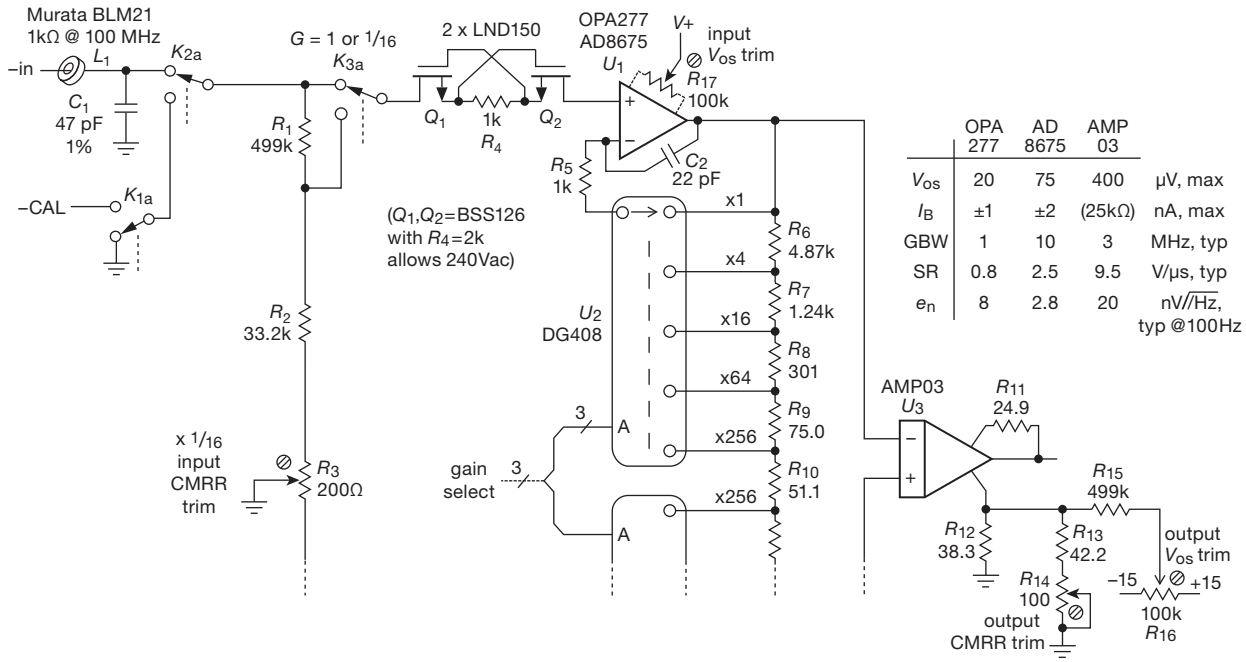


Figure 5.80. A "discrete" instrumentation amplifier design that combines precision, high CMRR, large common-mode voltage range, digital gain selection, and protection against ±250 V inputs. The symmetrical path to the noninverting input of U_3 is omitted to save paper.

of 464k, to accommodate input overvoltages without damage).

The curious Q_1Q_2 twins are a nice touch: these are *depletion-mode* high-voltage MOSFETs (see Table 3.6) from Supertex or Infineon,[78] used here as a ∼0.5 mA bidirectional current limiter to protect the op-amps (whose internal clamp diodes are unfazed by a few milliamps of input current; the AD8675, for example, specifies a maximum I_{in} of ±5 mA).[79] Here R_4 serves to reduce the saturation current from its maximum I_{DSS} of 3 mA (thus 750 mW dissipation at 250 V input, too much for a little SOT23 transistor), the tradeoff being an additional 1 kΩ of noise-adding input resistance in series with the unavoidable ∼1.7k on-resistance of the series pair of zero-biased MOSFETs. This circuit provides a reasonable level of overdrive protection; but an abrupt input step to +500 V, say, could well couple a large enough current transient (through the capacitance of Q_1 and Q_2) to damage the amplifier, or produce gate breakdown in the MOSFETs themselves.

The op-amps are precision bipolar types with input-current cancellation; note the tradeoff, for the two choices, of precision versus speed and bias current. These are laser trimmed for low offset voltage and are probably good enough without human intervention; but they include offset-trimming terminals to which you can attach a trimpot as shown. That sounds like a good idea – but be cautious, because the external trimming range you get when you do that is generally too much! For the AD8675, for example, the trimmer adjusts ±3500 µV (nearly 50 times the maximum untrimmed offset). So it may be a touchy (and unstable) adjustment to improve upon the already-trimmed V_{OS}.

The analog multiplexer selects a tap on the gain-setting string, which makes the gain insensitive to the on-resistance of the switches (here ∼100 Ω). That's the right way to do it; the *wrong* way would be to use a separate feedback resistor to each switch position, with R_{10} connected between the common terminals of the multiplexers. The small feedback capacitor C_2 ensures stability, given the lagging phase shift within the loop contributed by the ∼40 pF of switch capacitance to ground.

Finally, we arrive at our destination via the difference

[78] They're rated at 500 V, and come in three small package styles (TO-92, SOT23, and SOT89 with tab). The alternative BSS126 from Infineon is rated at 600 V, and costs less ($0.15 in small quantities). It comes only in the SOT-23 package, whereas the LND150 is available in three package styles, including a 1.5 W TO-243 small power package.

[79] Be sure to check that the op-amp does not suffer from phase reversal (see, e.g., §4.6.6), if you care about the output during input overdrive. A robust cure is to use a pair of input clamp diodes, as in Figure 5.81.

amplifier U_3, chosen for its combination of gain stability (an impressive $\pm 0.0008\%$ over temperature, typ), fast settling time (1 μs to 0.01%, typ), and low noise (20 nV/$\sqrt{\text{Hz}}$, typ). It's easy to add CMRR and offset trim as shown (see earlier discussion in §5.14.4), which matters most at low gain settings.

5.15.5 A riff on robust input protection

As we remarked above, a really nasty transient can ruin your whole day. Consider, for example, an input voltage step of 1 ns rise time and 350 V amplitude (the input accidentally touches a 240 Vac powerline): the 350 GV/s slew rate would force 350 mA through 1 pF! That's bad news for the op-amp U_1, as well as for the gate insulation of Q_1 and Q_2. How can the input protection in Figure 5.80 be improved to ensure robust protection against such egregious insults? The input choke L_1 helps, providing a series impedance that rises at high frequencies. But we can do better – see Figure 5.81.

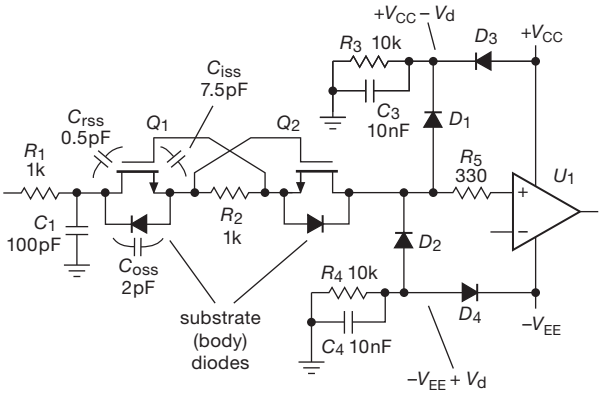


Figure 5.81. Bulletproofing the amplifier frontend with slew-rate control, diode clamping, and input-current limiting.

The first step is to limit the incoming slew rate, with R_1C_1: a 500 V step causes a maximum $dV/dt = 5$ V/ns slew rate across C_1,[80] with no significant degradation of bandwidth or noise. Looking at it most simply, that maximum slew rate can produce a transient current through the drain-source capacitance (C_{OSS}) of at most $I = C_{OSS}dV/dt = 10$ mA, thus a drop across R_2 of at most 10 V. That is well below the gate-source rating of ± 20 V; and it's a conservative estimate, because the actual gate-source voltage is reduced further by the relatively larger gate-source capacitance (C_{ISS}).

So Q_1 and Q_2 are safe. So is the amplifier, because the worst-case 10 mA transient current is clamped by diodes D_1 and D_2 to stay strictly within the amplifier's rails: those diodes clamp to bypassed voltages that are a diode drop inside the rails, set by diodes D_3 and D_4. Note the $\sim$1.5 mA dc forward bias of the latter (we're assuming ± 15 V rails), maintaining their diode drop even with a continuous dc input overdrive (with corresponding current limited to less than 1 mA, owing to Q_1Q_2).

Finally, the small input resistor R_5 is added for extra insurance, to limit any input current to the op-amp if the clamped voltage were to exceed the rails. This is more important with the usual arrangement of external clamp diodes (i.e., to the rails themselves), which allows the op-amp inputs to go a diode drop *beyond* the rails. Without the added series resistor it's then a current-divider contest between the external and the internal diodes.

This circuit looks pretty good, to us. Before shipping these to customers, though, you would be well advised to put it on the bench and beat it up, brutally. Circuits can surprise you.

5.16 Instrumentation amplifier miscellany

There's much to love about these amplifiers, whose nice variety is evident in the collection of Table 5.8 (with additional "programmable gain" types in Table 5.9 on pages 370–371). When you need some real performance, those tables are worth serious study! Here we collect some important advice on their care and feeding; refer often to Table 5.8 for enrichment as you read the following sections.[81]

5.16.1 Input current and noise

Instrumentation amplifiers (INAs) must condition the input signal without disturbing it; thus high input impedance Z_{in}, low input current I_{in}, and a low level of current noise i_n. There's the usual tradeoff – the lower voltage offset and voltage noise of BJTs versus the lower input current and current noise of FETs. Some bipolar-input INAs (for example the LT1167/8) use the trick of bias-current cancellation to achieve sub-nanoamp input currents. Conversely, the auto-zero INAs do best in terms of offset voltage, but they pay a price in current noise and chopping artifacts.

For some amplifiers the input current noise is close to

[80] And a momentary 500 V, 250 W surge in R_1, which should be a bulk-composition type, or several SMT resistors in series, to handle both the voltage and the energy transient; see Chapter *1x*.

[81] See also related material in the second edition, pp. 422-428.

Table 5.8 Selected Instrumentation Amplifiers

Part #	Circuit[q]	Supply range (V)	I_S typ (mA)	I_B max[a] (nA)	I_{OS} max[a] (nA)	V_{OS} max (µV)	ΔV_{OS} typ (µV/C)	ΔG max (%)[g]	Z_{in} (Ω)	CMRR G=100 typ (dB)	CMRR G=100 min (dB)	CMRR Unbal?	CMRR G=1 min (dB)	CMR Curve[z]	Noise[t] $V_{n(pp)}$[c] (µV)	Noise[t] e_n[x] (nV/√Hz)	Noise[t] i_n[x] (fA/√Hz)	Noise[t] R_n[x] (kΩ)	BW[d] –3dB (kHz)	Settle[d] 0.01% (µs)	Slew (V/µs)	Filter?	DIP avail?	Needs R_g?	swing to supplies? IN +	IN –	OUT +	OUT –	Gain, comments
high-voltage, BJT																													
INA103	A	18–50	9	12µA	1000	250	1.0	0.07	60M	125	100	0	72	2	-	1.0	2000	0.5	800	3.5	15	-	●	int	-	-	-	-	1, 100, adjustable
INA163	A	9–36	10	12µA	1000	250	1.0	0.1	60M	116	100	0	80	3	-	1.0	800	1.3	800	3.5	15	-	-	●	-	-	-	-	G = 1 + 6k/R_g
INA217	A	9–36	10	12µA	1000	250	1.0	0.2	60M	116	100	0	80	3	-	1.3	800	1.6	800	3.5	15	-	●	●	-	-	-	-	G = 1 + 5k/R_g
THAT1510	C	10–40	6	14µA	1400	250	-	6	30M	120	105	0	45	-	-	1.0	2300	0.43	7000	-	28	-	●	●	-	-	-	-	G = 1 + 10k/R_g
INA166	A2	9–36	10	12µA	1000	250	2.5	1	60M	120	100	0	-	4	-	1.3	800	1.6	450	3.5	15	p	●	-	-	-	-	-	G = 2000 + 60k/R_g
AD8428B	C	8–36	6.7	50	10	25	0.3	0.05	1G	153	140	0	-	27	0.05	1.3	1500	0.9	3500	0.75	50	p	-	-	-	-	-	-	G = 2000
AD627B	C2	2.2–36	0.06	10	1	150	0.1	0.25	20G	96	83	1k	-	16	0.56	38	50	760	3	290	0.05	-	●	●	-	●	●	●	G = 5 + 200k/R_g
INA128	A	4.7–36	0.70	5	5	50	0.2	0.5	10G	125	120	1k	80	1	0.2	8	300	27	200	9	4	-	●	●	-	-	-	-	G = 1 + 50k/R_g
INA126	B	2.7–36	0.18	25	2	250	0.5	0.1	1G	94	83	0	-	15	0.7	35	60	590	9	160	0.4	p	●	●	-	-	-	-	G = 5 + 80k/R_g
LT1789-10	C	2.2–36	0.12	40	0.2	750	0.2	0.27	1.6G	113	98	1k	-	14	1.1	48	62	775	12	190	0.06	-	-	●	-	●	-	n	G = 1 + 200k/R_g
AD8221B	C	4.6–36	0.90	0.4	0.4	25	0.3[d]	0.15	100G	-	110	1k	80	17	0.25	8	40	200	100	10	2	-	-	●	-	-	-	-	G = 1 + 49.4k/R_g
LT1167A	C	4.6–40	0.9	0.35[e]	0.35[e]	40	0.05	0.08	1T	125	120	1k	90	5	0.28	8	60	130	120	14	1.2	-	●	●	-	-	-	-	G = 1 + 49.4k/R_g
LT1168A	C	4.6–40	0.42	0.25[e]	0.3[e]	40	0.05	0.5	1T	135	120	1k	90	8	0.28	10	30	330	13	30	0.5	-	●	●	-	-	-	-	G = 1 + 49.4k/R_g
AD620B	C	4.6–36	0.9	1	0.5	50	0.3	0.15	10G	130	120	1k	80	5	0.28	9	100	90	120	15	1.2	-	●	●	-	-	-	-	inexpensive
AD8227	C	2.2–36	0.35	27	1.5	200	0.2	0.3	0.8G	-	105	1k	-	6	0.5	24	100	240	50	35	0.8	-	-	●	-	●	●	●	G = 5 + 80k/R_g
JFET																													
INA110B	A	12–36	3	0.1	0.05	250	2	0.1	5T	116	106	1k	80	7	1	15	1.8	8.3M	470	5	17	-	●	int	-	-	-	-	1, 10, 100, 200, 500
AD8220B	C	4.5–36	0.75	0.01	6pA	125	5[d]	0.2	10T	90	84	1k	80	8	0.8	14	1	14M	120	8	2	-	-	●	-	●	●	●	G = 1 + 50k/R_g
MOSFET																													
INA116	A	9–36	1	25fA[v]	25fA[v]	2000	20	0.5	10T[j]	94	86	1k	80	5a	2	28	0.1	300M	70	400	0.8	-	●	●	-	-	-	●	guard, G = 1 + 50k/R_g
low-voltage, BJT																													
AD8129A	E	4.5–25	10	2000	400	800	2	0.6	6M	110	90	0	-	18	-	4.5	1000	4.5	270M[s]	0.02[o]	930	p	-	●	-	-	-	-	250MHz, G≥10
AD8223B	C	3–24	0.65	25	2	100	1	0.3	2G	110	90	1k	-	20	0.6	30	70	430	50	18	0.2	-	-	●	-	●	●	●	low-cost, G=5(1+R2/R1)
MAX4194-7	A2	2.7–7.5	0.09	20	3	225	0.5	0.5	1G	115	95	1k	78	10a	0.6	8.7			1.5	40	-	-	-	●,-	-	n	●	●	G = 1, 10, 100, var
AD623B	A	5–12	0.38	25	2	100	0.1	0.35	2G	110	105	1k	77	13	1.5	35	1500	23	10	20	0.3	-	●	●	-	●	●	●	G = 1 + 100k/R_g
ISL28270	E[k]	2.4–5.5	0.12	2	1	150	0.7	0.5[t]	3M	110	90	0	90	12	3.5	60	370	160	240	-	0.5	p	-	●	●	n	●	●	G ≥ 100, dual
INA337	F2[k]	2.7–5.5	2.4	2	2	100	0.1	0.2	10G	120	106	0	106	10	0.8	33	150	220	10	160	filter	●	-	●	●	n	●	n	G = 2 R2/R1
CMOS																													
AD8236	B	1.8–5.5	0.04	10pA	5pA	3500	2.5	0.2	440G	110	100	0	-	25	4	76	15	5M	0.8	1000	0.01	-	-	●	●	●	●	●	G = 5 + 420k/R_g
INA321[y]	B2	2.7–7.5	0.04	10pA	10pA	500	7	0.1	10T	94	90	0	-	9	20	100	3	33M	20	12	0.4	p	-	●	-	-	●	●	G = 5 (1 + R2/R1)
ISL28272	E[k]	2.4–5.5	0.12	30pA	30pA	500	0.7	0.2[t]	1G	100	80	0	80	13	6	78	0.2	390M	100	-	0.5	-	-	●	●	●	●	●	dual, G ≥ 100[u]
auto-zero																													
AD8230	D	8–16	2.7	1	0.3	10	0.05	0.04	cap[w]	120	110	1k	110	19	3	240	-	-	2.5	slow	2	p	-	●	●	●	●	●	2kHz chopper noise
LTC2053	D	2.5–5.5	0.95	10	3	10	0.05	ext	cap[w]	120	105	1k	100	20	2.5[c]	170[b]	-	-	5	4000	0.2	p	-	●	●	●	●	●	3kHz chopper noise
MAX4209	E	2.9–5.5	1.4	0.001[t]	0.001[t]	20	0.05	0.25	2G	135	106	0	106	21	2.5	170	-	-	10	120	0.08	p	-	●	-	●	●	●	G=100, AZ, '08 var G
INA333	A	1.8–7	0.05	0.2	0.2	25	0.1	0.25	100G	115	100	0	80	22	1.0	50	100	500	3.5	400	0.2	-	-	●	n	n	●	●	G = 1 + 100k/R_g
AD8553	F	1.8–6	1.1	1	2	20	0.02	0.3	50M	140	120	0	-	23	0.7	30[h]	-	-	1	2400	slow	●	-	●	-	●	●	r	use 1kHz filter
AD8293	F	1.8–6	1	2	4	50	0.02	1	50M	140	110	0	na	24	0.7[c]	35	-	-	0.5	2400	filter	●	-	int	-	●	●	r	G = 80, 160 AZ-amp

Notes: (a) at 25°C. (b) at 100Hz. (c) 0.01–10Hz or 0.1–10Hz; at G=100 if available. (d) at G=100 max, not 1000. (e) superbeta + cancelled input current. (f) max. (g) gain error, at G=100. (h) plus high-freq AZ noise. (j) has input guard pins. (k) has charge pump to enhance CM performance. (m) plus filter stage and output clamp. (n) to within 50mV–200mV of rail. (o) to 0.1%. (p) can filter if gain greater than minimum. (q) see Figures 5.88 & 5.89. (r) V_{out} to V_{EE}, but V_{ref} 0.8V min. (s) G=10. (t) typical. (u) ISL28271 for G≥10; both are decomp. (v) *typical* bias, offset = 1fA, 3fA. (w) warning: high bias current. (x) at 1kHz. (y) dual=INA2321. (z) see Figure 5.82.

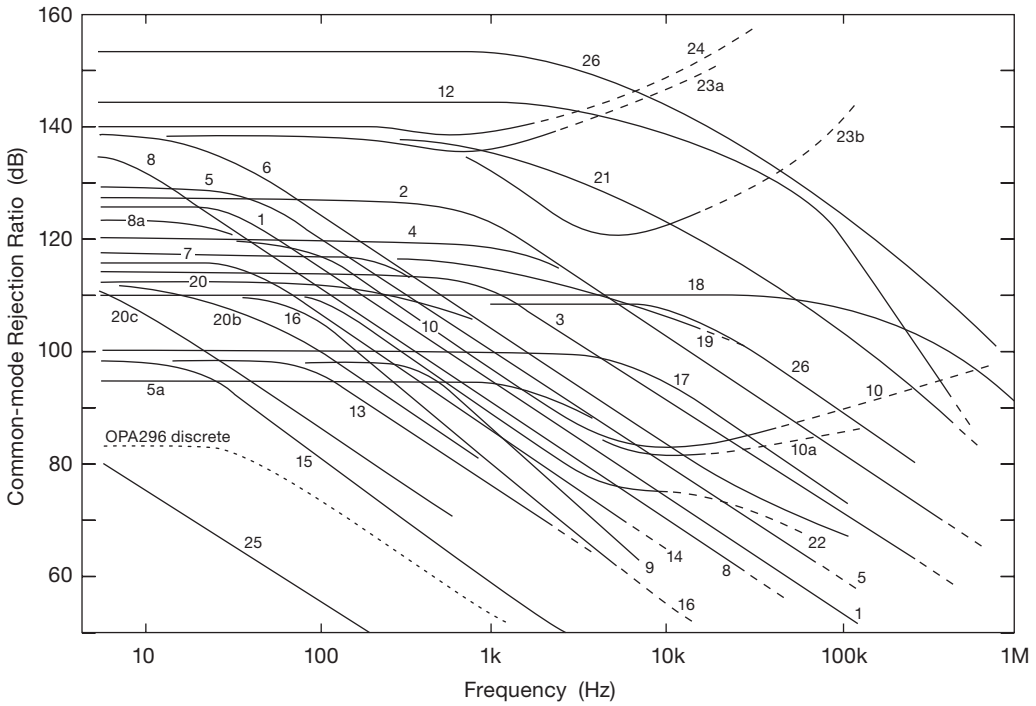


Figure 5.82. Common-mode rejection ratio versus frequency, for the instrumentation amplifiers listed in Tables 5.8 and 5.9. The "OP296 discrete" curve is measured data of a two-op-amp discrete implementation (Figure 5.88, circuit B) shown in the AD627 instrumentation amplifier (curve 16) datasheet as testimonial to the superiority of integrated amplifiers. Curves 23a and 23b show the effect of filter choices (1 kHz and 10 kHz, respectively) for the same amplifier (the AD8553). The dashed line extensions indicate a region well past the amplifier's cutoff frequency.

the shot-noise limit (as in §5.10.8), but this is greatly exceeded for auto-zero amplifiers, and BJT-input amplifiers that use bias-current cancellation.

5.16.2 Common-mode rejection

Instrumentation amplifiers must often deal with small difference signals riding on much larger common-mode voltages, requiring a high CMRR.

To get a sense of the problem, consider an INA used with a strain-gauge bridge powered from 5 V: the INA's common-mode signal input is 2.5 V, with a typical full-scale output of 10 mV (i.e., 2 mV/V). So a signal that is 0.1% of full-scale is just 10 μV. That's -108 dB with respect to the 2.5 V common-mode signal! Just seven of the INAs in Table 5.8 meet this spec for their minimum CMRR (though all but five meet it for their *typical* CMRR). An amplifier that fails to meet the CMRR spec will simply exhibit an output error larger than 0.1% of full scale, entirely down to insufficient CMRR. Keep this in perspective, though: most of the INAs in the table have more than 10 μV of input offset voltage, anyway. So we can say for many applications the listed INAs do reasonably well (and some exceptionally well), at least at dc.

A. CMRR versus frequency

A much more severe test of an INA's CMRR is its ability to reject common-mode signals at high frequencies. The plots in Figure 5.82 show this degradation, beginning somewhere in the range of 100 Hz to 5 kHz. In contrast to the (dc) strain-gauge application, imagine measuring the voltage across low-value current-sense resistors monitoring the windings of a three-phase motor. If the ac drive frequency is low enough (say 50 to 60 Hz), the INA may be up to the task. But if the motor windings are driven from a pulse-width-modulated (PWM) controller, with say 40 kHz pulses, high-frequency CMRR degradation may render an INA unusable for the task. So Figure 5.82's curves may be useful not only for choosing the best INA for the job, but in fact to determine whether an INA may be used at all.

Some of the curves of CMRR versus frequency level off after falling at the usual 6 dB/octave. This happens because

the INA includes provision for bandwidth filtering at an internal node (with an external capacitor) that attenuates the common-mode feedthrough by the same 6 dB/octave, thus cancelling out the input-stage CMRR degradation. For example, we see this for the AD8293 and AD8553 (curves 24 and 23a) with 1 kHz filtering, and for the latter and the INA337 (curves 23b and 10) with 10kHz filtering. This is seen also for some micropower parts, with their very limited bandwidth, for example the MAX4194 (curve 10a), which runs at 90 μA and has a 1.5 kHz bandwidth. That's where its response falls with $G = 100$. You can use this principle to advantage by adding a post filter in applications where fast response isn't necessary, for example the three-phase motor current monitor.

B. Case study: increasing CMRR with upstream gain

Here's a nice example from our research lab, where we needed to control the current through a magnet coil with a high degree of stability, to bring about a "Bose–Einstein condensate" of cold atoms, and (in a first experiment of its kind) slow the speed of light to that of a bicycle.[82] The currents ranged up to 875 A (!), and we wanted something close to ~10 ppm stability in the controlled current, which was sensed with a low-side 4-wire current shunt of 100 $\mu\Omega$ resistance (thus 87.5 mV full-scale signal[83]). With such high currents flowing, we had to deal with as much as a volt or so of common-mode signal, upon which the difference signal (87.5 mV maximum, but often much less) was riding. To control that current to 10 ppm thus required some 140 dB of CMRR, and of course very low input offset drift. It also required a low input noise voltage, ideally less than 0.1 μVpp integrated low-frequency noise in order to achieve 10 ppm.

The combination of very high CMRR and very low noise and offset is a tall order for any instrumentation amplifier. The solution here is to put some low-noise and low-drift gain upstream of the INA, as shown in Figure 5.83. A precision low-noise op-amp like the LT1028A (typical drift and low-frequency noise of 0.1 μV/°C and 35 nVpp) is wired as a $G = +51$ composite amplifier (§*4x.5*), within which conventional op-amp A_2 is configured with $G = 5$. The compensation capacitor C_C limits the bandwidth (and therefore the noise) to about 10 kHz. The precision op-amp is run at lower voltage to reduce thermal effects (offset-voltage drift, thermal-gradient errors), with its ±8 V rails regulated down (with floating 3-terminal linear regulators) from the ground-referenced ±15 V that powers A_2 and A_3.

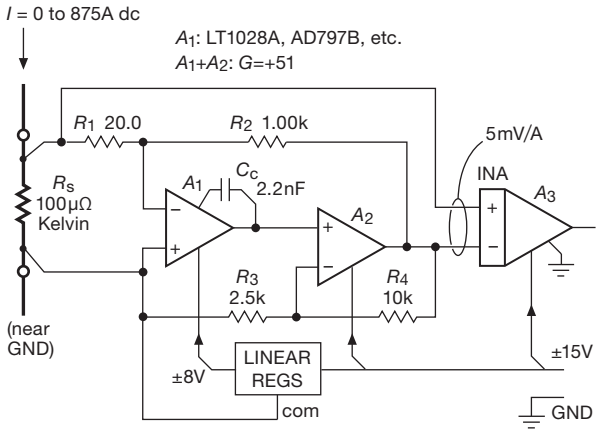


Figure 5.83. Current sensing with low drift and low noise. Op-amps A_1 and A_2 form a composite amplifier, with a precision low-noise first stage and a noncritical second stage. The output signal provides feedback to the high-current power supply.

5.16.3 Source impedance and CMRR

Instrumentation amplifiers excel in high input impedance, but that does not confer automatic immunity from the effects of mismatched signal source impedances (which so severely degrade CMRR in *difference* amplifiers, with their relatively low input impedance; see §5.14.3A). Most datasheets are bashful about displaying their dirty laundry in this regard, so we must applaud Analog Devices' candor, as shown in Figure 5.84. Note the greater effect at higher gain settings (where there's more to lose, because the CMRR is so good), and at higher frequencies (where the amplifier's input impedance is falling, owing to capacitance).

5.16.4 EMI and input protection

Whether you "roll your own" or use an integrated instrumentation amplifier, you've got to think about protection both from overloads and from electromagnetic interference (EMI). An example from real life: a colleague uses temperature-monitoring thermocouples in his lab, connected with the usual unshielded wire pairs to an INA front end running at high gain. All was well until a particular switching power supply was energized, at which point the thing went haywire.

The problem, of course, was common-mode EMI, coupled onto the long unshielded cable. And the solution, in

[82] L. Vestergaard Hau, S. E. Harris, Z. Dutton, and C. H. Behroozi, "Light speed reduction to 17 metres per second in an ultracold atomic gas," *Nature* **397** 594–598 (1999).

[83] A smallish voltage, but already an uncomfortably large power dissipation (~75 W), requiring a temperature stabilized oil bath.

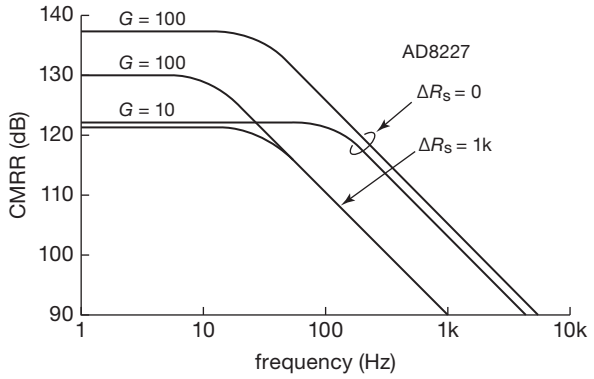


Figure 5.84. Instrumentation amplifier CMRR, best at high gain settings, is degraded by source-impedance mismatch.

a situation like this where you don't need bandwidth, is to filter the input aggressively, as in Figure 5.85. The diode clamps are optional, but probably a good idea if you want the amplifier to survive an unusual input event. In situations where you *do* need bandwidths that include interfering signals, you're unlikely to get away with unshielded cable; use shielded wiring, and pay attention to grounding paths.

It is hard to overstate the seriousness of EMI/RFI: RF signals leaking into your inputs cause rectification inside BJT op-amps, and thus dc offsets. Cables or PCB traces can exhibit narrowband (high-Q) resonances, enhancing these effects. If you notice changes of offset voltage when you grip a cable, or touch a circuit node with a pencil, or just wave your hands around, you're probably seeing RF coupling (the other possibility is a circuit oscillation). Lossy ferrite beads are excellent for attenuating RF, as well as for reducing the Q of unwanted wiring resonances; but they're not a panacea, and often you need to resort to additional filtering.

5.16.5 Offset and CMRR trimming

Instrumentation amplifiers that provide both SENSE and REF pins (e.g., the INA103) can be externally trimmed, if needed, for both offset voltage and CMRR, as shown previously for the difference amplifier (Figure 5.74). More often you're given only the REF pin, though. That's enough to trim the offset voltage, but note that any offset applied to the REF pin must have a source impedance of no more than $\sim 10^{-6} R_f$ (i.e., a few *milli*ohms) in order not to compromise the amplifier's 100 dB+ CMRR. This is best done with a precision op-amp, as in Figure 5.86.

Some INAs helpfully include output-stage offset trim terminals, to be used with an external trimpot. A few of these (e.g., the INA110 and others) even provide separate trim pairs for the front-end and output stages.

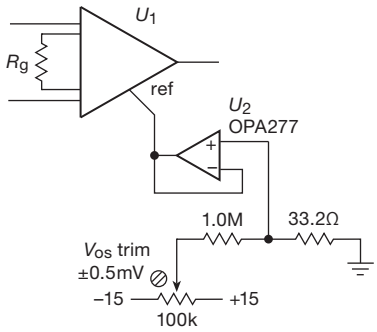


Figure 5.86. Trimming the offset of an INA that doesn't provide a SENSE pin, offset pins, or a buffered REF pin.

5.16.6 Sensing at the load

As with the difference amplifier, the REF and SENSE pins can be connected directly at the load, as in Figure 5.78(a), to eliminate errors from wiring resistance and unrelated ground currents.

5.16.7 Input bias path

Instrumentation amplifiers earn honest bragging rights from their very high input impedance, but, just as with op-amps, you've got to provide a dc return path. If you don't, the amplifier will saturate. This occurs naturally in circuits like the strain gauge of Figure 5.64A, but not in something like a thermocouple (Figure 5.85). For the latter you can use a resistor from one of the inputs to ground (or to mid-supply, for a single-supply amplifier), or you can use a bias resistor from each input to ground to preserve symmetry.

5.16.8 Output voltage range

If an INA is operated at low gain and with its common-mode input near the rails (but legally within the specified operating range), the internal amplifier may saturate, causing the INA's output to go to an incorrect voltage. For example, look at the "Maximum Output Voltage vs. Common-Mode Input" graph in the datasheet for the AD623 (this amplifier is like type A in Figure 5.88, except with *pnp*-input emitter followers). If the common-mode input is at 0 V with G=10 (which is legitimate), the maximum output capability is only 1.0 V! *You have been warned!*

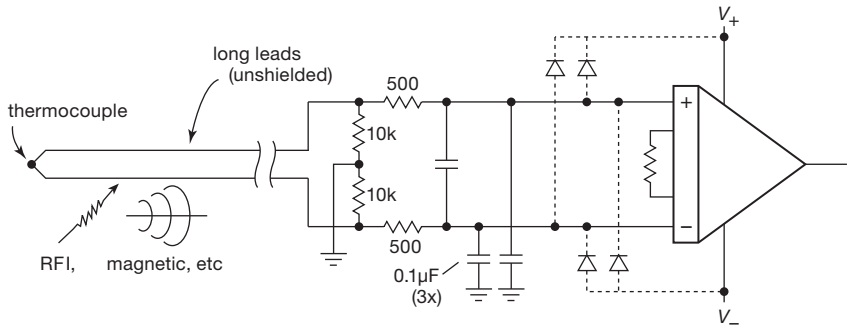


Figure 5.85. Electromagnetic interference couples nicely into long unshielded cables, which you can think of as *antennas*. Use lowpass filtering at high-gain inputs, with optional diode clamping to the rails. Note the 10 kΩ resistor pair, to set the dc level of the floating differential input.

5.16.9 Application example: current source

The excellent CMRR of instrumentation amplifiers, combined with very low input capacitance (typically ∼2 pF), lets you design an active current source in which the current sense resistor rides on the high side of the output, with a "flying" INA converting its voltage drop to a ground-referenced output. Such a circuit is shown in Figure 5.87A, exploiting the >80 dB CMRR of the AD8221B out to 50 kHz, where the stabilizing compensation network R_1C_C rolls off. The MOSFET, configured as a common-source transconductance stage, has inherently high output impedance, which makes it possible to maintain good current-source behavior at higher frequencies. This improves upon a current-source circuit like that in Figure 5.69 (§5.14.2D), in which the performance degrades with frequency owing to the op-amp's compensation and limited slew rate. In this circuit the source-degeneration resistor R_3 acts to reduce the MOSFET transconductance, to enhance stability (some breadboard testing, with anticipated load impedances, would not be inadvisable). See Table 3.4 for selected *p*-channel MOSFETs.

For this circuit the op-amp output must be able to swing to the positive rail (the LT1490 is RRIO), and the AD8221B needs a few volts of negative supply voltage (−3 V is adequate) because its common-mode input voltage does not extend to the negative rail.[84]

The AD8221B's low offset voltage (25 μV max) provides a large dynamic range, corresponding to an output error of just 0.25 mA out of the full-scale range of 5 A (20,000:1).

For lower output currents you can substitute a smaller bipolar transistor, as in Figure 5.87B; its lower capacitance permits greater loop bandwidth (we've configured it here as a follower). The AD8221B's low input

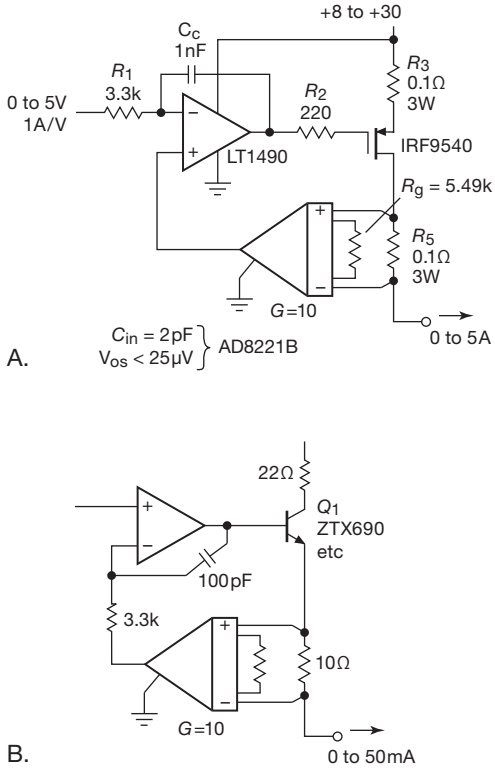


Figure 5.87. Precision current source with flying instrumentation amplifier. A. Power MOSFET for 5 A full-scale current, B. a small high-gain ($\beta \sim 500$ over 0.1–50 mA) BJT provides greater bandwidth.

[84] The AD8227 variant allows V_{CM} to the negative rail, so you could run it single supply; but you pay a price in larger V_{OS} and I_B, and its CMRR degrades at a lower frequency.

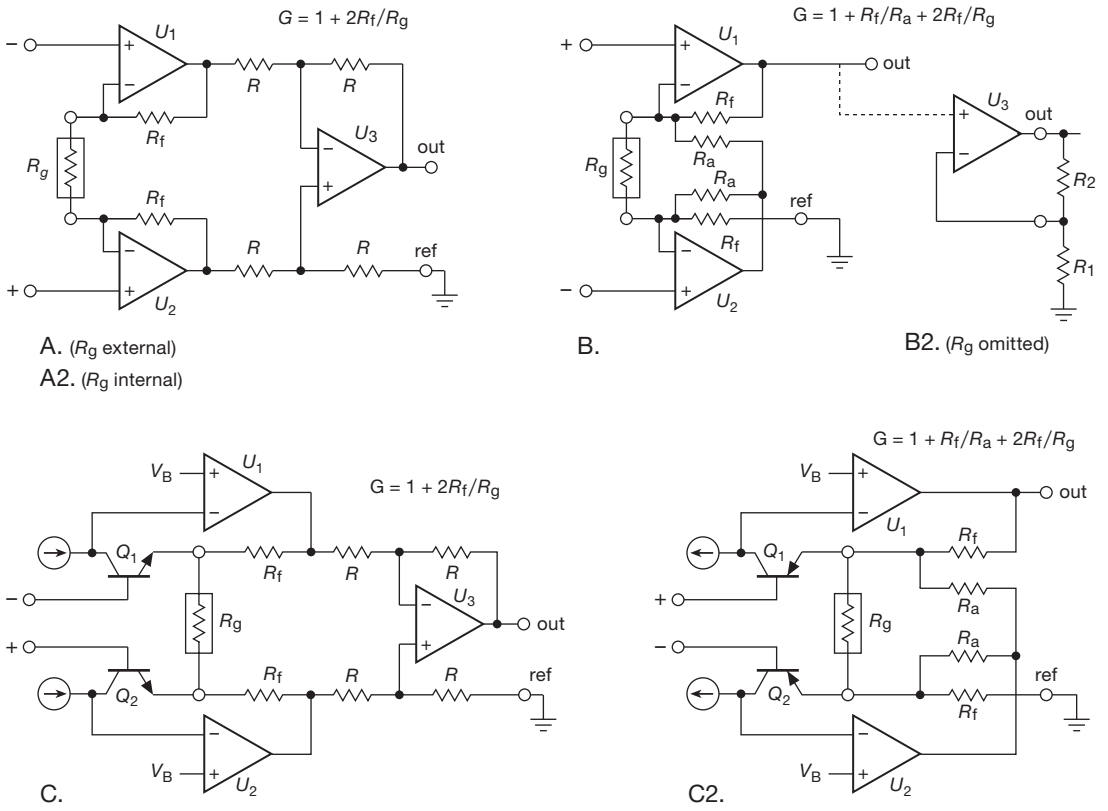


Figure 5.88. Instrumentation amplifier configurations A–C, as listed in Table 5.8 (page 363). For these and other differencing circuits the "ref" pin need not be grounded.

current (0.4 nA max) means that you can scale down the full-scale output current range, say to 100 μA full scale.

5.16.10 Other configurations

The classic 3-op-amp circuit of Figure 5.77 is widely used, notably in the INAs offered by Burr–Brown/TI (recognizable by their part numbering "INA*nnn*"); but you'll see other circuit configurations (if you get far enough into the datasheets) representing different tradeoffs among the various performance parameters and cost. Although you can get along fine without "looking under the hood"(most of what you need to know comes from the tabulated data), some of these configurations have unusual quirks that can catch you off-guard. For example, amplifiers with configuration E (Table 5.8 on page 363, and Figures 5.88 and 5.89) can be damaged by input voltage differentials greater than ± 0.5 V (!),[85] and the amplifiers with configuration F do not operate with the REF tied to ground (even though intended for low-voltage single-supply operation). Quite apart from these kinds of worries, curiosity impels a brief look at these circuits.

There's a general principle in play in nearly all of these circuits (D and E excepted): (a) the voltage across a gain-setting resistor R_g is precisely the same as the input difference voltage, creating a current $I_g = \Delta V_{in}/R_g$; and (b) that current is used to generate an accurately proportional output voltage $V_{out} \propto I_g$. Classic configuration A puts this in clear view: the input op-amps (or equivalent – they need not *be* fully featured op-amps) enforce a matched ΔV_{in} across R_g, with the resultant current flowing through the two R_f's; thus a differential output $\Delta V_{out} = (\Delta V_{in}/R_g)(R_g + 2R_f)$. The

[85] Some of these have back-to-back clamp diodes across the inputs (true also of some op-amps and comparators), for which the damage is caused by excessive input current; others tolerate greater differential inputs (as much as a few volts), though most of the E-configuration parts are considerably more restrictive than those with the other circuit topologies. What matters more, from the user's point of view, is the maximum differential input without performance degradation.

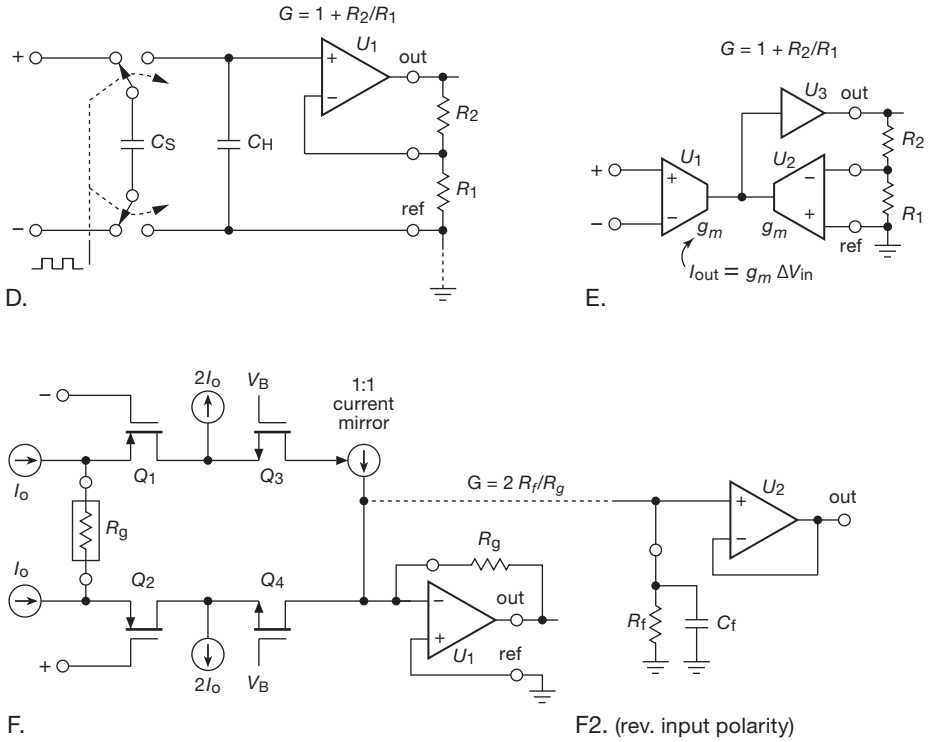


Figure 5.89. Instrumentation amplifier configurations D–F, as listed in Table 5.8 on page 363.

unity-gain difference amplifier converts this to a single-ended output, with gain $G = 1 + 2R_f/R_g$.

Configuration C works similarly, but here the matched emitter followers Q_1Q_2 create the ΔV_{in} replica across R_g, with the op-amps serving to ensure equal emitter currents (and thus no contribution to the differential output).[86] In this circuit and following, V_B is a reference "bias" voltage.

One nice thing about the C and C2 configurations is that a small RFI-defeating capacitor can be placed across base-to-emitter of the input transistors, because they're exposed pins. Keep those capacitors small – 100 pF or less – so the amplifier's bandwidth and stability are not degraded.[87]

The clever configuration B is different: it is more economical, requiring only two op-amps and fewer trimmed resistors, but its performance suffers, with poorer CMRR (particularly at higher frequencies). (The reader also suffers, trying to figure out how this tangle of a circuit works.) Configuration C2 is the discrete differential-pair analog of B (just as C was for A), with similarly underwhelming specifications.

Configuration F continues the theme of replicating ΔV_{in} across R_g, with the resultant unbalanced current at U_1's summing junction being converted to a single-ended output. In this circuit Q_3 and Q_4 form a "folded cascode," pinning the drains of Q_1 and Q_2 while passing their currents through (offset by twice the quiescent current, thus sinking). This circuit requires accurate matching of the current sources and of the current mirror (stated more precisely, it requires constancy of the current sources and mirror over common-mode variations). Evidently that can be accomplished with good design (and the assistance of circuit tricks like the cascode), given the impressive 140 dB (typ) CMRR specification.[88]

[86] To achieve low input currents with the C configuration, LTC uses superbeta BJTs with base-current cancellation in some of the listed parts ($I_B \approx 50$ pA); Analog Devices does even better using JFETs, but with greater offset and noise. Some of the TI/Burr–Brown INAs listed as A types may in fact use the C configuration; their datasheets are silent on the circuit details.

[87] BJT-input amplifiers are prone to RFI upset, because their inputs are forward-biased base–emitter (diode) junctions. And RFI is a real problem in these low-level circuits with inputs from remote sensors. Better to use a JFET-input amplifier if you are bedeviled by RFI.

[88] Because they contain only a few MOSFETs, current sources, and current mirrors, devices of configuration F can be quite inexpensive. For example, the AD8293 (an AZ with fixed G=80 or 160) sells for only $0.97 (qty 100).

Table 5.9 Selected Programmable-gain Instrumentation Amplifiers

Part #	Input device	circuit	Supply Voltage min (V)	Supply Voltage max[e] (V)	I_S typ (mA)	shutdown	Input Current[a] typ (nA)	Offset Voltage typ[g] (μV)	Offset Voltage max (μV)	ΔV_{OS} typ (μV/C)	Gain Error[d] max (%)[w]	Z_{in} (Ω)	CMRR[x] G=100[d] typ (dB)	CMRR[x] G=100[d] min (dB)	curve	Noise[t] V_{npp}[t] <10Hz[c] (μV)	Noise[t] e_n[t] 1kHz (nV/√Hz)
AD8250	BJT	A2	10	34	4.1	-	5	90	260	1.7	0.04	5G	110	98	4	1	18
AD8251	BJT	A2	10	34	4.1	-	5	95	275	1.8	0.04	5G	110	98	4	1.2	18
AD8253	BJT	A2	10	34	4.6	-	5	-	160	1.2	0.04	4G	120	100	26	0.5	10
PGA204B	BJT	A2	9	36	5.2	-	0.5	10	50	0.1	0.024	10G	123	110	8a	0.4	13
PGA202B	JFET	C	12	36	6.5	-	0.01	500	1000	12	0.15	10G	120	92	3	1.7	12
PGA207	JFET	C	9	36	12	-	0.002	500	1500	2	0.05	10T	100	95	3	1	18
PGA280 *low-voltage*	BJT	A2	10	40[f]	0.75	-	0.3	50[q]	250	0.2	0.15	1G	140	120	5	0.42	22
LTC6915	switch	D	2.7	11	0.9	x	5	3	10	0.05	0.5	cap[k]	125	105	1	2.5	225
AD8231	BJT	A2	3	6	4	x	0.25	4	15	0.01	0.8	high	-	110	3	0.7	39[h]
LMP8358	CMOS	E	2.7	6	1.9	x	0.006	1	10	0.05	0.1	50M	139	110	2	0.6	27[h]
PGA309	CMOS	A2	2.7	7	1.2	x	0.1	3	50	0.2	1	30G	105	-	-	4	210

Notes: (a) at 25°C. (b) at 100Hz. (c) 0.01–10Hz or 0.1–10Hz, at *G*=100 if available. (d) at *G*=100 or max, not 1000; no source imbalance. (e) abs max. (f) separate LV output rails. (g) at *G*=max. (h) plus HF auto-zero noise. (k) caution: high bias currents. (m) plus filter stage and output clamp. (n) to within 50mV–200mV of rail. (o) see Fig. 5.82. (p) rail-to-rail input, provided AGND away from rails. (q) 250kHz chopper. (r) V_{out} to V_{EE}, but V_{ref}=0.8V min. (s) 2V/μs at *G*=1000.

Parts using configurations D and E are unlike the others. In D a flying capacitor C_S periodically samples and conveys to the "hold" capacitor C_H the differential input voltage, thus creating a single-ended (ground-referenced) replica. This sounds good in principle; but the resulting noise is high, and the slow commutating rate (3–6 kHz, for the two exemplars in Table 5.8 on page 363) limits the bandwidth and extends the settling time. This technique is also vulnerable to aliasing at half the chopping frequency. However, these parts are inexpensive, and they may be well suited for some dc applications.

Finally, in E the output currents from a pair of differential-input transconductance amplifiers are combined and forced to equality: one amplifier sees the input signal pair, and the other sees a divided fraction of the output, producing a single-ended output voltage, as shown. This low-cost configuration (no laser-trimmed resistor pairs, etc.) is limited to small differential input signals (thus high gain), and generally exhibits relatively poor gain accuracy.[89] Interestingly, the fastest amplifier in Table 5.8, by far, uses this configuration.

5.16.11 Chopper and auto-zero instrumentation amplifiers

The same auto-zero techniques that are used in "zero-offset" CMOS op-amps (§5.11) are used in some CMOS low-voltage instrumentation amplifiers. These are recognizable by their very low offset-voltage specifications, down in the tens of microvolts, where non-AZ CMOSs do not venture (see Table 5.8 on page 363). These amplifiers attain excellent CMRR also, but they pay the price in broadband noise, switching-frequency noise,[90] and (sometimes) input bias and noise currents. The amplifiers are useful particularly in low-frequency applications, for example as input stages for integrating ADCs (see Figures 13.67), or combined with lowpass filtering.

5.16.12 Programmable gain instrumentation amplifiers

You set the voltage gain of a simple op-amp circuit with external resistors. In contrast, *difference* amplifiers (Table 5.7 on page 353) ordinarily are configured with fixed gain, set by a precision internal matched resistor network. And the gain of *instrumentation* amplifiers (Table 5.8 on page 363) is ordinarily set by a single gain-setting resistor R_g. Note, though, that some of the amplifiers in Table 5.8 include several internal R_g gain-setting resistors, allowing accurate gain selection with only an external jumper wire. Taking this one step further, you can get *programmable-*

[89] Some E types (e.g., the AD8130, a variant of the AD8129 in Table 5.8) are specified and characterized only for *G*=1. These are especially useful as differential line receivers, etc., but they are generally limited in swing, typically in the range of 3 to 4 Vpp (the AD8237, with its flying switched capacitors, is an exception). See also §12.10 for a discussion of differential signaling in the *digital* context.

[90] Which may not be evident from the datasheets, which sometimes omit spectral noise plots.

Table 5.9 Selected Programmable-gain Instrumentation Amplifiers (cont'd)

Part #	Gain choices	BW[d] –3dB (MHz)	Slew Rate (V/μs)	Settle[d] 0.01% (μs)	Swing to Supplies? IN +	IN –	OUT[v] +	OUT[v] –	mux?	filter?	out sense	Interface	Comments	Cost ($ US)
AD8250	1,2,5,10	3	25	0.65	-	-	-	-	-	-	-	pins	fast settle to 10ppm	7.78
AD8251	1,2,4,8	2.5	25	0.68	-	-	-	-	-	-	-	pins	fast settle to 10ppm	7.78
AD8253	1,10,100,1000	0.55	20 s	1.5	-	-	-	-	-	-	-	pins	fast settle to 10ppm	7.68
PGA204B	1,10,100,1000	0.01	0.7	100	-	-	-	-	-	-	●	pins	PGA205 for 1,2,4,8	17.09
PGA202B	1,10,100,1000	1	20	2	-	-	-	-	-	●	●	pins	PGA203 for 1,2,4,8	14.34
PGA207	1,2,5,10	0.6	25	3.5	-	-	-	-	-	-	●	pins	PGA206 for 1,2,4,8	18.36
PGA280	1/8–128, by x2	0.05	2	40	-	-	f	f	2	-	-	SPI	output supply 2.7–5V	6.46
low-voltage														
LTC6915	1,2,4...4096	slow	0.2	5ms	x	x	x	x	-	●	●	SPI,pins	chopper, high-gain	4.58
AD8231	1,2,4…128	0.07	1.1	slow	n	n	x	x	-	-	-	pins	auto-zero	3.08
LMP8358	10–1k, by 1-2-5	0.68	6.5	4	-	x	x	x	-	●	-	SPI,pins	auto-zero, prog filter	5.27
PGA309	4–128, 2.7–1152	0.4	0.02	slow	-	n	n	x	●	-	-	SPI	sensor interface[u]	7.05

(t) typical. (u) 148-page user manual; includes ADC for sensor span, offset tempco cal, etc.; parameters stored in external SOT23 EEPROM. (v) caution: the RR output parts often don't allow the REF pin to be close to V_{EE}; be sure to check the datasheet! (w) part with large gain error assume you'll perform gain calibration. (x) the CMRR spec is typically at 60Hz; check the plots if you care about performance at higher frequencies.

gain amplifiers (PGAs) in which that selection is made by a digital input code (either applied as a parallel logic-level code to a set of pins, or as a multibit serial code through a serial port like SPI or I^2C; see Chapters 14 and 15). These are, in essence, integrated versions of the discrete digitally programmable instrumentation amplifier of Figure 5.80.

Some examples of stand-alone instrumentation PGAs are the PGA204/5, LMP8358, and PGA280, with more listed in Table 5.9. The PGA202/3 (JFET) and PGA204/5 (BJT) are traditional "high-voltage" (to ± 18 V) parts that accept a 2-bit parallel code (on two pins) to select gains of 1, 10, 100, or 1000 (PGA202/4) or 1, 2, 4, or 8 (PGA203/5). The more recent LMP8358 is a low-voltage single-supply part (2.7–5.5 V) with auto-zero and gains from 10 to 1000 in a 1–2–5 sequence (i.e., 10, 20, 50, 100, 200, 500, 1000), and with ambidextrous programming – the three pins can be used either as a 3-bit gain-setting parallel port or a 3-wire SPI serial port that programs both the gain and some additional parameters such as input polarity reversal, fault detection, and bandwidth. It is fast (8 MHz) and accurate ($V_{OS} = 10\,\mu$V, max).

Finally, the very elegant PGA280 addresses the need for a PGA whose input signals can range over ± 10 V or more, but with a separately powered output stage matched to contemporary low-voltage single-supply ADCs and microcontrollers. The inputs can range over ± 15.5 V (with ± 18 V supplies), with the output supplied from the same +2.7 V to +5 V supply that powers the ADC or μC.[91] This solves the messy problem of protecting the inputs of a low-voltage part when it is driven from an IC running from higher voltages. There's a REF-like pin that sets the output mid-span voltage; the output is actually *differential*, with a pair of complementary outputs, but you can ignore one and treat it as single-ended (with a slight loss of accuracy).

This amplifier has excellent performance: it programs through a digital serial port, with selectable gains from 1/8 to 128 by factors of two. It combines low offset voltage (auto-zeroing: $\pm 15\,\mu$V max, at $G = 128$), high input impedance (>1 GΩ typ), low gain drift (± 3 ppm/°C max), and excellent CMRR (gain-dependent: 130–140 dB typ). Among its other tricks are an on-chip 2-input multiplexer (two differential-input pairs), an uncommitted byte-wide bidirectional digital port, and various signal conditioning and fault-detection options.

Figure 5.90 shows the sort of application for which this part would be ideal: our colleagues have developed an experimental robotic grasping hand[92] that is driven by a single torque motor through a set of passive linkages and couplings. For control you'd like to know the applied

[91] Or you can power the output from a split supply, staying within that total supply range.

[92] See, for example, Dollar and Howe, "The Highly Adaptive SDM Hand: Design and Performance Evaluation", *International Journal of Robotics Research* **29**, (5), 585–597 (2010), available at the web page of The Harvard BioRobotics Laboratory: `biorobotics.harvard.edu`.

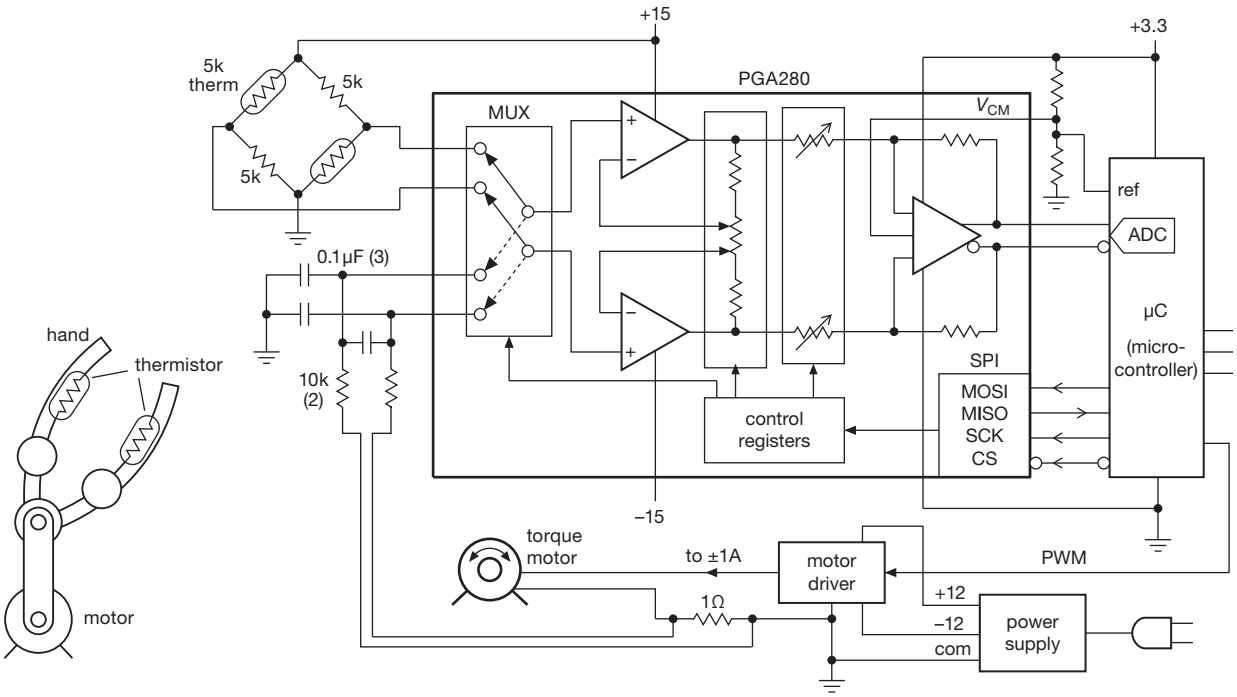


Figure 5.90. Two-channel level-shifting PGA application: reading out the applied torque and thermal response of a grasping robotic hand. This chopper-stabilized (250 kHz) instrumentation PGA works from ±15 V rails, but delivers its output to a 3.3 V ADC – *that's* being helpful! Recommended.

torque (from the motor current), and something about the contact pressure (if any) with the object being grasped. The PGA280's wide input compliance and range of gains, combined with 2-channel differential inputs, make this an easy task: we imagine a pair of thermistors, self-heated a few degrees above ambient, that sense the temperature at the grasping points, and a low-side sense resistor in the motor return line. The low-voltage output stage connects to the on-chip ADC of a microcontroller, which controls the channel switching and gain of the PGA. We've rigged the thermistors so that first contact makes an upward step in differential voltage, followed by a second step in the same direction when the other "finger" makes contact. (The step size provides further information about the object's material.) Note the differential "4-wire" connection at the motor-current-sense resistor, to eliminate errors from ground connection resistance; we've filtered that signal, because the use of pulse-width modulation tends to create high-frequency noise.

When looking around for a good instrumentation amplifier, be sure to consider specialized PGA variants that are intended to serve as front ends for low-level sensors and the like, for example the PGA309[93] or PGA2310.[94]

PGAs are quite popular as portions of more complex ICs, for example within ADCs (Figure 13.67 again, and Figures 13.70 and 13.71) and microcontrollers (Figure 15.10).

5.16.13 Generating a differential output

Both instrumentation amplifiers and difference amplifiers are used to convert a differential input signal to a single-ended output. That's fine, if that's what you want (and it usually is). There are some situations, though, in which you need a differential output signal, for example when

[93] Whose 148-page User's Guide describes it thus: "The PGA309 is a smart programmable analog signal conditioner designed for resistive bridge sensor applications. It is a complete signal conditioner with bridge excitation, initial span and offset adjustment, temperature adjustment of span and offset, internal/external temperature measurement capability, output over-scale and under-scale limiting, fault detection, and digital calibration."

[94] "The PGA2310 is a high-performance, stereo audio volume control designed for professional and high-end consumer audio systems."

driving several varieties of analog-to-digital converter (ADC, a major subject of Chapter 13). Most simply, this can be done by adding a unity-gain inverter to the single-ended output, as in Figure 5.91A.[95] This works, for sure, but the gain accuracy will be degraded unless the resistor pair is matched to at least the precision and stability of the driving amplifier. This pitfall is circumvented in circuit B, where the unity-gain inverter forces the SENSE output pin to a symmetrical voltage relative to ground (or some other reference voltage). With this circuit the gain accuracy is retained; the effect of any resistor mismatch is simply to offset the symmetry of the outputs about ground (or reference voltage), which is generally of less importance because of the differential-input nature of the device being driven.

Both circuits, however, share the drawback of introducing a time delay (or phase shift), owing to the finite bandwidth of the inverting stage. One solution is to use a pair of matched amplifiers, as in Figure 5.66F. But a better way, particularly when plenty of bandwidth and rapid settling is needed (as with fast ADCs), is to use a *differential amplifier* (or *fully differential amplifier*, to emphasize the distinction), a term that has come to mean an amplifier with differential inputs *and* outputs; see §5.17. The PGA280 of the previous section is such an amplifier (though its designers officially call it an instrumentation amplifier).

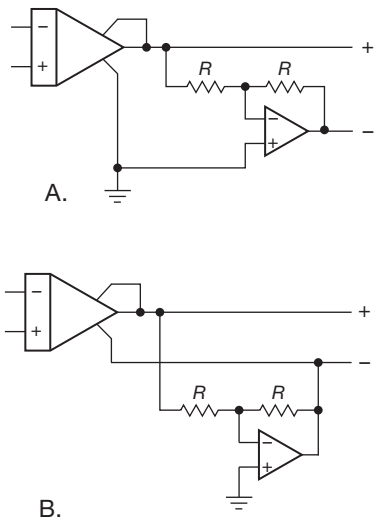


Figure 5.91. Generating a differential output from an instrumentation amplifier or difference amplifier. Method B maintains gain accuracy.

[95] See also the parts listed under "single-ended to differential" in Table 5.10 (page 375).

5.17 Fully differential amplifiers

The term "fully differential amplifier" (or sometimes "differential in/out amplifier," or just "differential amplifier") is used to describe an amplifier with differential input *and* differential output, along with an additional input pin ("V_{OCM}") that sets the common-mode voltage of the output pair. We favor the "fully" term, to distinguish clearly from difference amplifiers and instrumentation amplifiers, both of which have single-ended outputs.

For some important applications you need to create a balanced differential *output*, from either a differential or a single-ended input signal. This is often the case with ADCs that have complementary inputs; see Figure 13.65 (a "charge-redistribution" ADC), Figure 5.102 (pipeline flash), Figure 13.28 (flash), Figure 13.37 (SAR), and Figure 13.68 (delta–sigma). For that application the important performance parameters are likely to be settling time, gain accuracy and stability, the ability to set the common-mode output voltage, and the ability to drive a rail-to-rail swing into a low-voltage ADC.

Other applications where differential signals are widely used include analog signaling over twisted pairs (e.g., through existing Cat-5-style network cable); telecom applications such as ADSL and HDSL links; oscilloscope input stages; and RF communications subcircuits such as IF and baseband blocks.

You can, of course, create a differential-output signal pair by using single-ended amplifiers (op-amps, difference amplifiers, and instrumentation amplifiers), as illustrated in Figures 5.66F, 5.70, 5.91, and 13.37. But you do better, particularly in terms of speed and noise, with an integrated differential amplifier, which also lets you set the common-mode output voltage (i.e., the midpoint of the output swing); this capability is particularly useful when driving differential-input ADCs powered from a single supply, because of their fussy insistence on common-mode input voltages.

Table 5.10 on page 375 includes a good selection of currently available differential amplifiers, keyed to the circuit diagrams in Figures 5.94–5.98. These illustrate the unflagging creativity of the human species, and some nice circuit tricks as well.

Let's take a quick tour. Circuit A, intended for single-ended inputs, is simply a differential amplifier "kit," with an input op-amp whose gain you set with external resistors. You can wire it as a noninverting amplifier (therefore high input impedance), or you can configure it as an inverting amplifier (for example, to handle a large input swing by setting the gain to less than unity). The noninverting

input of A_2, conveniently of high impedance, lets you set the output common-mode output voltage. The LT6350 is a low-noise, low-distortion amplifier of this configuration, with the added feature of rail-to-rail outputs.[96] Figure 5.92 shows how you would use it to drive an ADC, in this case the LTC2393 that cooperates by providing a mid-scale dc reference output (V_{CM}).[97] The amplifier runs from the same +5 V and ground supply voltages, which eliminates the frequent worry about driving the ADC's input clamp diodes into conduction. The input lowpass filter to the ADC ($R_1R_2C_1$) serves two functions: (a) it is an anti-alias filter, limiting the input bandwidth to ∼150 kHz; and (b) it provides the recommended shunt input capacitance to suppress the effects of the ADC's internal switching transients (which afflict many ADCs, including "charge redistribution SAR" converters, and delta–sigma converters – see Chapter 13).

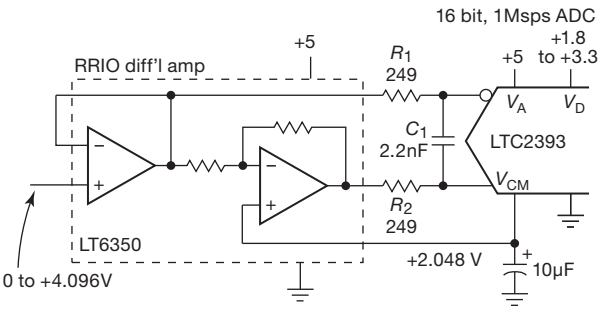


Figure 5.92. ADC driver circuit using a configuration A differential amplifier. The ADC provides a mid-span V_{CM} output reference, here used to set the amplifier's common-mode level. See also the better-performing AD4922.

Circuit B is a symmetrical balanced configuration optimized for pro-audio: well-balanced high-level drive (>15 Vrms) with low distortion into balanced pairs, and stability into the load capacitances you get with long cables (to 10,000 pF or more). Figure 5.93 shows a typical application, in this case generating the high-level low-distortion balanced output needed for pro-audio cable driving, first discussed in §5.14.2E. Of note here is the very high *common-mode* output impedance, which preserves signal balance by allowing the receive end to override the driver's default (symmetrical about ground at the driver end, set weakly by the 10k resistors). In fact, it's even OK to ground one side at the receive end, in order to generate a single-ended signal there. We'll have more to say about this in §5.17.1.

Circuit C is the configuration of Figure 5.66F, with high-impedance (buffered) common-mode voltage-setting inputs (V_{OCM}). The table entries with this configuration are low-noise wideband amplifiers, good for video and communications applications.

Circuit D is a popular configuration, sometimes with internal gain-setting resistors, sometimes external. The differential amplifier within the feedback loop consists of a symmetrical pair of transconductance amplifiers (voltage-to-current) generating a voltage across a resistive load, with voltage followers to generate the low-impedance output pair. The V_{OCM} input lets you assert the common-mode output voltage, which otherwise defaults to mid-supply (in which case it's good to attach a bypass capacitor). The V_{OCM} input bandwidth typically is comparable to that of the amplifier.

Circuit E continues the theme of differential transconductance amplifiers, but here they are configured in a feedback arrangement, the differential-output version of the analogous circuit E in Figure 5.89. Circuit F, used in the fastest amplifiers in the table, blends the output configuration of D with an input configuration like that of the classic instrumentation amplifier (Figure 5.65), again with differential transconductance amplifiers as the gain element.

Finally, circuit G is a completely different animal, a pair of offset-cancelled followers biased from the V_{OCM} input pin. This configuration, used in the table's 2 GHz entry, is intended for ac-coupled (or transformer-coupled) inputs.

5.17.1 Differential amplifiers: basic concepts

A. Gain The differential voltage gain is unity for most configurations in which the gain is set by matched pairs of feedback resistors $R_f = R_g$. The gain range column in Table 5.10 (page 375) identifies parts with fixed gains, minimum gains, or a set of gain selections. In some cases the exact gain is affected by source-impedance and termination-matching issues; see §5.17.4 and the formulas in Figure 5.104.

[96] The ADA4922-1 is a faster, 0.05% version, with a fixed unity-gain input stage.

[97] The ADC has an internal reference of reasonable accuracy (0.5%), but it lets you attach an external reference of better performance (e.g., the LT1790–4.096, with accuracy and drift of ±0.05% and 10 ppm/°C, max). Sounds good – but it would not accomplish much here, because the gain accuracy of the system is limited by that of the LT6350 amplifier (±0.6%, max). You could use the ADA4922-1 instead.

Table 5.10 Selected Differential Amplifiers

Part #	Circuit	Gain[a] Range	Gain[a] Set by	ΔG (%)	Bandwidth –3dB (MHz)	Bandwidth 0.1dB (MHz)	Speed Slew Rate (V/µs)	Speed Settle 0.05% (ns)	V_{out} diff'l max (Vpp[b])	Offset Voltage typ (mV)	Offset Voltage max (mV)	Input bias (µA)	Z_{in}[h] diff (Ω)	Noise e_n, typ @1MHz (nV/√Hz)	Supply Range (V)	I_s typ (mA)	RRO	Cost qty 25 ($US)	Comments
high voltage																			
PGA280	q	1/8–128	PGA	0.15	1.5	-	1	40µs	10	0.05	0.25	0.3nA	1G	22	10-40	0.8	•	6.46	INA, SPI, low V_{out}
THAT1606	D3	2.0	fixed	2	10	-	15	-	67	4	15	-	5k	25	8-40	4.9	-	3.91	audio[t], 7ppm dist
AD8270	C	0.5,1,2	fixed	0.08	15	-	30	700	55	0.2	0.75	0.5	20k	26[d]	5-36	2.3	-	3.56	precision resistors
LME49724	D	1.0 up	*Rf*/*Rg*	-	50	-	18	200	52	0.2	1	0.06	*Rg*[h]	2.1	5-36	10	-	3.53	audio, 0.3ppm dist
OPA1632	D	1–10	*Rf*/*Rg*	-	180	40	50	200	52	0.5	3	2	*Rg*[h]	1.3	5-33	14	-	4.39	pro audio
low voltage																			
LMP7312	D3	0.1-2	PGA	0.04	0.53	-	1.4	-	RR	-	0.1	c	160k	7.5	±5	2	y	6.43	2-ch-input, SPI
LTC1992	D	1–10	*Rf*/*Rg*	-	3.2	-	1.5	-	RR	0.25	2.5	2pA	30k	45[d]	2.7-12	0.7	w	2.80	R_f = 30k to 50k
LTC1992-x	D3	1–10	p/n	0.3	3.2	-	1.5	-	RR	0.25	2.5	2pA	30k	45[d]	2.7-12	0.7	w	5.23	-x for *G*=1,2,5,10
AD8137	D	1.0 up	*Rf*/*Rg*	-	76	10	450	100	RR	0.7	2.6	0.5	*Rg*[h]	8[d]	2.7-12	2.6	•	1.98	R_f=1k-10k, cheap
THS4521	D	1–10	*Rf*/*Rg*	-	145	20	490	13[k]	RR	0.2	2	0.65	*Rg*[h]	4.6[d]	2.5-5.5	1.1	w	2.81	R_f=1k; '4522 dual
EL5170	E3	2.0	2.0	1	100	12	1100	20	6.0	6	25	6	300k	28[d]	5-12	7.4	-	1.44	cheap, '5370 triple
LTC6605-14	D3	1, 4, 5	1,4,5	0.3	14[e]	f	f	f	RR	0.25	1	12	400/*G*	2.6	2.7-5.5	16	w	9.81	dual, 16 bit ready
LT6600-x	D2	1–8	402/*Rg*	7	10[g]	f	f	f	4.8	5	25	30	*Rg*[h]	10[d]	2.7-11	36	-	3.90	4th-order filters
LTC6601	D3	1–7	1 to 7	3	6-27[i]	f	f	f	RR	0.25	1.5	12	400/*G*	2.2	2.7-5.5	16	w	5.58	pin-select filter
AD8138	D	1.0 up	*Rf*/*Rg*	-	320	30	1150	16	15.6	1	2.5	3.5	*Rg*[h]	5	3-10	20	-	4.12	R_f=500Ω
LMH6551	D	1.0 up	*Rf*/*Rg*	-	370	50	2400	18	15.6	0.5	4	4	*Rg*[h]	6[d]	3.3-10	13	-	3.50	R_f=365Ω
EL5173	E3	2.0	2.0	0.5	450	60	900	10	6.0	3	30	11	150k	25[d]	4.8-12	12	-	2.80	EL5373 triple
EL5177	E	1.0 up	*Rf*,*Rg*	1.5	550	120	1100	10	6.0	1.4	25	14	150k	21	4.8-12	12	-	1.85	G=1+R_f/R_g, cheap
AD8139	D	1.0 up	*Rf*/*Rg*	-	410	45	800	45	RR	0.15	0.5	2.3	*Rg*[h]	2.2	4.5-12	25	-	6.00	R_f=200Ω
AD8132	D	1.0 up	*Rf*/*Rg*	-	350	90	1200	15	14.4	1	3.5	3	*Rg*[h]	8[d]	2.7-11	12	-	3.07	R_f=348Ω
LT6402-20	C	10	fixed	10	300	30	400	8[k]	7.0	1	6.5	c	100	1.9	4.0-5.5	30	-	3.90	opt 75MHz filter
LTC6404-4	D	4 up	*Rf*/*Rg*	-	600	450	1200	13	RR	0.5	2	23	*Rg*[h]	1.5	2.7-5.5	30	w	4.91	R_f=402; ver -1,-2[j]
THS4520	D	1–10	*Rf*/*Rg*	-	620	30	570	7	RR	0.25	2.5	6.5	*Rg*[h]	2	3-6	14	•	4.07	R_f=499Ω
PGA870	s	0.3-10	PGA	4	650	100	2900	5	4.8	5[z]	30[z]	c	150	6	4.8-6	143	-	9.70	0.5dB atten, pins
ADA4932-1	D	1.0 up	*Rf*/*Rg*	-	560	300	2800	9	15.0	0.5	2.2	2.5	*Rg*[h]	3.6[d]	3-10	10	-	4.72	R_f=499Ω; -2=dual
LT1993-10	C	10	fixed	12	700	50	1100	4	7	1	6.5	c	100	1.7	5	100	-	4.20	-2,-4 for G=2,4
ADA4950-1	D	1, 2, 3	pins	1.7	750	210	2900	9	15	0.2	2.5	c	500/*G*	5.5[d]	3-11	9.5	-	4.78	-2=dual
LMH6553	D	1.0 up	*Rf*/*Rg*	-	900	50	2300	10	15.4	-	-	50	*Rg*[h]	1.2	5-12	29	-	6.43	CFB, 274Ω; clamp
ADA4938-1	D	1–10	*Rf*/*Rg*	-	1000	150	4700	6.5	15.2	1	4	13	*Rg*[h]	2.6	4.5-11	37	-	6.06	R_f=200Ω
LMH6552	D	1.0 up	*Rf*/*Rg*	-	1500	450	3800	10	15.4	-	-	80	*Rg*[h]	1.1	5-12	23	-	8.14	CFB, 357Ω
THS4513	D	1.0 up	*Rf*/*Rg*	-	1600	700	5100	16	5.6	1	4	8	*Rg*[h]	2.2	3-6	35	-	10.34	R_f=348Ω
LMH6555	D3	4.84	4.84	3	1200	180	1300	2.2[k]	1.0	15	50	c	78	4[d]	3.3	120	-	10.55	assumes R_S=50Ω
THS4511	D	1–10	*Rf*/*Rg*	-	1600	620[r]	4900	3.3[k]	5.2	1	4	8	*Rg*[h]	2	3.5-5.5	39	v	9.13	R_f=349Ω
ADA4937-1	D	1–5	*Rf*/*Rg*	-	1900	200	6000	7	6	0.5	2.5	21	*Rg*[h]	2.2[d]	3-10	40	-	6.06	R_f=200Ω; -2=dual
THS4508	D	2–10	*Rf*/*Rg*	-	2000	400	6600	2[k]	5.2	1	4	8	*Rg*[h]	2.3	3.8-5.5	39	v	9.13	R_f=349Ω
THS4509	D	2–10	*Rf*/*Rg*	-	2000	300	6600	10	5.6	1	4	8	*Rg*[h]	1.9	3-6	38	-	9.13	R_f=349Ω
LTC6416	G	0.98	1.0	2	2000	300	3400	1.8[k]	4.2	0.5	5	5	12k	1.8	2.7-3.9	42	-	8.27	emitter follower
THS770006	D	2	fixed	3	2400	350	3100	2.2	4.9	0.5	5	c	100	1.7	5	100	-	10.45	Z_{out}=27Ω at 1GHz
AD8352	F	1–8	Rg	12	2500	190	8000	2[k]	6	-	6	5	3k	2.7	3.5-5	37	-	5.65	specs for *G*=10dB
ADL5561	D3	2,4,6	pins	0.5	2900	600	9800	2[k]	4.3	0.25	-	c	800/*G*	1.7	3-3.6	40	-	5.89	good for IF-strip
ADL5562	D3	2,4,6	pins	0.7	3300	270	9800	2[k]	4.3	0.25	-	c	800/*G*	1.6	3-3.6	80	-	5.88	good for IF-strip
ADA4960-1	F	1–10	*Rf*/*Rg*	6	5000	300	8700	1.4	3.5	-	36	20	10k	5.4	5	60	-	11.47	specs for *G*=6dB
single-ended to differential[m]																			
DRV134	B	2.0	fixed	2	1.5	-	15	2500	60	1	10	-	10k	35	9-40	5.2	-	4.21	audio[t], 5ppm dist
ADA4922-1	A2	2.0	fixed	0.05	38	-	260	580	43	0.18	0.55	3.5	11M	6	9-26	5.4	n	5.81	18-bit ADC driver
low voltage																			
LT6350	A	2.0 up	opamp	0.6	33	-	48	240	RRIO	0.1	0.7	1.2	4M	4.1[d]	2.7-12	4.8	•	3.42	precision
ADA4941-1	A	2.0 up	opamp	1	30	-	22	300	RR	0.2	0.8	3	24M	5.1[d]	2.7-12	2.5	w	4.45	odd f_{3dB} specs

Notes: (a) fixed gain, programmable gain (PGA), or gain set by input resistor *Rg*. (b) RR output means V_{out} diff = 2 x V_{supply} max. (c) included in V_{os} spec. (d) includes feedback resistor noise. (e) matched 2nd-order antialias filters; 7, 10MHz avail. (f) set by filter. (g) 6600-x specifies 2.5 to 20MHz 4th-order filters. (h) nominally Z_{in} = Rg, where $G = R_f/R_g$; but for "D" circuit types it's greater than *Rg*, see later section on Differential Amplifier Input Impedance. (i) filter, 6 to 27MHz strappable. (j) -4 version compensated for *G*<4. (k) settle to 1%. (m) most differential-to-differential amplifiers can convert single-ended input to differential output. (n) near. (o) instrumentation amplifier figures. (q) inst-amp input, differential-amp output. (r) *G*=2. (s) *R-2R* ladder input, differential-amp output. (t) high common-mode Z_{out}, like an isolation transformer. (v) inputs to $-V_{EE}$. (w) RRO and inputs to $-V_{EE}$ or within 0.2V of $-V_{EE}$. (y) RRO and input beyond rails, to ±15V. (z) RTO.

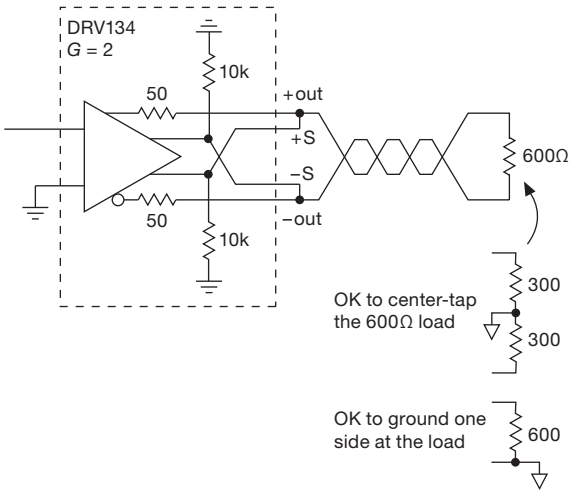
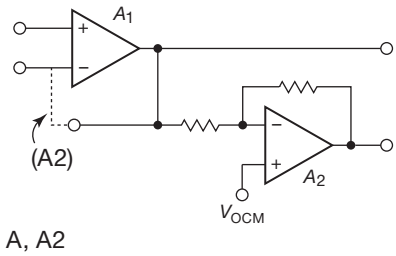
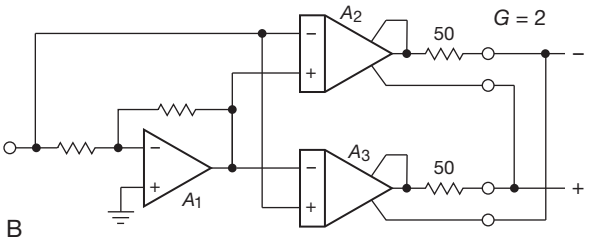


Figure 5.93. Balanced audio driver with a high common-mode output impedance, so that the receiver (load end) sets the common-mode voltage. $V_{CM(out)}$ defaults to 0 V if the load is left floating.



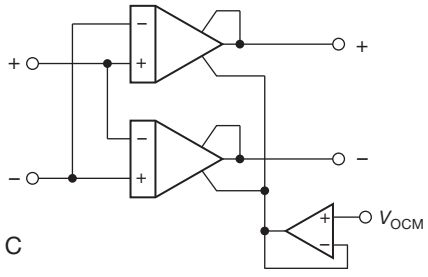
A, A2



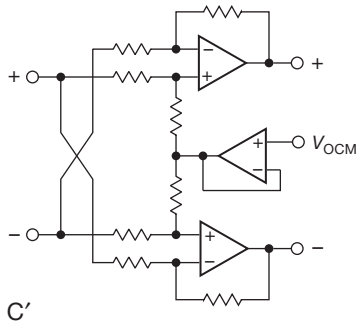
B

Figure 5.94. Differential-output amplifier configurations A and B, as listed in Table 5.10 on page 375. The uncommitted input op-amp can be configured as a noninverting amplifier (or follower), or an inverting amplifier.

B. Input impedance The input impedance of amplifiers with configuration D is equal to R_g, making them unsuitable for high gains because the input impedance becomes unmanageably low: the signal source is heavily loaded, the effective R_g is increased by the source impedance R_S, and the CMRR is degraded by source-impedance imbalance. The exact value of R_g (and therefore the input impedance) will be affected by source termination and loading considerations; see §5.17.4 and the formulas in Figure 5.104.



C



C′

Figure 5.95. Fully differential amplifier configuration C, as listed in Table 5.10 on page 375. The symmetry is evident in the redrawn version, C′, where the gain is $G = 2R_f/R_g$.

C. Single-ended input Most fully differential amplifiers work fine with single-ended inputs, i.e., with the "−" input grounded. But you may wish to use $G = 2$ or higher to achieve full peak-to-peak drive into a differential ADC.[98]

D. Common-mode rejection With differential pairs at both input and output, there are *two* measures of common-mode rejection: *differential* V_{out} versus common-mode V_{in}, which is usually quite good (e.g., 80 dB up to 1 MHz); and *common-mode* V_{out} versus common-mode V_{in}, which can be significantly poorer (e.g., 50 dB up to 1 MHz, degrading above that). But the latter is not terribly worrisome if the receiving device (e.g., an ADC) has good common-mode rejection itself. Resistor matching is important for configurations

[98] This configuration is also available with a voltage gain less than unity. In a burst of cuteness, Analog Devices calls their AD8475 (with gains of 0.4 and 0.8) a *funnel amplifier* (get it?). With it you can reduce a 20 Vpp signal to a differential pair of 4 Vpp signals, for input to a low-voltage ADC.

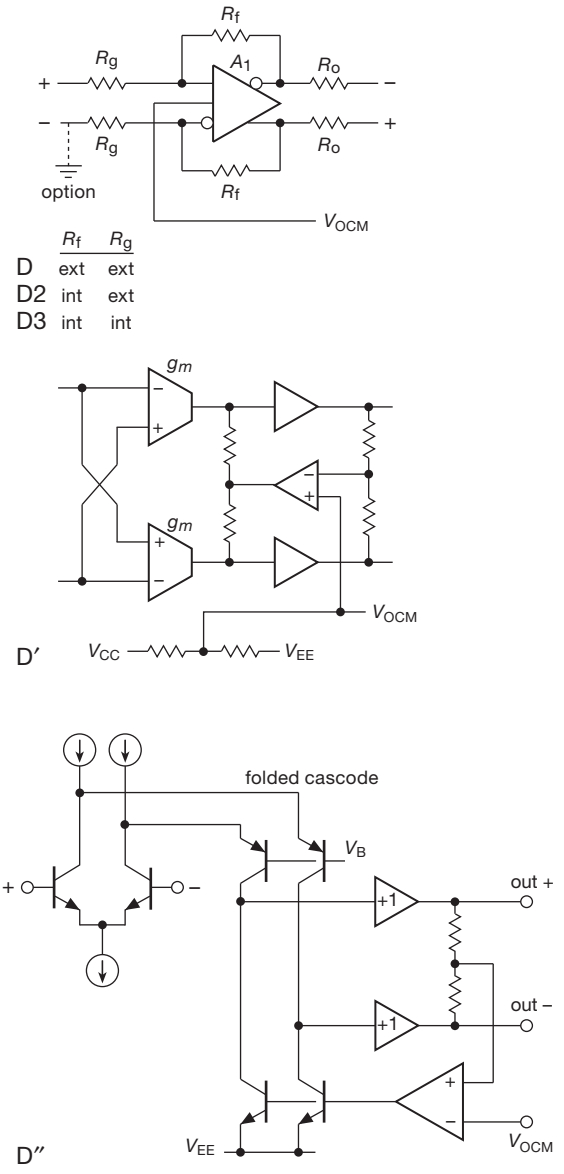


Figure 5.96. Fully differential amplifier configuration D, as listed in Table 5.10 on page 375; the gain is $G{=}R_f/R_g$. A typical configuration for output amplifier A_1 is shown in D′, and TI's version for their THS45xx series is D″; the THS4508/11/21, for example, use polarity complements (*pnp* input pair, etc.), permitting operation down to $V_{in}{=}V_{EE}{-}0.2$ V.

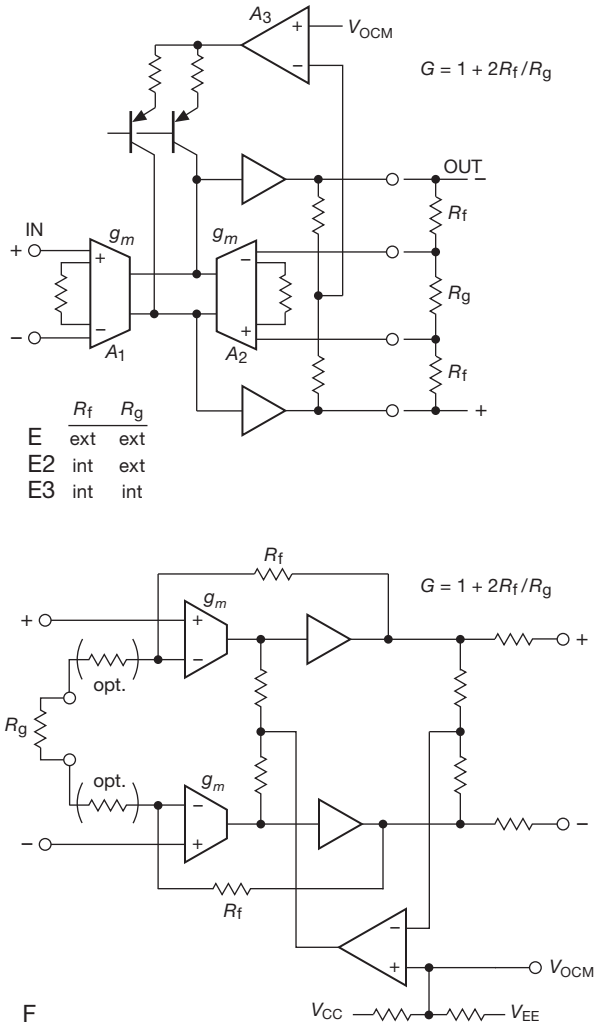


Figure 5.97. Fully differential amplifier configurations E and F, as listed in Table 5.10 on page 375.

with external R_f and R_g gain-setting resistors (configuration D); see §5.17.5 for more details.

E. Single-ended output The datasheets of some differential amplifiers describe operation with a single-ended output.[99] When it's operated in single-ended output mode, however, you care about the *output* offset voltage ΔV_{OCM} (i.e., the output error with respect to the V_{OCM} reference), which translates back to an input-referred error of $\Delta V_{OCM}/G$. For the LMP7312 the output offset is ± 20 mV, far larger than the maximum input offset of ± 0.1 mV. This is a low-gain amplifier ($G{=}0.1$ to 2), so this output offset looks like a corresponding worst-case

[99] For example, the datasheet for the "Precision SPI-Programmable AFE with Differential/Single-Ended Input/Output" LMP7312 says that "the output can be configured in both single-ended and differential modes with the output common mode voltage set by the user."

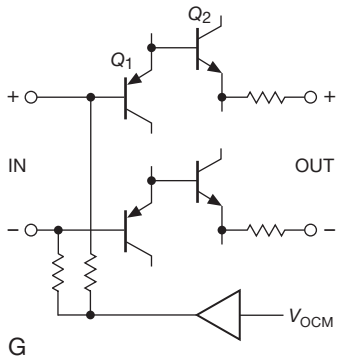


Figure 5.98. Fully differential amplifier configuration G, as listed in Table 5.10 on page 375. This type is intended for ac- or transformer-coupled inputs.

input error of ±200 mV to ±10 mV! This is hardly what one would call "precision."

F. V_{OCM} bias pin You set the output common-mode voltage by asserting a dc bias at this pin. Some devices buffer this input to get high R_{in}, but many parts present you with an input impedance in the tens of kilohms. Generally the operating V_{OCM} range does not extend to the negative rail. If this pin is left unconnected, most parts default to mid-supply. It's always a good idea to bypass this pin, because the fast signals associated with these wide-bandwidth amplifiers couple to the V_{OCM} node.

G. Input common-mode range The input common-mode voltage range of most fully differential amplifiers does *not* extend to the negative rail, which can seriously constrain a circuit that's running from a single positive supply. This isn't as bad as it sounds, though: the outputs, sitting up there around a positive common-mode output voltage (set by the dc you apply to the V_{OCM} input pin), bring the input terminals up by the voltage dividers formed by R_f and R_g. This effect is largest when operating at low gains; when operating at higher gains it's best to check that the input common-mode range is not violated. Assuming that there's plenty of loop gain (i.e., that $G_{OL} \gg G$) the (equal) voltage at the amplifier's inverting and noninverting pins is

$$V_{(+,-)} = \frac{V_{OCM} + GV_{in(CM)}}{G+1},$$

where the differential gain $G{=}R_f/R_g$, and $V_{in(CM)}$ is the common-mode voltage of the (differential) input signal source. If the input is single ended (with the other differential input grounded) then (substituting $V_{in}/2$ for $V_{in(CM)}$) you get

$$V_{(+,-)} = \frac{V_{OCM} + GV_{in}/2}{G+1}.$$

Note that with a balanced input signal source (i.e., fixed $V_{in(CM)}$), the voltages at the amplifier's (+) and (−) pins don't vary as the input differential signal does its thing. This is in contrast with the single-ended arrangement, in which the input signal amplitude causes a variation in input common-mode voltage. For the latter be sure to check for input common-mode violations at the extremes of the input signal.

Of course, you can do an end run around the problem by choosing an amplifier whose input common-mode range includes the negative rail, for example the THS4521 illustrated in §5.17.3.

H. Voltage feedback versus current feedback All of the "R_f/R_g" amplifiers in Table 5.10 (page 375) use conventional voltage-feedback amplifiers, with the exception of the LMH6552/3, which uses current feedback (CFB). Because they are VFB amplifiers, they work well with bandwidth-limiting capacitors across the feedback resistors when operated at higher gains (helpful for taming the integrated voltage noise $v_n{=}e_n\sqrt{\text{GBW}}$, which can be excessive owing to the large bandwidth of many of these amplifiers). To a decent approximation, current-feedback amplifiers have an f_{3dB} bandwidth independent of gain, whereas voltage-feedback amplifiers have a bandwidth inversely proportional to the closed-loop gain ($f_{3dB} = \text{GBW}/G_{CL}$). See the discussion in Chapter *4x*.

I. Gain-setting resistors Large R_f and R_g values can cause problems that are due to parasitic circuit-board capacitance. For example, R_f above 1k for the modest-capability 145 MHz THS4521 creates peaking (Figure 5.99). Dual and quad package options may also suffer from unavoidable peaking problems caused by leadframe issues, so it's generally better to choose fixed-gain types in multiple-amplifier packages. Large values of R_f and R_g also create (a) loss of speed, (b) increased input offset error from the relatively large bias currents characteristic of fast bipolar amplifiers, and (c) increased input-referred voltage noise, both from resistor Johnson noise and from the noise voltage developed across R_f by the amplifier's input noise current.

To put some flesh on these latter bones, consider the THS4521 again, with our (somewhat extreme) strawman $R_f = R_g = 100$k. From Figure 5.99 you can see the ×10 reduced bandwidth and peaking. This peaking (which occurs with VFB amplifiers but not CFB types)

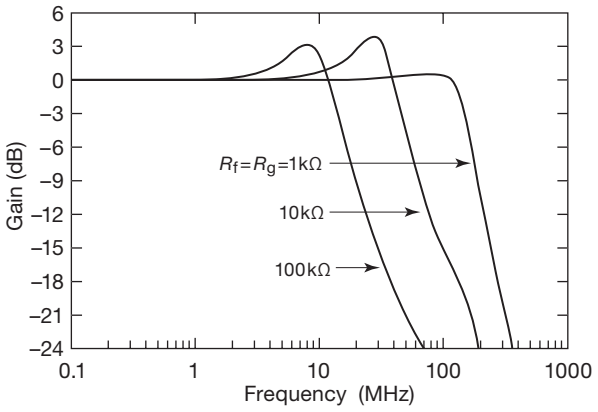


Figure 5.99. Large gain-setting resistor values create peaking in the frequency response, as shown in this datasheet plot for the THS4521 configured for unity gain ($R_f = R_g$).

can be tamed by putting a small capacitor across each feedback resistor, but you'll lose a bit more bandwidth in the process.[100] Turning this vice into a virtue, we note that you may wish to add feedback capacitors anyway, to reduce the bandwidth intentionally.

As for offset voltage, this part has an input bias current of 650 nA (typ), which would create a drop of 65 mV across 100k. But the bias currents are reasonably well matched, with an offset current spec of $\Delta I_B = \pm 50$ nA (typ), thus creating an input offset of 5 mV. That's much better, but it does seriously degrade the amplifier's typical $V_{OS} = \pm 0.2$ mV (± 2 mV max). You need to keep R_f and R_g less than 10k to preserve the amplifier's accuracy.

Finally, noise. There are two contributions, the resistors' Johnson noise ($e_n = \sqrt{4kTR} = 0.13\sqrt{R}$ nV/$\sqrt{\text{Hz}}$), and the noise voltage developed by the amplifier's current noise ($e_n = i_n R_f$). For $R_f = 100$k the Johnson noise voltage is 40 nV/$\sqrt{\text{Hz}}$, and the noise produced by the amplifier's $i_n = 0.6$ pA/$\sqrt{\text{Hz}}$ is 65 nV/$\sqrt{\text{Hz}}$. These disastrously degrade the amplifier's typical 4.6 nV/$\sqrt{\text{Hz}}$ (taking the usual root-sum-of-squares, the total added noise voltage is 76 nV/$\sqrt{\text{Hz}}$). The table below summarizes these figures, and also those corresponding to $R_f R_g = 10$k and $R_f = R_g = 1$k.[101]

Bottom line: compared with the nominal 1k value, the use of 100k gain-setting resistors reduces your bandwidth ×10, increases the typical offset voltage ×25, and increases the typical input-referred noise voltage ×16. You wouldn't want to do this. But you could reasonably use something like 2.49k, 4.99k, or perhaps 10k, buying increased input impedance at the expense of modest degradation of bandwidth, noise, and accuracy.

	−3 dB Bandwidth	Offset voltage*	Input-referred noise		
R_f,R_g	(MHz)	(mV, typ)	$\sqrt{4kTR}$ (nV/$\sqrt{\text{Hz}}$)	$i_n R_f$ (nV/$\sqrt{\text{Hz}}$)	Total* (nV/$\sqrt{\text{Hz}}$)
1k	150	±0.2	4	0.7	4.6
10k	45	±0.5	13	6.5	15
100k	15	±5	40	65	76

* Includes V_{OS} and e_n of the amplifier.

J. Common-mode output impedance The voltage asserted onto the V_{OCM} pin sets the common-mode output voltage. Put another way, differential amplifiers have a low common-mode output impedance. That's usually what you want; after all, that's the reason there *is* a V_{OCM} pin. But this can create difficulties if the output is sent to a remote load that needs to establish its preferred common-mode level. That is the case in balanced audio (or video), sent substantial distances over balanced twisted-pair cable.

Take a look at Figure 5.100B. By driving the V_{OCM} pin with the average output voltage, you create an amplifier that cooperates by letting the load lead the dance. In fact, the load can even unbalance the signal intentionally (by grounding one side), in which case the other output swings symmetrically around ground with the full desired differential output voltage.[102] There are a few differential amplifiers that are designed specifically for this kind of application, with an internal configuration that creates a high common-mode output impedance. We saw one example (the DRV134, similar to the SSM2142) in Figure 5.93. Another excellent one is the THAT1606, from the curiously named THAT Corporation.[103]

The traditional solution has been to use an isolating transformer, which also can do the job of converting between single-ended and balanced signals (this is known as a "balun," for *bal*anced–*un*balanced) as shown. But

[100] You can look at this another way: the manufacturer's recommended gain-setting resistor values are chosen such that a small amount of peaking is exploited to extend the amplifier's natural bandwidth.

[101] Many parts (the D and E configurations) let you add feedback capacitors to reduce the bandwidth. With some parts this may introduce instability at low gains, but with others it may improve the stability, especially when larger resistor values are used.

[102] Omit the small bypass capacitor shown, if this mode of operation is anticipated.

[103] A member of the hard-to-Google corporate name club, which includes *AND Displays*, and *ON Semiconductor*. (Try it: you'll get more than ten billion Google hits for "AND" or "ON.")

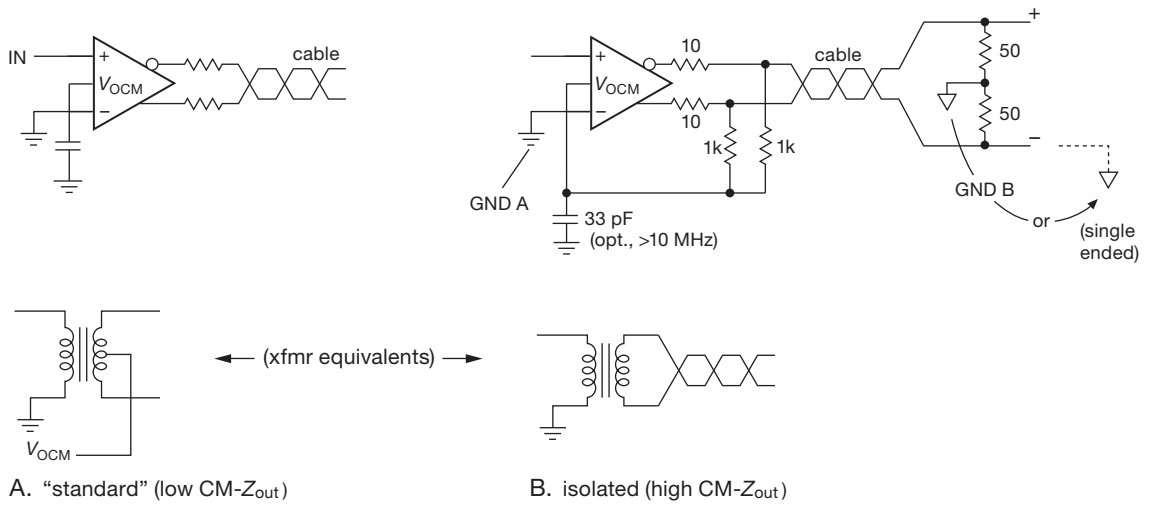


Figure 5.100. The common-mode output voltage of a differential amplifier is set by the V_{OCM} pin (usually defaulting to mid-supply if undriven), producing a low common-mode output impedance, as in A. But for balanced audio or video applications you want the (distant) *load* to be the boss. You can trick a differential amplifier, as in B, simulating the very high CM-Z_{out} of an isolating transformer or balun.

transformers are bulky, limited in bandwidth and linearity, and not inexpensive. High CM-Z_{out} differential amplifiers can be an attractive alternative.

5.17.2 Differential amplifier application example: wideband analog link

We conclude the discussion of differential amplifiers with several application examples: a wideband analog link over twisted-pair cable, and a riff on driving differential-input ADCs.

In §5.14.2F we illustrated the use of a difference amplifier as the receiving end of an analog link over differential twisted-pair cable. In that circuit R_4C_1 creates a rising response at high frequencies ("equalization"), to compensate for the increasing cable attenuation. To complete the link, of course, we need a differential driver.

Figure 5.101 shows the whole enchilada, here implemented with Intersil's EL5170/72 differential line driver/receiver pair. They make triple units (EL5370/72) also, with application to color video. Bandwidths to tens of megahertz are easily achievable over tens of meters of Cat-5 cable, with modest equalization at the receiver end. Coaxial cable is considerably better, and two 50 Ω coax lines can replace the 100 Ω differential pair. As always, the balanced signal combines with excellent receiver CMRR (here 95 dB typ) to provide high rejection of interference.

5.17.3 Differential-input ADCs

Many analog-to-digital converters require differential signal inputs. This is almost universally true of high-speed converters (e.g., pipelined flash converters), as well as the varieties known as "charge-redistribution SAR" and "delta-sigma" (ADCs, in all their glory, are the major subject of Chapter 13). And in many cases the input is hardly benign – the internal switched capacitors introduce charge transients at the input terminals, mandating some external shunt capacitance. An additional annoyance is the requirement that the driver must be able to swing the inputs over the full conversion range (which may include ground), but without driving them beyond the ADC's power rails (with the risk of damage from input clamp conduction and possible SCR latchup).

A. First iteration: single-supply ADC driver with V_{OCM} offset

Figure 5.102 shows two iterations of an input stage for fast single-supply ADCs with differential inputs. Our first design was based around the AD9225 12-bit 25 Msps pipelined flash converter, which runs from a single +5 V analog supply, and has a separate digital supply pin for interfacing to microcontrollers running from +3 V to +5 V. Its input span is programmable, either 0–2 V or 0–4 V, and it provides a mid-span dc output (+1 V or +2 V) that can be used to set the differential amplifier's common-mode output (via the V_{OCM} pin).

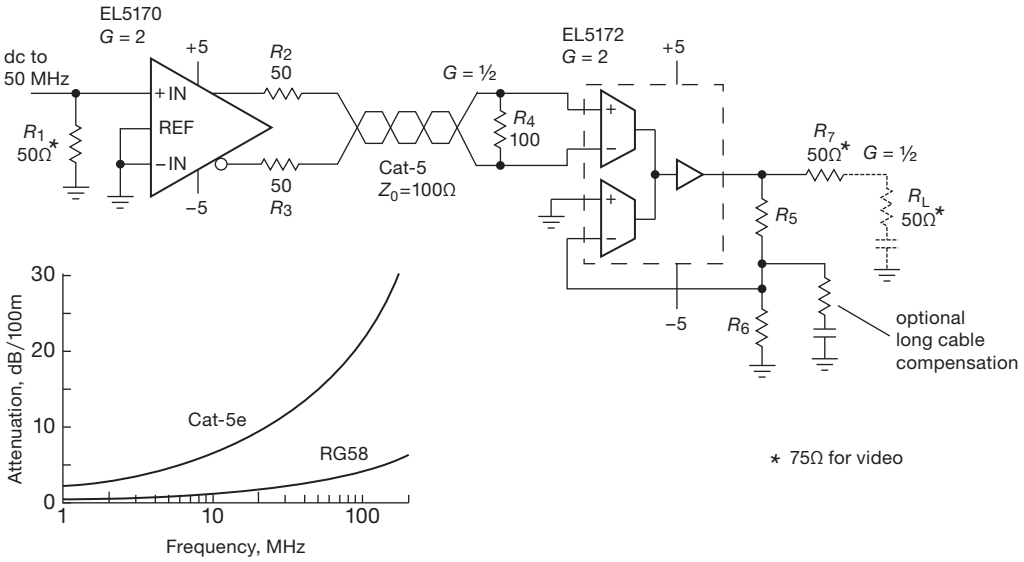


Figure 5.101. Wideband analog link over Cat-5 network cable. The EL5370/72 package three similarly performing drivers–receivers in a single IC, convenient for sending analog video (RGB, S-Video, or YPbPr component video) over a single cable (which has four twisted pairs). See also Figure 5.71.

By running the differential amplifier from the same +5 V and ground, we are assured that the ADC's inputs cannot be driven beyond the rails. We chose the AD8139 differential amplifier for its low noise ($2.2\,\text{nV}/\sqrt{\text{Hz}}$), ample bandwidth ($\sim$15 MHz at $G = 20$), and ability to swing its output rail-to-rail (to drive the ADC over its full-input span). We used the recommended pair of series resistors to suppress amplifier ringing from charge transients at the ADC input; and we added a shunt capacitor to both reduce these transients and also provide a second stage of antialias filtering.[104]

That's all good news. The bad news is that this amplifier, in common with most differential amplifiers, does not include ground in its input common-mode operating range: you've got to stay a volt away from either rail. So you cannot simply tie one input to ground and drive the other with a small signal around ground.[105] The amplifier *does* let you run from split supplies (e.g., $\pm$5 V), which solves the input signal-level problem; but then you have to worry about power-supply sequencing and the risk of driving negative current into the ADC's clamp diodes.

B. Second iteration: single-supply ADC driver with $V_{\text{in(CM)}}$ to ground

What to do? Find a single-supply amplifier that operates with inputs to the negative rail! That's what we did in the second circuit, whose THS4521's common-mode input range includes ground ("NRI" – negative rail input), and in fact guarantees proper operation with inputs to -0.1 V.[106] It also has the needed rail-to-rail output, but it is somewhat noisier and slower than the AD8139 ($4.6\,\text{nV}/\sqrt{\text{Hz}}$, and a gain of only $\times$5, to maintain 18 MHz bandwidth).

We teamed it with the ADC14L040, a more accurate and faster ADC (14-bit, 40 Msps) that runs on a single +3.3 V supply and uses less power (235 mW versus 335 mW). The ADC's span is $\pm$0.5 V, centered on an allowable midspan voltage of +0.5 V to +2.0 V. We could have used the ADC's reference-derived +1.5 V output to drive the amplifier's V_{OCM} pin, as before; but when that pin is not driven the amplifier defaults to mid-supply

[104] Repeating some important advice: when designing with high-speed ICs, it's particularly important to pay attention to special instructions in the datasheet; the AD9255, for example, devotes nearly a full page to a discussion of input R's and C's.

[105] Except when operating at low gain: here that would require $G \leq 1$, so that the AD8139's input terminals are brought up to 1 V or more by the $R_f R_g$ divider. See the discussion in §5.17.1.

[106] Other differential amplifiers whose feedback inputs can operate at or near ground are indicated with **w** or **v** in Table 5.10 (page 375), and include the LTC1992, LTC6605, LTC6601, LTC6404, THS4508, and THS4511. These parts span the bandwidth range from 3 MHz to 2000 MHz.

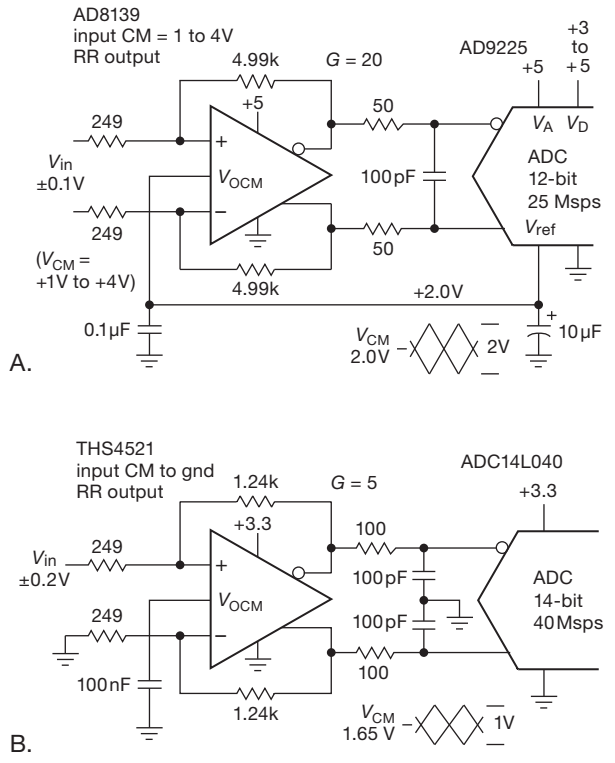


Figure 5.102. The care and feeding of single-supply differential-input ADCs. A. The AD9225 provides a midscale V_{ref} output to set the amplifier's common-mode output; but the AD8139's inputs cannot go closer than 1 V from either rail. B. The THS4521 is unusual in allowing input voltages to the negative rail (here ground).

(+1.65 V), which is fine. As before we added the recommended decoupling filter.

Given the ADC's higher resolution, it's worth checking to see how the noise voltage contributed by the amplifier and resistors compares with the converter's step size. Taking into account the input gain, the step size is $400\,\text{mV}/2^{14}$, or $25\,\mu\text{V}$. The amplifier's noise density ($4.6\,\text{nV}/\sqrt{\text{Hz}}$) combined with the (uncorrelated) resistor noise ($2.7\,\text{nV}/\sqrt{\text{Hz}}$) is about $5.3\,\text{nV}/\sqrt{\text{Hz}}$, or about $18\,\mu\text{V}$rms in the $\sim$12 MHz effective bandwidth of the amplifier and *RC* filter. In other words, the noise voltage is comparable to the converter's LSB step size. This is OK, though it would be nice to have it somewhat lower.[107] Perhaps a way to think about it is that the circuit's virtues of speed and resolution have made the relatively small noise contribution look bad. You can always throw away bandwidth (if you don't need it), or just look at the top 12 bits, if that makes you feel better.

C. Third Iteration: transformer coupling

If you don't need dc coupling, an easy way to drive differential-input converters is with a wideband transformer. They're widely used in radiofrequency applications, and you can get them in small surface-mount packages. Figure 5.103 shows how to do it. Use the ADC's mid-span reference output (suitably bypassed) to set the common-mode voltage, and use a resistive termination that matches the transformed impedance of the driving source. Here we used a 1:2 turns ratio transformer, which transforms impedances by that ratio squared, thus 50 Ω:200 Ω. This nicely finesses the problems of amplifier input- and output-voltage ranges, noise, and so on. But note that there's no intrinsic protection against ADC overdrive.

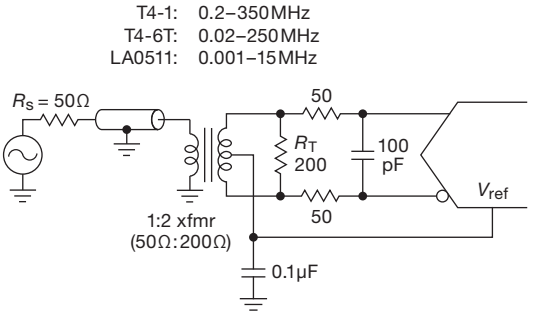


Figure 5.103. Wideband transformers can drive differential-input ADCs. They have excellent CMRR, and are available with frequency spans of 10,000:1.

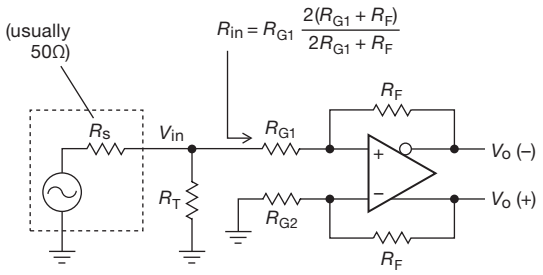
5.17.4 Impedance matching

Differential amplifiers are commonly used for wide-bandwidth applications, in which the (single-ended) input must be properly terminated to match the signal's source impedance (usually 50 Ω). This is particularly important when the signal arrives through a length of transmission line, in order to prevent reflections (see Appendix H). This is not difficult, as long as you keep your wits about you.[108]

Figure 5.104 shows the situation when differential amplifiers of the D type (Figure 5.96) are used. The extra resistor R_T is chosen so that the signal source sees an input impedance equal to R_S (that is, $R_T \parallel R_{in} = R_S$). Note

[107] But sometimes a little bit of noise can be a *good* thing, as it can improve ADC linearity and dynamic range by way of "dithering." See, for example, *The Art of Digital Audio* by John Watkinson (3rd ed., 2001); or Vanderkooy and Lipshitz "Dither in digital audio," *J. Audio Eng. Soc.*, **35**, (12), 966–975, (1987).

[108] And perhaps read the helpful Analog Devices MT-076 "Differential Driver Analysis."



for given R_S, G, and R_{G1},

$R_T \,||\, R_{in} = R_S$ (terminate)

$R_{G2} = R_{G1} + R_S||R_T$ (balance)

$R_F = G\,\frac{R_{G2}\,(R_S + R_T)}{2R_T}$ (gain)

$$\left(G \equiv \frac{V_o(+) - V_o(-)}{V_{in}} = \frac{2R_T}{R_T + R_S}\,\frac{R_F}{R_{G2}}\right)$$

Figure 5.104. Terminated single-ended to differential amplifier: design equations.

especially that the amplifier's noninverting input is not a virtual ground, so R_{in} is somewhat larger than R_{G1} alone, according to the equation in the figure. The differential amplifier has the usual equal-value feedback resistors R_F, but the gain-setting resistors R_G are unequal, to take account of the finite impedance at the drive point (marked V_{in}). That is, R_{G2} must be larger by the parallel resistance $R_S \parallel R_T$. Finally, the feedback resistors R_F must be adjusted upward to bring the gain back to the desired value.

Note that the gain is defined in terms of V_{in}, i.e., with respect to the *loaded* input signal amplitude (*not* the open-circuit source amplitude). This makes good sense, because signal amplitudes (from signal generators, etc.) are normally specified as their properly terminated amplitudes.

As an example, for a 50 Ω source, $G = 2$, and $R_{G1} = 200\,\Omega$, you would find (choosing nearest 1% standard resistor values) $R_T = 60.4\,\Omega$, $R_{G2} = 226\,\Omega$, and $R_F = 412\,\Omega$.

Unlike the situation at high frequencies, it is not necessary to terminate a signal source when operating at low frequencies (e.g., audio). In that case R_T is omitted and R_{G2} is simply $R_{G1} + R_S$. The gain, defined now in terms of the *open-circuit* source signal amplitude, is just $G = R_F/(R_{G2})$.

5.17.5 Differential amplifier selection criteria

All differential amplifiers are not created equal. There are plenty of subtleties lurking, associated with tradeoffs of bandwidth, accuracy, output drive capability, supply voltage, and the like. Here's a summary collection of such considerations, tied closely to the differential amplifiers listed in Table 5.10 on page 375.

A. Supply voltage, RR output capability

The first group in Table 5.10 lists the high-voltage differential amplifiers, with ±12 V to ±15 V supplies (though some can work down to ±5 V). These are usually used with bipolar (split) supplies, but most have the V_{OCM} common-mode output capability to drive single-supply ADCs. This common-mode capability distinguishes the parts on this table from other types. Most parts have internal rail-splitting divider resistors (which require a bypass capacitor) to establish the common-mode output voltage, but this can be overridden by a dc mid-span output provided from the ADC (see Figure 13.28 in §13.6.2). Be sure to check both datasheets – sometimes an op-amp will be required (in the manner of Figure 5.86).

Keep in mind that "V_{out} diff'l max (Vpp)" means $(V_{a+} - V_{b-}) + (V_{b+} - V_{a-})$; that is, twice the peak-to-peak output swing of any one output.

These parts have high differential-output capability, >50 Vpp (each output going ±12.5 V), and higher still for ±18 V supplies, thus well suited for line-driving applications. The differential THAT1606, OP1632, and LME49724, and the single-ended-input DRV134 are all intended for pro-audio (see §13.9.11D). As described before, the differential types can also be used with single-ended inputs. For lowest distortion, all four of these parts should be driven with a low-impedance source, such as an op-amp output.

Next in Table 5.10 (page 375) are the low supply-voltage amplifiers. Most high-frequency, low supply-voltage differential amplifiers are limited to a maximum of ±5 V supplies, or even less. Many cannot be used with total supply voltages greater than 5 V, or even 3.3 V in some cases. Some can run from a single supply as low as +2.7 V or +3.3 V, others need at least +5 V.

Many of the low- to mid-frequency low-voltage parts have rail-to-rail (RRO) outputs, convenient for single-supply ADCs, which don't allow signals beyond their supply rails: simply power the amplifier from the same supply rail as the ADC. But note especially that high-frequency RRO types may suffer degraded high-frequency capability when used near their supply rails. For example the LTC6404, with a featured 600 MHz bandwidth, reveals even at 10 MHz distortion that climbs dramatically when the output approaches within 400 mV of the rails.

An alternative to the use of low-voltage RR outputs to protect ADC inputs is to use an amplifier with output-voltage clamping. That's a nice feature of the LMH6553.

This part is also a CFB amplifier, good for wide bandwidth at high gains, but bad for noise (see the next subsection).

B. Common-mode input range, and the negative rail

Most of the parts have summing junctions that must be operated at least a volt or more above the negative supply rail (the low-power THS4521 is an exception). However, as explained in §§5.17.1G and 5.17.3), that does not necessarily prevent the input *signals* from going down to ground, most particularly when the amplifier is operated in a fully differential configuration at low gains ($G = 1$ or $G = 2$, for example).

Nine of the parts (marked **w** or **v** in Table 5.10's "RRO" column – see page 375) allow use of their summing junctions down to $-V_{EE}$. The datasheets will declare something like "Common-Mode Input Range Includes the Negative Rail" or "NRI" on the front page. In most cases performance is not degraded under this condition, in contrast to the loss of performance when RRO-capable amplifiers are run to the extremes of output swing.[109] This is especially useful when fully differential amplifiers are used as single-ended to differential converters, with the (−) input grounded, as in Figure 5.102B. But be careful – if either summing-junction input is taken further than the specified −0.2 V below $-V_{EE}$, polarity reversal at the output may occur, similar to the situation with the legacy (but still wildly popular) LM324/358 single-supply op-amps.[110]

C. Low Z_{in}

Most of these amplifiers present rather low input impedances to their signal sources, especially when configured for high gains, because the specified R_f gain-setting resistor values are low, and Z_{in} is roughly R_f/G (exceptions: LTC6416, and the EL5170 family). Most parts with higher input impedances are noisier, mostly because of resistor Johnson noise (exception: the AD8352, which uses the F rather than the D configuration).

Signal impedance matching is often a concern, especially at high frequencies, say 30–100 MHz and above, even with short PCB traces. An amplifier's low Z_{in} complicates the problem of matching the impedance of the signal source, and it also affects the amplifier's gain. See §5.17.4 for helpful formulas.

[109] Some parts (for example the THS4008 and THS4511) even specify "$V_{S-} = 0$" and "input referenced to ground" as the operating condition for their specifications.

[110] For the latter you can substitute the improved LT1013/1014, which eschew that nasty habit and provide better performance overall; but no such solution is available for differential amplifiers.

D. Offset voltage, CMRR

Many of the amplifiers in Table 5.10 (see page 375) have poor offset voltage and other dc specifications. Most of these are fixed-gain parts with internal resistors (e.g., with the D3 configuration). They appear to be suffering from offsets arising from the high V_{OCM} common-mode output voltage that's natural with differential amplifiers, combined with modest internal resistor mismatches; for example, a 1% resistor mismatch operating on a V_{OCM} of 1.5 V would produce an effective input offset of 15 mV. By contrast, most "bare" parts (e.g., the D configuration) in the table have attractive low offset-voltage specs. But they will surely develop high offsets if you were to use 1% gain-setting resistors to complete the amplifier. Note also that dc accuracy is degraded by mismatched external resistance when driving D-configuration differential amplifiers.

If we had room for a CMRR column in the table, we'd see a similar bare- versus fixed-gain dichotomy, and for the same reason. Taking two examples from their respective datasheets, the bare ADA4932 has a 100 dB typical CMRR, compared with 64 dB for the similar fixed-gain ADA4950. Likewise for the bare LTC1992 (90 dB) and the fixed-gain LTC1992-10 (60 dB).

In many fully differential applications CMRR isn't important. But if it matters in your design, use 0.1% resistors, or matched resistor arrays. And be sure to take special care with circuit-board wiring capacitances, which really matter at high frequencies: for example, to achieve −80 dB matching at 1 MHz with 500 Ω gain-setting resistors, you've got to match capacitances to a difficult 0.03 pF! And no good deed goes unpunished – the CMRR will degrade by 20 dB for each decade increase in frequency.

A better solution would be to use an E- or F-configuration part. Intersil's EL5170-series parts have good CMRR, e.g., 80 dB at 1 MHz (but poor 25 mV offset), and Analog Device's AD8352 boasts 60 dB at 100 MHz and 6 mV offset. Both are much less affected by unbalanced input resistances than D-configuration types.

E. Fixed gain, external resistor-settable gain

One good reason to select a fixed-gain amplifier is that some of them have better gain accuracy than can be easily and inexpensively achieved with discrete resistors – for example, attractive worst-case gain errors of ±0.04% for the NSC LMP3712, ±0.15% for the TI (Burr–Brown) PGA280 (both programmable-gain parts), and ±0.08% for the Analog Devices pin-programmed AD8270.

The simplicity of fixed-gain types may seem appealing, but some external-resistor types have their attractive aspects, e.g., TI's THS4520 and Analog's ADA4932

draw much lower supply currents than their competitors. The THS4520 can be used to make G=10 amplifiers with 120 MHz bandwidth.

Fixed-gain types are easier to use at high frequencies, because they avoid painful wiring and pin-capacitance problems. But most have poor absolute gain accuracy, ΔG in Table 5.10 (exception: the ADL5561). Most don't allow bandwidth-limiting filter capacitors to be added, and most restrict you to low gain values (exceptions: the PGA870 and LT1993-10, but note their high power consumption).

F. VFB, CFB, f_{3dB}, GBW, and filters

Few of the fixed-gain amplifiers offer special filtering capability. Three LTC offerings are an exception, especially the LT6600-x, with five fixed 4th-order filter frequencies (alliteration!) available from 2.5 MHz to 20 MHz.

All of the adjustable-gain types employ voltage-feedback-mode op-amps, with two exceptions (the LMH6552 and LMH6553). For gains $G \geq 4$ they follow the GBW rule, namely f_{-3dB} =GBW/G. But be aware that the "Bandwidth" value in the table is generally considerably higher (1.5× or more) than the part's GBW, because it is determined at unity gain, where the amplifier benefits from response peaking that extends its −3 dB rolloff frequency. (The manufacturer may want to show off its best-looking values, but the peaking for $G = 1$ may be so severe with some parts that you may not want to use them that way.) You may have to study datasheet response plots, etc., to determine the actual true (and very useful) GBW value. Because these are VFB op-amp types, they are stable with bandwidth-limiting filter capacitors added across the feedback resistors. You can increase the R_f value (which increases the input impedance) and add a small parallel C_f capacitor to bring peaking under control or a larger one to provide a bandwidth filter for your signal.[111]

G. Response peaking, GBW, and 0.1 dB bandwidth

Gain peaking is a primary killer of good "bandwidth-to-0.1 dB" ratings. The 0.1 dB bandwidth numbers may be much improved for higher gains, where low-gain peaking is eliminated. Take, for example, the ALD5561, which boasts a −3 dB bandwidth of 2900 MHz at its minimum gain ($G = 2$), but its −0.1 dB bandwidth is a disappointing 200 MHz (i.e., just 7% of its −3 dB bandwidth). However, at its maximum gain ($G = 6$), where the −3 dB is reduced somewhat (to 1800 MHz), its −0.1 dB bandwidth *improves* to 600 MHz (i.e., 33% of its −3 dB bandwidth). This behavior is displayed nicely in the datasheet's graphs (see Figure 5.105). It is possible that the settling time is also improved (from a lack of ringing), although this isn't specified.

Note that some of the parts in the table are available in dual configurations (noted in the Comments column in Table 5.10), which can be helpful in providing matched time-delay responses, important for many applications.

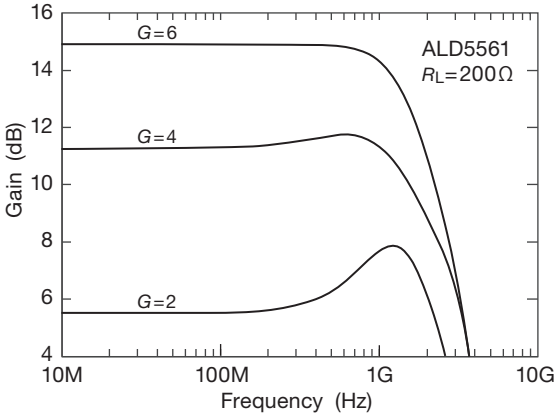


Figure 5.105. Amplifier peaking at low gain settings produces an extended "−3 dB bandwidth," at the expense of flatness of response.

H. Slew rate, settling time, large-signal bandwidth

Just as with some high-speed op-amps, the datasheets indicate a much wider bandwidth for small (∼100 mV) signals than for large (∼2 V) signals. This is a slew-rate issue: amplifier output-swing capabilities are reduced as their slew-rate limits are approached. For example, Analog Devices' low-power ADA4932 has a 2800 V/μs specified slew rate, which implies that 1 V amplitude sinewave outputs are possible to $f = S/2\pi A = 445$ MHz. Indeed the part's datasheet shows a −3 dB response to 560 MHz (or even 1 GHz with smaller R_f) for 100 mV output, but only to 360 MHz for 2 Vpp. Higher slew-rate parts are available,

[111] Taking the example of the LMH6553 and LMH6552 CFB amplifiers, these are specified with R_f=274 Ω and 357 Ω, at which the respective bandwidths are 900 and 1500 MHz, and the slew rates are 2300 and 3800 V/μs. These specs are for G=1, but with CFB op-amps you can dramatically increase their gain without losing too much bandwidth. For example the 1500 MHz LMH6552 claims to have 800 MHz of bandwidth still at G=4. For higher gain you may prefer not to reduce R_i much, but rather to increase R_f. With CFB amplifiers a primary effect of increasing R_f is to reduce the slew rate proportionally; but, hey, you had plenty to begin with! Increasing R_f with CFB does cause an increase in noise.

even to 10 kV/μs (the ALD5561), implying a 2 Vpp capability to 1.5 GHz.[112]

I. Distortion

Two of the high-voltage parts (the OP1632 and LME49724), often used for professional music applications, have their distortion performance plotted in Figure 5.43. We would hope that a fully differential circuit, with differential input signals, might have lower distortion than competing single-ended circuits, at least for the symmetrical 2nd-harmonic. And indeed the differential LME49724 does very well, in the −140 dB territory. However, the single-ended LME49990 and OPA134 op-amps do better on the graph.

Figure 5.106 plots distortion (from manufacturers' datasheets) for a few of the differential amplifiers in Table 5.10 (page 375) for frequencies to 100 MHz. As we cautioned earlier, the conditions for distortion data are not standardized, complicating direct comparisons. Consequently, many of these parts provide multiple plots, taken with different combinations of gain, load resistance, signal amplitude, and supply voltage, and in which 2nd- and 3rd-harmonic distortion curves are shown separately.

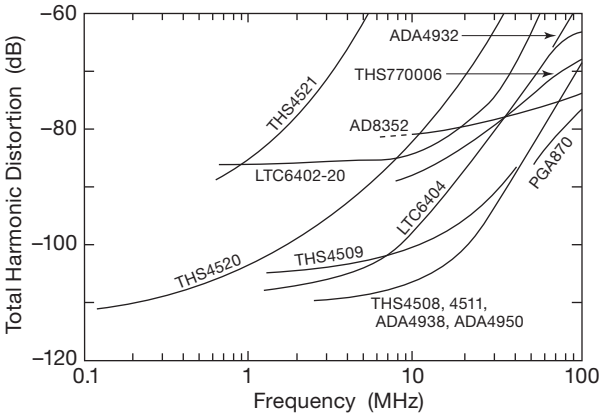


Figure 5.106. Total harmonic distortion (THD) versus frequency, taken from the respective datasheets, for a selection of the differential amplifiers in Table 5.10 on page 375.

Be careful when evaluating distortion (a parameter that we've not listed in the table). For example, the ADA4932 has 560 MHz (or 360 MHz) bandwidths, as discussed earlier, but the luster of its front-page claim to being a low-distortion amplifier (−90 dB at 20 MHz) is tarnished somewhat when you discover that it deteriorates by a factor of 10× by 50 MHz, well below its 360 MHz bandwidth.

As we saw with the op-amp distortion plots in Figures 5.43 and 5.44, there's a strong correlation between speed (high GBW, fast slew rate) and low distortion at high frequencies. This is especially clear in Figure 5.106 above 1 MHz. For example, the 145 MHz THS4521 that we've often mentioned performs poorly even in the under-5 MHz-region, compared with its TI relative, the 1.6 GHz THS4511 (note that both have NRI front ends, i.e., operating common-mode input range to the negative rail).[113] Four of the best parts in this class maintain better than −100 dB distortion out to 20 MHz, compared with just 7 MHz for the best-in-class AD8045 op-amp in Figure 5.44. In other words, at high frequencies (say above 10 MHz) fully differential amplifiers excel in low distortion compared with single-ended op-amps.

As we will see in Chapter 13, 16-bit ADCs are available with sampling rates to 250 Msps (e.g., the AD9467, see Table 13.4), justifying a need for better than 0.01% linearity (−80 dB distortion) at frequencies approaching 100 MHz. Happily, manufacturers of differential amplifiers are rising to this challenge.[114]

J. Noise, high 1/*f* noise corners

We conclude with a few comments about noise. We've tabulated amplifier *voltage* noise, but not *current* noise. But we do list the input bias current I_B, which is roughly predictive of current noise, which must equal or exceed the shot-noise contribution of $i_n=\sqrt{2qI_B}$. Note that CFB amplifiers, with their exceptionally high input currents, have much greater input current noise, generally as much as 10× greater than VFB amplifiers.

Looking at Table 5.10 (page 375), we see many amplifiers with noise densities in the range of 25–45 nV/$\sqrt{\text{Hz}}$. Assuming no increase above say 10 MHz (check the datasheet plots) this corresponds to a wideband input-voltage noise of $V_n=e_n\sqrt{\text{BW}}$, which evaluates to 175–700 μVrms for bandwidths of 50–250 MHz. This is quite a bit larger than the 30 μV LSB step size of a 16-bit ADC with a 2 Vpp full-scale differential input. While some

[112] The AD8351 (not in our table, but similar to the AD8352), offers 3 GHz bandwidth with 13 V/ns slew rate and 2 Vpp capability to nearly 2 GHz.

[113] In fairness, the THS4521 gets by on a miserly 1.1 mA of supply current and has RR outputs, whereas the THS4511 takes 39 mA and lacks RROs.

[114] But are they rising to the challenge posed by NSC's ADC12D1800, a 12-bit 3600 Msps converter? Or the even greater challenge posed by the even-speedier ADCs that surely will be available by the time this book's ink has dried? Such ADCs will likely have to be driven with transformers.

dither is a good thing, it's clear that even at $G = 1$ these amplifiers are too noisy for some applications.

There are many other amplifiers in the table with e_n specs down in the 1.1–5 $\text{nV}/\sqrt{\text{Hz}}$ range. But these are the bare amplifier summing-junction noise specs, without the necessary feedback resistors taken into account. Amplifier gains of 5 or 10 are generally assumed, not the least to overcome the amplifier's own output-stage noise. Many of the amplifiers specify R_f values of 350–500 Ω. For $G = 1$ the input resistor R_g would have the same value, and its Johnson noise of 2.4–2.8 $\text{nV}/\sqrt{\text{Hz}}$ would dominate the intrinsic noise of the quieter amplifiers. However, for $G = 10$ the 35–50 Ω resistor noise is under 1 $\text{nV}/\sqrt{\text{Hz}}$, which would not badly degrade the completed amplifier's noise.

Finally, many amplifiers have good looking noise specs, but we must warn you to examine the noise-versus-frequency curves on their datasheets; many have very high $1/f$ noise corners. This is especially true for current noise, with some $1/f$ corners way up at 1 MHz or beyond. An example of a troublesome spec might be the THS4508, with its *pnp* input transistors (for operation to GND) and 4.7 $\text{pA}/\sqrt{\text{Hz}}$ of current noise at 1 MHz. This creates 1.6 $\text{nV}/\sqrt{\text{Hz}}$ across a 349 Ω resistor, which is OK compared with the part's $e_n = 2.3\ \text{nV}/\sqrt{\text{Hz}}$. But if you were to use 1k resistors, the corresponding current-induced noise voltage would be 4.7 $\text{nV}/\sqrt{\text{Hz}}$, dominating the amplifier's e_n. Depending on the application, this might be of concern. Or it might not.

Review of Chapter 5

An A-to-M summary of what we have learned in Chapter 5. This summary reviews basic principles and facts in Chapter 5, but it does not cover application circuit diagrams and practical engineering advice presented there.

¶A. Precision and Dynamic Range.

A *precision* circuit is one that exhibits (through careful design and choice of components) both initial *accuracy*, and also *stability* (i.e., maintenance of accuracy over time and temperature). A precision circuit may (but need not) exhibit wide *dynamic range* (the ratio of signal amplitudes over which it operates); see §5.1.

¶B. Error Budget.

When designing a precision circuit, you need to keep track of numerous error contributions (from voltage offsets, current-induced offsets, component tolerances, and the like); this is best tallied in an *error budget*, which promotes design discipline, and which assists in helping you spot the dominant error sources; see the example in §5.5.

¶C. Strict versus Pragmatic.

Strict worst-case design mandates that all components be operated within their datasheet specifications, and that the effects of all their worst-case errors be added (as unsigned magnitudes) to determine the circuit's performance. The benefit of such conservatism is a circuit guaranteed to perform within specifications (assuming proper design); the downside is that it may be impossible to achieve design goals with available and/or affordable components, taking their worst-case specifications in a worst-case arithmetical combination (and noting that some critical performance parameters may be unspecified, or show only "typical" values). We favor a pragmatic approach (§5.3), taking with a grain of salt some of the published worst-case parameters (for example, input leakage current, where testing limits encourage highly conservative worst-case specs), or adopting reasonable estimates of unspecified parameters. It may be sufficient to establish that the circuit effects of the unspecified parameter are completely insignificant; or, if it's a closer call, you may have to set up a testing regime of incoming components to ensure that you meet specs. If have a situation in which there are many components contributing to an overfull error budget, you may have to validate the performance of the overall circuit, subassembly, or complete instrument at final test.

¶D. Component Errors – Resistors.

Taking a simple example, resistor accuracy (including the effects of tempco) sets a limit on the accuracy of an amplifier's gain. But it's not that simple, because amplifier gain usually depends on a resistance *ratio*; so the situation is greatly improved if you use a resistor pair with accurate ratio and matched tempco, and it's the *ratio* mismatch and the tempco of the *ratio* that are limiting. Likewise, the CMRR limit of a difference amplifier depends on resistor-pair matching, $\mathrm{CMRR(dB)} \approx 20\log(100/p)$, where p is the ratio mismatch in percent. To give scale to these statements, typical metal-film resistors come in accuracy ranges of 0.05%–1%, with a typical tempco spec of 25–100 ppm/°C; resistor arrays intended for high precision have accuracies in the range of 0.01%–0.05%, tempcos down to 1 ppm/°C or better, and matching accuracies and tempcos down to 0.01%–0.001% and 1–0.1 ppm/°C, respectively. Real resistors depart from the ideal also in *linearity*, i.e., they exhibit some resistance change with applied voltage. See §5.6.1; see also §*1x.2* for further discussion of resistors and their vicissitudes (e.g., long-term drift, and parasitic inductance and capacitance).

¶E. Component Errors – Capacitors and Switches.

Capacitors have several amusing traits that affect the accuracy of integrators and of sample-and-hold circuits; these include resistive leakage (thus exponential decay), and more seriously *dielectric absorption* (memory effect, see §5.6.2, §*1x.3*, and the dielectric-absorption plots in that section.) These application circuits include analog switches (for integrator reset, and sample-to-hold switching), which introduce errors through leakage and charge injection (§§3.4.2E and 5.6.3).

¶F. Amplifier Input Errors.

This is where most of your troubles are located. The primary qualification for membership in the category of *precision op-amps* is a small offset voltage V_{os}, and a correspondingly low tempco $\mathrm{TC}V_{OS}$ (sometimes called ΔV_{OS}). The offset voltage operates at the input, so the RTI (referred to the input) error is simply V_{OS}; at the output of an amplifier of voltage gain G_V the V_{OS}-induced error is $\Delta V_{out} = G_V V_{OS}$. In an integrator circuit with input resistor R_{in}, the input offset voltage is equivalent to an input current error of $\Delta I_{in} = V_{OS}/R_{in}$. To give an idea of scale, a typical precision op-amp has an offset voltage of 10–50 μV, and a tempco of 0.1–0.5 μV/°C. *Auto-zero* op-amps (see ¶I) improve on these figures by about a factor of ten; see Table 5.5 on pages 320–321 and Table 5.6 on 335. See also §5.10.5.

But there's more. Input bias current I_B flowing through

the source resistance R_S seen at the amplifier's input produces an RTI voltage error of $\Delta V_{in} = I_B R_S$. For precision *bipolar* op-amps, whose bias currents are of order 10 nA, this becomes serious for source resistances greater than a few kilohms (where $I_B R_S$ amounts to a few tens of microvolts, in the same ballpark as a precision op-amp's V_{OS}). High-R_S situations thus mandate low-bias op-amps, usually those with JFET or MOSFET inputs, or (for moderate source resistance) a bipolar precision op-amp with bias-cancellation (where I_B is of order 1 nA). *A warning*: the very low bias current of FET-input op-amps rises dramatically at higher temperatures (see Figure 5.6), where its input current may even exceed that of a bias-cancelled bipolar op-amp. See also §5.10.7.

Looking deeper, additional error sources at op-amp inputs include I_B variation with common-mode input voltage V_{CM} (§5.7.2, Figure 5.7), V_{OS} variation with V_{CM} (i.e., CMRR, §5.7.3, §5.10.9, and Figures 5.10, 5.29, and 5.30), PSRR, and input noise (e_n, i_n, §§5.10.6 and 5.10.8).

¶G. Amplifier Output Errors.

While much of precision analog design concerns dc and low frequencies, some applications require accuracy at higher speeds: audio and video, communications, scientific measurement, and so on. With falling op-amp loop gain, input errors are rising, output impedance is rising, and slew-rate limitations may come into play, along with reduced suppression of output-stage crossover distortion, gain nonlinearity, and phase error. And overshoot, ringing, and settling time are critical in dynamic applications.

These can be made quantitative. The gain error $\delta_G \equiv (G_{ideal} - G_{actual})/G_{ideal} = 1/(1 + AB)$, where B is the feedback fraction around open-loop gain A; see §5.8.5. The closed-loop bandwidth is $f_{3dB} = f_T/G_{CL}$, corresponding to a time constant of approximately $\tau \approx G_{CL}/2\pi f_T$, which (if well compensated, with good phase margin) produces a settling time of order 5–10τ; see §§5.8.2 and 5.10.10. An op-amp's *actual* settling time can be considerably longer, and there's no substitute for real data; see Table 5.5 on pages 320–321. A well-compensated op-amp exhibits a closed-loop phase error of $\phi \approx f/f_C$, where $f_C = f_T/G_{CL}$ is the frequency at which the closed-loop gain has fallen to unity.[115] Op-amp distortion depends strongly on the output-stage circuit; see the extensive plots in Figures 5.43 and 5.44.

[115] Throughout the book we use f_T as shorthand for the proper term *gain–bandwidth product* (GBP, GBW, or GBWP), which is the extrapolated unity-gain crossover frequency.

¶H. Rail-to-rail Op-amps.

It's tempting to choose op-amps with rail-to-rail common-mode input range (RRI), rail-to-rail output (RRO), or both (RRIO), especially for operation at low supply voltages.

RRI op-amps. But there are drawbacks associated with the duplicate complementary input stages that can seriously compromise precision, notably an abrupt change of both I_B and V_{OS} at the input crossover voltage (§5.9.1). Some RRIO op-amps (e.g., the OPA360-series) circumvent this problem by using an on-chip charge pump. If you don't need full rail-to-rail input, choose an op-amp whose input common-mode range extends only to one rail (usually the negative rail).

RRO op-amps. Op-amps that feature rail-to-rail output have their own issues, stemming from the use of a common-source (or common-emitter) output-stage topology, rather than the conventional source follower (or emitter follower). These include increased distortion, and higher output impedance (thus gain and phase shift are more susceptible to load impedance). However, many RRO op-amps mitigate this latter problem by using internal capacitive feedback to lower the open-loop output impedance at high frequencies, as seen for example in Figure 5.34.

¶I. Auto-zero Op-amps.

This subclass of op-amps includes on-chip offset-nulling circuitry that operates cyclically (with an on-chip oscillator) to trim the input offset voltage (§5.11). As the entries in Table 5.6 on page 335 demonstrate, this produces typical V_{OS} values around a microvolt, with tempcos around 10 nV/°C, roughly an order of magnitude better than the best conventional precision op-amps. Notable, too, is the absence of a $1/f$ low-frequency rising noise voltage (see for example Figures 5.53 and 5.54).

That's the good news. The bad news is that the switching action produces noise-spectra peaks at the switching frequency and its harmonics (Figure 5.51), superposed on a low-frequency noise voltage floor that is already considerably higher than that of conventional low-noise op-amps (compare Table 5.6 on page 335 with Table 5.5 on pages 320–321). The input switching circuitry also results in relatively high input noise *current*, compared with low-noise precision JFET-input op-amps.

Auto-zero op-amps are built with CMOS, and (except for some legacy parts) are generally limited to low supply voltages ($\leq$7 V total supply). A shining exception is the recent LTC2057HV (4.75 V to 60 V total supply!). Another caution: because of their internal voltage-storage

correction capacitors, auto-zero op-amps can exhibit slow recovery from saturation, as long as several milliseconds.

¶J. Difference, Differential, and Instrumentation Amplifiers: Taxonomy.

These share the property of accurate and stable differential gain, with high common-mode rejection: $V_{out}=G_V\Delta V_{in}=G_V(V_{in+}-V_{in-})$. In common usage, the terms are distinguished as follows.

Difference amplifier. Differential-in, single-ended out; op-amp plus two matched resistor pairs (Figure 5.65, §5.14); CMRR 90–100 dB; accurate but low gain (G_V=0.1–10); input impedance 25–100k, intended to be driven by a low impedance; inputs typically can go beyond rails.

Instrumentation amplifier. Differential-in, single-ended out; very high input impedance (10MΩ to 10GΩ), wide gain range (G_V=1–1000), and very high CMRR at higher gains (110–140 dB at $G_V = 100$); see §5.15 and Figure 5.77.

Fully differential amplifier. Differential or single-ended in, differential-out; most are low voltage, fast-settling, and wideband; ideal for twisted-pair cable drivers and fast differential-input ADCs; see §5.17.

¶K. Difference Amplifiers.

The classic *difference amplifier*, typified by the original INA105 (see Table 5.7 on page 353, §5.14), consists of an op-amp plus a pair of matched resistor dividers, with SENSE and REF inputs (at the bottom of the divider strings) in addition to the V_{in+} and V_{in-} signal inputs. With REF grounded and SENSE tied to the output, the gain is $G_{diff}\equiv V_{out}/\Delta V_{in}=R_f/R_i$, where R_f and R_i are the feedback and input resistors, respectively. Figure 5.66 shows circuit variations that exploit alternative connections of the SENSE and REF pins.

Difference amplifiers are simple, and good enough for many applications. Some permit common-mode inputs well beyond the rails (e.g., ±200 V for the INA117). But difference amplifiers suffer from relatively low input impedance (tens of kΩ), limited gain (typically in the range of G=1–10), relatively high noise voltage, unimpressive CMRR (typically $\lesssim$90 dB), and degradation of both gain and CMRR when driven with signals of non-zero source impedance.

¶L. Instrumentation Amplifiers.

These drawbacks are nicely remedied in the *instrumentation amplifier* configuration of three op-amps (Figure 5.77, §5.15). The input impedance is high (10 M–1 TΩ), and the gain (which can be as large as ×1000) is set with a single external resistor (or pin-selectable choice of internal resistors). When configured for high gain, most instrumentation amplifiers deliver a typical CMRR around 120 dB, and an input noise voltage e_n less than 10 nV/$\sqrt{\text{Hz}}$; they do not permit common-mode outputs beyond the rails. Instrumentation amplifiers are available in BJT, JFET, MOSFET, auto-zero, and programmable-gain types; see Tables 5.8 on page 363 and 5.9 on pages 370–371.

¶M. Fully Differential Amplifiers.

Unlike difference amplifiers and instrumentation amplifiers, fully differential amplifiers (§5.17) generate a balanced *differential output* centered on a settable common-mode voltage. This is useful for driving fast ADCs with complementary inputs, balanced transmission lines, and RF communication circuits. Befitting these applications, they tend to be wideband (up to 1 GHz or more), high slew-rate (1000 V/μs or more), fast settling (a few nanoseconds), and quiet (5 nV/$\sqrt{\text{Hz}}$ or less). There are many internal circuit topologies (labeled **A**–**G** in §5.17); see Table 5.10 for a representative selection.